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$\mathcal{R}(J/\psi)$ muonic Run-2 ANA

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Abstract

This analysis note details the measurement of the ratio of branching fractions between the decays $B_c^+ \to J\psi\tau^+\nu_\tau$ and $B_c^+ \to J\psi\mu^+\nu_\mu$ in the combined 2016, 2017, and 2018 datasets. A brief overview of necessary theoretical background is given, followed by a documentation of the methods and strategies used in the analysis. Results are discussed along with estimated uncertainties.

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Update log

v8 → v9

- Added descriptions of additional selections in 20, updated table in 5.
- Added subsection 8.2.1, further studies in 8.3, 8.5, and 8.6.2.
- Added subsection 10.1.
- Updated plots to reflect new selections in 15, 16.
- Updated 17 to reflect changes in systematic uncertainties.
- 28 now lives in the appendix, with an added section explaining why the lifetime acceptance correction is discontinued.
- Updated the misID background in subsection 29.1.
- Updated the double charm background in subsection 29.2.
- Added the feeddown background in subsection 29.3.
- Updated subsection 29.4.
- Added a new subsection on the study of the VELO error issue (subsection 29.5).
- Added a new subsection on B_c^+ lifetime correction (subsection 29.6), serving as a supplement to 28.

v7 → v8

- Added upper bound on Ξ_{bc}^+ contribution in 13.3.
- Added description of angular reweighting for charm combinatorial.

v6 → v7

- Added description of the mis-ID decay-in-flight efficiency correction (Section 8.6.2), along with a corresponding systematic (Section 17.8).
- Added further postfit validation using the $J/\psi K$ peak in Section 8.5.
- Fixed bug in efficiency ratio table.
- Fixed form factor efficiencies.
- Updated fit results and postfit plots.
- Updates systematic uncertainties to reflect final fit configuration and efficiency ratio.
- Updated toy studies plot for final fit.
- Updated systematic uncertainty fits in Appendix.

- Updated Appendix 29 event mixing method and misID background.
- Updated systematics table.
- Updated description of background modeling in mis-ID efficiency correction fit.
- Updated systematics with unfolding cross check.
- Changed correlation matrix visualization.

v5 $\rightarrow$ v6

- Reordered some subsections and moved some things to appendix.
- Added check on charm combinatorial systematic 17.7.
- Revised Z binning systematic strategy **??**.
- Revised $\mu\mu$ combinatorial systematic estimate **??**.
- Added description of double charm constraint 12.
- Small edits to language in response to RC first round comments.
- Improved the misID background section: we now provide a full description of the misID background 29.1.
- The double charm section remains mostly unchanged. We added clarification on how these yields are used in the normalization fit, especially the fixed parameters 29.2.
- Added a new section describing the event-mixing method used to estimate the combinatorial background from the $J/\psi + \mu$ combinations **??**.
- Added a new section describing combinatorial $(\mu^+ + \mu^-)\mu^+$ background **??**.
- Revised the title and introduction of Appendix 29(Appendix with different event-selection by UCAS) (comment now) to reflect its content more accurately.
- To do list
  - Continue waiting for the DiF effect on the misID rate.
  - The smearing sample for the misID background needs to be updated to include events selected with the muBDT.
  - The event-mixing method needs to be optimized by implementing the RC's suggestion: the bachelor muon in the mixed event should be rotated so that the three-momentum of the B_c^+ in the two events are aligned, making the mixed event more representative of a combinatorial background.

v4 → v5

- UCAS group, Added cross check for misID background 29.1 after aligned selections. To do list, waiting for DiF effect on misID rate.
- Added estimate of combinatorial charm yield from multiple candidates data sample.

v3 → v4

- Changed the nominal fit to include the combinatorial charm template.
- Added a systematic for the combinatorial charm modeling (Section 17.7).
- Corrected systematic uncertainties table which was computed with out-of-date efficiencies (Table 24).

v2 → v3

- Added Appendix 25 describing investigations of charm combinatorial background.
- Added studies on the potential background $X(3872)$ in Section **??**.
- Added a small section documenting the study following the $B \to \mu^+\mu^-\mu^+\nu_\mu$ analysis's treatment of the trimuon effect (Section 26).
- UCAS group, Added cross check for double charm background **??** after aligned selections.

1 Introduction

This study is carried out to measure the ratio of branching fractions

$$R(J/\psi) = \frac{Br(B_c^+ \to J/\psi\, \tau^+\, \nu)}{Br(B_c^+ \to J/\psi\, \mu^+\, \nu)}, \tag{1}$$

as a test of lepton flavor universality (LFU), which is a fundamental feature of the Standard Model. LFU is the principle that the electroweak couplings of leptons to gauge bosons are identical, with differences only in their couplings to the Higgs sector. Therefore, LFU predicts that any differences in branching fractions involving electrons, muons, and taus are due to explicitly mass-dependent effects, such as phase space and helicity supression. Any deviation from LFU would be unambiguous evidence for new physics.

The LHCb collaboration has measured $R(J/\psi)$ to be $0.71 \pm 0.17(stat) \pm 0.18(syst)$ with the LHCb Run-1 dataset collected in 2011 and 2012 [1]. The measured value is in excess of the Standard Model predictions [2–4] by approximately 2 standard deviations. The $B_c \to J/\psi \ell \nu$ decays are an example of one of the charged-current $b \to c\ell\nu$ transitions. Additionally, these ratios allow the cancellation of uncertainties dependent on $|V_{cb}|^2$. Another quantity,

$$\mathcal{R}(D^{(*)}) = \frac{\mathcal{B}(B^0 \to D^{(*)-}\tau^+\overline{\nu}_\tau)}{\mathcal{B}(B^0 \to D^{(*)-}\mu^+\overline{\nu}_\mu)}, \tag{2}$$

has been measured at BaBar, Belle, and LHCb. HFLAV reports that the combined measurement of $R(D)$ and $R(D^*)$ is 3.17σ away from the Standard Model prediction [5]. Measuring $R(J/\psi)$ allows us to investigate the same underlying partonic transition and indirectly probe any potential beyond the Standard Model physics.

The LHCb experiment has obtained the first measurement of $R(J/\psi)$ using data from Run I, by reconstructing tau with $\tau^+ \to \mu^+\nu_\mu\overline{\nu}_\tau$ decays [6]:

$$R(J/\psi) = 0.71 \pm 0.17(stat) \pm 0.18(syst)\ . \tag{3}$$

The most recent result for $R(J/\psi)$ in the muonic decay channel was reported by the CMS collaboration [7], with the measured value:

$$R(J/\psi) = 0.17^{+0.18}_{-0.17}\ (\text{stat})^{+0.21}_{-0.22}\ (\text{syst.})^{+0.19}_{-0.18}\ (\text{theo.}) = 0.17 \pm 0.33\ . \tag{4}$$

For the hadronic decay channel, the latest $R(J/\psi)$ measurement from the CMS collaboration [8] is:

$$R(J/\psi) = 1.04^{+0.50}_{-0.44}. \tag{5}$$

By combining the two CMS results, the overall measurement of $R(J/\psi)$ is obtained as:

$$R(J/\psi) = 0.49 \pm 0.09\ (\text{stat}) \pm 0.25\ (\text{syst})\ . \tag{6}$$

This result is consistent with the Standard Model prediction, 0.258 ± 0.004.

Before the publication of these experimental results, several theoretical calculations of $R(J/\psi)$ were performed. A summary of these results can be found in Figure 2 of reference [9], and these values generally fall within the range of 0.25 ± 0.05. After the publication of the experimental results, various theories based on the Standard Model have been used to recalculate $R(J/\psi)$, and the corresponding results are listed in Table 1. The average value and error of these results are determined to be 0.25 ± 0.02.

Table 1: Theoretical values of $R(J/\psi)$ based on the Standard Model.

Theory	$R(J/\psi)$ value
Quark model	0.24 ± 0.07 [9] [10]
Phenomenological	0.26 ± 0.01 [11]
QCD sum rule	0.23 ± 0.01 [12]
Perturbative QCD	0.27 ± 0.01 [13]
Salpeter equation	0.27 ± 0.01 [14]
Lattice QCD	0.258 ± 0.004 [15] [16]
Schwinger function	0.24 ± 0.05 [17]

2 Theoretical background

Some necessary theoretical background for this study is briefly reviewed, including semileptonic B_c^+ decays, the $B_c^+ \to c\bar{c}$ form factors, and the rest frame approximation.

The B_c is too heavy to be produced at the e^+e^- B factories, and thus the only measurements of $R(J/\psi)$ have been at LHCb and CMS. Therefore, a new measurement of $R(J/\psi)$ with the LHCb Run-2 dataset collected from 2016 to 2018 will be an important update to the tests of LFU.

2.1 Semileptonic B_c^+ decays

The B_c^+ meson is a bound state of an anti-beauty quark ($\bar{b}$) and a charm quark (c). It has a relatively large mass of 6.274 GeV, compared with lighter B mesons such as the B^+ meson (a bound state of $\bar{b}$ and u) which has a mass of 5.279 GeV. The B_c^+ meson's mean lifetime is 0.510 ps, which is shorter than those of lighter B mesons (such as B^+ with a 1.638 ps mean lifetime).

The SM semileptonic B_c^+ decays at the tree level are described by the Feynman diagram in Figure 1. The $\bar{b}$ quark decays via the weak interaction into a $\bar{c}$ quark and a virtual W^+ boson, which in turn decays into a charged lepton and a neutrino. The $\bar{c}$ quark and the spectator c quark form a $c\bar{c}$ bound state, such as the J/ψ meson. In beyond-the-SM (BSM) theories with a charged Higgs boson or a leptoquark, semileptonic B_c^+ decays can happen as described by the Feynman diagrams in Figure 2.

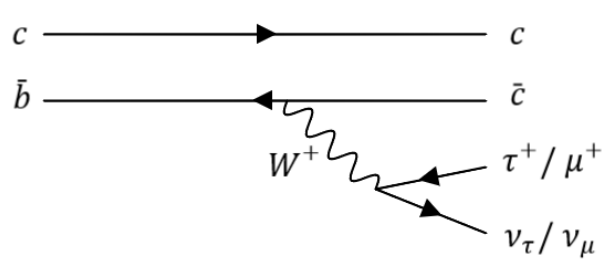


Figure 1: Feynman diagram for semileptonic B_c^+ decays in the SM.

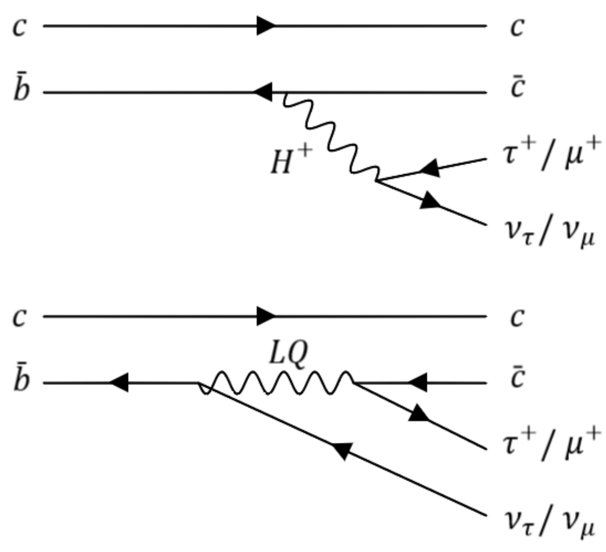


Figure 2: Feynman diagrams for semileptonic B_c^+ decays in BSM theories with a charged Higgs boson or a leptoquark.

2.2 Semileptonic form factors

For some general semileptonic decay of a particle with mass M into $X\ell\nu$, the matrix element can be written as a sum over helicity states. In the case of a $b \to c$ transition, the matrix element can be written as

$$\mathcal{M}_{\lambda_X}^{\lambda_\ell} = \frac{1}{\sqrt{2}} G_F V_{cb} \sum_{\lambda_W} \eta_{\lambda_W} L_{\lambda_W}^{\lambda_\ell} H_{\lambda_W}^{\lambda_X} \tag{7}$$

where the leptonic and hadronic tensors respectively are

$$L_{\lambda_W}^{\lambda_\ell} = \epsilon_\mu(\lambda_W) \langle \ell\overline{\nu} | \overline{\ell}\gamma^\mu(1-\gamma^5)\nu | 0 \rangle \tag{8}$$

$$H_{\lambda_W}^{\lambda_X} = \epsilon_\mu^*(\lambda_W) \langle X | \overline{c}\gamma^\mu(1-\gamma^5) b | B \rangle, \tag{9}$$

and η_{λ_W} is 1 for the ±1, 0 helicity states and -1 for the s helicity state.

For $b \to c$ decays, because of the weakness of the electroweak force, the tree level diagram is a good approximation. This allows us to describe the leptonic tensor by the lepton mass, the square of the momentum transfer to the lepton system q^2, and the angle $\cos\theta$ between the lepton and leptonic flight direction in the leptonic rest frame.

On the other hand, the initial and final states of the b and c mesons cannot be described perturbatively due to the role played by QCD. Due to conservation of azimuthal spin, $H_{\lambda_W}^{\lambda_X}$ is diagonal. With a spin-0 daughter X which only has the helicity state 0, H_0^0 and H_s^0 are non-zero. For a spin-1 X which has the helicity states ±1, we also have nonzero H_+^+ and H_-^-. A spin-2 X has ±2 helicity states, but these are not accessible from any of the W helicity states. We thus are able to unambiguously denote these helicity states as $H_0^0 \equiv H_0$, $H_s^0 \equiv H_s$, and $H_\pm^\pm \equiv H_\pm$. The tensor form of $H_{\lambda_W}^{\lambda_X}$ is constrained by various symmetries to be the linear combination of several terms, the form of which depend on the spin and parity of the daughter particle X. The scalar coefficients of each of these terms is called a form factor.

As a function of q^2 and the helicity angle θ, the differential decay rate can be written as

$$\begin{aligned}
\frac{d\Gamma}{dq^2 d\cos\theta} =& \frac{\left|\vec{p}_X\right|(q^2)}{256\pi^3 M^2}|\mathcal{M}|^2 && (10)\\
=& \frac{G_F^2\left|V_{cb}\right|^2\left|\vec{p}_X\right|(q^2)\,q^2}{256\pi^3 M^2}\left(1-\frac{m_\ell}{q^2}\right)^2 && (11)\\
&\times\Big[(1-\cos\theta)^2 H_+^2 + (1+\cos\theta)^2 H_-^2 + 2\sin^2\theta H_0^2 && (12)\\
&+\frac{m_\ell^2}{q^2}\left(\sin^2\theta\left(H_+^2+H_-^2\right)+2\left(H_s - H_0\cos\theta\right)^2\right)\Big], && (13)
\end{aligned}$$

where $|\vec{p}_X|(q^2)$ is the momentum of X in the B rest frame. Using conservation of energy, we can also write

$$|\vec{p}_X|(q^2) = \sqrt{(\frac{M^2+M_X^2-q^2}{2M})^2 - M_X^2}. \tag{14}$$

Using this and integrating the decay rate over $\cos\theta$, we have the differential decay rate as a function of q^2:

$$\frac{d\Gamma}{dq^2} = \frac{G_F^2\left\|V_{cb}\right\|^2\left\|\vec{p_X}(q^2)\right\|q^2}{96\pi^3 M^2}\left(1-\frac{m_\ell^2}{q^2}\right)^2\left[(1+\frac{m_\ell^2}{2q^2})[H_+^2+H_-^2+H_0^2]+\frac{3m_\ell^2}{2q^2}H_s^2\right]. \tag{15}$$

We proceed to briefly review the tensor forms and associated form factors for different daughter particle spins and parities. In the following, denote the momentum of the B as p^μ, the momentum of the daughter X as k^μ, $P = p+k$, and $Q = p-k$. A summary of the models used in EvtGen to generate the $B_c \to J/\psi$ MC is followed by a description of the form factor parameterization used in the calculation of the $B_c \to J/\psi$ form factors. The LQCD calculation strategy is summarized, and the models used to generate MC for $B_c \to \psi(2S)$ and $B_c \to \chi_{c1,2}$ backgrounds are described.

2.2.1 Scalar X

The matrix element of a scalar X has the tensor form

$$\langle X|\bar{c}\gamma^\mu(1-\gamma_5)b|B\rangle = A(q^2)P^\mu + B(q^2)Q^\mu. \tag{16}$$

Writing the helicity components in terms of the form factors gives

$$H_0 = \frac{2M|\vec{p}_X|(q^2)}{\sqrt{q^2}}A(q^2) \tag{17}$$

$$H_s = \frac{M^2-M_X^2}{\sqrt{q^2}}A(q^2) + \sqrt{q^2}B(q^2). \tag{18}$$

We can redefine the form factors as

$$\langle X|\bar{c}\gamma^{\mu}(1-\gamma_5)b|B\rangle = (P^{\mu} - \frac{M^2-M_X^2}{q^2}Q_{\mu})f_+(q^2) + (\frac{M^2-M_X^2}{q^2}Q_{\mu})f_0(q^2) \qquad (19)$$

in order to write H_0 and H_s as proportional to $f_+(q^2)$ and $f_0(q^2)$:

$$H_0 = \frac{2M|\vec{p}_X|(q^2)}{\sqrt{q^2}}f_+(q^2) \qquad (20)$$

$$H_s = \frac{M^2-M_X^2}{\sqrt{q^2}}f_0(q^2). \qquad (21)$$

This convention is used in `EvtGen`.

2.2.2 Vector X

In the case where the J/ψ is a vector or pseudovector, the tensor form of the matrix element must be

$$\begin{aligned}\langle X|\bar{c}\gamma^{\mu}(1-\gamma_5)b|B\rangle &= iA(q^2)\varepsilon^{\mu\nu\rho\sigma}\epsilon^*(\lambda_X)_{\nu}P_{\rho}Q_{\sigma} \\ &- B(q^2)\epsilon^*(\lambda_X)^{\mu} \\ &- C(q^2)(\epsilon^*(\lambda_X)\cdot Q)P^{\mu} \\ &- D(q^2)(\epsilon^*(\lambda_X)\cdot Q)Q^{\mu}.\end{aligned} \qquad (22)$$

A belongs to the vector current, and B, C, D belong to the axial vector current for vector X, and vice versa for pseudovector X. These form factors can be reparameterized as

$$\begin{aligned}\langle X\left|\bar{c}\gamma^{\mu}\left(1-\gamma_5\right)b\right|B\rangle &= \frac{2iV\left(q^2\right)}{M+M_X}\varepsilon^{\mu\nu\rho\sigma}\epsilon^*_{\nu}p_{\rho}k_{\sigma} \\ &- 2M_XA_0\left(q^2\right)\frac{\epsilon^*\cdot q}{q^2}Q^{\mu} \\ &- \left(M+M_X\right)A_1\left(q^2\right)\left(\epsilon^{*\mu} - \frac{\epsilon^*\cdot Q}{q^2}Q^{\mu}\right) \\ &+ \frac{1}{M+M_X}A_2\left(q^2\right)\left(\epsilon^*\cdot Q\right)\left(P^{\mu} - \frac{M^2-M_X^2}{q^2}Q^{\mu}\right)\end{aligned} \qquad (23)$$

which is also the parameterization used by `EvtGen`. This is the standard form factor parameterization, essentially a rescaled version of the A, B, C, D parameterization used for scalar form factors. In terms of these form factors, the hadronic tensor helicity components are

$$
\begin{aligned}
H_{\pm}\left(q^2\right) &= \left(M+M_X\right) A_1\left(q^2\right) \mp \frac{2MM}{M+M_X}\left|\vec{p}_X\right|\left(q^2\right) V\left(q^2\right) \\
H_0\left(q^2\right) &= -\frac{1}{2M_X\sqrt{q^2}}\left[\frac{4M^2\left|\vec{p}_X\right|\left(q^2\right)^2}{M+M_X} A_2\left(q^2\right) - \left(M+M_X\right)\left(M^2-M_X^2-q^2\right) A_1\left(q^2\right)\right] \\
H_s\left(q^2\right) &= \frac{2M\left|\vec{p}_X\right|\left(q^2\right)}{\sqrt{q^2}} A_0\left(q^2\right).
\end{aligned} \tag{24}
$$

2.2.3 Tensor X

For a tensor X, `EvtGen` uses a parameterization similar to the A, B, C, D parameterization, defined by

$$
\begin{aligned}
\langle X|\bar{c}\gamma^\mu(1-\gamma_5)b|B\rangle = {} & ih(q^2)\epsilon^{\mu\nu\rho\sigma}\epsilon^*_{\nu\alpha}P^\alpha P_\rho Q_\sigma \\
& - k(q^2)\epsilon^{*\mu\nu}P_\nu \\
& - b_+(q^2)\epsilon^*_{\alpha\beta}P^\alpha P^\beta P^\mu \\
& - b_-(q^2)\epsilon^*_{\alpha\beta}P^\alpha P^\beta Q^\mu.
\end{aligned} \tag{25}
$$

The relationship between $\{V, A_i\}$ and $\{A, B, C, D\}$ is

$$
\begin{aligned}
V(q^2) &= -M(M+M_X)h(q^2) \\
A_2(q^2) &= -M(M+M_X)b_+(q^2) \\
A_1(q^2) &= \frac{M}{M+M_X}k(q^2) \\
A_0(q^2) &= \frac{M}{2M_X}[q^2 b_-(q^2) + k(q^2) + (M^2-M_X^2)b_+(q^2)].
\end{aligned} \tag{26}
$$

2.2.4 $B_c^+ \to c\bar{c}$ form factors

Form factors can be estimated using experimental data, for example in the case of $B \to D^*$ decays. However, the $B_c^+ \to c\bar{c}$ form factors have been calculated using theoretical models, due to the lack of sufficient experimental input.

2.2.5 BGL parameterization

`Hammer`, the software used to reweight our MCC according to updated LQCD results, uses the BGL parameterization, which writes form factors in the helicity basis. The BGL parameterization is related to our previous parameterization by

$$
\begin{aligned}
g =& \frac{2}{M_{B_c}+M_{J/\psi}} V \\
f =& \left(M_{B_c}+M_{J/\psi}\right) A_1 \\
F_1 =& \frac{M_{B_c}+M_{J/\psi}}{M_{J/\psi}}\left[-\frac{2M_{B_c}^2\,|\overrightarrow{p}'|^2}{\left(M_{B_c}+M_{J/\psi}\right)^2}A_2-\frac{1}{2}\left(t-M_{B_c}^2+M_{J/\psi}^2\right)A_1\right] \\
F_2 =& 2A_0.
\end{aligned}
\tag{27}
$$

The LQCD results in BGL parametrization with z-expansion truncated at $n = 3$ are given by [18]. They are used to reweight the aforementioned simulations of $B_c^+ \to J/\psi$ decays, which is further described in Section 14.1.

2.2.6 $B_c \to J/\psi$ models

There are two form factor predictions for $B_c \to c\bar{c}$ decays used for EvtGen models at LHCb created by Kiselev and Ebert, Faustov, and Galkin (EFG).

The EFG model uses the parameterization given in 23. It is derived from a relativistic constituent quark model. In this model, a meson is treated as a bound state in the center of mass frame with a relativistic reduced mass μ_r and a relative momentum $p_{on-shell}$, with the two quarks having equal and opposite momenta. These quantities are functions of the quark masses and the meson masses. For some meson M in this off-shell relative momentum basis, the wavefunction is governed by

$$
\left(\frac{p^2_{on-shell}}{2\mu_r}-\frac{\mathbf{p}^2}{2\mu_r}\right)\psi_M(p)=\int\frac{d^3\mathbf{q}}{(2\pi)^3}V(\mathbf{p},\mathbf{q};M)\psi_M(\mathbf{q}), \tag{28}
$$

where the quasipotential is derived from the QCD scattering amplitude. The parameters of this model are derived from fits to experimental data and from comparisons to heavy quark effective theory. The derived wavefunction can then be used to extract the matrix element and thus the form factors [19].

The Kiselev model uses what is basically the A, B, C, D parameterization,

$$
\begin{aligned}
\langle J/\psi(p_{J/\psi})|\bar{c}\gamma^\mu b|B_c^+(p_{B_c})\rangle &= iF_V\varepsilon^{\mu\nu\rho\sigma}\epsilon^*_\nu p_\rho q_\sigma \\
\langle J/\psi(p_{J/\psi})|\bar{c}\gamma^\mu\gamma^5 b|B_c^+(p_{B_c})\rangle &= F_0^A\epsilon^{*\mu}+F_+^A(\epsilon^*\cdot p_{B_c}))p^\mu+F_-^A(\epsilon^*\cdot p_{B_c})q^\mu
\end{aligned}
\tag{29}
$$

The translation between the two parameterizations is

$$
\begin{aligned}
V(q^2) &= (m_{B_c}+m_\psi)F_V(q^2) \\
A_2(q^2) &= -(m_{B_c}+m_\psi)F_A^+(q^2) \\
A_1(q^2) &= \frac{F_A^0(q^2)}{m_{B_c}+m_\psi} \\
A_0(q^2) &= \frac{1}{2m_\psi}(q^2F_A^-(q^2)+F_A^0(q^2)+(m_{B_c}^2-m_\psi^2)F_A^+(q^2))
\end{aligned}
\tag{30}
$$

The Kiselev model is derived using a non-relativistic QCD (NRQCD) approach, in which the QCD Lagrangian is expanded in powers of Λ/m_b and v/c, where v is the relative velocity of the quarks. This model uses QCD sum rule techniques which allow QCD correlation functions to be written in terms of their poles, which are associated with hadronic resonances. This allows the usage of experimental data on hadronic properties in these computations [4].

The two models show good agreement for $B_c \to J/\psi$. The signal and normalization MC are generated using the EFG model, but reweighted to the most recent lattice results using `Hammer`.

2.2.7 Lattice results for $B_c \to J/\psi$

More recently, the $B_c^+ \to J/\psi$ form factors have been precisely determined for the full q^2 range using lattice QCD (LQCD), with the first calculation here [18], and the improved calculation which will be used as input to this analysis [20]. These results significantly improve the modeling of $B_c^+ \to J/\psi$ semileptonic decays.

The newest lattice calculation of the $B_c \to J/\psi$ form factors uses heavy quarks with masses between the b and c masses on lattices, covering the full q^2 range allowed by such a mass at each lattice spacing. Correlation functions computed on the lattice are used to calculate the form factors, which are then fitted as a function of the heavy quark mass to extrapolate their value at the b mass. This is done using the so-called z-expansion, which maps the physical q^2 range to the unit circle using a new variable z:

$$z(q^2, t_0) = \frac{\sqrt{t_+ - q^2} - \sqrt{t_+ - t_-}}{\sqrt{t_+ - q^2} + \sqrt{t_+ - t_-}}, \tag{31}$$

where $t_+ = (m_B + m_{D*})^2$ and the physical q^2 ranges from 0 to $t_- = (m_{B_c} - m_{J/\psi})^2$ [21]. When q^2 is mapped onto the z plane, the branch cut is mapped to the unit circle, t_+ is mapped to -1, and t_- is mapped to 0. The form factors are expanded into a power series in z, which in the BGL parameterization is given by

$$F(q^2) = \frac{1}{P(q^2)\phi(t, t_0)} \sum_{n=0}^{\infty} a_{in}^{BGL} z(t, t_0)^n. \tag{32}$$

The outer functions ϕ are defined in [21], and the pole functions $P(q^2)$ are calculated by

$$P(q^2) \prod_{M_{pole}} z(q^2, M_{pole}^2). \tag{33}$$

The pole masses represent $\bar{b}c$ states corresponding to each form factor, with $V(q^2)$ corresponding to 1^- states, $A_0(q^2)$ corresponding to 0^- states, and $A_1(q^2)$ and $A_2(q^2)$ corresponding to 1^+ states. The values of the resonance masses are given in Table 2.

The most recent lattice QCD calculations for $B_c \to J/\psi$ also include correlations between the z-expansion coefficients. These are given in Appendix A of [18], and have been used to implement the form factor correlations by creating a covariance eigenvector matrix in `Hammer`. These are given later in Section 14.1.1, Table 18.

0^-/GeV	1^-/GeV	1^+/GeV
6.275	6.335	6.745
6.872	6.926	6.75
7.25	7.02	7.15
	7.28	7.15

Table 2: Pole masses for pseudoscalar, vector, and axial vector B_c excited states, with pseudoscalar values taken from experiment, and others taken from theory estimations.

2.2.8 $B \to \psi$ form factors

For $B_c^+ \to \psi$ decays, the initial simulation for this study was produced using the Ebert-Faustov-Galkin (EFG) model [19] and the Kiselev model [4]. The two models have very different q^2 shapes. The Kiselev model for $\psi(2S)$ uses the same parameters as the $B_c \to J/\psi$ decays but with different numerical values. On the other hand, using the EFG model, there is a node in the wavefunction due to the fact that $\psi(2S)$ is an $n = 2$ radially excited charmonium state. This leads to a very different q^2 spectrum compared to the J/ψ distribution. The nominal fit uses the EFG model.

2.2.9 $B \to \chi_c$ form factors

For $B_c^+ \to \chi_c$ decays, the Wang, Wang, and Lu (WWL) model [22] is used. The model of vector form factors used by WWL is similar to the `EvtGen` model. This is a pseudovector transition, so the form factors are labelled as A, V_0, V_1, and V_2, and defined by

$$\begin{aligned}
\langle X(k)\,|\bar{c}\gamma^{\mu}\left(1-\gamma_5\right)b|\,B(p)\rangle = &\frac{-2A\left(q^2\right)}{M-M_X}\varepsilon^{\mu\nu\rho\sigma}\epsilon^*_{\nu}p_{\rho}k_{\sigma} \\
&-2M_X\frac{\epsilon^*\cdot q}{q^2}iV_0\left(q^2\right)Q^{\mu} \\
&-\left(M-M_X\right)iV_1\left(q^2\right)\left(\epsilon^{*\mu}-\frac{\epsilon^*\cdot Q}{q^2}Q^{\mu}\right) \\
&+\frac{1}{M-M_X}iV_2\left(q^2\right)\left(\epsilon^*\cdot Q\right)\left(P^{\mu}-\frac{M^2-M_X^2}{q^2}Q^{\mu}\right).
\end{aligned} \tag{34}$$

The relationship to the `EvtGen` form factors is

$$\begin{aligned}
iV(q^2) &= \frac{M+M_X}{M-M_X}A(q^2) \\
iA_2(q^2) &= \frac{M+M_X}{M-M_X}V_2(q^2) \\
iA_1(q^2) &= \frac{M-M_X}{M+M_X}V_1(q^2) \\
iA_0(q^2) &= V_0(q^2).
\end{aligned} \tag{35}$$

(a) B_c^+ rest frame (b) Lab frame

Figure 3: A semileptonic B_c^+ decay into $Y = J/\psi\mu$ and an invisible neutrino system. Taken from the Run-1 $R(J/\psi)$ analysis note [24].

2.3 *B* rest frame approximation

Analyses of semileptonic B decays rely on several key kinematic quantities in the rest frame of the B which are useful for separating out different contributions to the same final state: the invariant mass of the four-momentum transfer to the leptonic system $q^2 = (p_B - p_X)^2$, the charged lepton energy E_ℓ^*, and the missing mass of the neutrino system $m_{miss}^2 = (p_B - p_X - p_\ell)^2$, where p_B, p_X p_ℓ are the four-momenta of the B meson, X hadron, and charged lepton respectively. The missing mass is used to differentiate between the signal and normalization channels, with larger missing mass for signal due to additional neutrinos in the final state, and the lifetime is used to differentiate between B_c decays and other B decays using the relatively short lifetime of the B_c. In order to boost into the rest frame, we require a measurement of the B_c momentum.

The observables relevant to this measurement are the flight direction of the B_c and the energy and momentum of the visible final state $Y = J/\psi\mu$. The flight direction is measured by the tracking system, with the decay vertex determined by a fit to the three muon tracks. However, the momentum of the B_c is not uniquely determined by these variables. In the case where the semimuonic decay has the system Y with momentum parallel to the B_c, the neutrino could be pointing parallel or antiparallel to the B_c. In the case of the semitauonic decay, the three-neutrino subsystem could have a range of invariant masses from 0 to $m_{B_c} - m_Y$, leading to a range of possible solutions.

Although the B rest frame cannot be directly determined in proton-proton collisions, a rest frame approximation technique was developed at LHCb [23] to approximate the rest frames of semileptonic B decays and estimate the kinematic variables.

The technique uses the approximation that along the z-axis, the proper velocity $\gamma\beta$ of the B is equal to that of the visible final states (denoted as Y). To get to this approximation, let us start with a Lorentz transformation between the B_c rest frame and the lab frame,

$$p_{B_c} = \frac{m_{B_c}}{E_{Y,CM}} p_Y^{\parallel} - E_{B_c}\beta_{Y,CM}\cos\theta_{CM} \tag{36a}$$

$$p_Y^{\perp} = p_{Y,CM}\sin\theta_{CM} \tag{36b}$$

with θ being the angle between the B_c flight direction and the momentum of the Y system (see Figure 3).

Quantities labeled CM are in the B_c center of mass frame, all others are in the lab frame. Because the B_c is spin-0, the direction of the Y system's momentum is isotropically distributed and also uncorrelated with $\beta_{Y,CM}$ and E_{B_c} without taking acceptance and selection into account. So, $\cos\theta_{CM}$ is zero on average, or $\theta_{CM} = \frac{\pi}{2}$ on average. From this assumption it follows that $p_Y^{\perp} = p_{Y,CM}$, so

$$p_{est} \equiv p_{B_c}|_{\theta_{CM}=\frac{\pi}{2}} = \frac{m_{B_c} p_Y^{\parallel}}{\sqrt{M_Y^2 + (p_Y^{\perp})^2}}. \tag{37}$$

The B_c mesons produced at LHCb are highly boosted, so $\beta_{B_c} \approx 1$. Conservation of energy and momentum require

$$\beta_{Y,CM} < \beta_{max} = \frac{m_{B_c}^2 - (m_\psi + M_\mu)^2}{m_{B_c}^2 + (m_\psi + M_\mu)^2} \approx 0.587. \tag{38}$$

The error from taking this approximation is

$$\frac{p_{est} - p_{B_c}}{p_{B_c}} = \frac{\beta_{Y,CM}}{\beta_{B_c}} \cos\theta_{CM} + A(1 + \frac{\beta_{Y,CM}}{\beta_{B_c}} \cos\theta_{CM}), \tag{39}$$

where

$$A = \frac{1}{\sqrt{1 - \beta_{Y,CM}^2 \cos\theta_{CM}}} - 1 \approx \frac{1}{2}\beta_{Y,CM}^2 \cos\theta_{CM} + \mathcal{O}(\beta_{Y,CM}^4). \tag{40}$$

The first term is on average zero, and roughly bounded by β_{max}. The second term is non-zero on average, but is a higher order correction in $\beta_{Y,CM}$ and thus less significant.

$p_Y^{\perp}$ is small and the B_c flight direction is nearly parallel with the z axis in LHCb's acceptance, so therefore, the B momentum along the z-axis can be estimated as

$$(p_B)_z = \frac{m_B}{m_Y}(p_Y)_z, \tag{41}$$

where m_B is the B_c^+ mass, and m_Y and p_Y are the reconstructed invariant mass and momentum of the $(J/\psi)\mu^+$ system in this study. The total B_c^+ momentum is then calculated using the B_c^+ flight direction, which is determined from the reconstructed B_c^+ decay vertex and its associated PV:

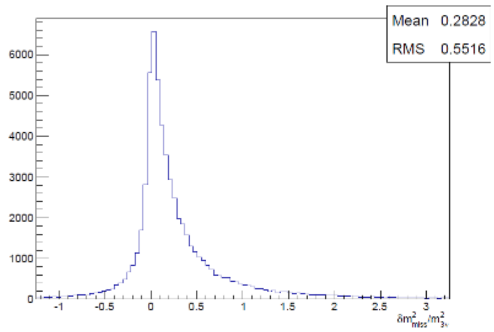
$$\|p_B\| = (p_B)_z \sqrt{1 + \tan^2\theta} = \frac{m_B}{m_Y}(p_Y)_z \sqrt{1 + \tan^2\theta}, \tag{42}$$

where θ is the angle between the B_c^+ flight direction and the z-axis.

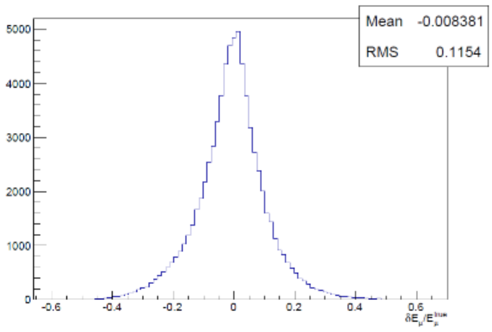
2.4 Fit variables

The rest frame kinematic variables can be then calculated using the estimated B_c^+ momentum, including the missing mass $m_{miss}^2 = (p_{B_c} - p_{J/\psi} - p_\mu)^2$, the momentum transfer to the lepton system $q^2 = (p_{B_c} - p_{J/\psi})^2$, the muon's energy E_μ^* in the B_c rest frame, and the B_c^+ candidate's proper decay time. The resolutions on the estimated m_{miss}^2, E_μ^*, and q^2 in simulation are shown in Figure 4, with figures taken from $R(D^{(*)})$. Figure 5 is provided which shows the distribution of the absolute value of the shift in the rest frame variables as a function of the true rest frame variable value, taken from the signal MC, and Figure 6 shows candle plots where the y-axis represents the signed shift in the true versus estimated rest frame variable.

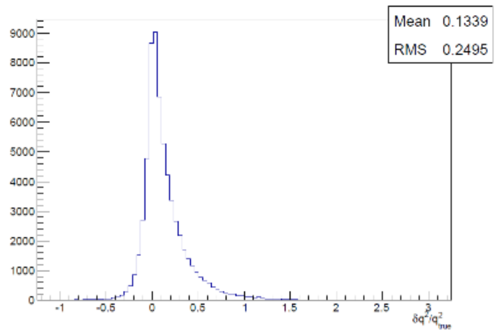
The distributions of the estimated rest frame variables for the simulated $B_c^+ \to J/\psi\,\tau^+\,\nu$ and $B_c^+ \to J/\psi\,\mu^+\,\nu$ decays are shown in Figure 7. Because the $B_c^+ \to J/\psi\,(\tau^+ \to \mu^+\nu_\mu\bar{\nu_\tau})\,\nu_\tau$ decay has three unobserved neutrinos, the missing mass is non-zero. The muon from the $\tau^+ \to \mu^+\nu_\mu\bar{\nu_\tau}$ decay is likely to have lower energy than that from the $B_c^+ \to J/\psi\,\mu^+\,\nu$ decay. The semitauonic decay tends to have a larger momentum transfer to the leptonic system than the semimuonic decay. In the case of tauonic decays, the energy of E_τ is not measurable due to the missing mass from the neutrino.



(a) m^2_{miss} resolution



(b) E^*_μ resolution



(c) q^2 resolution

Figure 4: The relative resolutions on the estimated rest frame variables m^2_{miss}, E^*_μ, and q^2 in simulation, taken from the $R(D^{(*)})$ ANA note [25].

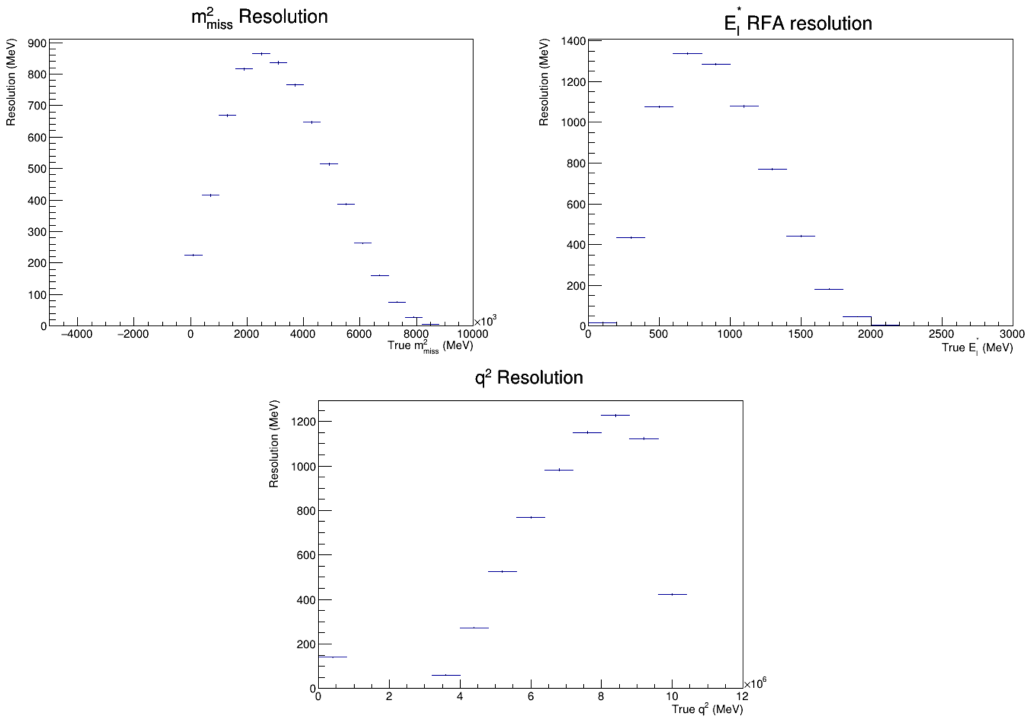


Figure 5: The size of the shift in the RFA estimated vs. true rest frame variables m^2_{miss}, E^*_μ, and q^2 as a function of the true variables in simulation, taken from $B_c \to J/\psi\tau\nu$ simulation.

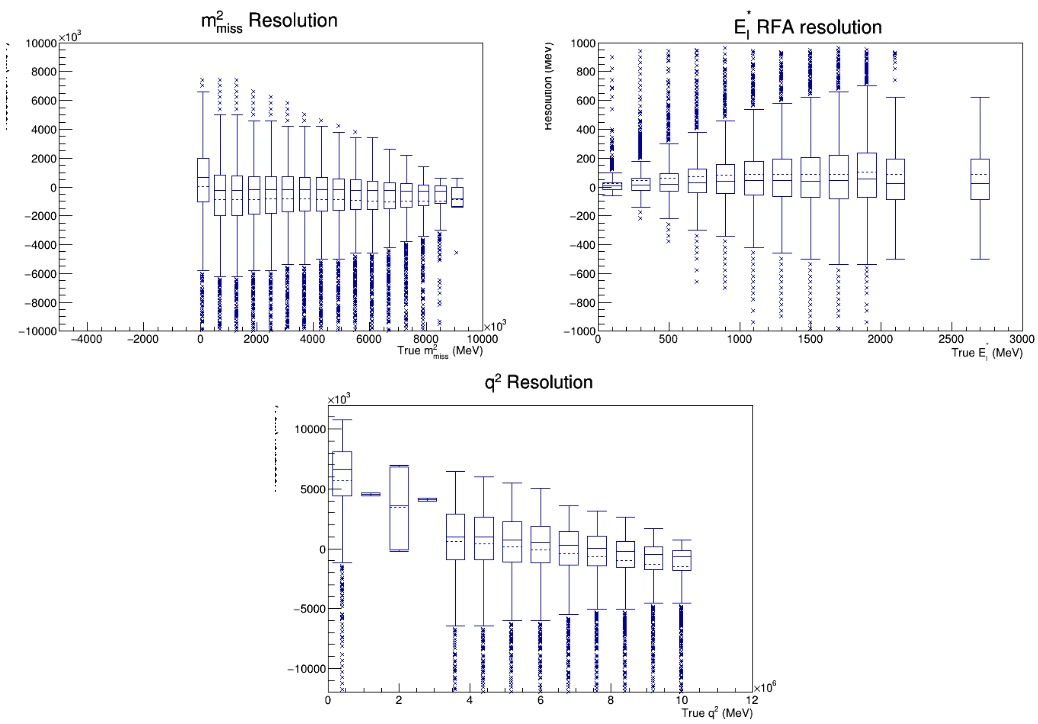


Figure 6: Candle plots used for determining the resolution and bias on the estimated rest frame variables m^2_{miss}, E^*_μ, and q^2 as a function of the true variables in simulation, taken from $B_c \to J/\psi\tau\nu$ simulation. The y-axis shows the shift in the rest frame variable, signed.

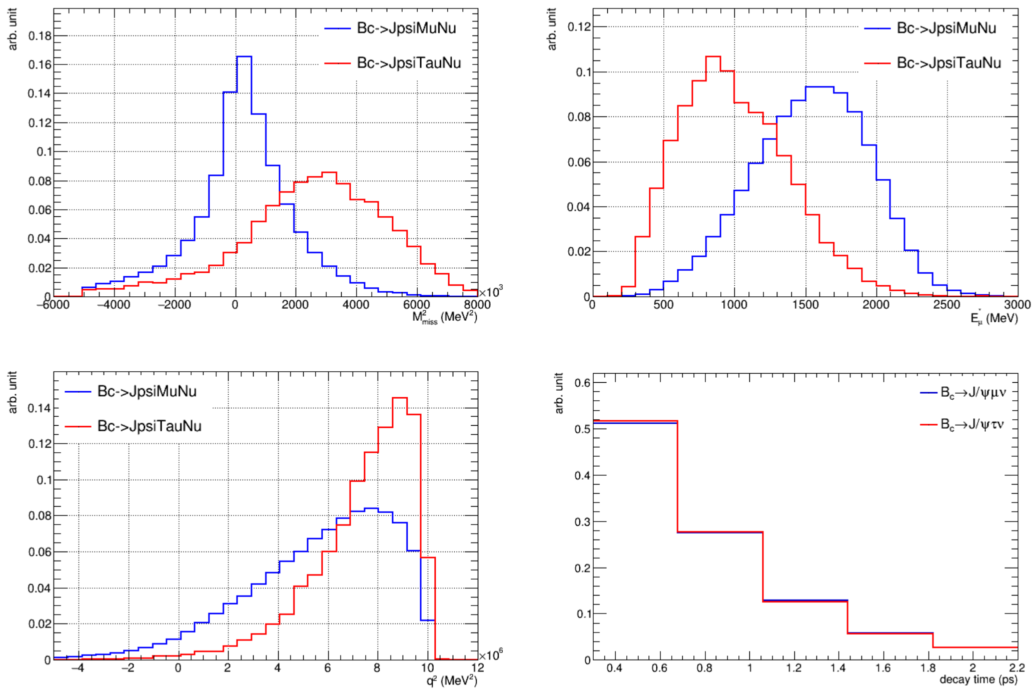


Figure 7: The distributions of the estimated rest frame variables m^2_{miss}, E^*_μ, and q^2 for the signal and normalization simulations, along with the lifetime of the B_c.

3 Analysis overview

The study analyzes data collected by the LHCb experiment from 2016 to 2018, selecting for a sample containing events from the signal mode $B_c^+ \to J/\psi\, \tau^+ \nu$, normalization mode $B_c^+ \to J/\psi\, \mu^+ \nu$, and various backgrounds.

With muonic decays of $\tau^+ \to \mu^+ \nu_\mu \bar{\nu_\tau}$ and $J/\psi \to \mu^+\mu^-$, the signal mode $B_c^+ \to J/\psi\, \tau^+ \nu$ and normalization mode $B_c^+ \to J/\psi\, \mu^+ \nu$ share the same visible trimuon final states (two paired muons and one unpaired muon). They are differentiated in m^2_{miss}, q^2, E^*_μ, and the B_c^+ candidate's proper decay time. m^2_{miss} has 25 bins from -5 GeV^2 to 10 GeV^2, the lifetime of the B_c^+ candidate is binned into 5 bins from 0.3 ps to 5 ps. The 3rd variable Z is a categorical variable of E^*_μ and q^2, with 8 bins corresponding to $E^*_\mu = [0, 0.68, 1.15, 1.64, \infty)$ GeV and $q^2 = [0, 7.15, \infty)$ GeV^2.

The relative yields of signal, normalization, and backgrounds are determined with a binned maximum likelihood fit to the distributions of m^2_{miss}, Z, and B_c^+ decay time, using templates derived from MC and data control samples. Differences between data and MC in areas such as lifetime acceptance and production multiplicity are corrected for, along with form factor reweighting to account for newer lattice QCD results, and a final reweighting to improve any remaining data/MC discrepancies.

The leading background is from inclusive decays of B^0, B^+, and B_s mesons into J/ψ and hadrons, from which a charged particle (e.g. pion, kaon, proton, or electron) is misidentified as a muon. The muon misidentification rate is relatively low due to the powerful particle identification system at LHCb. But because the production cross sections of the lighter B mesons are at least two orders of magnitude larger than that of the B_c^+, the muon misidentification background is a significant component in the selected B_c^+ candidate sample. The lighter B mesons, however, have lifetimes about three times longer than that of the B_c^+ meson, therefore the B_c^+ candidate decay time is useful in separating the contributions of different B mesons in the data.

The misidentification background is modeled using a data-driven approach using a control sample of events with J/ψ and a track not identified as muon, as described in Section 8.

There are several backgrounds with trimuon final states from B_c^+ decays, which are modeled with MC simulations of various decay modes. The feed-down backgrounds include B_c^+ semileptonic decays with a heavy charmonium state such as $\psi(2S)$ or χ_c which further decay down to J/ψ. The other backgrounds from B_c^+ decays include $B_c^+ \to J/\psi\, D\, X$ decays, in which the D meson decays leptonically or semileptonically. The fit strategy for these backgrounds is described more in Section 12.

Additionally, there are two different combinatorial backgrounds. The $(J/\psi + \mu)$ combinatorial background is formed between a J/ψ from a b-hadron decay and a muon from a separate decay. This is modeled with MC simulations of the inclusive $B_{d,s} \to J/\psi X$ decays, where X represents the recoiling hadron system, this is described in greater detail in Section 9. The other combinatorial background is due to random muon pairs forming combinatorial J/ψ candidates, and it is modeled using data from the upper sideband of the J/ψ candidate's invariant mass distribution. The modeling is further described in Section 11.

Table 3: The simulation samples produced for the analysis, adjusted for detector running conditions with subsample size proportional to the yearly integrated luminosity.

Type	Decay mode	Num. of events	EventType	Notes
Signal	$B_c^+ \to J/\psi\,\tau^+\,\nu$	11 M	14643408	EFG model
Normalization	$B_c^+ \to J/\psi\,\mu^+\,\nu$	15 M	14543013	EFG model, full simulation
	$B_c^+ \to J/\psi\,\mu^+\,\nu$	40 M	14543013	tracker-only simulation
Feed-down	$B_c^+ \to \psi(2S)\,\mu^+\,\nu$	2 M	14845007	Kiselev model
	$B_c^+ \to \psi(2S)\,\tau^+\,\nu$	0.04 M	14845017	Kiselev model
	$B_c^+ \to \psi(2S)\,\mu^+\,\nu$	3 M	14845008	EFG model
	$B_c^+ \to \psi(2S)\,\tau^+\,\nu$	3 M	14845018	EFG model
	$B_c^+ \to \chi_{c1}\,\mu^+\,\nu$	2.4 M	14543221	WWL model
	$B_c^+ \to \chi_{c2}\,\mu^+\,\nu$	2.4 M	14543222	WWL model
Double-charm	$B_c^+ \to J/\psi\,D_x$	3 M	14873607	
	$B_c^+ \to J/\psi\,D_s^{**+}$	3 M	14873620	cocktail
	$B_c^+ \to J/\psi\,D_x\,K_x$	3 M	14873430	
Combinatorial	$B^0 \to J/\psi\,X$	12 M	11442012	cocktail
	$B_s \to J/\psi\,X$	2 M	13442012	
Correction	$B^0 \to J/\psi K^*$	12 M	11144001	

4 Data sets

The study uses data that the LHCb experiment collected from 2016, corresponding to an integrated luminosity of 1.67 fb^{-1}, in proton-proton collisions at the center-of-mass energy of 13 TeV, along with data from 2017 and 2018, which recorded an integrated luminosity of 3.90 fb^{-1}. The 2016, 2017, and 2018 datasets use stripping versions 28r2, 29r2, and 34, respectively. The initial fit was developed using half of the 2016 data.

The Monte Carlo (MC) simulations of the signal, normalization, and various backgrounds have been produced. Pythia [26] is used to generate the hard-scattering, parton shower, hadronization, and decay processes in high-energy proton-proton collisions. EvtGen [27] simulates the decays of heavy flavor particles including B mesons. Bcvegpy [28] is the dedicated generator for the hadronic production of B_c^+ mesons. Geant4 [29, 30] is used to simulate long-lived particles' transportation and interactions within the LHCb detector.

Both data and simulation samples are processed on the Worldwide LHC Computing Grid before offline analysis. The simulation samples are summarized in Table 3. The samples are adjusted for detector running conditions, with subsample size proportional to the yearly integrated luminosity.

5 Selection

5.1 Stripping

We require the events to pass the `StrippingFullDSTDiMuonJpsi2MuMuDetachedLineDecision` stripping line in the dimuon stream.

This line selects for two opposite-sign muon candidates with $p_T > 550$ MeV, impact parameter (IP) $\chi^2 > 4$ with respect to any primary vertex (PV), and a distance of closest approach (DOCA) $\chi^2 < 30$. These muons must form a vertex with a fit $\chi^2 < 25$, and have a parent J/ψ whose mass is within 100 MeV of the known J/ψ mass and who is separated by the closest PV by a decay length greater than three times its uncertainty.

5.2 Trigger

The J/ψ candidate has to trigger `L0DiMuonDecision`, either `Hlt1DiMuonHighMass` or `Hlt1DiMuonLowMass`, and `Hlt2DiMuonDetachedJpsiDecision`. We do not trigger on the bachelor muon because of the reduced signal efficiency caused by the soft p_T spectrum of muons produced in τ decays.

5.3 Pre-selection

To select candidates of $B_c^+ \to J/\psi\, \ell^+\, \nu$ decays, the events are required to first pass the LHCb trigger and stripping selections for $J/\psi \to \mu^+\mu^-$ candidates that are detached from the PV. Both muon candidates' tracks should not originate from the PV and have significant impact parameters with respect to the PV. This is done by calculating the increase in the primary vertex fit χ^2 when including this track. The IP χ^2 with respect to any PV must be larger than 4. The J/ψ candidate is required to have $p_T > 2$ GeV and is selected from `StdLooseJpsi2MuMu`. There is also a detached J/ψ requirement which vetoes the background where a prompt J/ψ is combined with a muon.

The two opposite-charged muon candidates are required to have $p_T > 500$ MeV and to pass a loose particle identification (PID) selection for muons `PIDmu` > 0. The log-likelihood difference between muon and pion hypotheses, DLLmu, is required to be larger than 0.

There are also additional preselections placed on the normalization MC at the generator level. These require the bachelor muon candidate to have $1.6 < \eta < 5.1$, $p > 2.5$ GeV, the J/ψ candidate to have $p_T > 1$ GeV, and the J/ψ daughter muon candidatesto have $p > 3$ GeV. The J/ψ daughter muon candidate with higher p_T must have $p_T > 1$ GeV and the lower p_T muon must have $p_T > 450$ MeV. These are summarized in Table 4.

Table 5 contains a summary of the preselections in this analysis.

5.4 Main selection

For simulation, we perform truth-matching for the bachelor muon candidate, the J/ψ candidate, and the B_c candidate. We truth-match the parent (and grandparent, when applicable for certain backgrounds) of the J/ψ to its expected particle ID along with requiring it to be identified as the parent (grandparent) of the J/ψ candidate and the grandparent (great-grandparent) of the bachelor muon candidate.

The vertex fit from the track parameters for the opposite-sign muon pair should have $\chi^2 < 9$ to reduce the combinatorial background. With the J/ψ mass resolution of

Object	Selection
$\mu^+\mu^-$	$p > 3$ GeV
$\mu_{p_T>}$	$p_T > 1$ GeV
$\mu_{p_T<}$	$p_T > 450$ MeV
J/ψ	$p_T > 1$ GeV
bachelor μ (μ_B)	$1.6 < \eta < 5.1$
	$p > 2.5$ GeV

Table 4: Summary of filter requirements on different objects only for the normalization MC.

Object	Selection
L0	L0DiMuon
HLT1	`Hlt1DiMuonHighMass` or `Hlt1DiMuonLowMass`
HLT2	`Hlt2DiMuonDetachedJpsiDecision`
Stripping	`StrippingFullDSTDiMuonJpsi2MuMuDetachedLineDecision`
$\mu^+\mu^-$	$p_T > 500$ MeV, `DLLmu` > 0; selected from `StdLooseMuons`
J/ψ	$p_T > 2$ GeV; selected from `StdLooseJpsi2MuMu`
bachelor μ (μ_B)	$p_T > 300$ MeV, `DLLmu` > 0, IP $\chi^2 > 4$

Table 5: Summary of preselection requirements on different objects.

approximately 13 MeV, the measured invariant mass of the J/ψ candidate is required to be within 55 MeV of the true J/ψ mass. The candidate is also required to be associated with signatures for the hardware and software dimuon triggers, which identify $J/\psi \to \mu^+\mu^-$ candidates.

The J/ψ candidates are then combined with muon candidate tracks to form B_c^+ candidates. The invariant mass of $(J/\psi + \mu^+)$ combination must be less than the B_c^+ mass, due to the missing the neutrino energy in the reconstruction of signal and normalization decays.

To reduce combinatorial background, the J/ψ and μ^+ candidates are required to have a small DOCA and not be in opposite directions in the XY plane. This selection has an efficiency of about 80% for combinatorial and 90% for signal. There is a small probability for a muon track from the J/ψ candidate to be misreconstructed as two same-sign muons and lead to a fake trimuon candidate. This background is rejected by requiring the angle between the unpaired muon and the same-sign muon to be not close to zero in the lab frame, i.e. $\cos\theta < 1 - e^{-8}$. The wrong pair of opposite-sign muons can be chosen in the decay reconstruction, when it has an invariant mass consistent with the J/ψ. These ambiguous cases are removed by requiring the unpaired muon and the opposite-sign muon to have an invariant mass which is outside the J/ψ mass window by more than 50 MeV. The bachelor muon candidate has a minimum p_T in order to exclude the region where the misID background is poorly modeled. The momentum of the bachelor muon candidate is also restricted to the range in which there is PIDCalib information. A selection requiring that the bachelor muon candidate's `NShared` value to be 0 which ensures that no other muon tracks are reconstructed using the hits in the muon chamber associated with the muon candidate of interest. This reduces backgrounds which are false muon candidates, as the likelihood of two muons passing very close together and sharing a hit is small.

The combinatorial background disproportionately occupies the low missing mass region. To reduce the contribution in this region, a new selection were introduced in Run-2 requiring large $\cos\theta$, where θ is the angle between the visible $Y = J/\psi + \mu$ system and the B_c candidate. Expanding the missing mass squared gives

$$\begin{aligned} m^2_{\text{miss}} &= (p_{B_c} - p_Y)^2 \\ &= (E_{B_c} - E_Y)^2 - (\vec{p_{B_c}} - \vec{p_Y})^2 \\ &= m^2_{B_c} + m^2_Y - 2E_{B_c}E_Y + 2\,\|p_{B_c}\|\,\|p_Y\|\cos\theta, \end{aligned} \tag{43}$$

meaning this cut would reduce contributions to the low missing mass region. The DOCA cut was also tightened in Run-2 to reflect a sign dependence in the data, difficulty in obtaining data/MC agreement, and to most efficiently reduce combinatorial background (positive being J/ψ ahead of the B_c, and negative being the J/ψ behind the B_c).

A selection on the component of the bachelor muon momentum perpendicular to the B_c flight direction was added in Run-2. This cut was designed to reduce the amount of mis-ID background and improve postfit agreement in certain variables in mis-ID enriched regions.

Finally, the lifetime fit region was tightened to exclude events with $\tau < 0.45$ ps. This was due to observance of poor MC replication of the signed DOCA asymmetry observed in data in the low lifetime region, potentially related to the VELO error parameterization issues in Run-2. The cut also drastically reduces the charm combinatorial background (see Section 10 for further descriptions), which is difficult to model.

Additional motivation and checks for the new Run-2 selections are provided in Appendix 20.

In the case of multiple candidates passing the selection, a random one is chosen. Events with a candidate passing the selection have a single candidate passing the selection 90% of the time. Most of the multiple candidates are bachelor muon candidates rather than J/ψ candidates. The main selection is summarized in Table 6. Truth matching was performed on all particles, as well as on the parents and grandparents if applicable. A few alternative selection improvements were considered but ultimately not used, as described in Appendix 22.

5.4.1 Differences with Run-1

This analysis uses a uBDT for the bachelor muon PID (motivated in Section 5.6) along with DLLmu, whereas in Run-1 ProbNNmu is used. In Run-1, the variable `NShared` was used, but it was bugged so as to be basically useless. This has been fixed in Run-2. There is an additional IP χ^2 cut on both daughters of the J/ψ candidate. The veto in Run-1 which excluded $\eta \geq 3.6$ and $p \leq 15\text{GeV/c}$ in order to improve fit agreement at $Z = 0$ has also been removed, since this discrepancy does not exist in Run-2. As described above in greater detail in Section 5, a new selection was added on the angle between the $J/\psi\mu$ system and the B_c candidate, as well as selections on the lifetime, the component of the

Table 6: Main selection.

Object	Selection
Event	nTracks <600
J/ψ	mass within 55MeV of nominal $m_{J/\psi}$(3095MeV)
-	masses within 95MeV retained for sideband
-	Vertex $\chi^2 < 9$
μ_B	momentum in range 3GeV to 100GeV
-	$p_T > 750\text{MeV}/c$
-	p_{perp} to $B_c^+ > 500\text{MeV}/c$
-	pseudorapidity η in the range 2 to 5
-	PIDmu > 2.0, PIDe < -1.0, muBDT > 0.65, NShared $= 0$,`DLLmu`>0
-	IP $\chi^2 > 4.8$
OS, SS muon	IP $\chi^2 > 4$
$J/\psi + \mu_B$	vertex $\chi^2 < 25$
-	inv. mass lies between nominal $m_{J/\psi} + M_\mu$ and m_{B_c}
-	-0.13 mm $<$ DOCA $(J/\psi, \mu) < 0.05$ mm
-	$\cos(J/\psi, \mu)$ in XY plane > -0.8
-	max. isolation BDT < 0.2
-	missing mass squared $> -5\text{GeV}^2/c^4$
-	lifetime > 0.45ps
$\mu_B +$ OS muon	mass not within 50MeV of nominal m_ψ
$\mu_B +$ SS muon / OS muon	$\cos(\theta) < 1 - e^{-8} \approx 0.9997$
$Y + B_c$	$\cos(\theta) > 0.9998$

bachelor muon momentum perpendicular to the B_c flight direction, and a tighter DOCA cut. These updated selections are summarized in Table 7.

5.5 Isolation algorithm

To reduce the backgrounds from partially reconstructed b-hadron decays which produce more than three charged tracks, such as $B_c \to J/\psi n\pi\ell\nu$ and $B_c \to J/\psi H_c$, a multivariate isolation algorithm is used for identifying tracks associated with the signal B meson originally developed for the $R(D^*)$ analysis [23]. The algorithm is a boosted decision tree (BDT) trained with simulated samples. Associated tracks from $B \to (D^{**} \to D^*)\,\mu^+\,\nu$ events and unassociated tracks from $B^0 \to D^{*-}\,\mu^+\,\nu$ events after excluding the signal decay are used for the training. The input variables include the track's IP χ^2 with respect to the PV, secondary $(J/\psi\,\mu^+)$ vertex, transverse momentum, angle between the track and the $J/\psi\,\mu^+$ momentum, and significance of separation between the PV and the $J/\psi\,\mu^+$ vertex or the $J/\psi\,\mu^+$ + track vertex. Figure 8 shows the distributions of the

Table 7: New selections from Run-2.

Object	Selection
J/ψ	Vertex $\chi^2 < 9$
μ_B	PIDmu > 2.0, PIDe < -1.0, muBDT > 0.65, NShared $= 0$
-	IP $\chi^2 > 4.8$
-	p_{perp} to $B_c^+ > 500\text{MeV}/c$
$J/\psi + \mu_B$	lifetime > 0.45ps
-	-0.13 mm $<$ DOCA $(J/\psi, \mu) < 0.05$ mm
OS, SS muon	IP $\chi^2 > 4$
$Y + B_c$	$\cos(\theta) > 0.9998$

maximum isolation BDT score for the normalization mode and the muon misidentification background. The selected candidate events have a maximum isolation BDT score of 0.2.

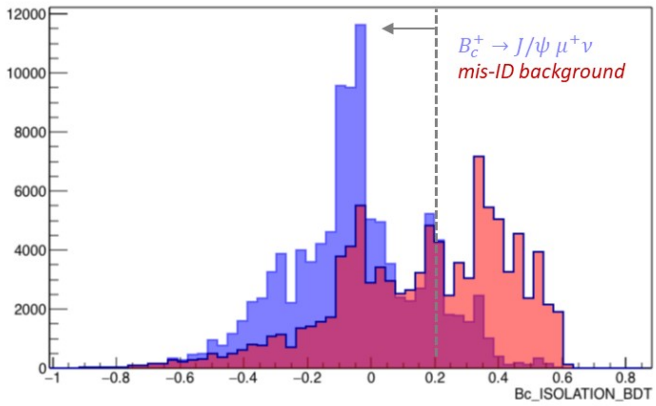


Figure 8: The distributions of the maximum isolation BDT score for the normalization mode and the muon misidentification background [24].

5.6 Muon PID

Because the leading background is due to muon misidentification, a tight PID requirement on the unpaired muon is important to the analysis. But the standard PID algorithms, such as DLLmu and neural-network-based ProbNNmu, have been found to strongly depend on the transverse and total momentum of the track [25]. With a softer muon p_T distribution from the signal's tau decay, a tight requirement using standard PID would be much less efficient for the signal decays than the normalization channel. The resulting distortion in the p_T spectrum also makes modeling harder with finite data control samples.

A custom multivariate algorithm for muon PID was developed by [25] to improve background rejection without strong dependence on the track p_T in semileptonic B decays. Details of this algorithm are given in the reference, and the variables used in training are given in Table 8.

TrackChi2PerDof	RichAboveMuThres	InAccHcal
TrackNumDof	RichAboveKaThres	HcalPIDe
TrackLikelihood	RichDLLe	HcalPIDmu
TrackGhostProbability	RichDLLmu	InAccPrs
TrackFitMatchChi2	RichDLLk	PrsPIDe
TrackFitVeloChi2	RichDLLp	InAccBrem
TrackFitVeloNDoF	RichDLLbt	BremPIDe
TrackFitTChi2	MuonBkgLL	VeloCharge
TrackFitTNDoF	MuonMuLL	isMuonTight
RichUsedAero	MuonNShared	TrackP
RichUsedR1Gas	InAccEcal	TrackPt
RichUsedR2Gas	EcalPIDe	

Table 8: Training variables used for the muon BDT, table taken from [25].

The uBoost method [31] implemented in the TMVA package [32] is used to train a BDT classifier with a constant selection efficiency as a function of muon p_T and P. The uBDT classifier is trained using $B \to D^{(*)}\,\tau\,\nu$ simulation, with muons tracks as the signal and other tracks from the simulation as the background.

The threshold for the uBDT score to be used in the selection is optimized using simulation samples. The figure of merit (FoM) used for optimization is defined as

$$\text{FoM} = \frac{\epsilon(\text{signal})}{\sqrt{\epsilon(\text{misID})}} \tag{44}$$

where the efficiency ϵ(signal) is evaluated from the signal and normalization simulations, and the efficiency ϵ(misID) is evaluated using an inclusive J/ψ simulation with misidentified accompanying tracks. The FoM as a function of the muon uBDT cut is shown in Figure 9. Therefore, a uBDT score greater than 0.65 is chosen as the unpaired muon's PID requirement, and a selection of DLLmu>2 and DLLe<-1.

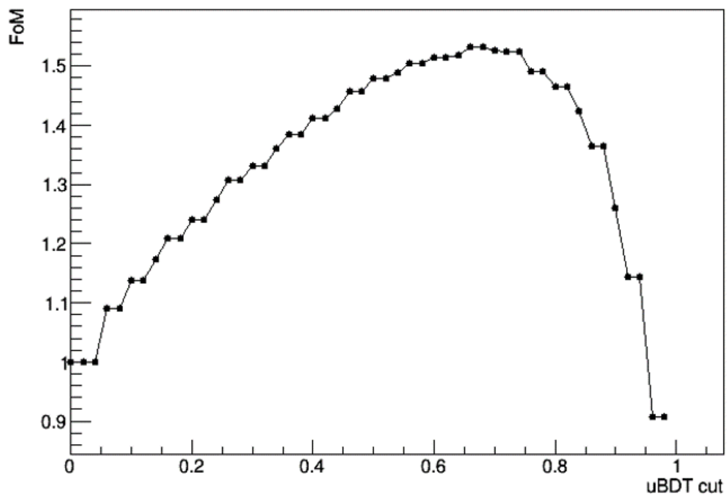


Figure 9: The optimization figure of merit (FoM) as a function of the muon uBDT cut.

6 Efficiency ratio

To measure the true ratio between the signal and normalization events, the efficiency differences between the two modes must be accounted for. These discrepancies are the differences in detector acceptance efficiency, measured at the generator level (GEN); the filter efficiency (FIL); the reconstruction and preliminary selection efficiency (REC); the final selection efficiency without isolation and PID (SEL); the isolation efficiency (ISO); the bachelor muon PID selection (PID); the L0 and HLT1 triggers (L0); and the HLT2 trigger (HLT2). The overall efficiency ratio is denoted as

$$\frac{\epsilon_\tau}{\epsilon_\mu} = \frac{\epsilon_\tau^{\mathrm{GEN}}}{\epsilon_\mu^{\mathrm{GEN}}} \times \frac{\epsilon_\tau^{\mathrm{FIL}}}{\epsilon_\mu^{\mathrm{FIL}}} \times \frac{\epsilon_\tau^{\mathrm{REC}}}{\epsilon_\mu^{\mathrm{REC}}} \times \frac{\epsilon_\tau^{\mathrm{SEL}}}{\epsilon_\mu^{\mathrm{SEL}}} \times \frac{\epsilon_\tau^{\mathrm{ISO}}}{\epsilon_\mu^{\mathrm{ISO}}} \times \frac{\epsilon_\tau^{\mathrm{PID}}}{\epsilon_\mu^{\mathrm{PID}}} \times \frac{\epsilon_\tau^{\mathrm{L0}}}{\epsilon_\mu^{\mathrm{L0}}} \times \frac{\epsilon_\tau^{\mathrm{HLT2}}}{\epsilon_\mu^{\mathrm{HLT2}}}. \tag{45}$$

The efficiencies are measured separately for the different years and magnet polarities, and summarized in Table 9, which gives the efficiencies relative to each previous stage. The generator efficiencies are taken from the documentation for Sim09 generator statistics. The filter efficiencies are calculated by finding the rejected statistics in bookkeeping. The other efficiencies are taken from MC. The PID efficiencies are taken from MC as well, due to the good agreement shown in postfit comparisons of the bachelor muon momentum (Section 16.1) and the trimuon effect potentially causing issues with PID modeling in systems without J/ψ (Section 17.12).

The total efficiency ratio is $(45.4 \pm 0.6)\%$, the efficiencies were combined with relative weight given by the number of events in each year and polarity. This is slightly smaller than the Run-1 efficiency of $(51.4 \pm 0.6)\%$. The change in efficiency is primarily a result of the new cuts added to reduce background. Assuming Standard Model predictions of $R(J/\psi)$, the signal yield in the combined datasets for all 3 years is expected to be 496.

Sample	$\frac{\epsilon_\tau^{\mathrm{GEN}}}{\epsilon_\mu^{\mathrm{GEN}}}$	$\frac{\epsilon_\tau^{\mathrm{FIL}}}{\epsilon_\mu^{\mathrm{FIL}}}$	$\frac{\epsilon_\tau^{\mathrm{REC}}}{\epsilon_\mu^{\mathrm{REC}}}$	$\frac{\epsilon_\tau^{\mathrm{SEL}}}{\epsilon_\mu^{\mathrm{SEL}}}$	$\frac{\epsilon_\tau^{\mathrm{ISO}}}{\epsilon_\mu^{\mathrm{ISO}}}$	$\frac{\epsilon_\tau^{\mathrm{PID}}}{\epsilon_\mu^{\mathrm{PID}}}$	$\frac{\epsilon_\tau^{\mathrm{L0}}}{\epsilon_\mu^{\mathrm{L0}}}$	$\frac{\epsilon_\tau^{\mathrm{HLT2}}}{\epsilon_\mu^{\mathrm{HLT2}}}$	$\frac{\epsilon_\tau}{\epsilon_\mu}$
2016 MagDown	1.348(3)	2.325(1)	0.3329(3)	0.466(1)	0.997(2)	0.937(4)	1.007(6)	0.9997(4)	0.46(1)
2016 MagUp	1.339(3)	2.334(1)	0.3317(3)	0.467(1)	1.000(2)	0.940(4)	0.995(7)	1.0000(6)	0.45(1)
2017 MagDown	1.350(3)	2.325(1)	0.3315 (3)	0.478(1)	1.007(2)	0.947(3)	0.979(6)	0.9995(5)	0.46(1)
2017 MagUp	1.342(3)	2.336(1)	0.3222(3)	0.470(1)	1.008(2)	0.933(3)	0.997(6)	1.0000(4)	0.45(1)
2018 MagDown	1.344(3)	2.326(1)	0.3288(2)	0.4739(8)	1.006(1)	0.942(2)	0.995(5)	1.0000(4)	0.46(1)
2018 MagUp	1.348(3)	2.338(1)	0.3291(2)	0.4734(8)	1.002(2)	0.940(2)	0.985(5)	0.9993(4)	0.46(1)
Combined	1.3451(4)	2.3306(5)	0.32942(3)	0.471(1)	1.003(1)	0.938(2)	0.992(3)	0.9998(1)	0.454(6)

Table 9: Table of efficiency ratios.

7 Signal and normalization modeling

The signal $B_c^+ \to J/\psi\, \tau^+\, \nu$ and normalization $B_c^+ \to J/\psi\, \mu^+\, \nu$ decays are modeled with MC simulation. The simulation is produced with the Ebert-Faustov-Galkin (EFG) model [19] for $B_c^+ \to J/\psi$ form factors, and is later reweighted according to lattice QCD results [18]. The details of the form-factor reweighting are described in Section 14.1.

The signal and normalization have different topologies and kinematics which lead to different characteristics in the rest frame variables, as shown in Figure 7. The normalization mode $B_c^+ \to J/\psi\, \mu^+\, \nu$ has only one neutrino, so a small missing mass peak around zero is expected. The signal mode $B_c^+ \to J/\psi\, \tau^+\, \nu$ with $\tau^+ \to \mu^+\, \nu_\mu\, \bar{\nu_\tau}$ has three neutrinos, which lead to a non-zero and broader missing mass distribution in the reconstructed events. The normalization mode tends to have lower q^2 and higher E_μ^* than the signal. The decay time distributions of the candidates from both modes follow the exponential distribution of B_c^+ decay time.

Based on the LQCD prediction for $R(J/\psi) = 0.2582(38)$ [2] and the world-average branching fraction for $Br(\tau^+ \to \mu^+\, \nu_\mu\, \bar{\nu_\tau}) = 0.173937(384)$ [33], the expected ratio of signal to normalization events is approximately 0.045, before corrections for differences in detection efficiency.

8 Misidentification background

The production cross sections of the lighter B mesons (B^+, B^0, B_s) are more than two orders of magnitudes larger than that of the doubly-heavy B_c^+ meson. The largest background comes from the inclusive decays of lighter B mesons to J/ψ + hadrons, where a pion or kaon is misidentified as a muon. This background is referred to as the misidentification (misID) background. Although the muon misidentification rate is relatively low due to the powerful particle identification system at LHCb, the large production cross sections of lighter B mesons result in a significant number of misidentified events passing the selection for the nominal data sample.

A data control sample of inclusive $B \to J/\psi\, X$ decays is used to model this background. The sample is created by combining J/ψ candidates with a track that fails muon identification (`isMuon`, which is defined by various requirements of hits in the muon stations at different momenta) while repeating the signal reconstruction. This sample is distributed with higher missing mass, high B_c candidate lifetime, and low q^2. This control sample has unpaired candidate tracks enriched in pions, kaons, protons, electrons, and ghost tracks. A ghost track is wrongly reconstructed from hits in the detector that are aligned but not formed by a real particle.

To represent the misID background in the nominal data sample, the data control sample needs to be reweighted according to the probabilities of different tracks passing the muon PID requirement. Weights are produced through an unfolding procedure that identifies tracks by their probabilities of passing the muon PID requirement using the PID rates derived from calibration samples. The procedure is described in more detail in Section 8.2.

The PID calibration samples and tools have been developed by the LHCb collaboration for analyses that require PID calibration input [34]. Events are selected from data to build high-purity samples of muons, electrons, pions, kaons, and protons, e.g. a $J/\psi \to \mu^+\mu^-$

Quantity	Thresholds
p(MeV)	$3000, 6000, 10000, 15600, 27000, 60000, 100000$
η	$1.5, 3.6, 5$
nTracks	$0, 200, 600$

Table 10: Binnings for PID samples in p, η, and nTracks, used for both PIDCalib and the inclusive J/ψ sample from which ghost and electron misID efficiencies are derived.

sample for muons, a $J/\psi \to e^+e^-$ sample for electrons, a $D^0 \to K^-\pi^+$ sample for pions and kaons, and the $\Lambda^0 \to p\pi^-$ and $\Lambda_c^+ \to pK^-\pi^+$ for protons. There is no data-driven ghost sample, and the electron sample is not compatible with the uBDT, so for these samples we use the inclusive J/ψ MC with the same selections as in the nominal fit, with the exception of allowing events that both pass and fail `isMuon`. The sPlot method [35] is used to statistically subtract background contributions from these events, which are weighted to reproduce the kinematic and PID variables of particles of interest. These samples are then used to measure the efficiencies for various PID selections and misidentification rates.

The PID samples are binned in the variables p, η, and nTracks, shown in Table 10.

8.1 PID sorting

The candidate tracks in the control sample are sorted into six different categories: pion-enriched, kaon-enriched, proton-enriched, electron-enriched, muon-enriched, and ghost-enriched. This is done by applying PID sorting criteria to the candidate tracks.

First, the tracks identified by ProbNNghost as ghost candidates are sorted into the ghost-enriched category. Second, the remaining tracks which are identified by ProbNNe as electron candidates are sorted into the electron-enriched category. Third, the remaining tracks which have a significant log-likelihood for the muon hypothesis are sorted into the muon-enriched category. Fourth, the remaining tracks which pass a ProbNNk requirement for kaon candidates and have a significant log-likelihood difference of kaon over proton hypotheses are sorted into the kaon-enriched category. Fifth, the remaining tracks which are identified by ProbNNp as proton candidates and have a significant log-likelihood difference of proton over kaon are sorted into the proton-enriched category. Lastly, all remaining tracks are sorted into the pion-enriched category.

These PID sorting criteria have been optimized to produce relatively pure enriched sub-samples, although cross-contamination between subsamples is later handled by the unfolding procedure. The selections are summarized in Table 11, in the form of a filter ordered from top to bottom.

8.1.1 Trimuon corrections

In Run-1, it was observed that the mis-ID efficiency of `IsMuon` for events with a J/ψ is higher than that computed from PIDCalib samples at high momentum. This effect was re-examined in Run-2 using a $B \to J/\psi K^*$ sample in order to model kaon and pion mis-ID efficiencies, and a $\Lambda_b^0 \to J/\psi p^+K^-$ sample to model the mis-ID efficiency of the proton. Further studies suggest that the change in efficiency at high momentum may not be due to this effect alone, and thus no correction was implemented to correct for this effect

Table 11: The PID sorting criteria for categorizing the misID data control samples, in the form of a filter from top to bottom.

Category	PID sorting criteria
ghost-enriched	ProbNNghost > 0.2
electron-enriched	ProbNNe > 0.1
muon-enriched	DLLmu > 2
kaon-enriched	ProbNNk > 0.1 & (DLLk-DLLp) > 2
proton-enriched	ProbNNp > 0.1 & (DLLp-DLLk) > 2
pion-enriched	None of the above

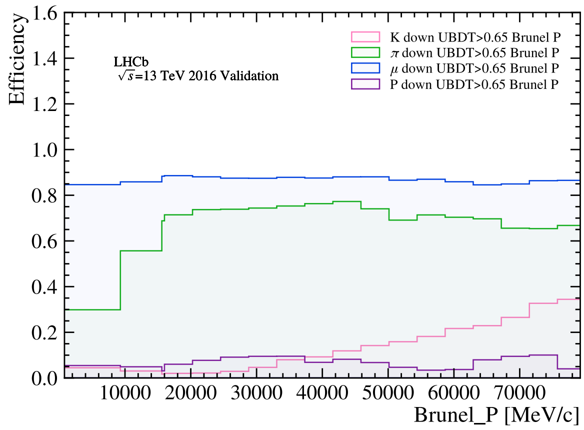


Figure 10: Efficiency of the selection requiring muon BDT > 0.65 as a function of the track momentum p from the 2016 MagDown PIDCalib samples. The efficiency curves for the kaon, pion, and muon are depicted in red, green, and blue respectively.

in the final fit (see details in Appendix 26. A systematic uncertainty was evaluated to account for this effect (Section 17.12.

8.2 Unfolding

The LHCb PID calibration samples and tools are used to measure the probabilities of identifying and misidentifying a given track [34], in the form of efficiency histograms binned in total momentum, pseudo-rapidity, and track multiplicity, simultaneously. The uBDT efficiencies are given for the kaon, pion, proton, and muon as a function of track p and p_T in Figures 10 and 11, respectively, and the efficiency of the total PID selection is given in Figure 12.

The average number of tracks in the tagged categories, $N_{\hat{h}}$, should be related to the true number of each species, N_h, by the probability matrix in Equation 46, where the probability that a track of species h fails `isMuon` and passes the selection for $\hat{h}'$ is denoted

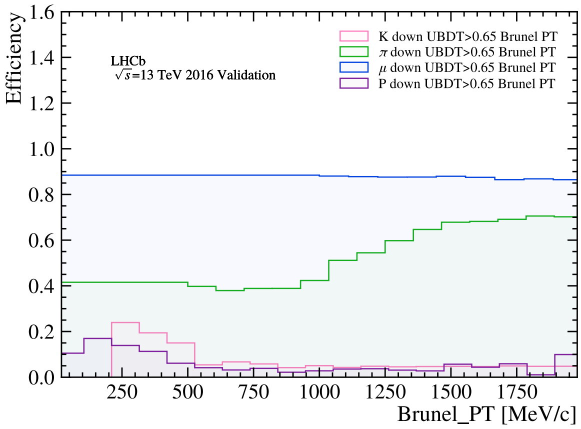


Figure 11: Efficiency of the selection requiring muon BDT > 0.65 as a function of the track momentum p_T from the 2016 MagDown PIDCalib samples. The efficiency curves for the kaon, pion, and muon are depicted in red, green, and blue respectively.

as $P(\hat{h}'|h)$:

$$\begin{pmatrix} N_{\hat{\pi}} \\ N_{\hat{K}} \\ \vdots \\ N_{\hat{g}} \end{pmatrix} = \begin{pmatrix} P(\hat{\pi}|\pi) & P(\hat{\pi}|K) & \cdots & P(\hat{\pi}|g) \\ P(\hat{K}|\pi) & P(\hat{K}|K) & \cdots & P(\hat{K}|g) \\ \vdots & \vdots & \ddots & \vdots \\ P(\hat{g}|\pi) & P(\hat{g}|K) & \cdots & P(\hat{g}|g) \end{pmatrix} \begin{pmatrix} N_{\pi} \\ N_{K} \\ \vdots \\ N_{g} \end{pmatrix}. \tag{46}$$

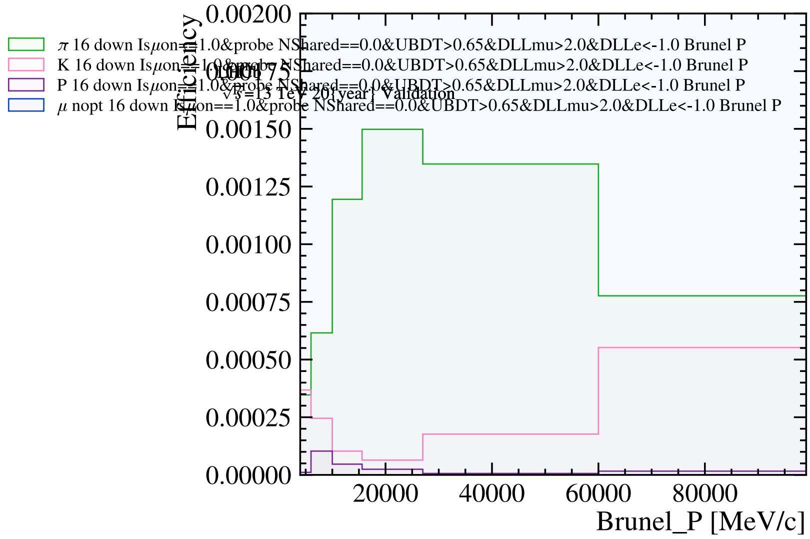
The number of fake muons is a function of $P(\hat{\mu}|h)$ and the true numbers of other species in the sample N_h, which can be solved for by using 46. A naive matrix inversion may not be possible if the matrix is singular. In some cases negative terms in the matrix inverse can lead to nonsensical negative numbers of unfolded events, or have large statistical fluctuations [36]. Instead, an iterative unfolding method using the RooUnfoldBayes tool [37] provides a more stable solution. `RooUnfoldBayes` repeatedly applies Bayes' theorem in order to invert the response matrix, stopping before reaching the "true" but unstable inverse. We use four iterations, which is standard. With the control sample sorted into six enriched categories and the matrix of probabilities from PID calibration samples for identifying or misidentifying six species of tracks, the unfolding decomposes the control sample into components representing each species.

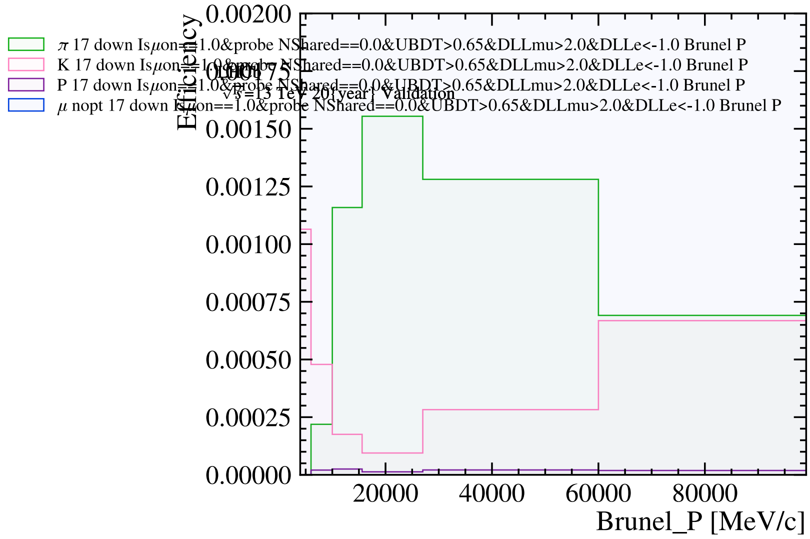
The probabilities $P(\hat{g}|h)$ are obtained by subtracting from 1 the probabilities that a given species is not identified as a ghost, i.e.

$$P(\hat{g}|h) = 1 - \sum_{\hat{h}' \neq \hat{g}} P(\hat{h}'|h). \tag{47}$$

Table 12 gives the estimated proportions of each species before and after unfolding. We see a decrease in the number of pions and muons and an increase in the numbers of all other species.

Equation 46 can be solved for the true number of species h, N_h, which we then use to calculate the a priori probability that a random track is species h, given by





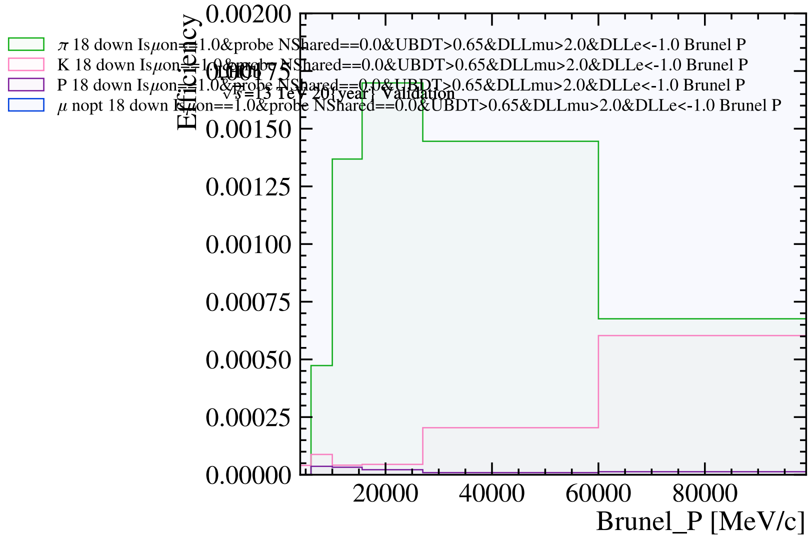


Figure 12: Efficiency of the selection requiring muon BDT > 0.65, IsMuon, NShared$== 0$, DLLmu> 2.0, DLLe<-1, which is the nominal PID selection, as a function of the track momentum p_T from the MagDown PIDCalib samples for 2016, 2017, and 2018 (top, middle, and bottom). The efficiency curves for the kaon, pion, and muon are depicted in red, green, and blue respectively.

Table 12: Proportions of species before and after misID unfolding from 2016.

Species	Before Unfolding	After Unfolding
π	0.544	0.755
K	0.303	0.162
p	0.013	0.004
μ	0.045	0.000
e	0.010	0.000
g	0.084	0.080

Year	UMD	UCAS
2016	1.017	1.016
2017	1.090	1.063
2018	1.051	1.008

Table 13: A comparison of the χ^2/DOF fits using the UMD and UCAS mis-ID templates by year.

$P(h) = N_h / \sum_{h'} N_{h'}$. With the conditional probabilities $P(\hat{h}'|h)$, we can calculate the probability $P(h|\hat{h}')$ that a track marked as $\hat{h}'$ is actually h using Bayes' theorem:

$$P(h|\hat{h}') = \frac{P(h) \times P(\hat{h}'|h)}{\sum_{h''} P(h'')P(\hat{h}|h'')}. \tag{48}$$

Along with the fake rates $P(\hat{\mu}|h)$, this gives the probability that a track marked $\hat{h}'$ is falsely classified as a muon. Using this information a weight can be computed, representing the probability that a given track marked as $\hat{h}'$ is falsely classified as a muon–

$$w_{h'} = \sum_h \frac{N_h}{\sum_{h'} N_{h'}} \times P(h|\hat{h}') \times P(\hat{\mu}|h). \tag{49}$$

This weight is applied to all tracks in the inclusive $J/\psi X$ sample in order to generate the mis-ID fit template.

8.2.1 Mis-ID cross check

An alternative method for calculating the relative fractions of the mis-ID components and obtaining their distributions is described in Appendix 29.1. The two methods give different shapes, as can be seen in Figures 13 and 14. The major differences in the shape seems to arise from the difference in kaon estimates, where the UCAS method results in far fewer kaons than the UMD unfolding. The UCAS template does not account for ghosts. A fit using the UCAS mis-ID templates resulted in a significant improvement in fit quality for 2017 and 2018 according to the χ^2 values from Table 13.

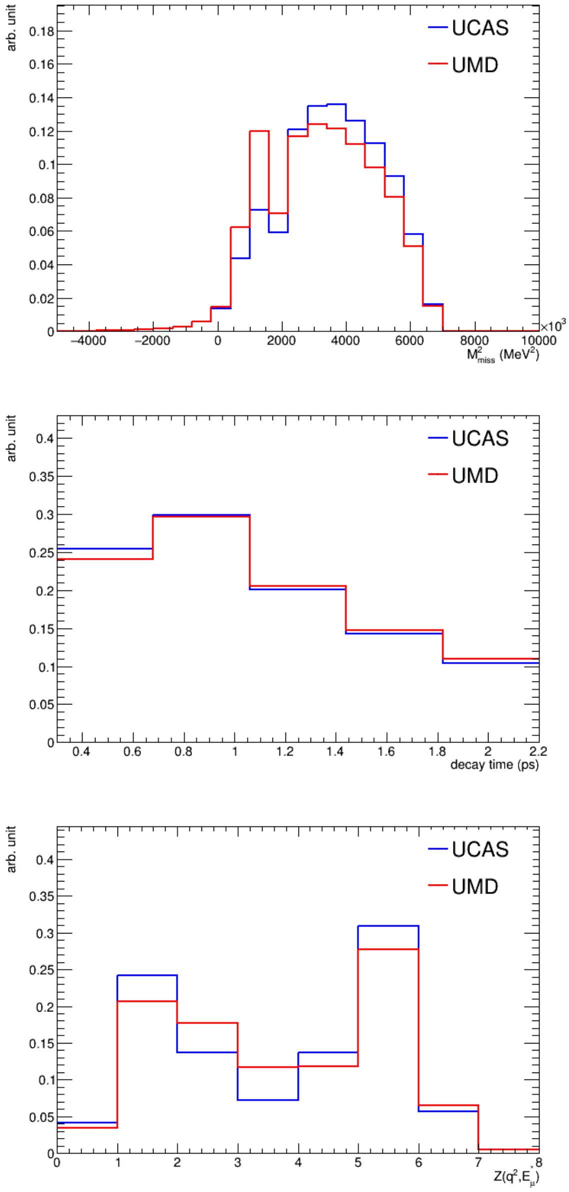
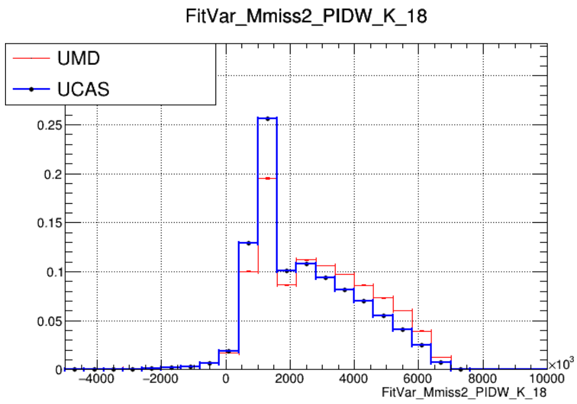


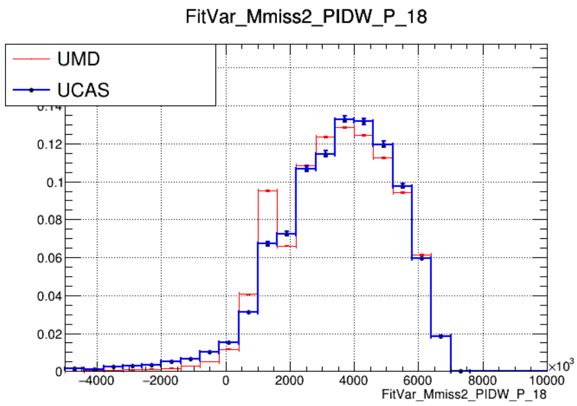
Figure 13: Comparison of the resulting templates using the two mis-ID methods.

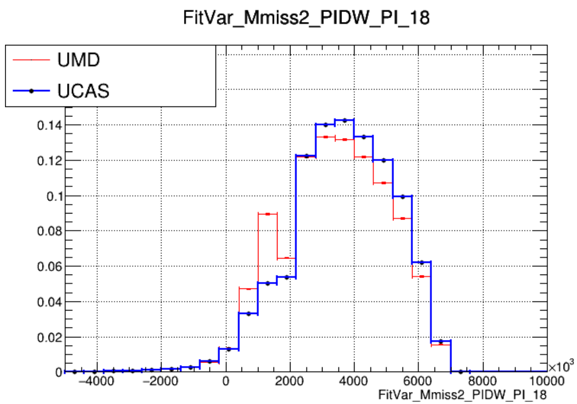
The two methods give compatible fit results within the statistical uncertainties, with more details given in Section 17.5. The final result averages the two values.

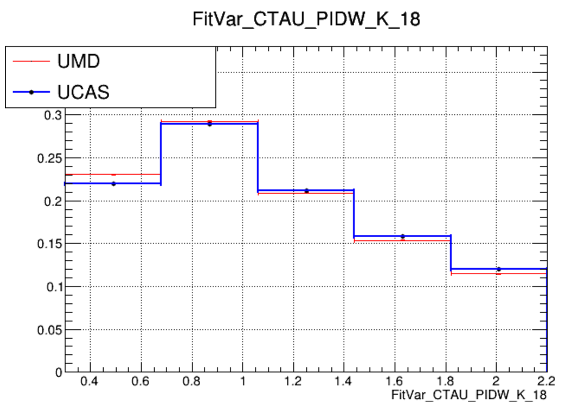
8.3 Ghost tracks

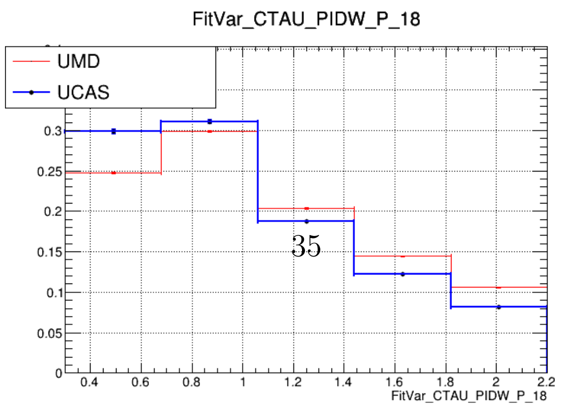
The J/ψ + hadron control sample contains some fraction of "ghost" tracks, which are tracks reconstructed by combining segments of different tracks and do not correspond to real particles. This class of tracks needs to be accounted for during the unfolding and weighting procedure. Without a `PIDcalib` sample of ghost tracks to select from data, we model the PID efficiencies $P(\hat{h}'|g)$ for ghost tracks by using inclusive J/ψ simulation with tracks whose hits fail to be truth-matched with simulated particles, i.e. with MC `TRUEID` $= 0$. From these candidates we extract the efficiency with which ghosts pass the


FitVar_Mmiss2_PIDW_K_18
UMD
UCAS
×10^3
FitVar_Mmiss2_PIDW_K_18


FitVar_Mmiss2_PIDW_P_18
UMD
UCAS
×10^3
FitVar_Mmiss2_PIDW_P_18


FitVar_Mmiss2_PIDW_PI_18
UMD
UCAS
×10^3
FitVar_Mmiss2_PIDW_PI_18


FitVar_CTAU_PIDW_K_18
UMD
UCAS
FitVar_CTAU_PIDW_K_18


FitVar_CTAU_PIDW_P_18
UMD
UCAS
FitVar_CTAU_PIDW_P_18

muon PID selection. We also generate efficiencies for the ghosts passing the other PID sorting requirements with the part of the sample that fails `isMuon`. Similarly, the electron sample is not compatible with uBDT, so we also model the electron misID efficiencies using the inclusive J/ψ MC, with MC `TRUEID` required to be that of an electron instead.

As another source of background which is modeled in an ad hoc way using MC, the ghost contribution to the misID background was another source of scrutiny. Additionally, the lack of ghosts is the main discrepancy that the nominal mis-ID template has with the UCAS mis-ID template, which does not explicitly allow for a separate ghost contribution. Thus, it is important to understand the effect of ghosts (or lack thereof) on the fit quality and the value of $R(J/\psi)$.

A few different ghost configurations were considered. The most extreme was to remove the ghosts at the unfolding stage, this shifted $R(J/\psi)$ by less than 10% of the statistical uncertainty and did not significantly change the fit quality. Other less extreme methods included scaling the ghosts down by 75%, 50%, assuming ghosts can only be misidentified as muons and not as any other species, and floating the ghost component. All these methods resulted in a smaller shift in $R(J/\psi)$ than the fit configuration with ghosts removed, and all methods preserved similar fit quality.

8.4 Muon uBDT efficiencies

Many of the input variables (listed in Table 10 from [38]) used in the muon uBDT algorithm are not included in the standard PID calibration samples. These additional variables are normally used as input to PID algorithms, whose outputs are saved in the standard processed samples. To evaluate the probabilities of misidentifying muons using the muon uBDT, the PID calibration samples were re-processed to include all input variables. The muon uBDT is then applied offline to the re-processed samples. With the muon uBDT output available, the re-processed samples are then used to produce the misID efficiencies histograms for the unfolding.

8.5 The $B^+ \to J/\psi K^+$ peak

The misID background contains the fully reconstructed $B^+ \to J/\psi K^+$ decays in which the kaon is misidentified as a muon. Because these decays have no neutrinos and are fully reconstructed, their distributions peak in invariant mass, missing mass, and E^*_μ. The invariant mass of the J/ψ + track system can be recomputed after assuming the kaon hypothesis for the unpaired muon candidate track. The invariant mass distribution can be subsequently drawn in a stacked histogram with the unfolding weight for each species. The peak from $B^+ \to J/\psi K^+$ decays is prominently seen around the B^+ mass. Since the peak is due to true kaons, it is used as an important reference to check the unfolding results. The stacked contributions of the unfolding-weighted invariant mass distributions is shown in Figure 15, and most of the peak is shown to be the kaon component in red. This reference is used to optimize the PID sorting criteria and correct the over-estimation of ghost tracks from previous iterations of unfolding.

A postfit comparison is done between the data and model to validate the amount of $B^+ \to J/\psi K^+$ seen in data (more descriptions of the postfit procedure are given in 16.1). The comparison of visible mass distributions of the stacked templates against data in the high lifetime region is shown in Figure 16 and shows good agreement, validating

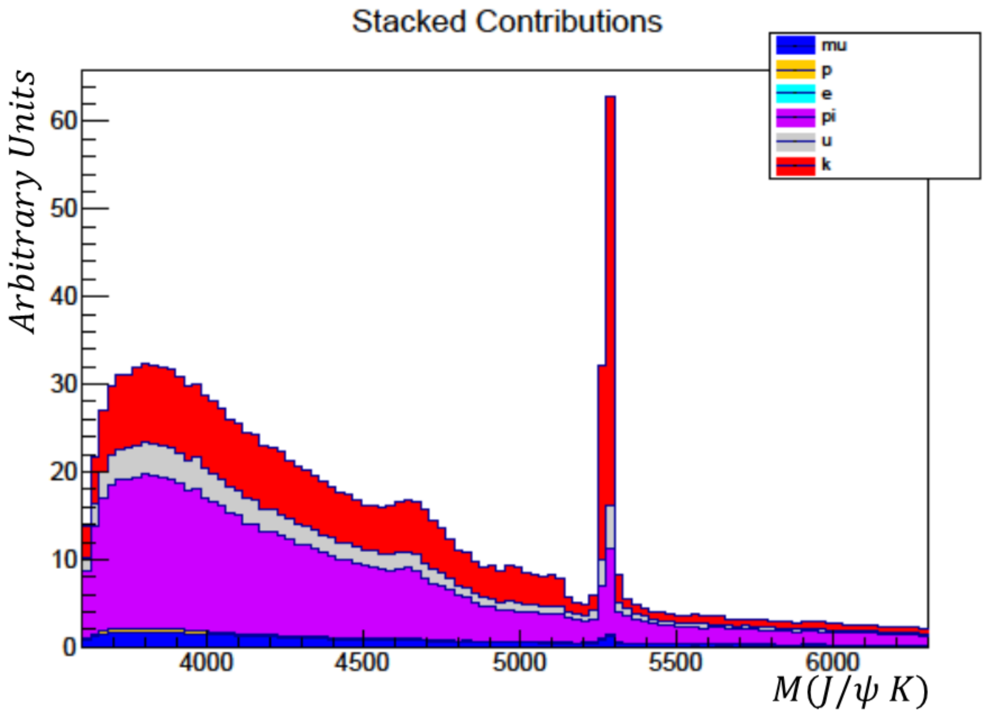


Figure 15: The stacked contributions of unfolding-weighted invariant mass distributions for each species: kaon (red), pion (magenta), ghost (grey), muon (blue), proton (yellow), and electron (cyan).

the fit procedure and the modeling of the mis-ID shape. To calculate the visible mass, it is assumed that the muon has the mass of the kaon and the energy is recomputed accordingly.

This figure further validates that the amount of smearing chosen by the fit, as the mis-ID template generated with default smear does not perform as well in this test.

Figure 17 shows that the kaons have a longer lifetime distribution than the other mis-ID components. Moreover, we note that the UCAS template has fewer kaons than the nominal template, and that a double smearing is required to bring the $J/\psi K$ peak height from the model into agreement with that from the data, possibly pointing to an excess of the kaon component. Though the size of the low lifetime enhancement in mis-ID due to the bug is large compared to the difference in the distribution of the kaon lifetime relative to that of other mis-ID components and thus probably cannot be solely explained by a different kaon fraction, further investigations into the effect of changing the kaon fraction in the mis-ID component on the fit were performed since this is *a* way the mis-ID lifetime distribution may vary.

Reducing the amount of kaons by a fixed amount of 30% did not change the fit quality or the blinded value of $R(J/\psi)$. Floating the kaon component with $\sigma = 0.3$x the total number of estimated kaons as a constraint resulted in a large decrease in the amount of kaons from about 20% of the total mis-ID component in the nominal fit to about 10% of the total mis-ID component when floated. This, however, had a negligible effect on both the value of $R(J/\psi)$ and the fit quality.

Ultimately, our modeling of the mis-ID quantity and shape is validated by the postfit plot of the $J/\psi K$ visible mass peak, which in Figure 18 shows good agreement with data

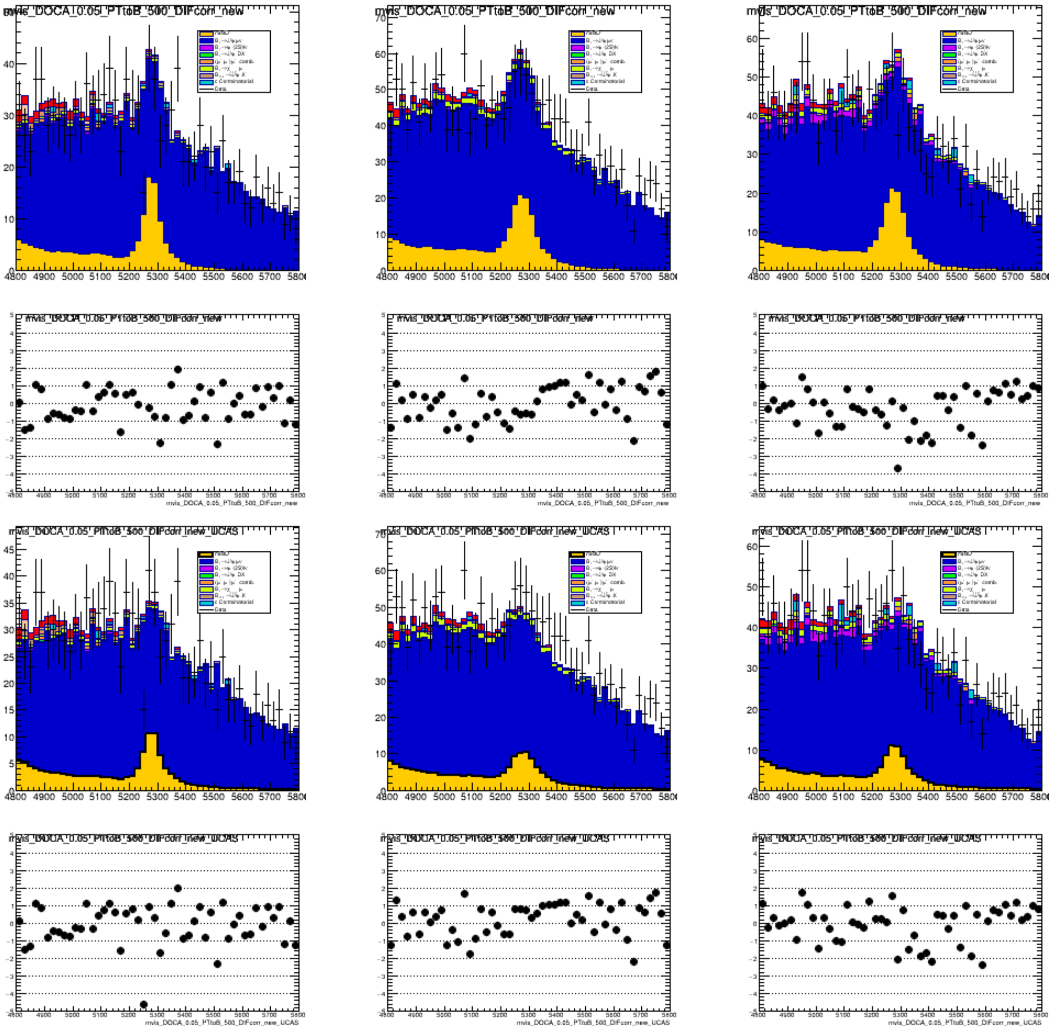


Figure 16: Mass of the $J/\psi K$, using fitted yields. The plot is done in the region of lifetime longer than 1 ps, which is enriched in mis-ID background. The top row uses the UMD mis-ID template, and the bottom row uses the UCAS mis-ID template. The plots are ordered by year going from 2016 to 2018 from left to right.

in all years. As discussed above, any decreases in the fraction of kaons in the mis-ID template do not change the measurement of $R(J/\psi)$ or the fit quality appreciably.

8.6 Decay in-flight contribution

Pions and kaons can decay in-flight with a small change of angle in the lab frame, and lead to real muons registering signatures in the muon stations. Approximately half the rates of muon misidentification from pions and kaons are found to be due to decays in-flight [25]. When this happens between the tracking systems upstream and downstream of the magnet, tracks are misreconstructed as muons, but with incorrect momentum.

Because the misID data control sample is selected from tracks that fail muon identification, all tracks in this sample have no signature in the muon stations. Therefore, the data control sample is not affected by decay-in-flight misreconstruction and does not fully model the nominal data sample.

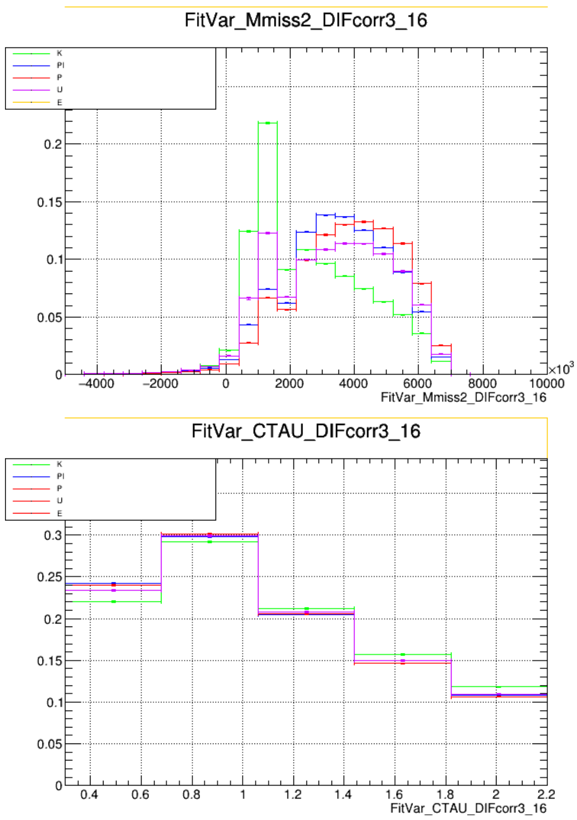


Figure 17: Separated misID components.

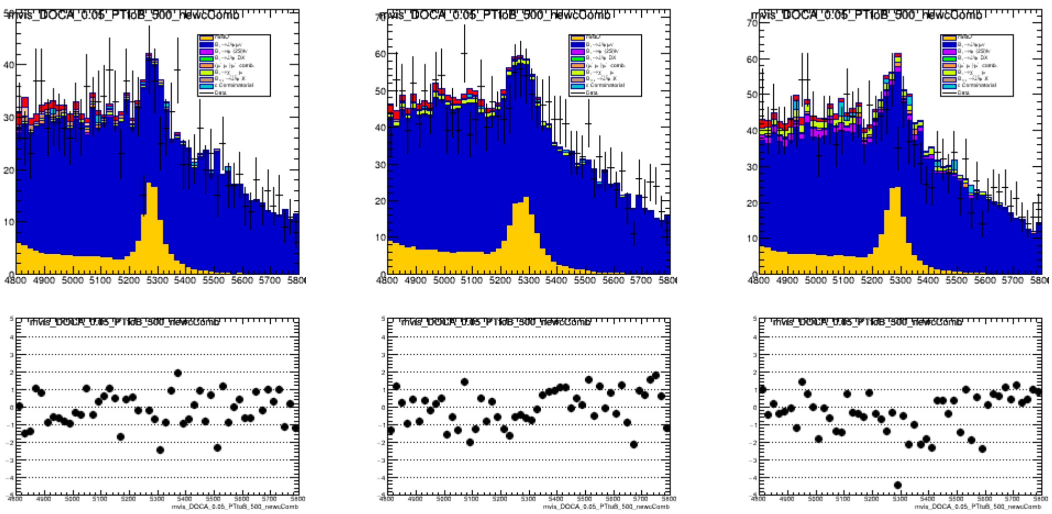


Figure 18: $J/\psi K$ visible mass peak in mis-ID enriched region.

8.6.1 Decay in-flight smearing

A smearing procedure was developed as in reference [25] to model the momentum misreconstruction of decay in-flight pions and kaons. Simulated samples of pions and kaons which are misidentified as muons are used to build a correction map of true momentum vs misreconstructed momentum ratios. The pion and kaon components from the unfolding are then randomly modified by the momentum ratios, and the rest frame variables are recomputed with the smeared momentum. This creates a smeared misID template to be used in the fit. The fit is allowed to extrapolate between the unsmeared template, the smeared template, and a "double-smeared" template which is created by doubling the weight of the smeared template and using negative weights for the unsmeared template. Figure 19 shows the distributions of missing mass and muon energy in comparison between smeared (in red) and unsmeared (in blue) misID background models from the misID data control sample. The peaks from $B^+ \to J/\psi K^+$ decays are shown to be broadened by the smearing.

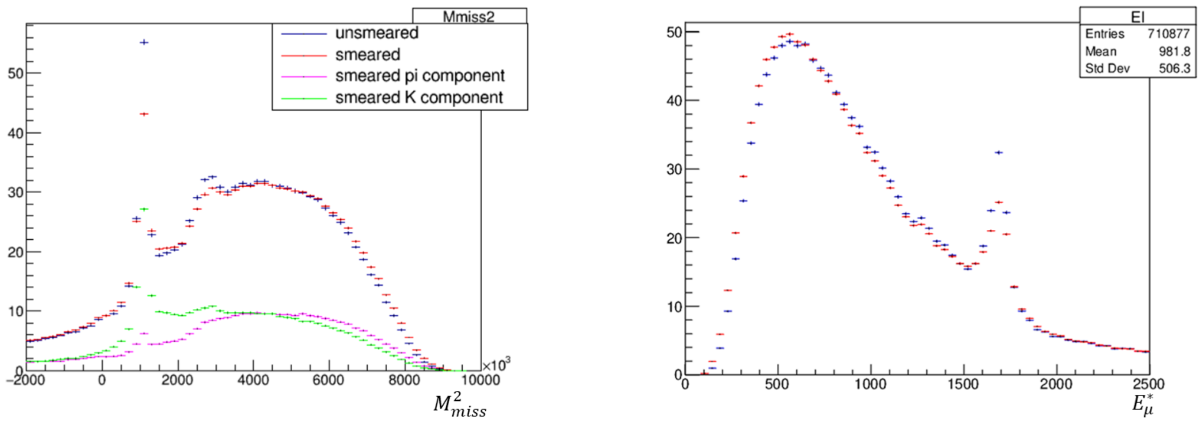


Figure 19: The distributions of missing mass and muon energy for the smeared (in red) and unsmeared (in blue) misID background models from the misID data control sample. The peaks from $B^+ \to J/\psi K^+$ decays are shown to be broadened by the smearing.

8.6.2 Decay in-flight efficiency corrections

The presence of decay in-flight candidates additionally results in PIDCalib underestimating pion and kaon mis-ID efficiencies. PIDCalib samples are generated with a strict mass window cut, and the sPlot technique assumes there is no correlation between the discriminating and control variables. However, this is false for the case of muon PID selections in pions and kaons, since most decay in-flight mis-ID candidates live in the low mass tail region.

The data is divided into subsamples which pass or fail the PID cuts, and the efficiency of the mass window cut is computed for each of these groups which have passed or failed the PID cuts and are in a certain kinematic bin. In each kinematic bin, a simultaneous fit to the D^0 mass and $\delta m = m_{D^*} - m_{D^0}$ is performed in order to estimated the correction to the PID efficiency from the decay in-flight events, taking into account events outside the mass window using an MC sample. The fits are performed accounting for several sources of background (Table 14).

Background	D^0 mass modeling	δm modeling
Partially reconstructed D^0s	Exponential	MC
Combinatorial background from soft πs	MC	Power law
Full combinatorial background	Exponential	MC

Table 14: Backgrounds and their respective modeling strategy in the fits to the D^0 mass and δm mass.

The corrective factor is

$$N_{data,corr} = N_{data}(1 + \frac{N_{tail,MC}}{N_{core,MC}}), \tag{50}$$

where "core" and "tail" refer to the region within and without the mass window, respectively.

The fits in each kinematic bin are shown in Appendix 27.

9 Combinatorial ($J/\psi + \mu$) background

The ($J/\psi + \mu$) combinatorial background is formed between a J/ψ from a b-hadron decay and a muon candidate from a separate decay. This background is modeled with Monte Carlo simulations of $B_{d,s} \to J/\psi X$ decays, since the three muon signature has no wrong sign combination that could be used to model the background in a data-driven way. Because the proton-proton collisions produce $b\bar{b}$ pairs, one b-hadron decay is accompanied by its partner anti b-hadron. With the requirement of a significant impact parameter from the PV, the combinatorial muon is generally produced by the partner b-hadron.

By isospin symmetry, $B_u \to J/\psi X$ decays should have the same kinematics as the B_d decays. Therefore, the $B_{d,s}$ simulations are sufficient for modeling this combinatorial background.

This analysis uses the same `DEC` files as in Run-1, with `12442012` modeling B_d decays and `13442012` modeling B_s decays. These `DEC` files focus on possible decays to J/ψ, $\psi(2S)$, and $\chi_{c1,2}$, using measured branching fractions from the PDG when possible [39]. When measurements were not available, a phase space factor along with $SU(3)$ flavor symmetry guidelines were used to estimate branching fractions.

To create these cocktails, we selected MC samples using a custom full event cut tool `JpsiLeptonInAcceptance`, which required a $J/\psi\mu$ pair with a DOCA of less than 30 μm. This event cut has about a 2% efficiency.

The simulation includes a cocktail of many B decays to J/ψ, $\psi(2S)$, and $\chi_{c1,2}$. The relative fractions in the nominal fit of $B_d \to J/\psi X$ and $B_s \to J/\psi X$ is fixed according to the production cross sections of B^+, B^0 and B^0_s taken from the PDG to the value

$$\begin{aligned} \frac{f_s}{f_u + f_d} &= \frac{(10.0 \pm 0.8)\%}{(40.8 \pm 0.7)\% + (40.8 \pm 0.7)\%} \\ &= 0.12 \pm 0.01.1001[39] \end{aligned} \tag{51}$$

An investigation to improve the $J/\psi\mu$ combinatorial background modeling is described in Section 17.9. This approach was ultimately not used in the nominal fit, but was used to estimate the systematic error associated with the modeling of this background. A further description of the validation of this correction is described in Appendix 19.

The sub-component of the $(J/\psi + \mu)$ combinatorial background with the muon originating from a prompt D is fitted separately. Initially, this was due to investigations of disagreements in the low lifetime region, since this background appears to have shorter lifetime. The proportion of this background relative to other $(J/\psi + \mu)$ events from simulation is difficult to model, and so the fit is allowed to choose the relative proportion.

10 Combinatorial charm background

Backgrounds from spurious combinations in real B_c events could be enhanced due to kinematic correlations between the B_c meson and the associated charm hadron produced in the fragmentation process from the other half of the $c\bar{c}$ pair providing the charm half of the B_c^+ meson [40]. The contribution from this background is referred to as the combinatorial charm background hereafter.

When the associated charm hadron decays (semi)muonically, the resulting muon can be wrongly combined with a J/ψ from the associated B_c^+ decay, hence the term combinatorial charm background. The topology of this background is shown in Figure 20.

The shape of the fit variables for the combinatorial charm background can be derived from the simulation. After reconstruction, the MC sample is truth matched to require the bachelor muon's parent to be a D, D^0, D_s, Λ_c, and to not have a B_c as an ancestor or to be related to the J/ψ. In order to enhance the statistics of the simulation samples, the normalization, signal and $B_c \to \chi_{c1,2}\mu\nu$ MC samples for the 2016, 2017, and 2018 productions were combined. The template distributions are shown in Figure 21. Notably, the missing mass distributions of the combinatorial charm and that of the $B_d \to J/\psi X$ combinatorial share similarities, while the lifetime distribution of the charm combinatorial is heavily shifted towards low lifetime, far lower than any other template used in this fit. This feature appears robust against different production kinematics (see Appendix 25), and arises from the fact that the bachelor muon and the D have low p_T and are produced

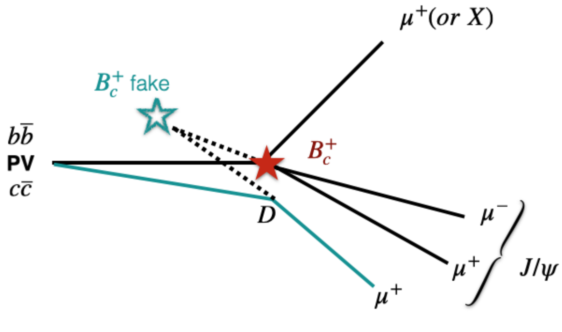


Figure 20: Depiction of the topology of a combinatorial charm background event. The real B_c^+ vertex is depicted in red, and the B_c^+ can decay into any $J/\psi X$. The D emerges from the primary vertex with trajectory shown in blue with momentum similar to that of the B_c^+, and the produced muon and the J/ψ are wrongly vertexed to from a false B_c^+ candidate shown in blue.

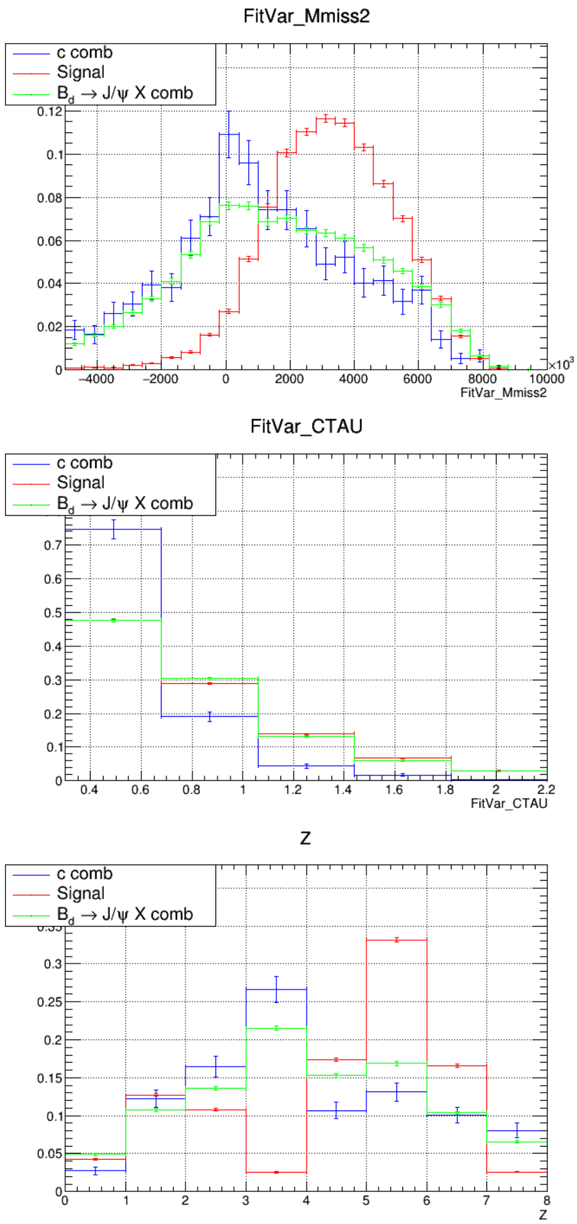


Figure 21: The distributions of the rest frame variables of the combinatorial charm (blue), signal (red), and $B_d \to J/\psi X$ combinatorial (green).

relatively colinearly to the B_c^+, resulting in a B_c^+ vertex which points towards the primary vertex and hence a shorter reconstructed lifetime.

10.1 TrackerOnly

A few key postfit plots in key kinematic variables were produced in the low ($m^2_{\mathrm{miss}} < 2\,\mathrm{GeV}^2$), middle ($m^2_{\mathrm{miss}} > 2\,\mathrm{GeV}^2$), and high ($m^2_{\mathrm{miss}} > 5\,\mathrm{GeV}^2$) missing mass regions. Despite overall improvements in the fit after the new lifetime selection and good agreement in the low to middle missing mass regions, a small excess in data remained unexplained by the model in the low lifetime distribution in the high missing mass region (defined as $m^2_{miss} > 5\,\mathrm{GeV}^2$), which is shown in the bottom row of Figure 23. Due to these issues, the

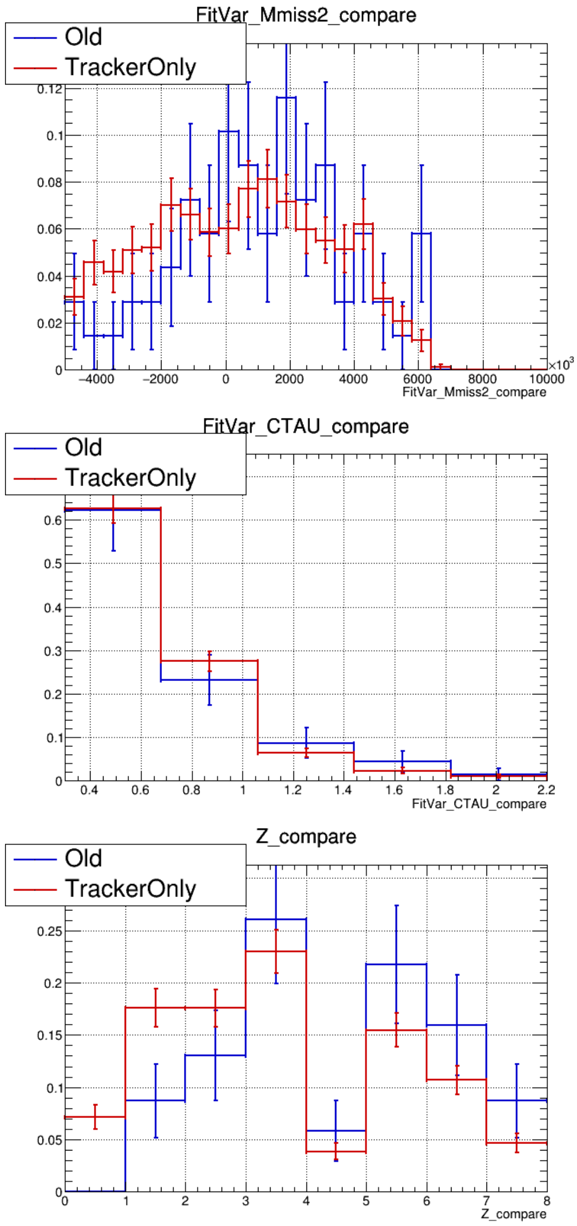


Figure 22: Comparison of old and new charm combinatorial templates with TrackerOnly statistics.

charm combinatorial, which is the shortest-lived by far of all the fit components, came under scrutiny. The old c combinatorial template suffered from very low statistics, and therefore was potentially affected by a sparse binning issue resulting in the underfitting of this background.

Having exhausted all our produced B_c full MC tuples, only the TrackerOnly normalization MC sample remained as a viable addition. Previously, an L0 emulation was performed using a BDT with some validation already performed, and the PID emulation was done using PIDCalib efficiencies. The addition of the TrackerOnly increased statistics from about 300 events shared between all years to about 40k events per year.

Adding TrackerOnly brings the fitted charm combinatorial yield to about 20-50 events per year. The overall fit quality and the value of $R(J/\psi)$ do not change significantly. The

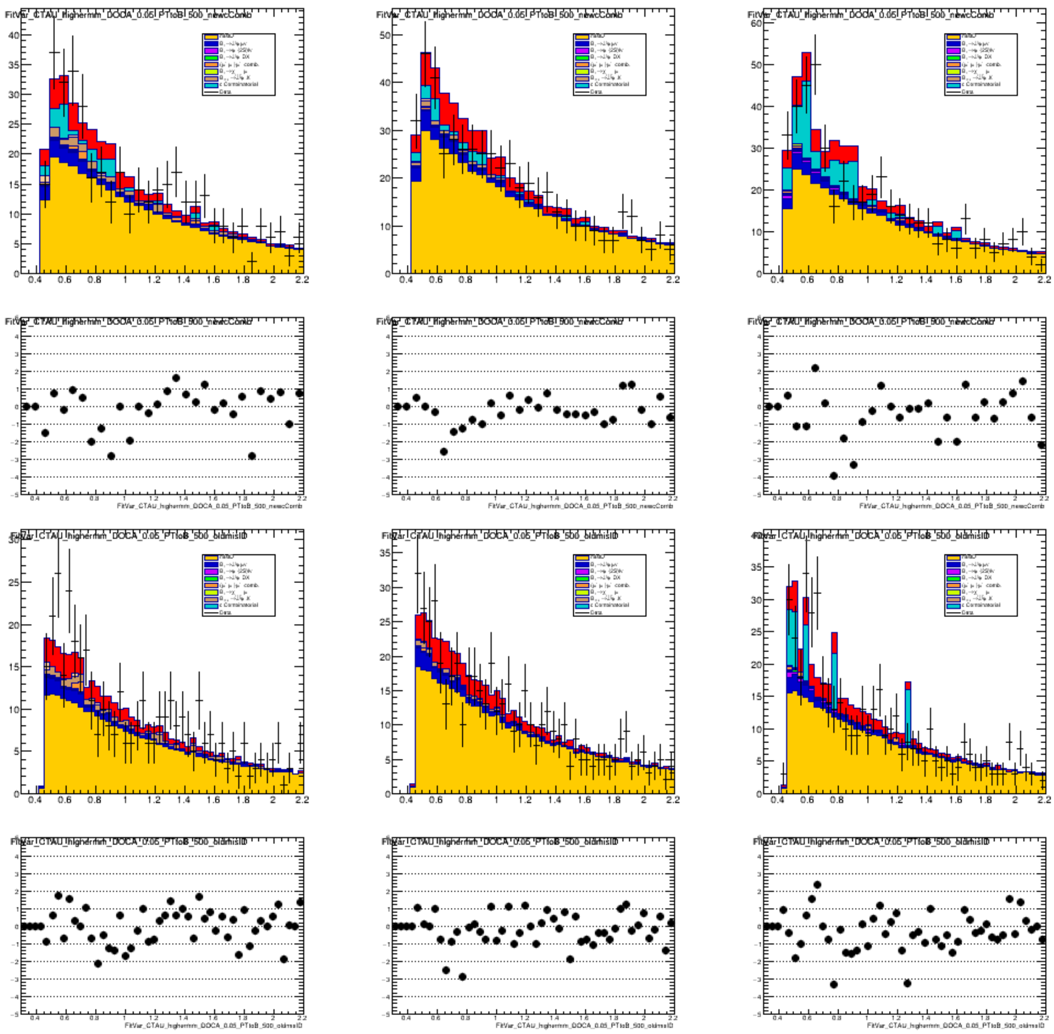


Figure 23: Improvements to low lifetime region in with postfit. Updated charm combinatorial on top, old charm combinatorial on the bottom. From left to right, 2016, 2017, and 2018.

postfit plots in Figure 23 shows improvement in the low lifetime excess.

10.2 Angular reweighting

An angular reweighting derived from data/MC comparisons is applied to the template:

$$w = e^{-0.17+1.61\phi} \tag{52}$$

where ϕ is the angle in the XY plane between the mother of the bachelor muon and the B_c. The effects of this reweighting are further explored in the form of a systematic uncertainty in Section .

11 Combinatorial $(\mu^+ + \mu^-)\mu^+$ background

The $(\mu^+ + \mu^-)\mu^+$ combinatorial background is due to random combinations of muon candidates forming fake J/ψ candidates. Because the randomly combined opposite-charge muons do not have correlated kinematics or share the same parent particle, their invariant mass distribution has no resonance structure. We can therefore model this background using data from the upper sideband of the J/ψ candidate invariant mass, assuming that the fit variables are uncorrelated with the J/ψ mass, which is true in the J/ψ mass window. While the signal selection requires J/ψ candidate mass to be between 3040 MeV and 3150 MeV, the upper sideband sample is selected from 3150 MeV to 3190 MeV. The lower sideband is not used to avoid selecting real J/ψ decays with large final-state radiation from muons.

The J/ψ candidate invariant mass distribution is fit with a double-sided Crystal Ball function for the J/ψ signal component

$$f\left(m; m_0, \sigma, \alpha_L, n_L, \alpha_R, n_R\right) = \begin{cases} A_L \cdot \left(B_L - \frac{m-m_0}{\sigma_L}\right)^{-n_L}, & \text{for } \frac{m-m_0}{\sigma_L} < -\alpha_L \\ \exp\left(-\frac{1}{2} \cdot \left[\frac{m-m_0}{\sigma_L}\right]^2\right), & \text{for } \frac{m-m_0}{\sigma_L} \leq 0 \\ \exp\left(-\frac{1}{2} \cdot \left[\frac{m-m_0}{\sigma_R}\right]^2\right), & \text{for } \frac{m-m_0}{\sigma_R} \leq \alpha_R \\ A_R \cdot \left(B_R + \frac{m-m_0}{\sigma_R}\right)^{-n_R}, & \text{otherwise} \end{cases} \tag{53}$$

$$\begin{aligned} A_i &= \left(\frac{n_i}{|\alpha_i|}\right)^{n_i} \cdot \exp\left(-\frac{|\alpha_i|^2}{2}\right) \\ B_i &= \frac{n_i}{|\alpha_i|} - |\alpha_i| \end{aligned} \tag{54}$$

and an exponential decay function for the combinatorial J/ψ background component. The misID background also contains combinatorial J/ψ candidate backgrounds. These are already accounted for in the misID template, and so in order to not double count we also do an estimate of the number of expected combinatorial J/ψ candidates in the misID template. The combinatorial J/ψ candidates that are also misID background candidates are about 20% of all combinatorial J/ψ candidates. In order not to double count, the misID-combinatorial events are subtracted from the scaling of the combinatorial J/ψ contribution. The pulls in Figure 25 are large, but the error on the number of background events is small compared to the error on background events in Figure 24. Any remaining systematic uncertainty due to choice of function for modeling is covered by Section **??**.

The fitted parameters are given in Figure 24 for data and Figure 25. The yield of the combinatorial J/ψ background in the signal region is estimated from the number of background events within the J/ψ candidate mass window. An alternative method using a double Gaussian rather than a Crystal Ball gives an alternative yield. The difference between the two yields is used as a constraint on the fitted yield, which floats between the two. The analysis was originally developed with intent to fix the yield, but observations of fit quality dependence on the estimate led to the choice to float the yield.

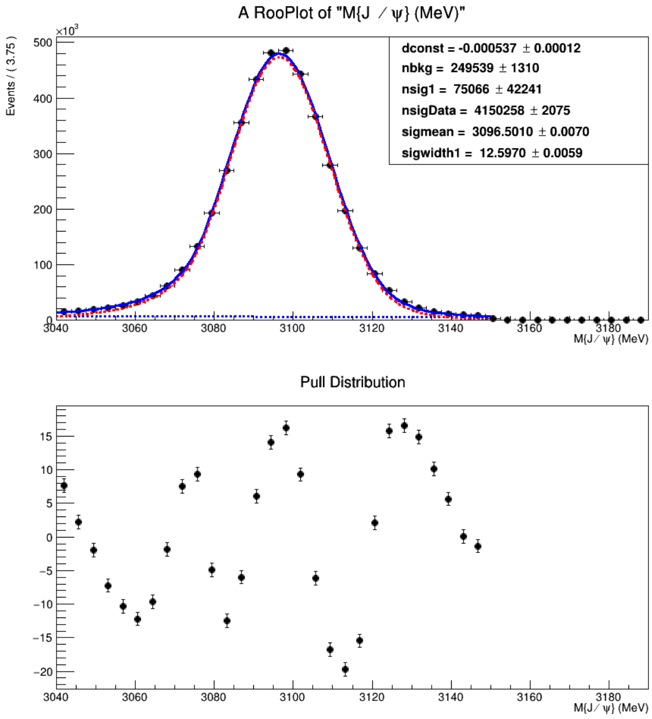


Figure 24: Fit to the J/ψ candidate invariant mass spectrum in data. The J/ψ signal component is in red, the combinatorial J/ψ background is in blue, and the data is in black. The lower panel shows the residual distribution. `dconst` is the exponential decay constant; `alphaL`, `alphaR`, `sigmean`, and `sigwidth1` are Crystal Ball function parameters. `nbkg` and `nsigData` are the total amount of estimated signal and background in the fit region.

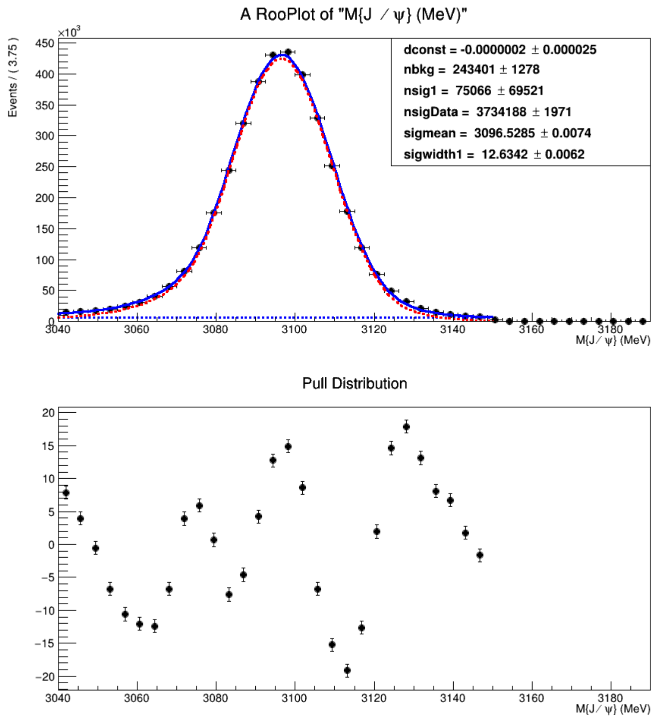


Figure 25: Fit to the J/ψ candidate invariant mass spectrum in the misID data sample. The J/ψ signal component is in red, the combinatorial J/ψ background is in blue, and the data is in black. The lower panel shows the residual distribution. `dconst` is the exponential decay constant; `alphaL`, `alphaR`, `sigmean`, and `sigwidth1` are Crystal Ball function parameters. `nsigData` and `nbkg` are the total amount of estimated signal and background respectively in the fit region, and reflect the number of events available in the misID data sample.

12 Double charm background

The other backgrounds from B_c^+ decays include $B_c^+ \to J/\psi\, D^+ X$ decays, in which the D meson decays leptonically or semileptonically. They are modeled with MC simulations of two-body $B_c^+ \to J/\psi\, D^{(*)}$ decays and quasi-two-body $B_c^+ \to J/\psi\, D_s^{**}$ decays. Additionally, the three-body $B_c^+ \to J/\psi\, D\, K$ decays are simulated with a phase space model. The compositions of the MC cocktails are shown in Table 16. The distributions of rest frame variables for these decays are shown in Figure 26.

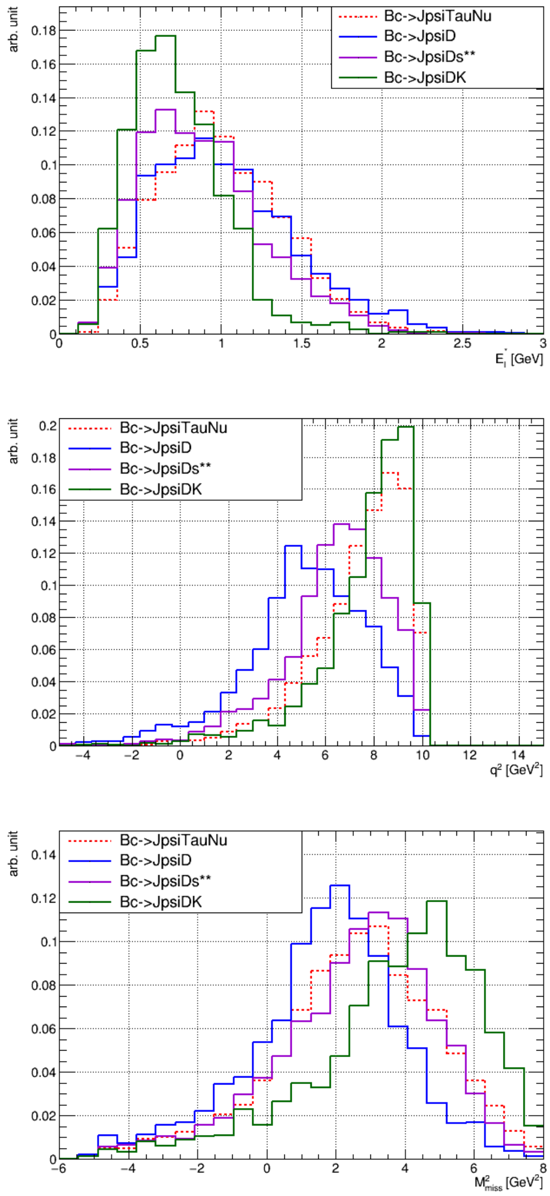


Figure 26: The distributions of rest frame variables for $B_c^+ \to J/\psi\, D\, X$ decays, in comparison with the signal mode (dashed red). The two-body mode is in blue, the quasi-two-body mode is in violet, and the three-body mode is in green.

The heaviest contribution covered by the quasi-two body MC is that of the D_{s2}^*, which has a mass of 2.57 GeV (Table 16). The higher mass contributions to this background

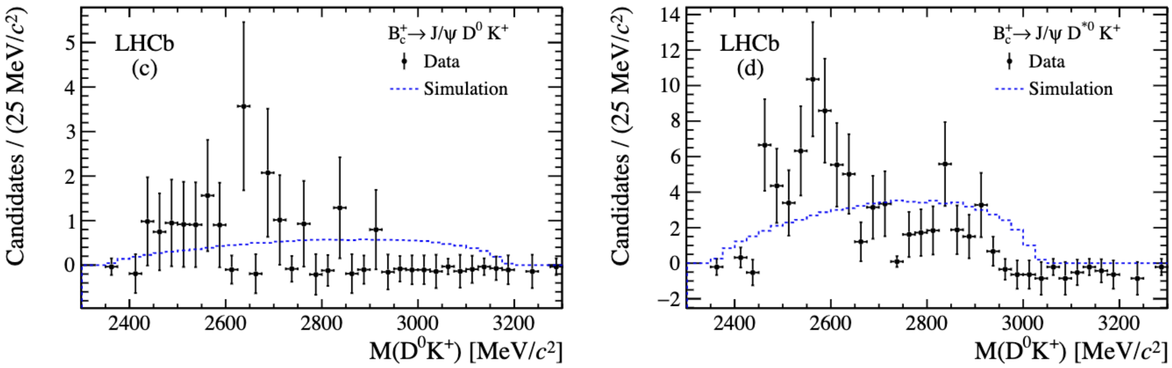


Figure 27: Figure taken from Figures 3(c,d) from [41]. These figures show the invariant mass of D^0K^+ reconstructed from $B_c^+ \to J/\psi D^0 K^+$ for the left figure, and $B_c^+ \to J/\psi D^{*0} K^+$ for the right figure.

Year	Constraint
2016	85^{+11}_{-12}
2017	78^{+9}_{-10}
2018	107^{13}_{-14}

Table 15: Constraints derived from $J/\psi D_s^{(*)+}$ fit in Figure 193.

also need to be accounted for–for example, there exists another peak at around 2.8 GeV in $m(D^0K^+)$, which is shown in Figure 27. We use the high mass region defined by $2.6 < m(DK) < 3$ GeV of the $B_c^+ \to J/\psi D\, K$ MC in order to model this region.

12.1 Constraints

The yield of the two-body component is fixed using results from Appendix 29.2.4. The estimation from fitting the $J/\psi D_s^{(*)+}$ peak is transformed into a constraint on the two-body decays using the composition of the two-body backgrounds, with only $D_s^{*+} \to D_s^+\gamma$ and D_s^+ decays are considered to be contributors to the peak in Figure 197, giving a total constraint of

$$\begin{aligned} N_{\text{twobody}} &= N(J/\psi D_s^+)/[\text{BF}(J/\psi D_s^+)_{\text{twobody}} + \text{BF}(J/\psi D_s^{*+})_{\text{twobody}} \times \text{BF}(D_s^{*+} \to D_s^+\gamma)] \\ &= 1.10 N(J/\psi D_s^+). \end{aligned} \tag{55}$$

The expected yield per year in the two body decays is shown in Table 15, with the asymmetric error bars accounting for possible contributions to $N(J/\psi D_s^+)$ from quasi-two body decays as well.

The relative fraction of $B_c^+ \to J/\psi D^{**}$ to $B_c^+ \to J/\psi D$ and the uncertainty of the fraction is derived from the values in [41] and [42]. Using

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D_s^+)}{\mathcal{B}(B_c^+ \to J/\psi \pi^+)} = 2.90 \pm 0.57 \pm 0.24 \tag{56a}$$

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D_s^{*+})}{\mathcal{B}(B_c^+ \to J/\psi D_s^+)} = 2.37 \pm 0.56 \pm 0.10 \tag{56b}$$

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D^0 K^+)}{\mathcal{B}(B_c^+ \to J/\psi \pi^+)} = 0.432 \pm 0.136 \pm 0.028 \tag{56c}$$

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D^{*0} K^+)}{\mathcal{B}(B_c^+ \to J/\psi D^0 K^+)} = 5.1 \pm 1.8 \pm 0.4 \tag{56d}$$

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D^{*+} K^{*0})}{\mathcal{B}(B_c^+ \to J/\psi D^0 K^+)} = 2.10 \pm 1.08 \pm 0.34 \tag{56e}$$

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D^{+} K^{*0})}{\mathcal{B}(B_c^+ \to J/\psi D^0 K^+)} = 0.63 \pm 0.39 \pm 0.08 \tag{56f}$$

and summing the contributions and multiplying by the ratio of isolation efficiencies, we have

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D)}{\mathcal{B}(B_c^+ \to J/\psi D^{**})} = 2.61 \pm 0.59 \pm 0.04. \tag{57}$$

The uncertainty is used as a constraint in the fit. The contribution of $B_c^+ \to J/\psi DK$ is constrained by the area under the peak at 2800 MeV/c^2 in Figure 27, which is ultimately estimated to give

$$\frac{\mathcal{B}(B_c^+ \to J/\psi D^{**})}{\mathcal{B}(B_c^+ \to J/\psi DK)} = 10.4. \tag{58}$$

Decay	14873607	14873430	14873620
$J/\psi D_s^+$	0.15696		
$J/\psi D_s^{*+}$	0.79608		
$J/\psi D^+$	0.01503		
$J/\psi D^{*+}$	0.03193		
$J/\psi D^+ K^0$		0.2197	
$J/\psi D^0 K^+$		0.0717	
$J/\psi D^{*+} K^0$		0.3847	
$J/\psi D^{*0} K^+$		0.3239	
$J/\psi D_{s0}^{*+}$			0.1000
$J/\psi D_{s1}^+$			0.8153
$J/\psi D_{s1}'^+$			0.0651
$J/\psi D_{s2}^{*+}$			0.0197

Table 16: Relative branching fractions of different contributers to $B_c^+ \to J/\psi DX$ decays from `EvtGen` models.

13 Feed-down $\psi(2S)\,\ell\,\nu$ and $\chi_c\,\ell\,\nu$ backgrounds

The feed-down backgrounds include B_c^+ semileptonic decays with a heavy charmonium state such as $\psi(2S)$ or χ_c, which further decays down to J/ψ. The significant feed-down channels include $\psi(2S) \to J/\psi\pi\pi$, and $\chi_{c0,1,2}(1P) \to J/\psi\gamma$.

13.1 $\psi(2S)\ell\nu$ backgrounds

For $B_c^+ \to \psi(2S)\,\mu^+\,\nu$ and $B_c^+ \to \psi(2S)\,\tau^+\,\nu$ decays, simulations are produced using both the EFG and Kiselev form factor models to build templates for this background. In this analysis, we use the EFG model and use the tuple produced with the Kiselev model for a systematic uncertainty (see 17.2). The distributions of rest frame variables for both models are shown in Figure 28, in comparison with the normalization mode.

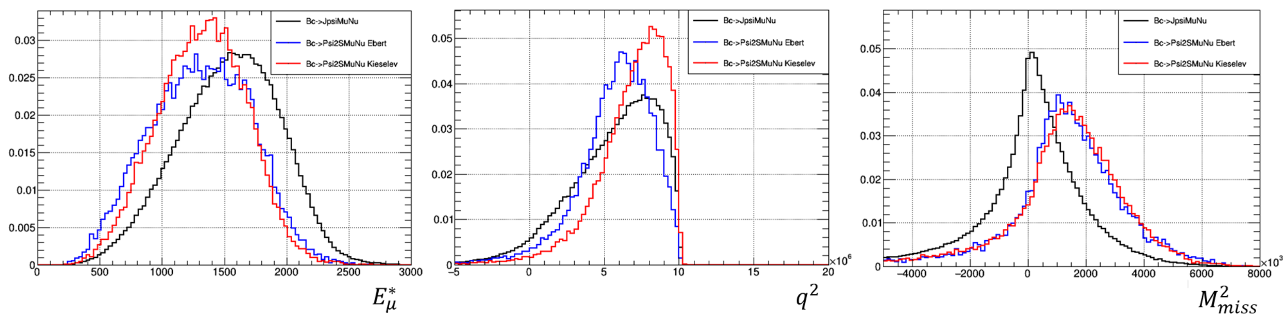


Figure 28: The distributions of rest frame variables for EFG (blue) and Kiselev (red) models of $B_c^+ \to \psi(2S)\,\mu^+\,\nu$ decays, in comparison with the normalization mode (black)

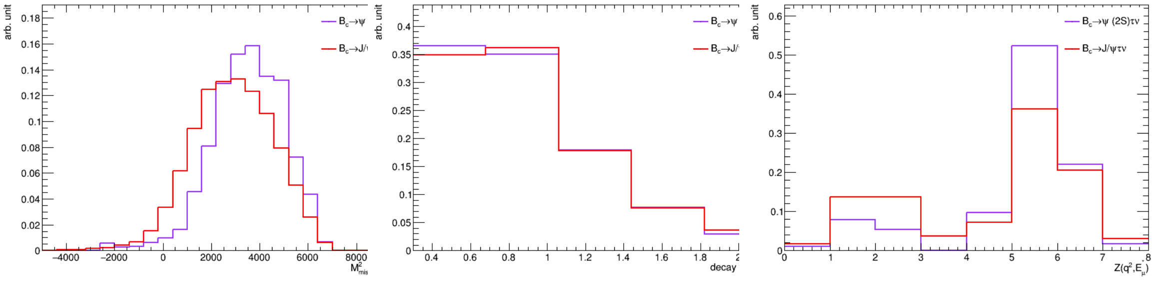


Figure 29: The distributions of rest frame variables for the $B_c^+ \to \psi(2S)\tau^+\nu_\tau$ decays (purple) compared with the signal mode (red).

The branching fraction ratio

$$\frac{\mathcal{B}(B_c^+ \to \psi(2S)\mu^+\nu_\mu)}{\mathcal{B}(B_c^+ \to J/\psi\mu^+\nu_\mu)} \tag{59}$$

is predicted by the EFG model to be about 2.5% [19], and the Kiselev model to be about 4.9% [4]. The branching fraction of $\psi(2S) \to J/\psi X$ is 61%, giving an expected contribution of $1.5 - 3\%$ relative to the normalization channel.

The ratio of $B_c^+ \to \psi(2S)\,\mu^+\,\nu / B_c^+ \to \psi(2S)\,\tau^+\,\nu$ is fixed in HistFactory by the ratio of the phase space integrals

$$\int_{q^2_{\min}}^{q^2_{\max}} dq^2 q^2 (1 - \frac{m_\ell^2}{q^2})^2 (1 + \frac{m_\ell^2}{q^2}) \sqrt{[q^2 - (m_{B_c} - m_{\psi(2S)})^2][q^2 - (m_{B_c} + m_{\psi(2S)})^2]} = 0.8635, \tag{60}$$

and the branching fraction of $\tau \to \mu$ ($0.8635 \times 0.1739 = 0.1208$). In the fit, the branching fraction ratio is fixed to the predicted value. The distributions of the rest frame variables for $B_c^+ \to \psi(2S)\tau^+\nu_\tau$ are shown in Figure 29. The total contribution of $\psi(2S)\ell\nu$ is allowed to float in the nominal fit with a log normal constraint of two around the theoretical prediction, with further investigations for systematic uncertainties (Section 17.11).

13.2 $\chi_c\ell\nu$ backgrounds

For $B_c^+ \to \chi_{c1}\,\mu^+\,\nu$ and $B_c^+ \to \chi_{c2}\,\mu^+\,\nu$ decays, simulations are produced using the WWL form factor model [22]. According to [33], the branching fraction ratios are predicted to be

$$\frac{\mathcal{B}(B_c^+ \to \chi_{c(0,1,2)}(1P)\mu^+\nu_\mu)}{\mathcal{B}(B_c^+ \to J/\psi\mu^+\nu_\mu)} = (7.1\%, 4.3\%, 8.6\%). \tag{61}$$

Using the prediction of $\mathcal{B}(B_c^+ \to J/\psi\mu^+\nu_\mu)$, WWL predicts branching fraction ratios of

$$\frac{\mathcal{B}(B_c^+ \to \chi_{c(0,1,2)}(1P)\mu^+\nu_\mu)}{\mathcal{B}(B_c^+ \to J/\psi\mu^+\nu_\mu)} = (13.6\%, 9.1\%, 11.2\%). \tag{62}$$

The $\chi_{c(0,1,2)}(1P) \to J/\psi X$ branching fractions are $((1.27\pm0.06)\%, (33.9\pm1.2)\%, (19.2\pm 0.7)\%)$ [33]. The χ_{c0} mode contributes about $0.1\% - 0.2\%$ relative to normalization, and can therefore be ignored. The χ_{c1} and χ_{c2} modes are expected to contribute about $1.5\% - 3.1\%$ and $1.7\% - 2.2\%$ relative to normalization, respectively. These are similar enough that in the fit, we fix the proportions of χ_{c1} and χ_{c2} to be equal. The total expected contribution of $\chi_{c1,2}$ is then expected to be about $3.2\% - 5.3\%$. In the nominal fit, this is allowed to float, with further investigations for systematic uncertainties in Section 17.6.

Hernandez et al. predict $R(\chi_{c1,2}(1P)) = (12\%, 11\%, 7\%)$ while WWL predict $R(\chi_{c1,2}(1P)) = (11\%, 11\%, 6\%)$. The muonic branching fraction of the τ is 17.7%, so therefore the expected contribution of the tauonic channel of the $B_c \to \chi_c$ decays is about $0.05\% - 0.09\%$, which is negligible. In the fit, the $\chi_c \ell \nu$ background floats with a log normal constraint of 2 around the predicted theoretical yield relative to normalization.

The distributions of rest frame variables for both decay modes are shown in Figure 30, in comparison with the normalization mode.

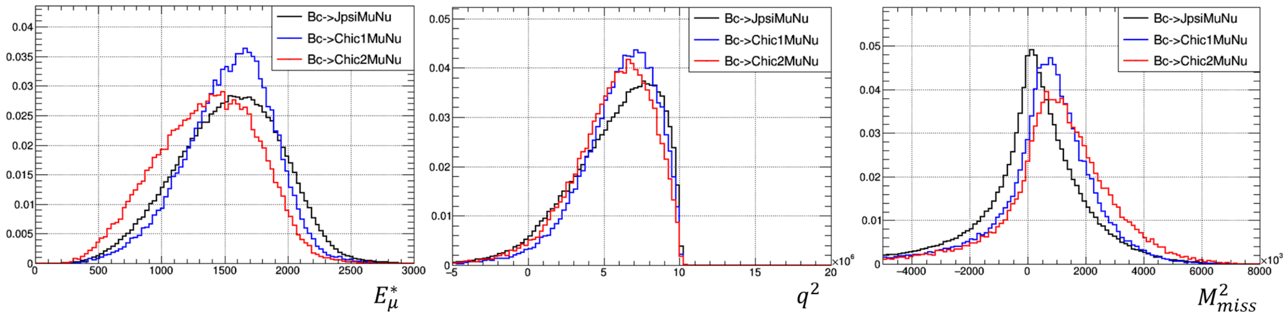


Figure 30: The distributions of rest frame variables for $B_c^+ \to \chi_{c1,2}\, \mu^+ \,\nu$ decays with the WWL form factor model, in comparison with the normalization mode (black). The decay with $\chi_{c1}(1P)$ is in blue, and the decay with $\chi_{c2}(1P)$ is in red.

13.3 Exotic feed-down backgrounds

Charmonium states above the DD threshold rarely decay to $J/\psi X$ and thus contribute negligibly to the background. The branching fraction of $\psi(3770) \to J/\psi X$ is about 0.002. In the B system, the widths of charmonium states can be used to conservatively estimate the charmonium branching fraction ratios, for example

$$\frac{\mathcal{B}(B_c^+ \to \psi(2S)X)}{\mathcal{B}(B_c^+ \to J/\psi X)} \sim \frac{\Gamma(J/\psi)}{\Gamma(\psi(2S))} \sim \frac{1}{3}. \tag{63}$$

For the $\psi(3770)$, this ratio of widths is about 1/300, and if each of these were misidentified as a signal event, the increase in $R(J/\psi)$ would still be very small. Note that this estimate is extremely conservative, as Ebert predicts $3 \times 10^{-}5$ for this ratio.

The $X(3872)$ is a special case, with an extremely small width but large branching fraction to $J/\psi X$. It is similar to the $\psi(2S)$ in terms of mass and inclusive decay modes to $J/\psi X$, and therefore its contribution be covered by the same template. A study was done to investigate the need for a systematic uncertainty for this mode. A sample was prepared which was anti-isolated, with selection defined in Table 17, inspired by the selections used to develop the 2OS control sample in [25]. The first two tracks are required to be anti-isolated, with the third track having an isolation requirement. The first two tracks

Table 17: Anti-isolated sample selections, with 1, 2, and 3 referring to the track numbers.

Selections
Isolation BDT (1, 2)$>$0.2
Isolation BDT(3) $<$ 0.1
1, 2 oppositely charged
ProbNNk(1,2)$<$0.2
Track 1 is a long track
$p_1 > 5$ GeV
$p_{T1} > 150$ MeV

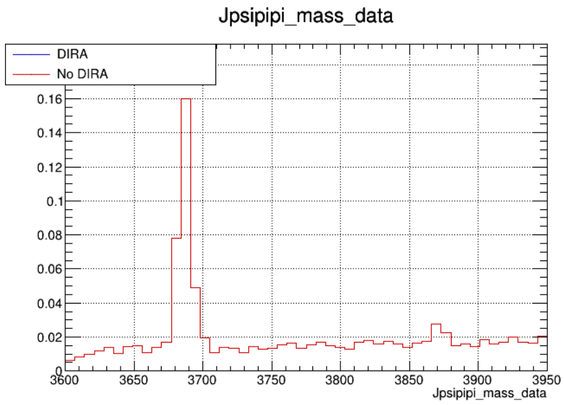


Figure 31: $m(J/\psi\pi^+\pi^-) - m(J/\psi)_{reco} + m(J/\psi)_{true}$ for nominal data sample, with both the $\psi(2S)$ and $X(3872)$ peaks visible.

are required to be oppositely charged, and to not be kaons. Momentum cuts are added to clean up nonphysical backgrounds.

With this sample, we plot $m(J/\psi\pi^+\pi^-) - m(J/\psi)_{reco} + m(J/\psi)_{true}$, and are able to see peaks at the $\psi(2S)$ and $X(3872)$ masses in the nominal data sample (Figure 31) and in the mis-ID data sample (Figure 32, zoomed in to show the $X(3872)$ peak).

In the nominal data sample, a gaussian and quadratic were fitted in order to estimate the contribution of $\psi(2S)$ and $X(3872)$ events. The fits are given in Figure **??**, which estimates the $X(3872)$ contribution to be about 10% of the $\psi(2S)$ contribution. Because the predicted $X(3872)$ contribution is smaller than the fit uncertainty on the $\psi(2S)$ yield (see Table 21), we elect not to add an additional systematic for potential background events from $X(3872)$.

Potential other backgrounds arising from cascades such as Ξ_{bc}^0, can produce backgrounds with similar final states with $\Xi_{bc}^0 \to J/\psi\Xi_c^0$, where $\Xi_c^0 \to X\mu\nu$. From [1], we know that the upper bound on this cascade contribution relative to the $B_c^+ \to J/\psi D_s^+$ is

$$\mathcal{R} = \frac{\sigma\left(\Xi_{bc}^+\right) \times \mathcal{B}\left(\Xi_{bc}^+ \to J/\psi\Xi_c^+\right) \times \mathcal{B}\left(\Xi_c^+ \to pK^-\pi^+\right)}{\sigma\left(B_c^+\right) \times \mathcal{B}\left(B_c^+ \to J/\psi D_s^+\right) \times \mathcal{B}\left(D_s^+ \to K^+K^-\pi^+\right)} = 0.015 \qquad (64)$$

when the Ξ_{bc}^0 mass is in the region favored by theoretical predictions. Additionally,

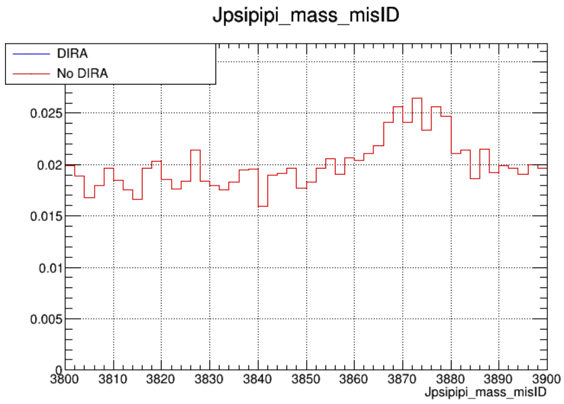
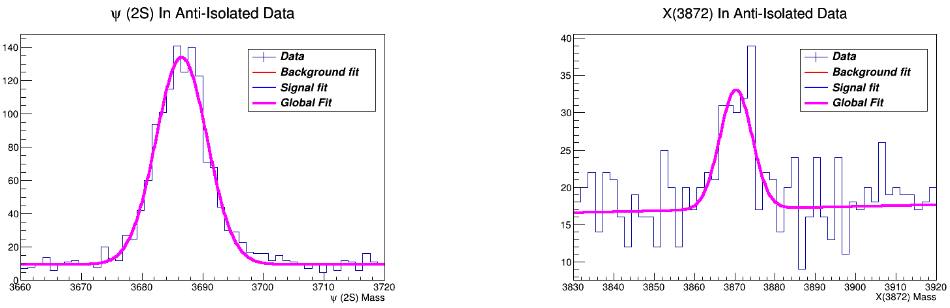


Figure 32: $m(J/\psi\pi^+\pi^-) - m(J/\psi)_{reco} + m(J/\psi)_{true}$ for the mis-ID data sample, zoomed in on the $X(3872)$ mass.



(a) Gaussian and quadratic fit to the $\psi(2S)$ mass peak in anti-isolated data sample, yielding 1315 events.

(b) Gaussian and quadratic fit to the $\psi(2S)$ mass peak in anti isolated data sample, yielding 131 events.

$$\frac{\mathcal{B}\left(\Xi_c^+ \to pK^-\pi^+\right)}{\mathcal{B}\left(D_s^+ \to K^+K^-\pi^+\right)} = 0.115, \tag{65}$$

and

$$\frac{\mathcal{B}\left(\Xi_c^+ \to X\mu\nu\right)}{\mathcal{B}\left(D_s^+ \to X\mu\nu\right)} \approx \tau_{D_s^+}/\tau_{\Xi_c^+} \approx 1, \tag{66}$$

leaving us with an upper bound of 13% of the $J/\psi D_s^+$ contribution. Per year, this translates to expecting no more than 10 events.

14 Corrections to simulation

Corrections are applied to the simulated data to improve their modeling of real data. They include form factor corrections to signal and normalization modes, lifetime acceptance corrections, production multiplicity corrections, and a final reweighting to all simulations.

14.1 Form factor corrections

The MC data for the signal mode $B_c^+ \to J/\psi\,\tau^+\,\nu$ and normalization mode $B_c^+ \to J/\psi\,\mu^+\,\nu$ are originally simulated with the EFG model for $B_c^+ \to J/\psi$ form factors. Recently published lattice QCD results provide significant improvement to the modeling of form factors [20]. The Hammer software library is an efficient way to reweight MC samples to account for any new physics effects, or in our case, using the newest LQCD results. The LQCD form factor results provided in the BGL parameterization [43] have been implemented in the Hammer tool [44]. The Hammer tool is used to reweight simulation samples to any implemented description of the hadronic matrix elements, using the true four-momenta of the particles as input. The form factors are given by

$$\begin{aligned} g &= [0.02617, -0.12680, -0.25080, 0.30282], \\ f &= [0.01755, -0.10597, 0.28943, 0.16756], \\ F_1 &= [-0.01520, -0.04259, 0.51217], \\ P_1 &= [0.03977, -0.27245, 0.05169, 0.95991]. \end{aligned} \tag{67}$$

The distributions of q^2 and missing mass for normalization MC in the original EFG model and reweighted LQCD results are in Figure 34, where the corrections are shown to range up to approximately 20%.

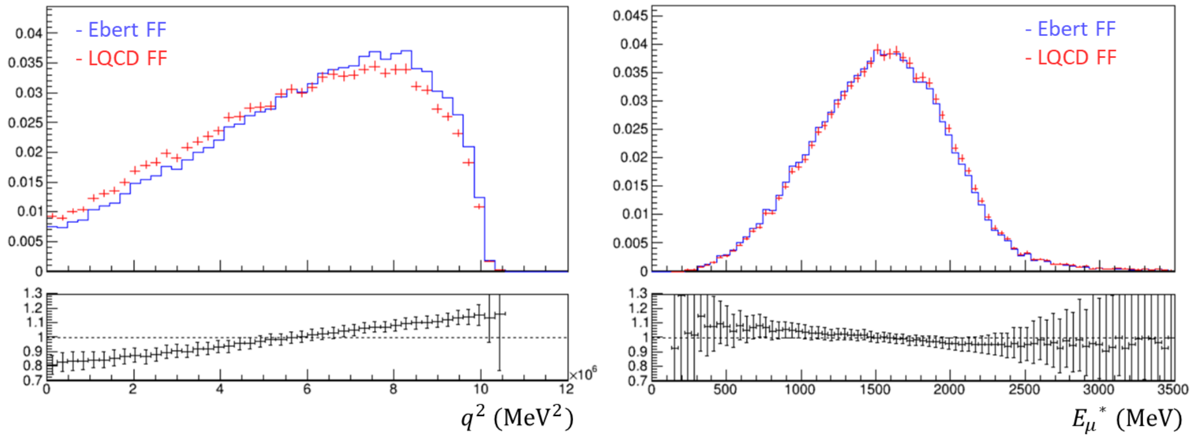


Figure 34: The distributions of q^2 and missing mass for normalization MC in original EFG model (blue) and reweighted LQCD results (red). The lower panels show the ratios.

With the form factor variations turned off in the fit, and using the final version of the fit complete with final reweighting (which is detailed in Section 16.2), the fit is in Figure 35. The pulls in the Z distribution appear worse, especially in the higher bins in 2016 and 2018 and in the middle bins of 2017.

14.1.1 Form factor parameter correlations

The correlations between the z-expansion coefficients (given in the appendix of [18]) have been implemented in `Hammer` as well. The covariance eigenvector matrix is given in Table 18, where each eigenvector e_i is normalized to the square root of its eigenvalue λ_i. This covariance matrix uses the BGL parameterization.

	α_0	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	α_{11}	α_{12}	α_{13}	α_{14}
f_0	2.9189e-07	1.5932e-04	-5.5159e-07	-1.7821e-06	-5.8163e-06	-4.1753e-06	1.9930e-04	-2.1432e-06	2.8321e-04	-1.6547e-05	-1.9473e-06	-1.1465e-12	3.8510e-06	-6.4378e-06	-8.3235e-04
f_1	-2.3304e-05	-1.4214e-02	2.3830e-04	4.7879e-06	3.8300e-04	2.9694e-04	-1.3323e-03	-1.8493e-04	-2.5504e-02	1.2071e-03	8.3034e-05	-5.7053e-14	7.7673e-08	9.4451e-08	-9.2591e-06
f_2	1.9655e-03	2.2442e-01	2.5472e-04	1.0708e-04	9.6790e-04	3.8121e-03	-1.4556e-01	1.0070e-05	-3.4890e-05	2.1383e-05	2.7817e-06	-2.4941e-15	3.2590e-09	2.5581e-09	-8.9937e-07
f_3	4.0673e-03	5.4116e-01	-5.4640e-04	-2.5689e-04	-8.7821e-04	-1.6607e-03	6.0323e-02	-7.9099e-06	-6.5543e-04	2.4714e-05	1.1335e-06	-3.4666e-16	-4.2549e-10	4.8021e-09	3.7478e-07
g_0	4.2553e-06	-1.4729e-08	2.0210e-05	1.5496e-05	-3.7133e-08	1.2850e-06	6.3618e-07	2.0186e-05	1.2461e-06	2.4599e-05	3.6240e-05	1.6004e-06	-1.6747e-04	-4.4833e-05	-4.6503e-06
g_1	-1.1958e-04	-8.1240e-06	-3.7269e-03	2.9317e-04	-2.8606e-04	-5.8212e-03	-1.8497e-04	2.0447e-04	-2.6957e-04	-3.9835e-04	-8.9191e-03	2.5613e-11	-2.4504e-06	-8.2689e-06	6.6687e-08
g_2	2.5599e-03	4.1291e-04	8.0115e-02	-7.0342e-03	1.0101e-02	1.6765e-01	3.7806e-03	-4.5831e-05	3.1517e-05	-3.0451e-05	-3.1454e-04	-1.5537e-13	-7.8387e-08	-2.0093e-07	2.2480e-08
g_3	4.5985e-04	5.2411e-04	3.6727e-01	-1.9149e-02	-5.9698e-03	-3.6646e-02	-6.5108e-04	-7.5884e-06	3.6857e-06	-2.3137e-05	-2.1763e-05	1.8091e-13	2.3452e-08	2.2611e-08	2.7121e-09
F_{10}	-4.5582e-04	4.1713e-06	-1.6862e-04	-1.2546e-03	-2.2219e-04	-2.2606e-05	-1.4097e-05	-2.1091e-03	-1.1099e-04	-1.5562e-03	-7.5675e-05	5.2682e-07	1.3648e-04	8.9423e-04	-4.7049e-07
F_{11}	7.3798e-03	2.9402e-05	2.1093e-03	2.0072e-02	2.4254e-02	-3.1346e-04	1.2407e-04	5.8859e-02	3.1544e-04	5.1843e-03	-2.2104e-05	2.8571e-08	7.8656e-06	4.6600e-05	-9.0308e-08
F_{12}	2.4388e-02	-5.8739e-04	1.1859e-02	2.0875e-01	-2.1964e-01	5.9558e-03	-3.0166e-04	1.3418e-03	-1.9880e-05	5.4720e-05	-1.3519e-06	1.5486e-09	4.2964e-07	2.6471e-06	-3.6626e-09
P_{10}	5.0923e-05	-1.4167e-06	1.8363e-05	-2.1353e-05	-3.3717e-05	-4.6383e-06	2.9069e-06	-7.5281e-05	1.8966e-05	2.4822e-04	1.2058e-05	-7.7649e-07	-2.5161e-04	5.1885e-04	-9.9063e-06
P_{11}	-1.0098e-02	9.3592e-05	-8.9057e-04	-5.8755e-03	-3.6978e-03	3.2414e-05	-1.5649e-04	1.6064e-02	-1.4756e-03	-1.9542e-02	1.8740e-04	-4.2094e-08	-1.2100e-05	-5.1969e-05	1.2533e-08
P_{12}	1.4276e-01	-9.0845e-05	2.0171e-02	2.3317e-01	1.8479e-01	-3.2910e-03	4.3776e-04	-5.6535e-03	-4.9177e-05	-1.0383e-03	7.6739e-06	-2.2828e-09	-6.5137e-07	-2.8852e-06	4.0803e-09
P_{13}	6.0921e-01	-4.2937e-03	-5.8538e-03	-6.3295e-02	-3.4900e-02	-1.4203e-04	-4.3230e-05	8.2307e-04	-1.2722e-05	-1.4693e-04	1.1462e-06	-1.2300e-10	-3.6276e-08	-2.3062e-07	1.3614e-09

Table 18: Covariance eigenvector matrix in `Hammer`, normalized to the square root of eigenvalues. Each column is an eigenvector, and each eigenvector has an element corresponding to an element of g, f, F_1, and P_1, in order.

The eigenvectors given in Table 18 are all normalized to the square root of the corresponding eigenvalue. The leading F_1 coefficient c_0 is set by the constraint $F_1(q^2 = q^2_{max}) = (M_b - M_c)f(q^2 = q^2_{max})$.

The BGL form factor parameterization is linearized with respect to a basis of parameters. Call the parameter set $\{\mu\}$. We want to linearize $\{\mu\}$ around the best fit point $\{\mu_0\}$, such that

$$F_i(\{q\};\{\mu\}) = F_i(\{q\};\{\mu^0\}) + \sum_a F'_{i,a}(\{q\};\{\mu^0\})e_a \tag{68}$$

where a is one of the variational indices, and e_a is the variation. $F'_{i,a}$ is the perturbation of F_i along the principal component e_a, which is the ath principal component of the covariance matrix [44].

In detail, these correlations have been implemented as follows. First, we normalize each eigenvector to the square root of its corresponding eigenvalue. Index the form factors and their elements in order from f^i, g^i, F^i_1, P^i_1 as 0 to 14, call this index k. The elements of the eigenvectors in 18 corresponding to f^i, g^i, F^i_1, P^i_1 are in the same order, i.e. the rows are indexed by k. To create the $\pm\sigma$ variations in some variable k, we must modify it while taking into account the covariances with other eigenvectors. To select the eigenvector corresponding to variable k, we take the kth row and choose the eigenvector with the largest element in the kth row. Finally, we iterate through each eigenvector and modify each form factor by adding the corresponding elements in the eigenvector we have chosen multiplied by however many σ we want the form factors to vary by (so for example, f gets index 0 added to the first element, index 3 added to its last element, and g has index 4 added to its first element, etc.).

14.1.2 Form factor parameter uncertainties

The form factor variations are allowed to float up to 2σ in the nominal fit. Previously, a 1σ fit was attempted. However, the discrepancy between LQCD predictions and experimental measurements of the form factors of $B \to D^*$ (see Figure 36 from [45]) motivated an investigation into loosening the form factor constraints. Since it appears that the lattice error bars may have been underestimated by about $2 - 3\sigma$, we ran the fit allowing HistFactory to extrapolate between templates with $\pm 2, 3\sigma$. Letting the fit vary between $\pm 2\sigma$ did indeed improve the fit. 3σ variations were also studied, but these did not appear to change the fit from 2σ variations. The nominal fit uses 2σ variations. The effect of changing form factor constraints are further investigated as a source of systematic uncertainty, see Section 17.4.

14.2 Production multiplicity correction

The MC simulation is known to mismodel the production multiplicity of events, with a smaller average number of tracks produced than in the real data. A boosted decision tree reweighter [46] is trained using the simulation sample of the normalization mode and a normalization-rich data control sample. The samples are required to pass selections designed to enrich normalization events in data along with the normal selections, see Table 19 for a definition of the normalization-rich region.

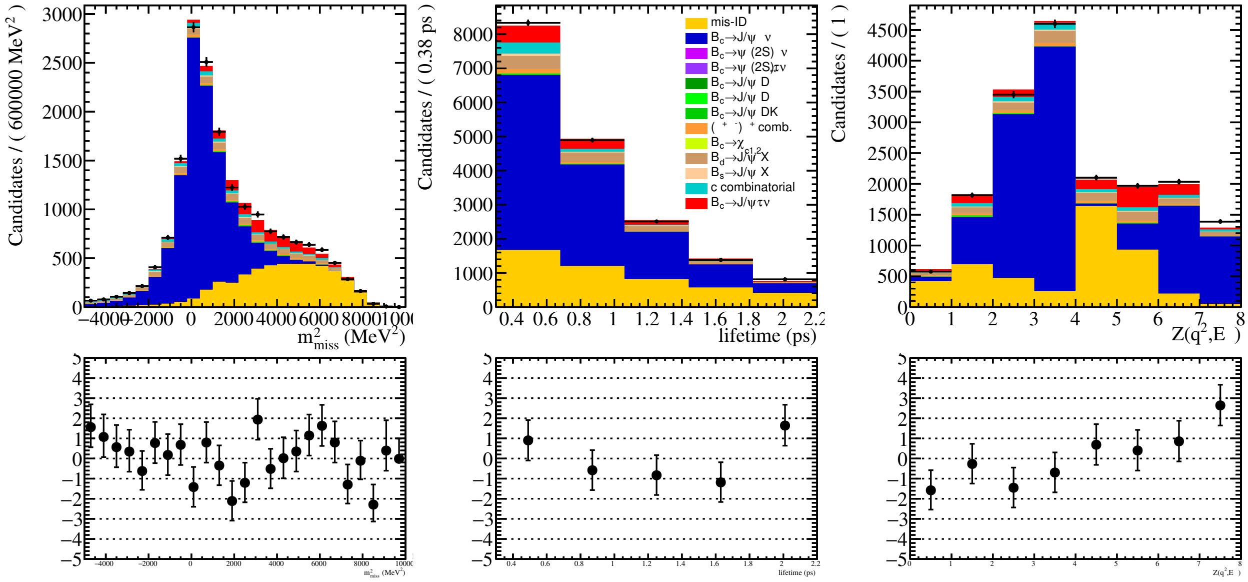
Species	Selection
Event	$m^2_{miss} < 0$
	Isolation BDT score < 0
B_c	$\tau_{B_c} < 0.5$ ps

Table 19: Selection for the normalization-enriched region.

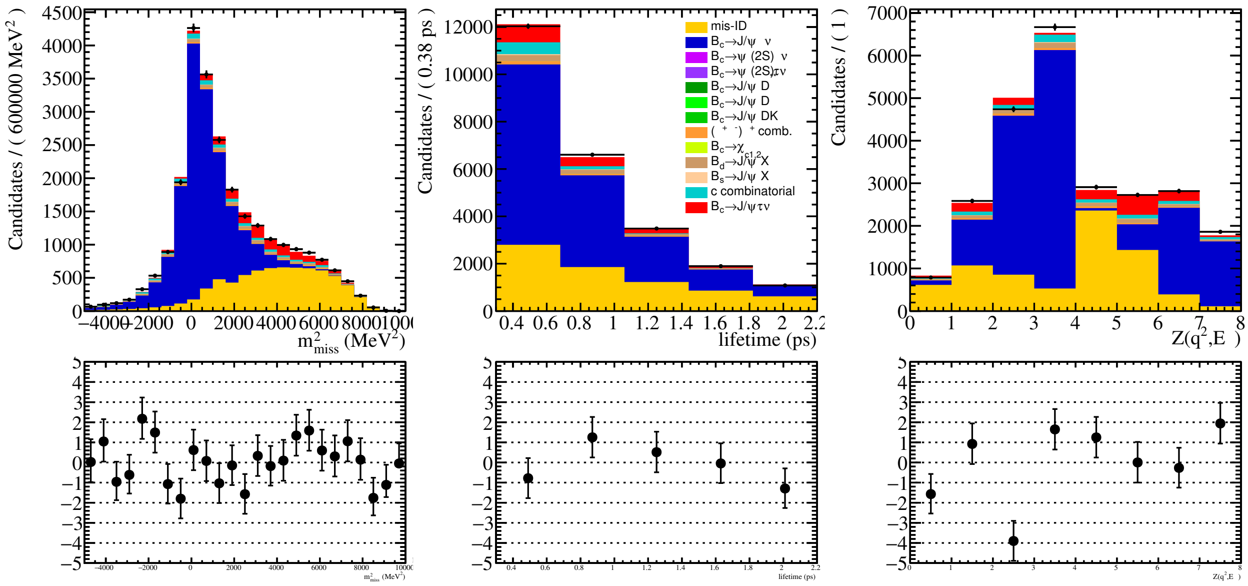
The BDT reweighter optimizes the splitting of multi-dimensional regions to produce weights instead of simply binning data into histograms. The multiplicity reweighter uses hyperparameters `n_estimators=500`, `learning_rate`$=0.1$, `max_depth`$=3$, `min_samples_leaf=100`, and `subsample=0.4`. `n_estimators` is the number of trees used for the reweighter. `learning_rate` controls how conservative the boosting process is, a larger learning rate requires fewer trees but is also less stable. `max_depth` is the maximum depth of the tree. `min_samples_leaf` is the minimum number of input data points that can be assigned to each leaf, and `subsample` is the proportion of the sample used for training, with larger values used to prevent overtraining.

The corrections are made as a function of the event track multiplicity (nTracks) and the number of degrees of freedom for fitting the primary vertex (PVNDOF). The comparisons before and after reweighting are shown in Figure 37.

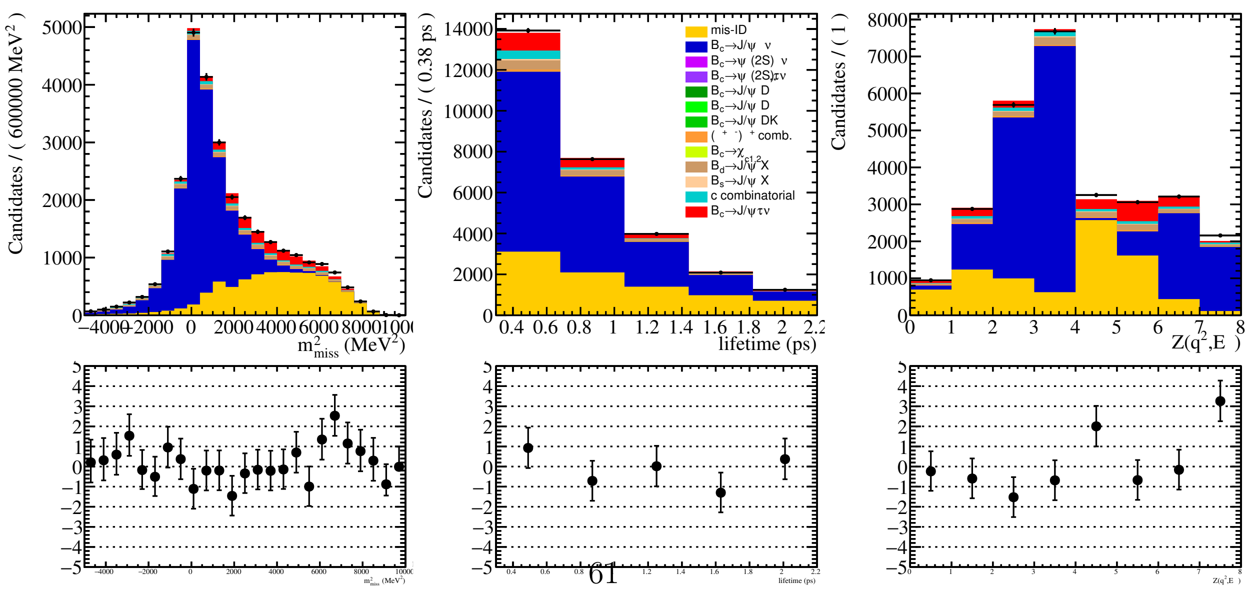
Additionally, we examined the potential need to correct the B_c^+ production kinematics. The LHCb Collaboration has performed a measurement of B_c^+ production using data recorded in proton-proton collisions at the center-of-mass energy $\sqrt{s} = 8$ TeV with $B_c^+ \to J/\psi\pi^+$ decays [47]. The B_c^+ transverse momentum and rapidity distributions of the background-subtracted data are found to be in good agreement with those of the simulation produced by the B_c^+ generator BCVEGPY. Therefore, no major correction is expected for the B_c^+ production kinematics.



(a) 2016 fit.



(b) 2017 fit.



(c) 2018 fit.

Figure 35: The results of the fit to the full data, projected in the fit variables, without form factor variations. The lower panels show the residual distributions.

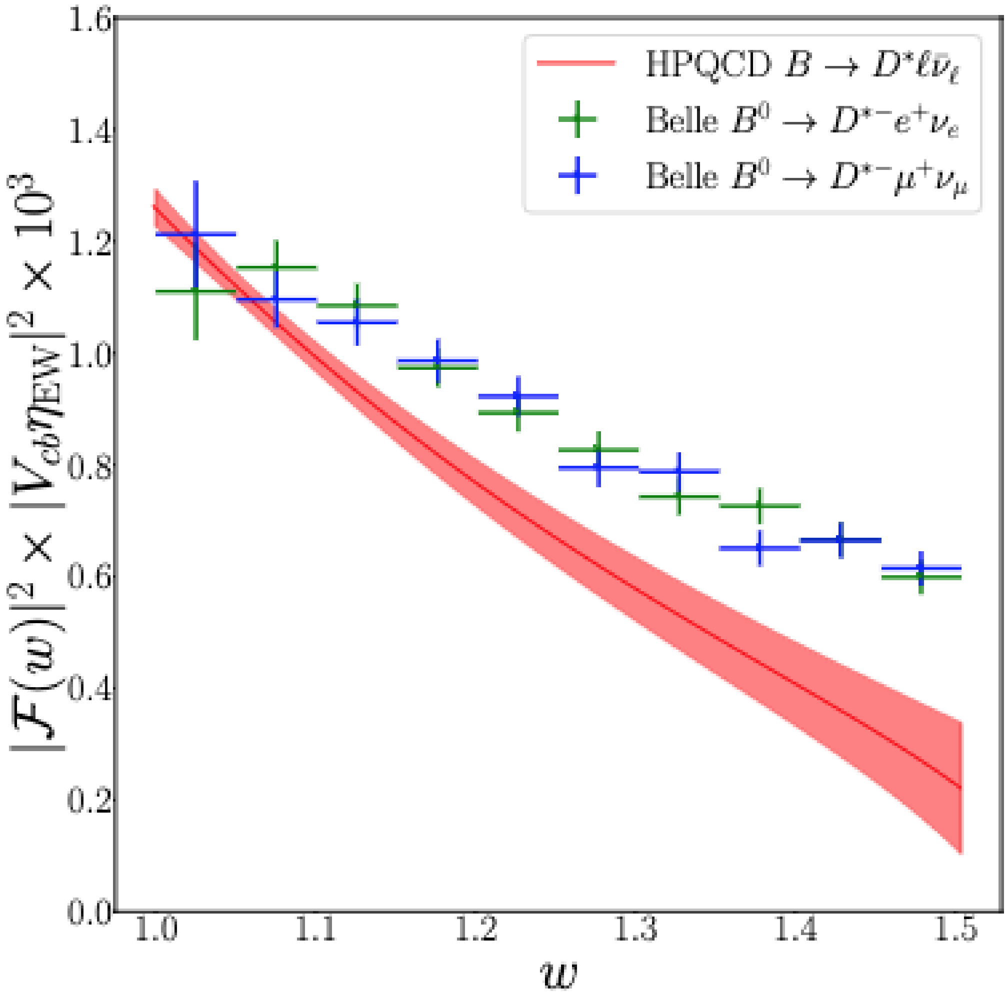


Figure 36: This plot is taken from Figure 11 in [45]. It depicts the value of $\|\mathcal{F}(w)\eta_{EW}V_{cb}\|^2$, which is proportional to the differential decay rate, as a function of w, which is the product of the D^* and B four velocities. Note the discrepancy between the lattice result (red) and the Belle II result (blue and green).

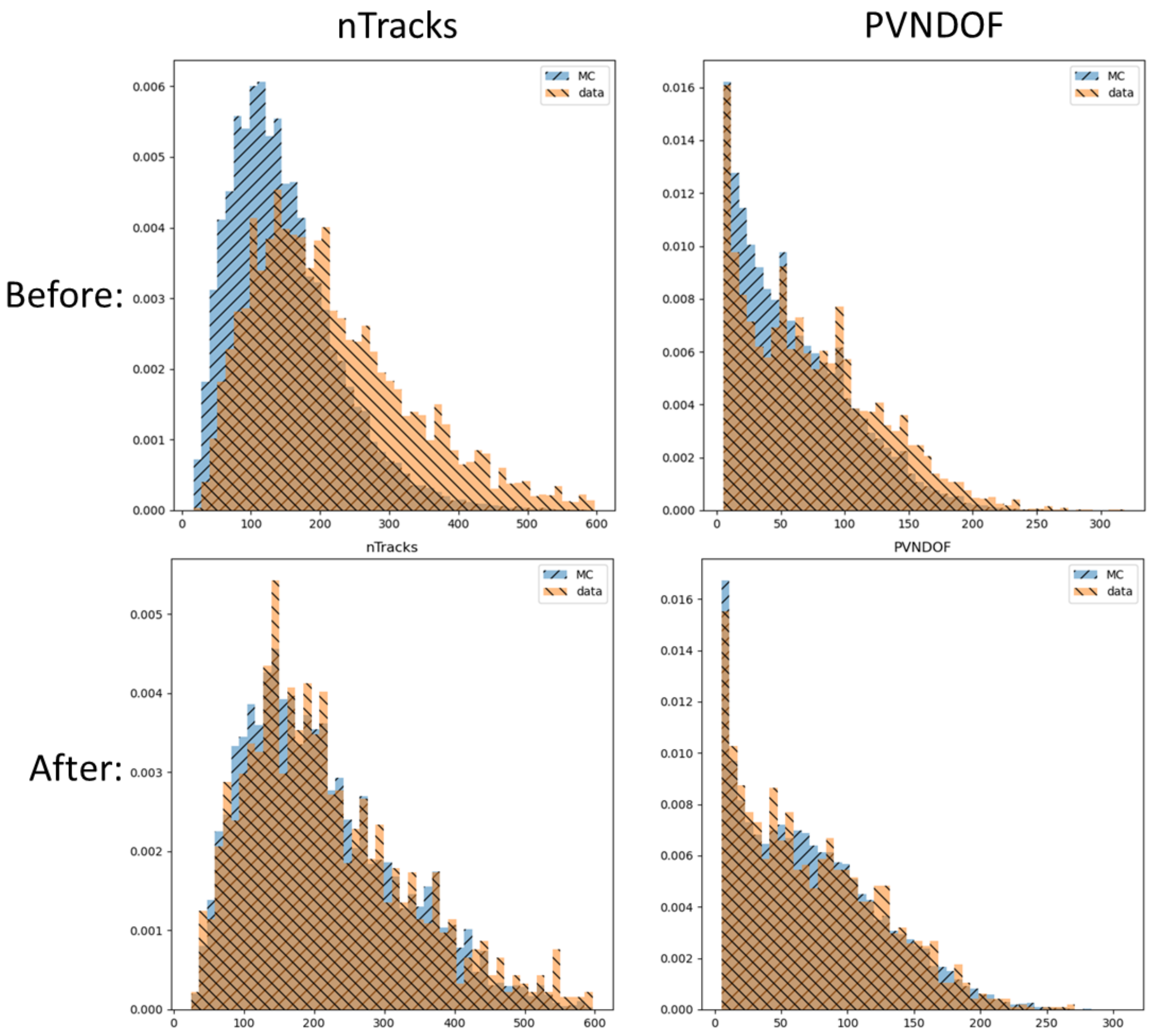


Figure 37: The comparisons in nTracks and PVNDOF between simulation (in blue) and data (in orange), before and after reweighting. The data shown in the before plots is from the training sample, and the data shown in the after plots is from the testing sample.

15 Fit strategy and results

Here we reiterate and summarize the fit procedure. A binned maximum likelihood estimation fit is used to measure the parameter $R(J/\psi)$, with the probability density function (PDF) templates derived from simulations and data control samples. The distributions are binned in missing mass, candidate decay time, q^2, and E^*_μ, for data and simulation samples passing the selections described in Section 5. The relative fractions of different components are allowed to vary in the fit with some constraints, to maximize the likelihood function with Poisson distributions in every bin. A systematic can be added associated with the a constraint on the normalization of the component–this type of systematic is Gaussian constrained. Shape systematics are also added for various parts of the fit, such as form factor reweighting, smearing, and several reweighting corrections. The different HistFactory elements and their constraints and interpolation are further described in Appendix 23.

The fit is performed using the HistFactory package as part of the RooFit and RooStats framework [48–50]. The numerical minimization for the fit is performed using the MINUIT library [51].

15.1 Fit variables binning

The fit variables (missing mass, decay time, q^2, and E^*_μ) allow for discrimination between the signal, normalization, and backgrounds. Because these four-dimensional distributions cannot be easily modeled with analytical functions, histogram templates are good representations for the underlying PDFs. The fit is therefore performed in a binned maximum likelihood estimation procedure using the HistFactory package (with more detail on HistFactory given in Appendix 23).

Because the ROOT data analysis framework allows for histograms in up to three dimensions, the q^2 and E^*_μ are binned into a categorical variable $Z(q^2, E^*_\mu)$ to handle the effectively four-dimensional distributions. The binning defined for $Z(q^2, E^*_\mu)$ is shown in Figure 38.

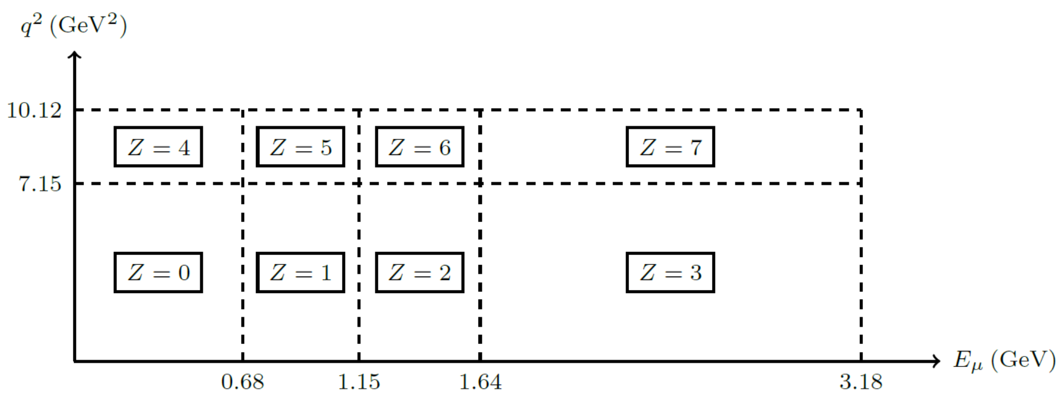


Figure 38: The binning defined for the categorical variable $Z(q^2, E^*_\mu)$ [24].

Table 20: Summary of characteristics of different fit templates in the fit variables.

Mode	Topology	Signature
	$B_c^+ \to J/\psi\tau^+\nu_\tau$	J/ψ dimuon, bachelor μ
Signal	$\Rightarrow J/\psi \to \mu\mu$	Large missing mass (3ν)
	$\Rightarrow \tau \to \mu\nu\nu$	Short B_c^+ lifetime, low E_{ℓ^*}, high q^2
	$B_c^+ \to J/\psi\mu^+\nu_\mu$	J/ψ dimuon, bachelor μ
Normalization	$\Rightarrow J/\psi \to \mu\mu$	Smaller missing mass (1ν)
		Short B_c^+lifetime
	$B_c^+ \to (c\bar{c})^*\mu\nu$	J/ψ dimuon, bachelor μ
feed-down	$\Rightarrow (c\bar{c})^* \to J/\psi X$	Large missing mass ($1\nu + X$)
		Short B_c^+lifetime, high E_{ℓ^*}, low q^2
	$B_c^+ \to (c\bar{c})DX$	J/ψ dimuon, bachelor μ
		μ channels have similar Z to normalization
		τ channels have similar Z to signal
$J/\psi DX$	$\Rightarrow (c\bar{c})J/\psi(X)$	Small to large missing mass ($1\nu, (X), (Y)$)
	$\Rightarrow D \to (\mu, \tau)\nu(Y)$	Short B_c^+lifetime
		Low q^2 for 2-body, high q^2 for 3-body
	$B \to J/\psi X, B \to \mu X$	J/ψ dimuon, 1 real OS μ
$J/\psi\mu$ combinatorial	$\Rightarrow J/\psi \to \mu\mu$	Mimics B_c^+lifetime
		Broad missing mass distribution
MisID	$B \to J/\psi\mu_{fake}X$	Large missing mass (X)
	$\Rightarrow J/\psi \to \mu\mu$	Long B lifetime
$(\mu^+ + \mu^-)\mu^\pm$ combinatorial		Broad missing mass distribution
c combinatorial	$B_c^+ \to J/\psi X$, $D \to \mu X$	Very short B_c^+ lifetime
	$D \to \mu X$	Broad missing mass distribution

15.2 Templates projections

The fit templates projected in the fit variables missing mass, decay time, and $Z(q^2, E_\mu^*)$ for signal and each background are shown in Figures 39, 40,41,42,43,**??**, and 44.

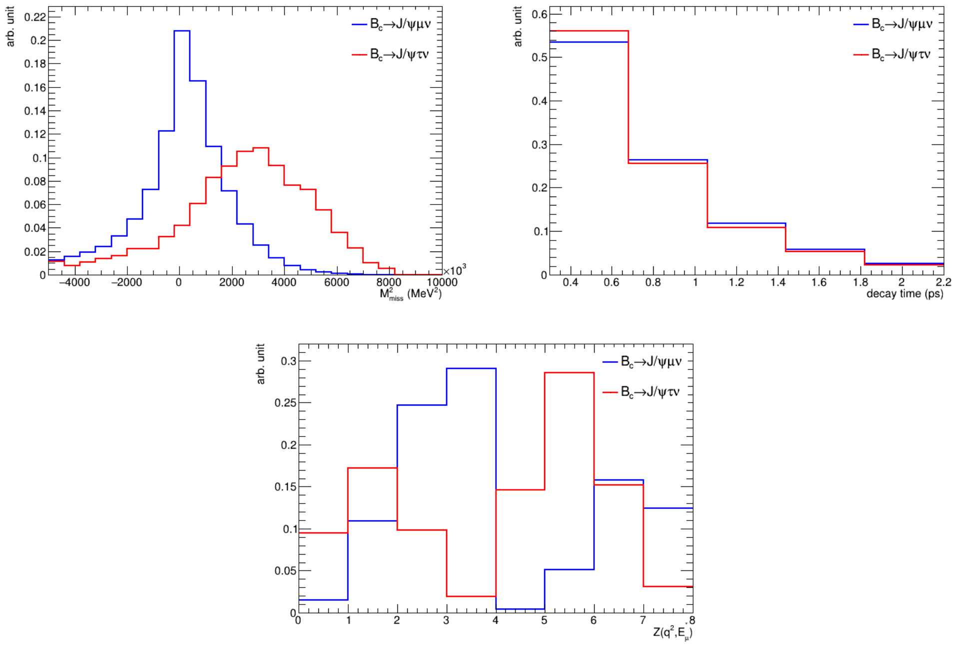


Figure 39: Fit template projections for the signal mode (red) compared with the normalization mode (blue).

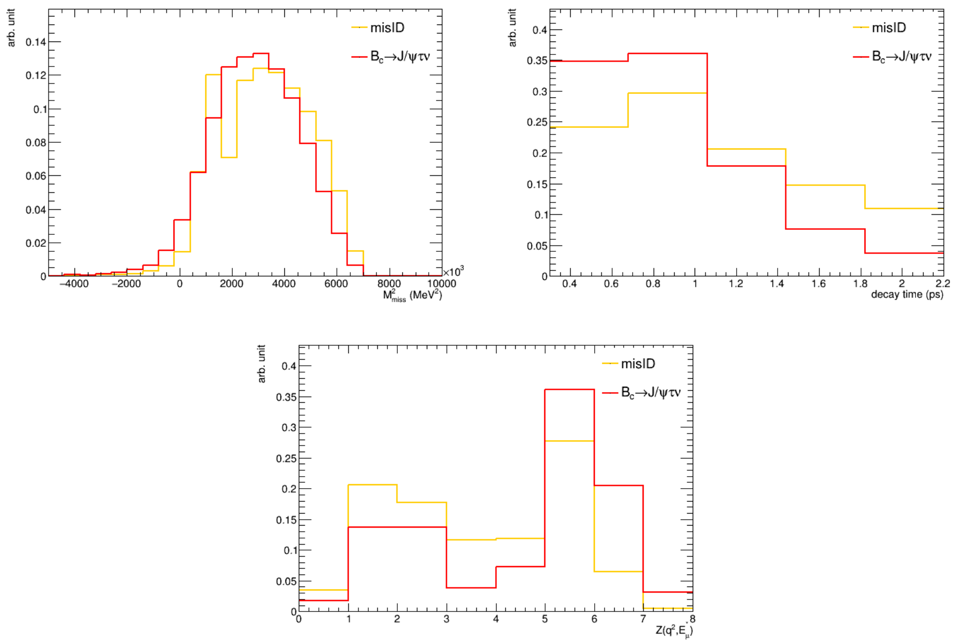


Figure 40: Projections of the fit templates for the misID background (yellow) compared with the signal mode (red).

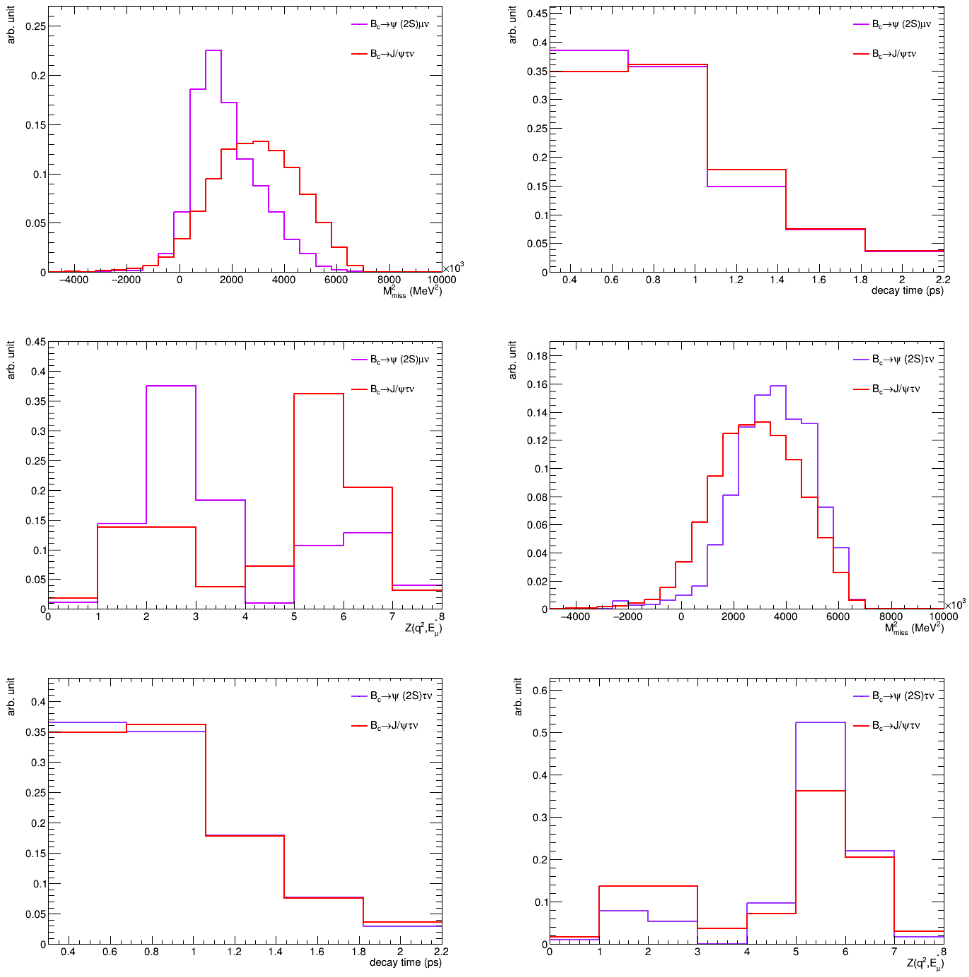


Figure 41: Projections of the fit templates for the feed-down $B_c^+ \to \psi(2S)\,\mu^+/\tau^+\,\nu_{\mu/\tau}$ background (violet, top and bottom respectively) compared with the signal mode (red).

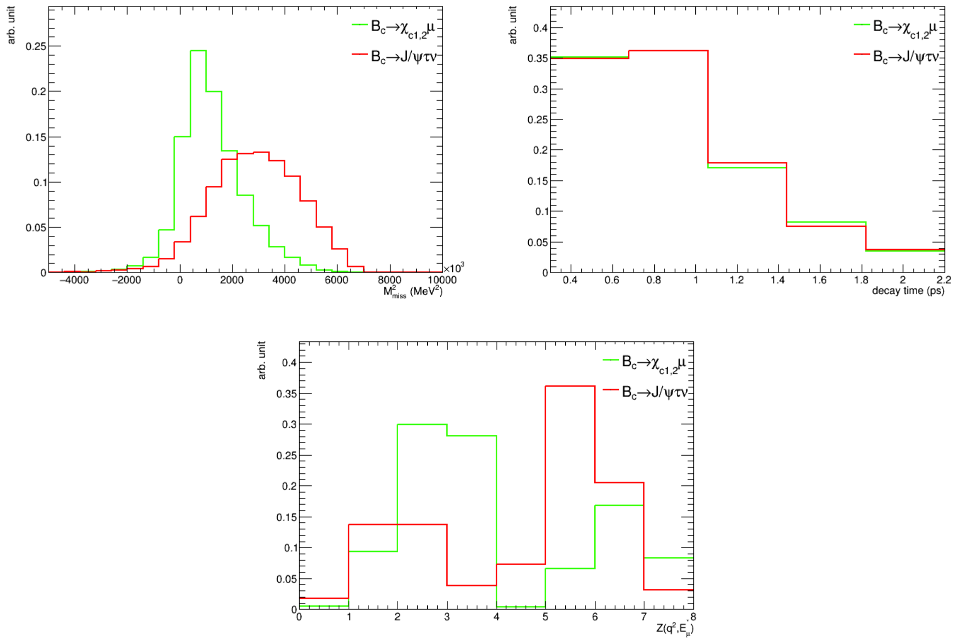


Figure 42: Projections of the fit templates for the feed-down $B_c^+ \to \chi_{c1,2}(1P)\,\mu^+\,\nu$ background (green) compared with the signal mode (blue).

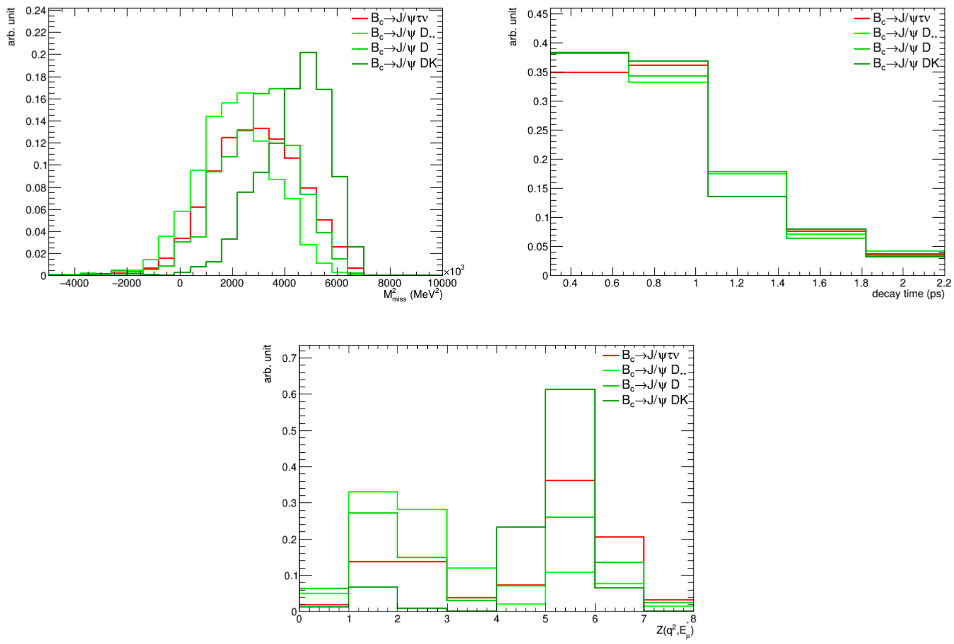


Figure 43: Projections of the fit templates for the double-charm $B_c^+ \to J/\psi\,D\,X$ background (green), modeled with a combination of two-body and quasi-two-body decays, along with the signal mode (red).

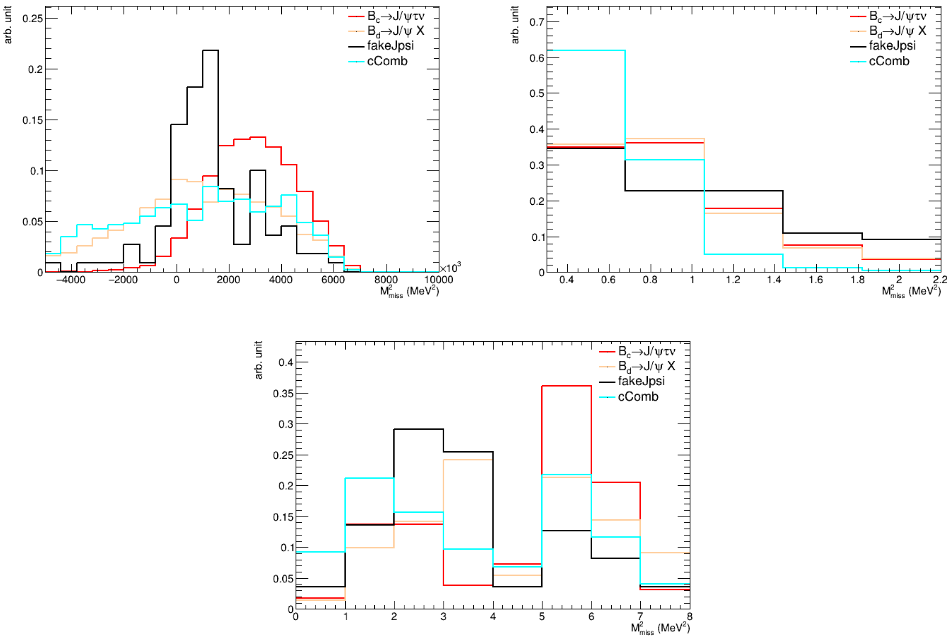


Figure 44: Projections of the fit templates for the $(\mu^+ + \mu^-)\mu^+$ combinatorial background (black), the combinatorial background from $B_d \to J/\psi X$ (tan), and the combinatorial background from the associated charm (teal), compared with the signal mode (red).

15.3 Fit results

The fit has been performed on the full 2016, 2017, and 2018 data combined, with projections of the fit variables shown in Figures 45 and 46 for the two alternative nominal fits using different mis-ID templates. The form factor variations, misID decay in-flight smearing, relative fraction between two body and quasi-two body backgrounds, and relative fraction between the B_d and B_s candidate contributions to the combinatorial background are shared between years. A final reweighting is applied to the nominal fit in order to improve data/MC agreement in variables identified after a postfit investigation, which is described in more detail in Sections 16.1 and 16.2. All results shown have undergone the final reweighting procedure.

The raw numbers of events for all components from the fit are summarized in Table 21, and the fitted nuisance parameters are given in Table 22. The correlations between the different nuisance parameters are shown in Figure 47. The yields for the double charm backgrounds given in Table 21 and calculated through the normalization factors and the ratio parameter from 22 are discrepant, but this is due to an issue with how the nuisance parameter for the ratios between these backgrounds was defined, and is not a cause for concern. The proper yields for the double charm backgrounds are in Table 21.

The study remains blind for the $R(J/\psi)$ value to avoid introducing bias before finalizing the measurement by blinding the yield of the signal in the fit. The blinding procedure adds a fixed blinded offset to the value of $R(J/\psi)$ measured in HistFactory.

Parameter	Value	Constraint

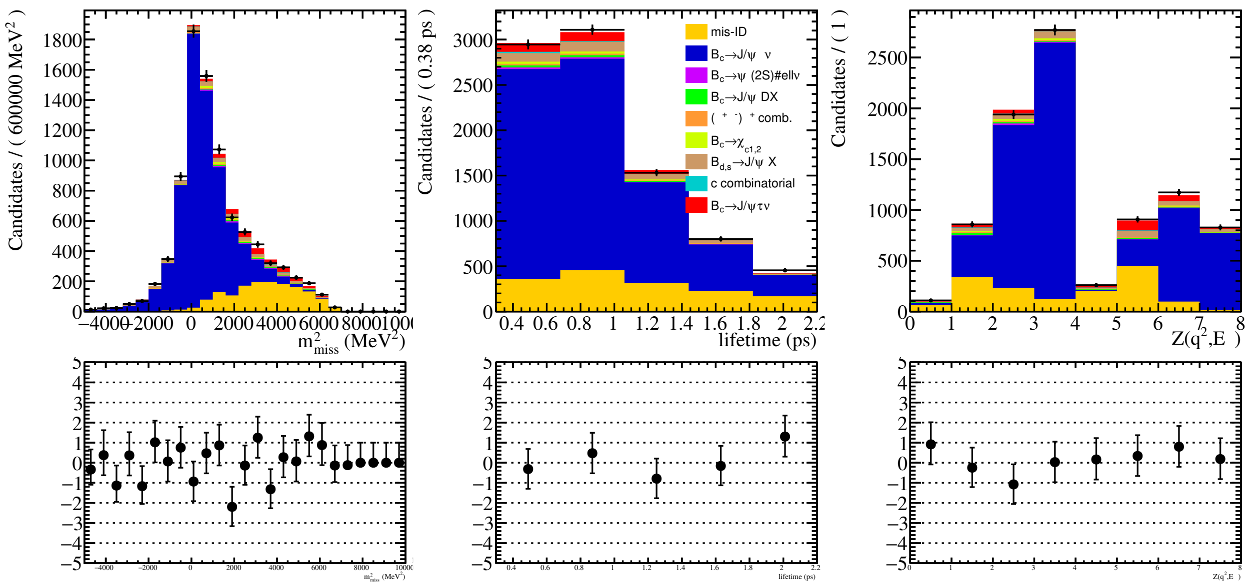
Raw $R(J/\psi)$	N/A	none
Num. of $J/\psi\mu\nu$ (2016)	6425 ± 118	none
Num. of $J/\psi\mu\nu$ (2017)	9240 ± 171	none
Num. of $J/\psi\mu\nu$ (2018)	8521 ± 163	none
Num. of $B_{d,s} \to J/\psi X$ comb. w/ μ from prompt D (2016)	0 ± 14	none
Num. of $B_{d,s} \to J/\psi X$ comb. w/ μ from prompt D (2017)	17 ± 45	none
Num. of $B_{d,s} \to J/\psi X$ comb. w/ μ from prompt D (2018)	0 ± 125	none
Num. of $B_d \to J/\psi X$ comb. (2016)	265 ± 65	none
Num. of $B_d \to J/\psi X$ comb. (2017)	267 ± 78	none
Num. of $B_d \to J/\psi X$ comb. (2018)	87 ± 63	none
Ratio of B_s/B_d	0.1 ± 1.0	0.123 ± 0.0115
Num. of c comb. (2016)	24 ± 27	none
Num. of c comb. (2017)	54 ± 36	none
Num. of c comb. (2018)	91 ± 37	none
$\chi_c\mu\nu$ cons. (2016)	-0.3 ± 0.4	$1e-6$
$\chi_c\mu\nu$ cons. (2017)	-0.3 ± 0.5	$1e-6$
$\chi_c\mu\nu$ cons. (2018)	-0.3 ± 0.5	$1e-6$
$J/\psi D$ (2016)	-0.1 ± 1.0	± 0.14
$J/\psi D$ (2017)	0.1 ± 1.0	± 0.14
$J/\psi D$ (2018)	0.1 ± 1.0	± 0.14
Ratio of D^{**}/all double charm	0 ± 1	0.75 ± 0.20
Ratio of D/D^{**}	-0.1 ± 1.0	0.38 ± 0.09
D_s cons. (2016)	0 ± 1	± 0.65
D_s cons. (2017)	0 ± 1	± 0.68
D_s cons. (2018)	0 ± 1	± 0.38
Fake J/ψ cons. (2016)	-1.0 ± 0.5	
Fake J/ψ cons. (2017)	-1.5 ± 0.6	
Fake J/ψ cons. (2018)	-0.9 ± 0.7	
$\psi(2S)$ cons. (2016)	-0.4 ± 0.8	
$\psi(2S)$ cons. (2017)	-0.3 ± 0.8	
$\psi(2S)$ cons. (2018)	1.2 ± 0.8	
χ_c cons. (2016)	-0.6 ± 0.8	
χ_c cons. (2017)	0.0 ± 0.7	
χ_c cons. (2018)	0.0 ± 0.9	

$\psi(2S)\mu\nu$ cons. (2016)	-0.3 ± 0.5	$1e-6$
$\psi(2S)\mu\nu$ cons. (2017)	-0.3 ± 0.5	$1e-6$
$\psi(2S)\mu\nu$ cons. (2018)	-0.2 ± 0.1	$1e-6$
$R(\psi(2S))$	0.021	fixed
Num. of misID (2016)	1612 ± 224	none
Num. of misID (2017)	2480 ± 338	none
Num. of misID (2018)	2556 ± 251	none
Amt. of K misID smear (2016)	0.6 ± 0.5	Gaussian (-1 unsmeared, 1 double-smeared)
Amt. of K misID smear (2017)	1.0 ± 0.4	Gaussian (-1 unsmeared, 1 double-smeared)
Amt. of K misID smear (2018)	0.9 ± 0.4	Gaussian (-1 unsmeared, 1 double-smeared)
Amt. of π misID smear (2016)	0.0 ± 0.7	Gaussian (-1 unsmeared, 1 double-smeared)
Amt. of π misID smear (2017)	1.1 ± 0.6	Gaussian (-1 unsmeared, 1 double-smeared)
Amt. of π misID smear (2018)	0.9 ± 0.7	Gaussian (-1 unsmeared, 1 double-smeared)
MisID DIF corr. (2016)	0.00 ± 0.05	
MisID DIF corr. (2017)	0.00 ± 0.03	
MisID DIF corr. (2018)	-0.6 ± 0.4	
α_0	0 ± 1	Gaussian
α_1	0 ± 1	Gaussian
α_2	0 ± 1	Gaussian
α_3	0 ± 1	Gaussian
α_4	0 ± 1	Gaussian
α_5	0 ± 1	Gaussian
α_6	0 ± 1	Gaussian
α_7	-0.4 ± 0.9	Gaussian
α_8	0.5 ± 0.8	Gaussian
α_9	0.6 ± 0.7	Gaussian
α_{10}	0.5 ± 0.8	Gaussian
α_{11}	0 ± 1	Gaussian
α_{12}	0 ± 1	Gaussian
α_{13}	-0.4 ± 0.9	Gaussian
α_{14}	0.2 ± 0.7	Gaussian

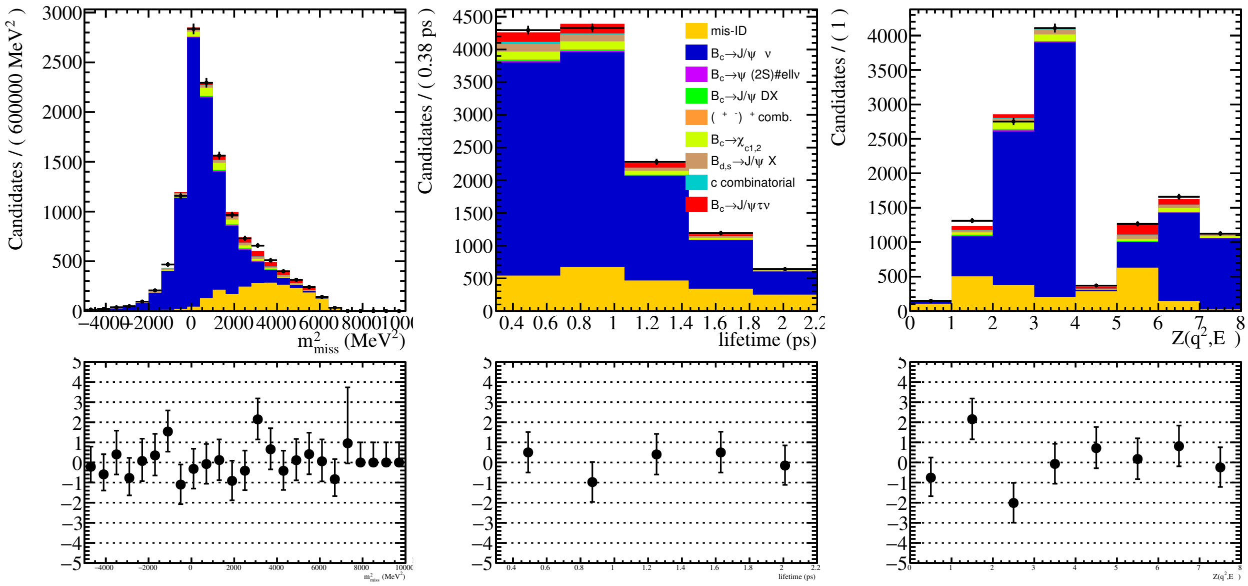
Table 22: Nuisance parameters in nominal fit. Uncertainties are statistical with Beeston-Barlow enabled, constraints are Gaussian unless stated otherwise. See Section 14.1 for details on α_i form factor variation constraints and Section 17.10 for details on mis-ID smearing constraints.

Table 21: The raw numbers of events for each component from the fit to data for each year with Beeston Barlow enabled. Uncertainties are statistical.

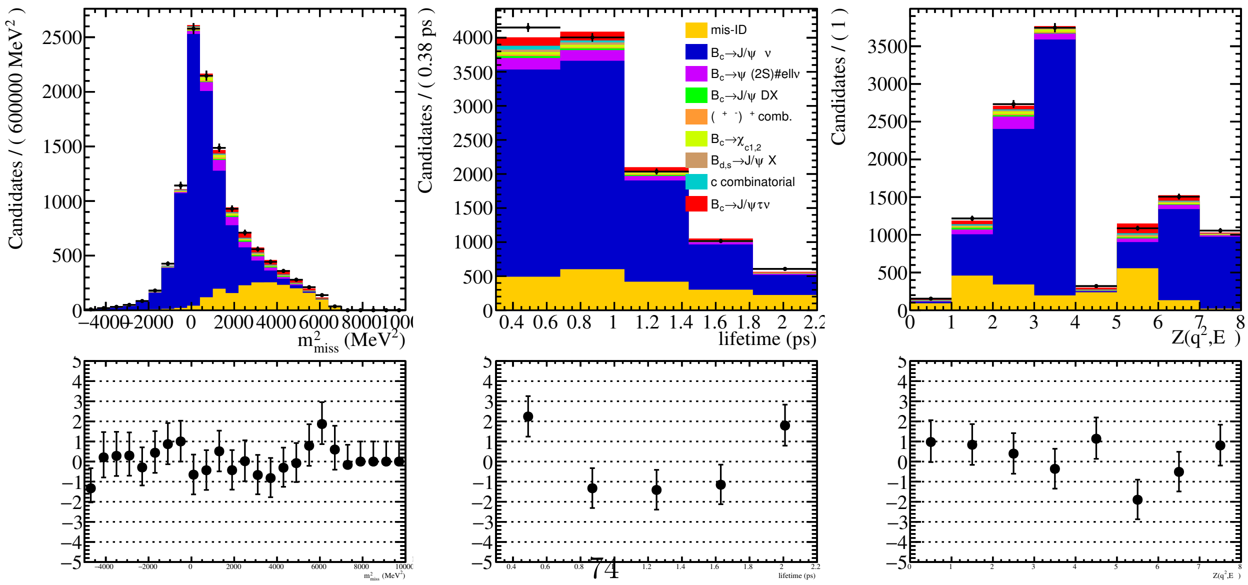
Component	2016	2017	2018
$B_c^+ \to J/\psi\, \tau^+\, \nu$	N/A	N/A	N/A
$B_c^+ \to J/\psi\, \mu^+\, \nu$	6425 ± 118	9240 ± 171	8520 ± 163
$B_c^+ \to J/\psi\, D$	44 ± 6	40 ± 6	55 ± 8
$B_c^+ \to J/\psi\, D^{**}$	13 ± 5	12 ± 4	17 ± 6
$B_c^+ \to J/\psi\, D\, K$	4 ± 1	4 ± 1	5 ± 2
$B_c^+ \to \psi(2S)\, \mu^+\, \nu$	64 ± 27	94 ± 39	260 ± 229
$B_c^+ \to \psi(2S)\, \tau^+\, \nu$	0 ± 0	1 ± 0	2 ± 0
$B_c^+ \to \chi_c\, \mu^+\, \nu$	134 ± 77	292 ± 75	277 ± 181
misID bkg.	1518 ± 100	2273 ± 119	2074 ± 251
Comb. $(J/\psi + \mu^+)$ bkg. $(B_{u,d} \to J/\psi X)$	265 ± 65	284 ± 78	87 ± 63
Comb. $(J/\psi + \mu^+)$ bkg. $(B_s \to J/\psi X)$	33 ± 11	33 ± 12	10 ± 8
Comb. c bkg.	24 ± 27	55 ± 36	90 ± 37
Comb. $(\mu^+ + \mu^-)\mu^+$ bkg.	48 ± 17	25 ± 10	60 ± 16



(a) 2016 fit.

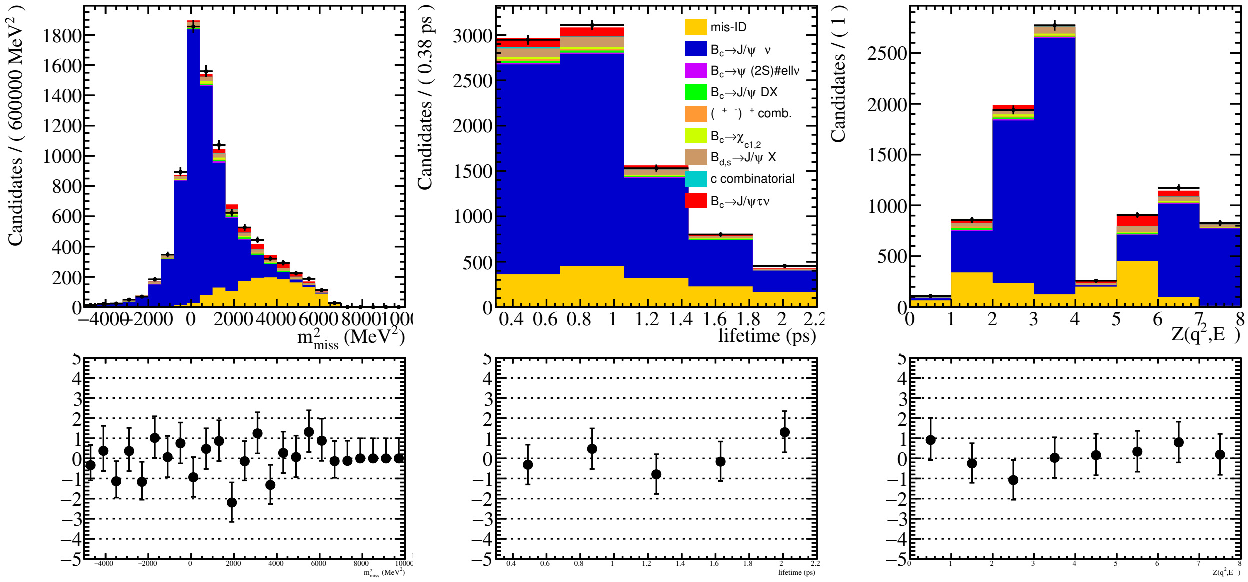


(b) 2017 fit.

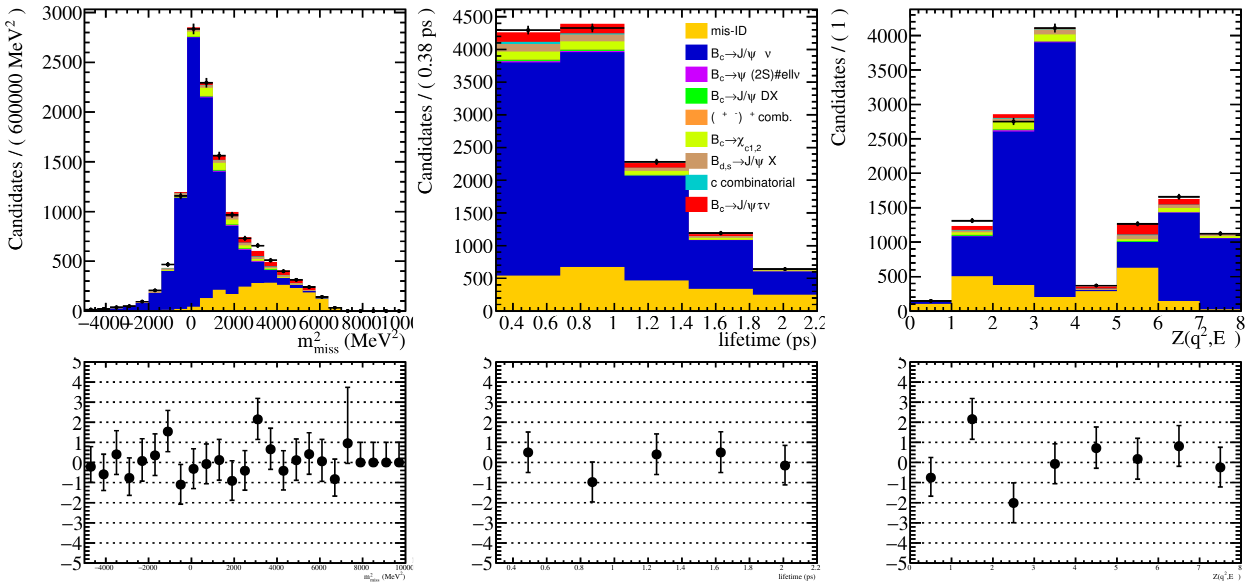


(c) 2018 fit.

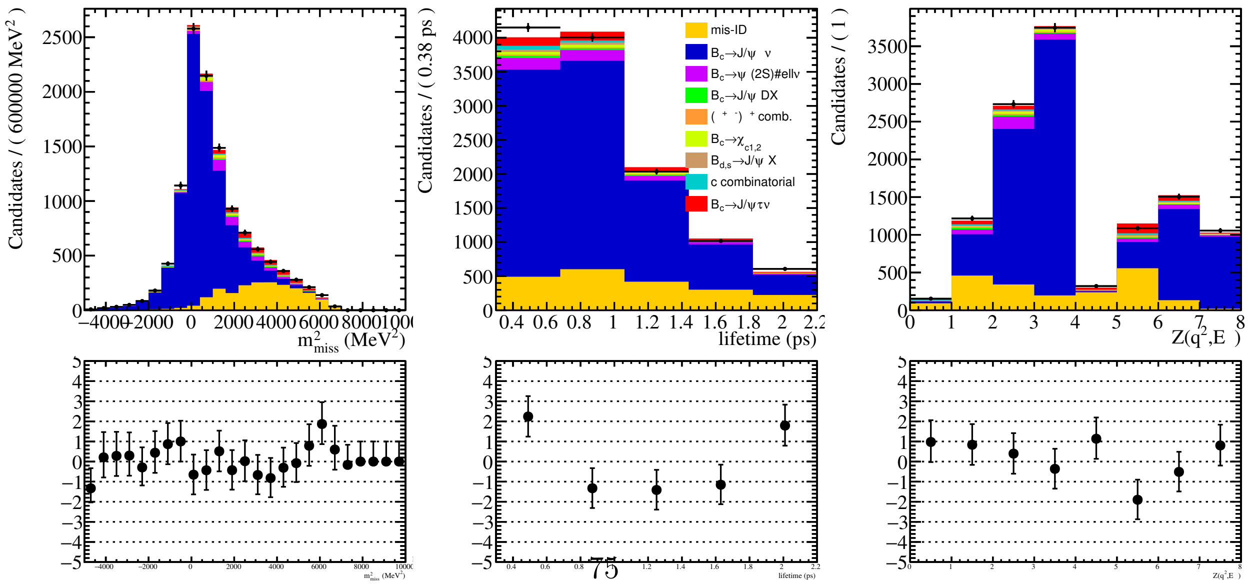
Figure 45: The results of the fit to the full data using the UMD mis-ID method, projected in the fit variables. The lower panels show the residual distributions.



(a) 2016 fit.



(b) 2017 fit.



(c) 2018 fit.

Figure 46: The results of the fit to the full data using the UCAS mis-ID method, projected in the fit variables. The lower panels show the residual distributions.

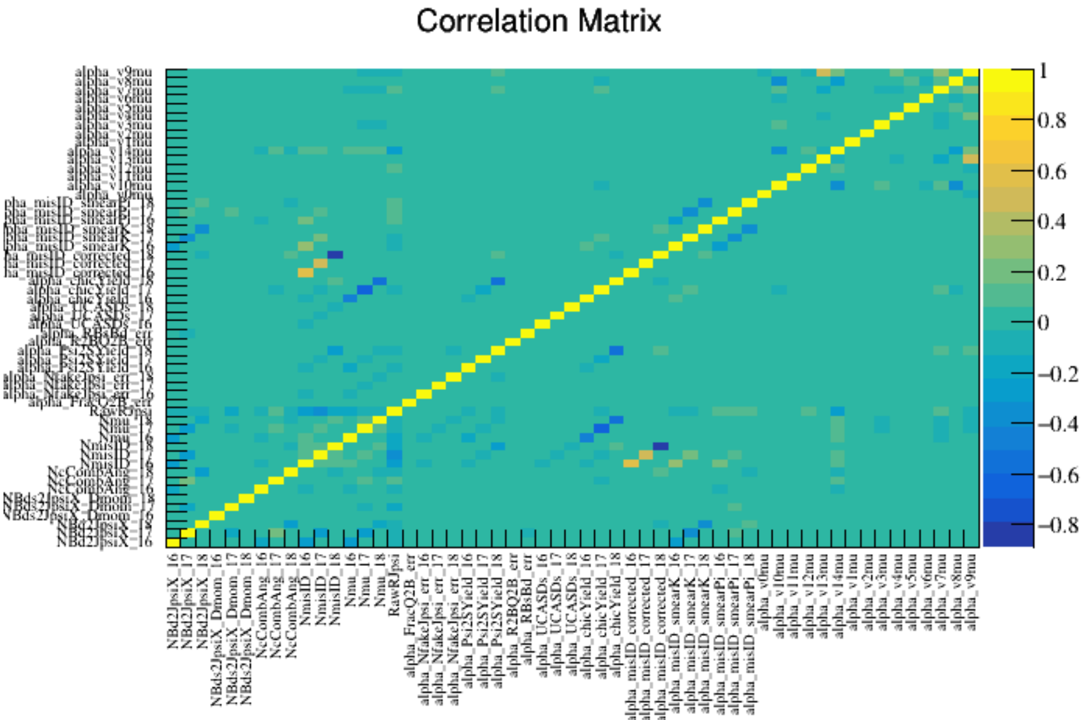


Figure 47: Correlation matrix for the nominal fit with nuisance parameters labeled on the axes.

16 Fit validation

16.1 Data and simulation comparison

The agreement between data and simulation is further studied with the cocktail of simulations weighted by their relative fractions determined by the fit. This comparison also accounts for form factor variations following the procedure indicated in the HistFactory manual, using linear extrapolation for when the variation $\|\alpha_i\| > 1$, where the variation is simply the difference between the nominal value of the weight and the value of the weight given by the variation of the ith form factor variation, multiplied by α_i. In the case of $\|\alpha_i\| < 1$, the Histfactory manual gives us that

$$w = w_{nom} + \sum_i \alpha_i \times (S_i + \alpha_i A_i \times (15 + \alpha_i^2 \times (-10 + 3\alpha_i^2))), \tag{69}$$

where

$$\begin{aligned} S &= 0.5 \times (w_{i+} - w_{nom}) \\ A &= 0.0625 \times (w_{i+} - w_{i-}), \end{aligned} \tag{70}$$

and the weights $w_{i\pm}$ are the weights given by varying the ith form factor element by $\pm$ its uncertainty, taken directly from HAMMER [48].

The postfit template does not account for decay in flight smearing. The distributions of various tracking and vertexing variables in the normalization-rich region (defined in Section 14.2) are found to be in good agreement between data and simulation (see Figure 37). Because the fit remains blinded for $R(J/\psi)$, to normalize the signal template in these postfit plots we used the value of $R(J/\psi)$ given by an initial unblinding on 2016 data using a preliminary fit strategy.

The post-fit variable comparisons in this section use 2018 data and MC. Plots for 2016 and 2017 are available in Appendix 24. The $J/\psi\,\mu^+$ system's vertex fit χ^2 and flight distance from the PV are shown in Figures 50. The J/ψ candidate vertex fit χ^2 and IP χ^2 with respect to the PV are shown in Figures 49, 53. Both the muons from the J/ψ decay and the unpaired muon's IP χ^2 with respect to the PV are shown in Figures 52, 51, and 55.

Figure 48 is the plot of data and the MC cocktails with yields determined by the fit–as a check of the functionality of the postfit code, this plot should be very similar to the plots in Figure 131c, with the exception of the lack of mis-ID smearing. With this check completed, other relevant variables have been plotted to check the data/MC agreement in Figures 51, 50, 49, and 52.

16.2 Final reweighting

After the initial fit, a comparison was done between data and MC using a cocktail of MC with the proportions of different channels fixed by the results of the fit, as seen in Section 16.1. Some variables showed disagreement between data and MC, most notably p, p_T, η, and the IP χ^2 of the best fit PV. These disagreements were present in the reconstructed

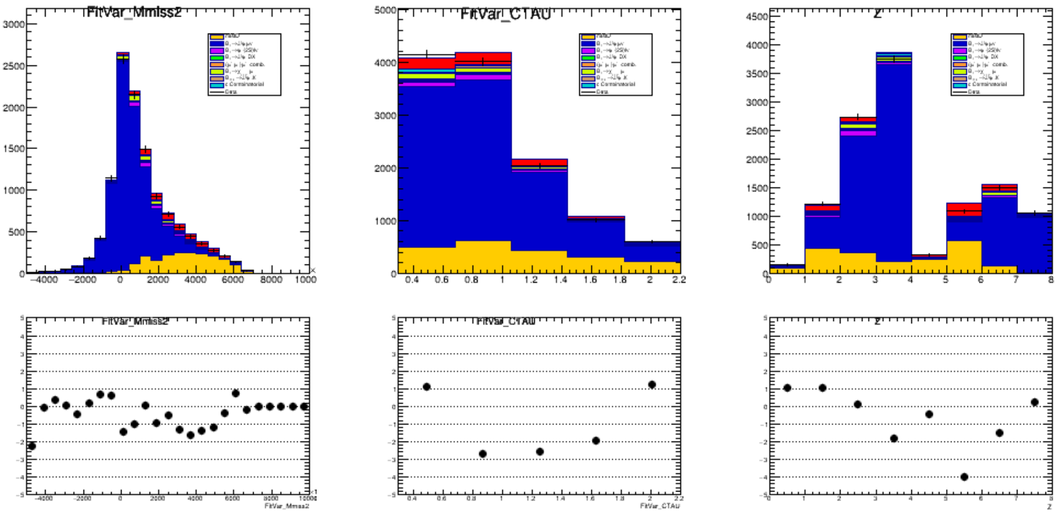


Figure 48: Plots for the fit variables with MC yields fixed from fit in 2018.

J/ψ, bachelor muon, B_c, and the dimuon pair candidates. There was also a discrepancy in the χ^2 of the end vertex of the B_c candidate (and equivalently, of the J/ψ candidate).

To correct for these discrepancies, we reweight the MC tuples using a `GBReweighter`. The original sample was constructed by creating an MC cocktail, with each sample weighted to the expected proportions given by the fit. The initial weights (e.g. lifetime, multiplicity, and form factor corrections) were also provided as input. For the target sample, we used data with the expected contributions from data-driven backgrounds (misID and the fake J/ψ combinatorial) weighted out. The reweighting was performed using samples in the normalization-rich region.

The input variables to the reweighter are the p, p_T, η, $\log(\mathrm{IP}\chi^2)$ of the primary vertex, and the $\sqrt{\mathrm{IP}\chi^2}/\mathrm{IP} \approx 1/\sigma_{IP}$ (since $\mathrm{IP}\chi^2 \approx (\mathrm{IP}/\sigma_{\mathrm{IP}})^2$) of the primary vertex for the B_c, J/ψ, bachelor muon candidates, and J/ψ daughter muon candidates sorted by higher and lower p_T. For the J/ψ and B_c candidates, we also use the χ^2 of the end vertex.

Finally, the distance of closest approach of the track with respect to the beam axis (DOCA(z)) was used. The first reconstruction step in the VELO is $R-\phi$ projection, which results in tracks not being reconstructed in a straight line when they don't intersect with the beam axis. These tracks are thus reconstructed with less efficiency. To account for this, the DOCA(z) variable is added to the final reweighting. The location of the beam axis, or z, is calculated separately for each year and polarity by averaging the location of primary vertex with respect to time. This variable significantly improves the lifetime fit.

The $\log(\mathrm{IP}\chi^2)$ was the most significant input to the final reweighting in improving fit quality overall. This is suspected to be related to detector acceptance effects.

Following some testing of alternative parameters, we used `n_estimators`=100, `learning_rate`=0.1, `max_depth`=3, `min_samples_leaf`=100, and `subsample`=0.4.

After reweighting, the disagreements between data/MC show significant improvement, see Figures 58, 59, 60, 61, and 62. Postfit plots of the relevant variables were performed before and after the final reweighting in order to validate its intended effect (see more in Section 15.2).

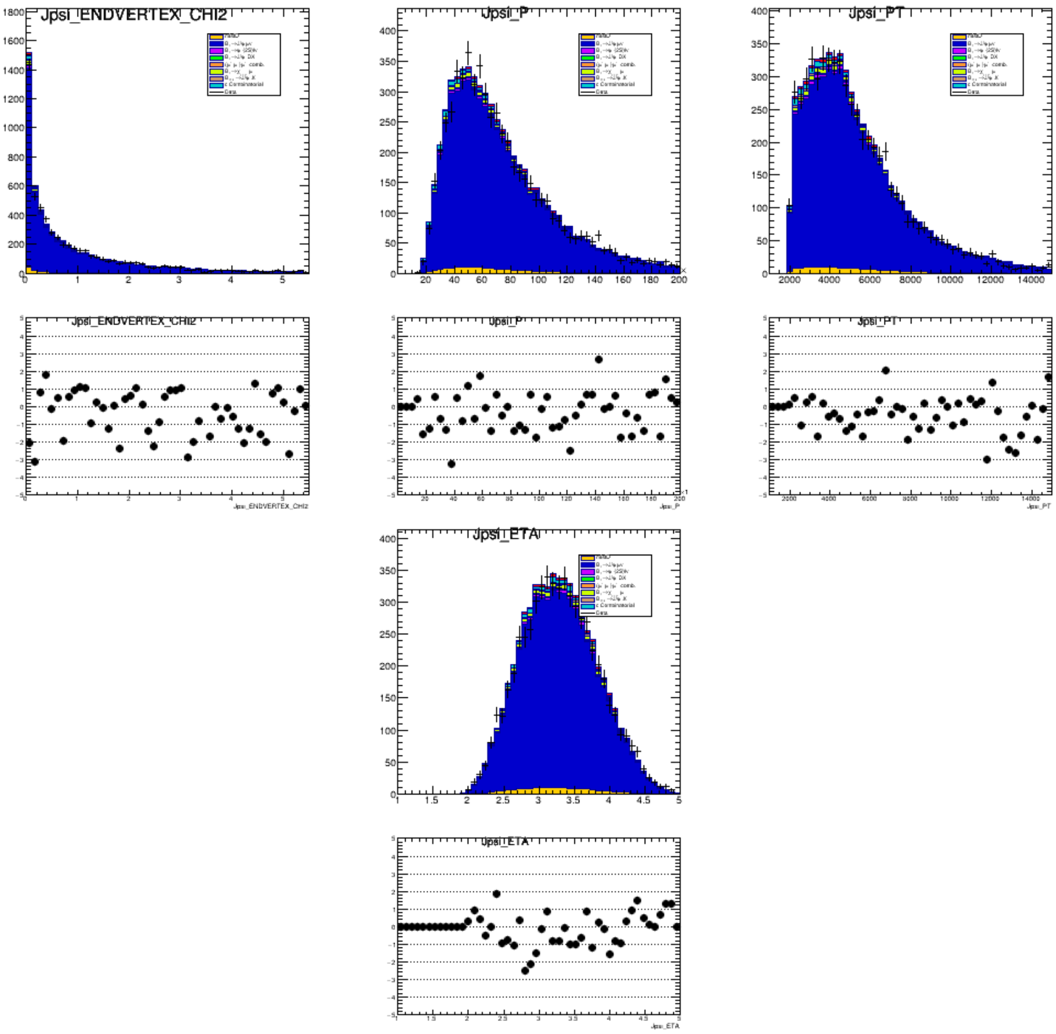


Figure 49: Comparison between data and simulation for the J/ψ candidate's vertex χ^2 along with the momentum, transverse momentum, and eta in 2018.

16.3 Toy studies

A maximum likelihood estimate (MLE) of a physical parameter produced using a finite dataset may be biased. Take for example the MLE estimate of the variance of a normal distribution fit to N events, the factor of $1/N$ must be replaced with $1/(N-1)$ in order to produce an unbiased estimate. A maximum likelihood fit could underestimate certain processes and overestimate others, resulting in a biased measurement.

The bootstrap procedure is used to estimate this possible bias. In an unbinned maximum likelihood fit, the parametric bootstrap samples values around the best fit point from the different analytic PDFs to produce a set of pseudo-datasets. The parameter of interest is measured in each of these pseudo-datasets using the fit procedure. The bias is then estimated by comparing the generated versus measured values of the parameter of interest.

A binned maximum likelihood fit samples from the total histogram template instead of an analytic PDF. Bias investigations using 5000 generated toy templates are shown in Figure 64. The fit configuration using the UMD mis-ID template shows a bias of

-0.03 ± 0.01, and the fit configuration using the UCAS mis-ID template has a bias of 0.1 ± 0.01. There no significant bias observed in the pull of $R(J/\psi)$. The biases between the two alternative fit configurations are in opposite directions, again providing justification for averaging the two values.

An additional bias study was performed on the different fit nuisance parameters. The results of these studies are reported in Table 23. No problematic biases which affect the outcome of $R(J/\psi)$, as judged by the mean of the pulls, the uncertainty of the pulls, and the correlation with $R(J/\psi)$.

Parameter	Mean pull	Pull error	Correlation with $R(J/\psi)$
NBd2JpsiX 16	0.0240	0.9088	-0.0978
NBd2JpsiX 17	-0.0083	0.9345	-0.0922
NBd2JpsiX 18	0.0366	0.8610	-0.0795
NBds2JpsiX Dmom 16	0.2442	0.3709	-0.0256
NBds2JpsiX Dmom 17	0.5474	1.7204	-0.0847
NBds2JpsiX Dmom 18	0.3297	0.4764	-0.0378
NChicMu 16	0.4295	0.7312	0.0025
NChicMu 17	-0.4415	1.0594	-0.0309
NChicMu 18	0.4009	0.5686	-0.0267
NPsi2Mu 16	0.0487	0.4873	-0.0427
NPsi2Mu 17	0.8096	1.2866	-0.0843
NPsi2Mu 18	-0.3157	0.7256	-0.0552
NcCombAng 16	0.0541	0.5453	-0.0436
NcCombAng 17	-0.0446	0.9326	-0.1074
NcCombAng 18	-0.0371	1.1514	-0.0662
NmisID 16	0.1768	0.4207	-0.0771
NmisID 17	0.7902	2.0209	-0.0791
NmisID 18	-0.0606	0.8838	-0.2672
Nmu 16	-0.5812	1.0148	-0.1144
Nmu 17	-0.0231	0.9598	-0.0452
Nmu 18	-0.4287	0.7442	-0.0933
alpha FracQ2B err	-0.0433	0.0953	0.0191
alpha NfakeJpsi err 16	0.3525	0.5474	-0.0177
alpha NfakeJpsi err 17	1.3431	0.6806	-0.0304
alpha NfakeJpsi err 18	0.7881	0.5721	-0.0181
alpha R2BQ2B err	0.1313	0.0633	0.0208
alpha RBsBd err	-0.0659	0.1280	-0.0407
alpha UCASDs 16	-0.1046	0.0910	-0.0205

alpha UCASDs 17	-0.0582	0.0676	-0.0172
alpha UCASDs 18	-0.0979	0.0926	0.0042
alpha misID corrected 16	-0.1053	0.3866	-0.0028
alpha misID corrected 17	-0.4179	2.0922	-0.0560
alpha misID corrected 18	0.3861	1.2121	0.0849
alpha misID smearK 16	-0.1548	0.8554	-0.0455
alpha misID smearK 17	-0.1011	0.8770	-0.0198
alpha misID smearK 18	-0.1823	0.7955	-0.0350
alpha misID smearPi 16	0.0977	0.7385	0.0611
alpha misID smearPi 17	-0.5290	0.7622	0.0777
alpha misID smearPi 18	-0.5197	0.7399	0.0182
alpha v0mu	0.0242	0.0437	0.0474
alpha v10mu	-0.1668	0.3958	-0.1311
alpha v11mu	0.0006	0.0012	-0.1869
alpha v12mu	-0.0561	0.1184	0.2355
alpha v13mu	-0.0273	0.1819	0.1970
alpha v14mu	0.1377	0.4904	-0.3145
alpha v1mu	-0.0558	0.0325	0.0224
alpha v2mu	-0.0193	0.0779	0.0169
alpha v3mu	0.0986	0.1523	0.1156
alpha v4mu	0.0092	0.0808	0.2444
alpha v5mu	-0.0398	0.1219	0.0674
alpha v6mu	-0.0820	0.1197	0.0845
alpha v7mu	0.1492	0.2653	0.1889
alpha v8mu	-0.2827	0.3261	-0.1566
alpha v9mu	0.0579	0.3461	-0.1589

Table 23: Nuisance parameter pulls in nominal fit (first column), pull uncertainties (second column), and correlations with $R(J/\psi)$ (third column).

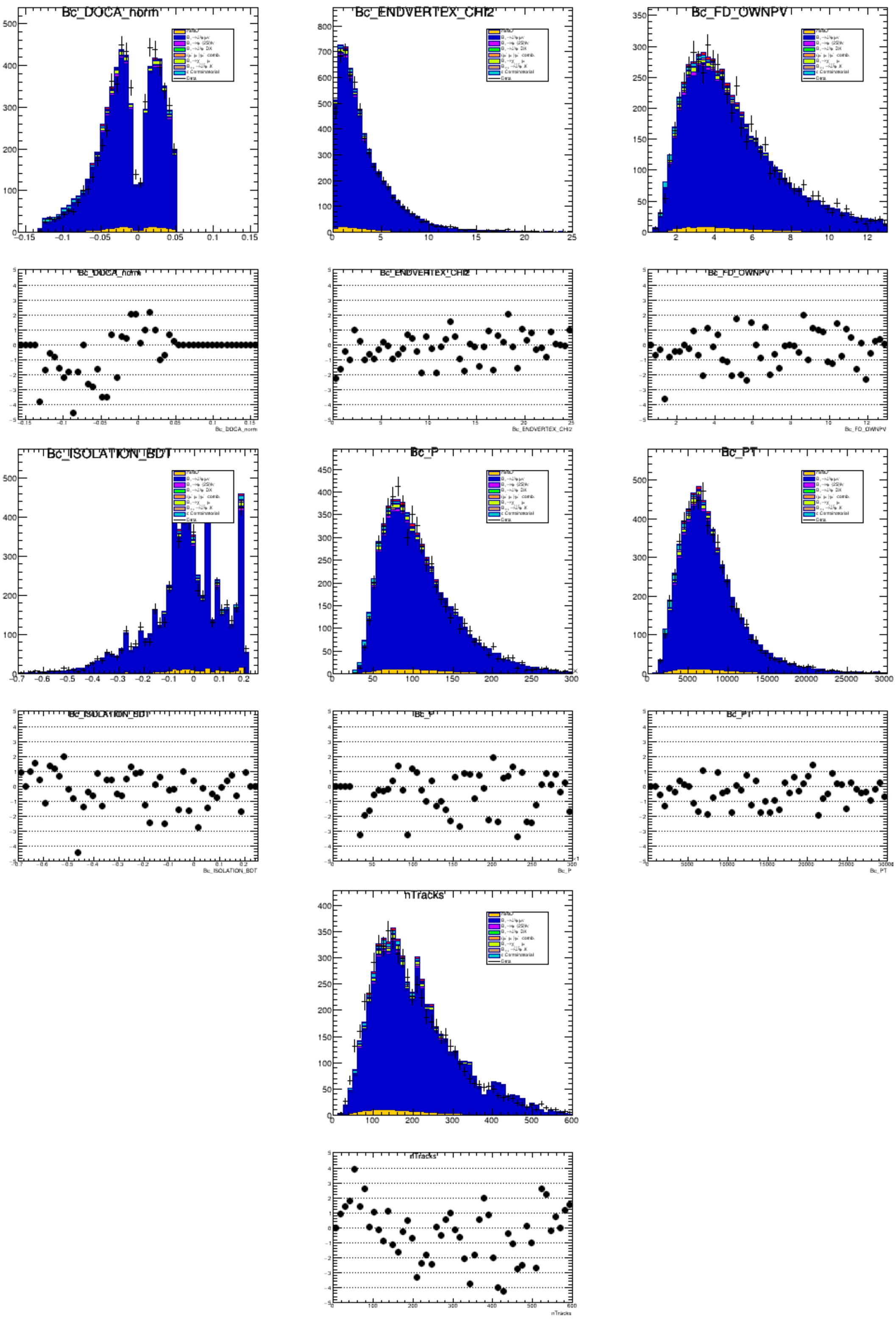


Figure 50: Comparison between data and simulation for the $J/\psi\mu^+$ system's DOCA, vertex fit χ^2, flight distance from the PV, momentum, transverse momentum, and nTracks in 2018.

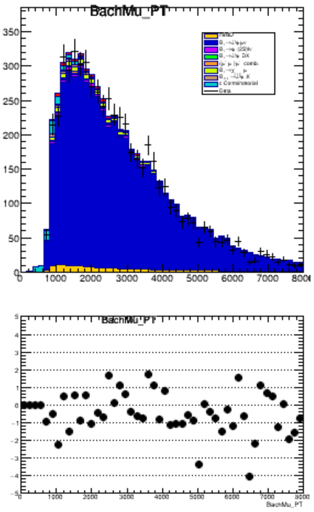


Figure 51: Comparison between data and simulation for the unpaired muon's momentum and transverse momentum in 2018.

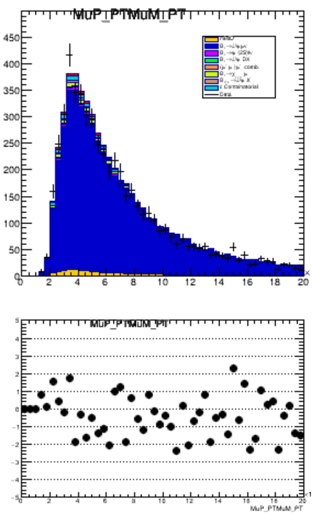


Figure 52: Comparison between data and simulation for the product of the PT of the dimuon system in 2018.

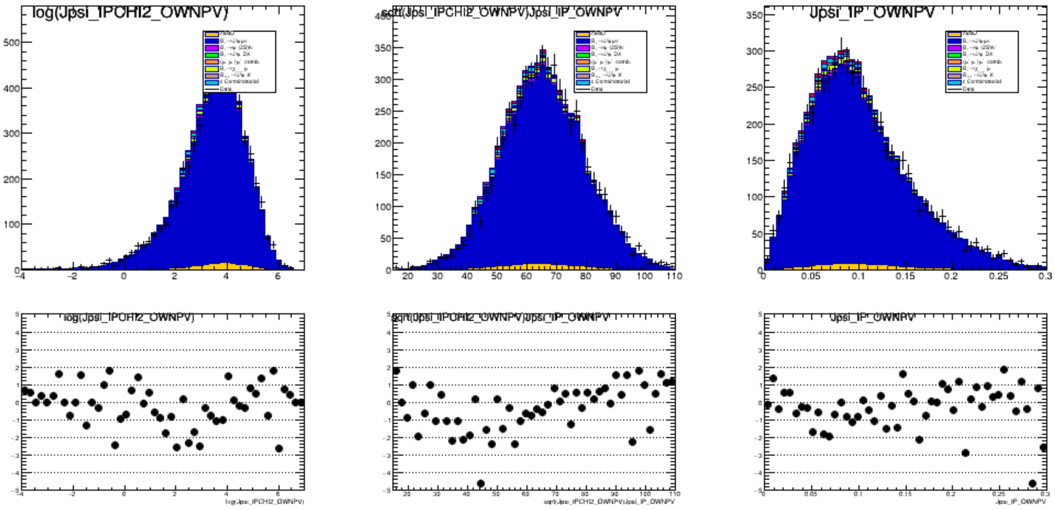


Figure 53: Comparison between data and simulation for the J/ψ's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2018.

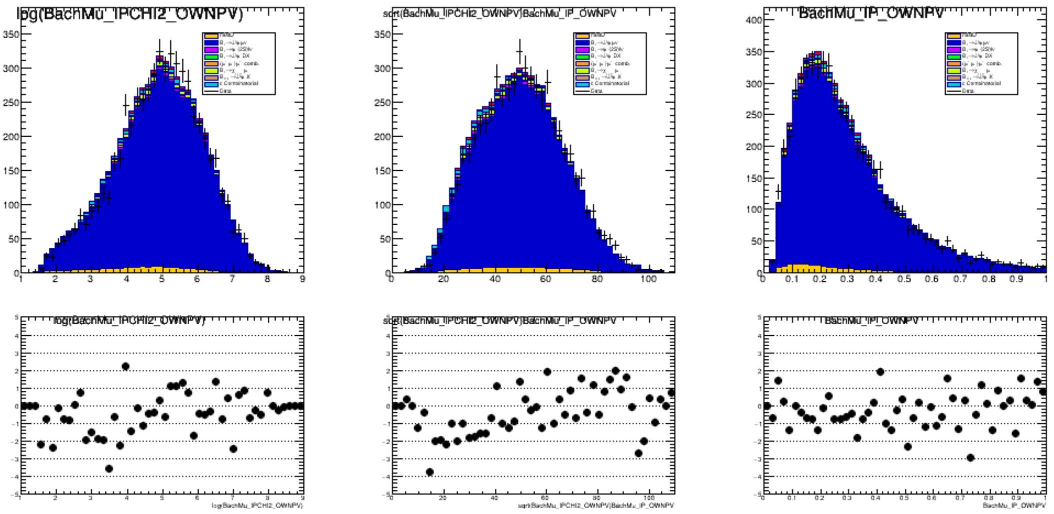


Figure 54: Comparison between data and simulation for the unpaired muon's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2018.

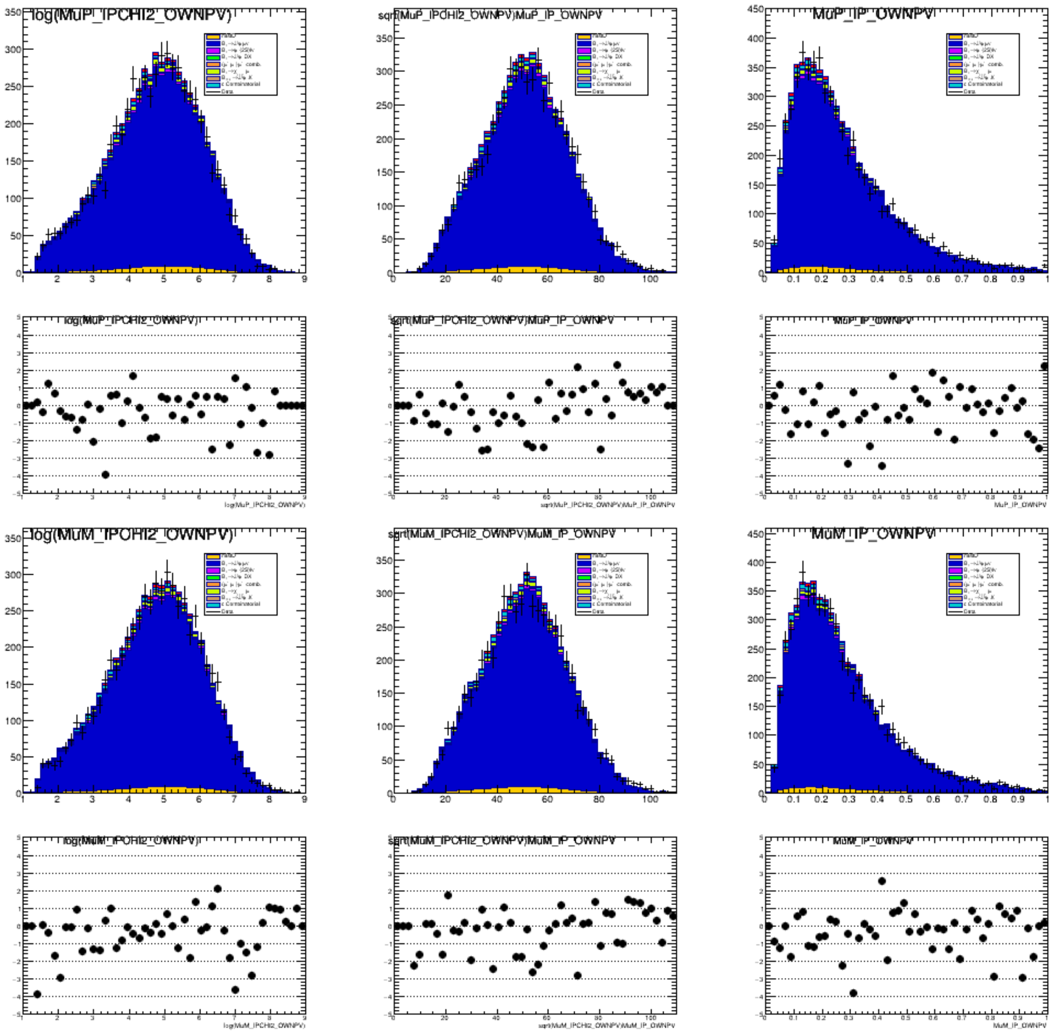


Figure 55: Comparison between data and simulation for the dimuon system's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2018.

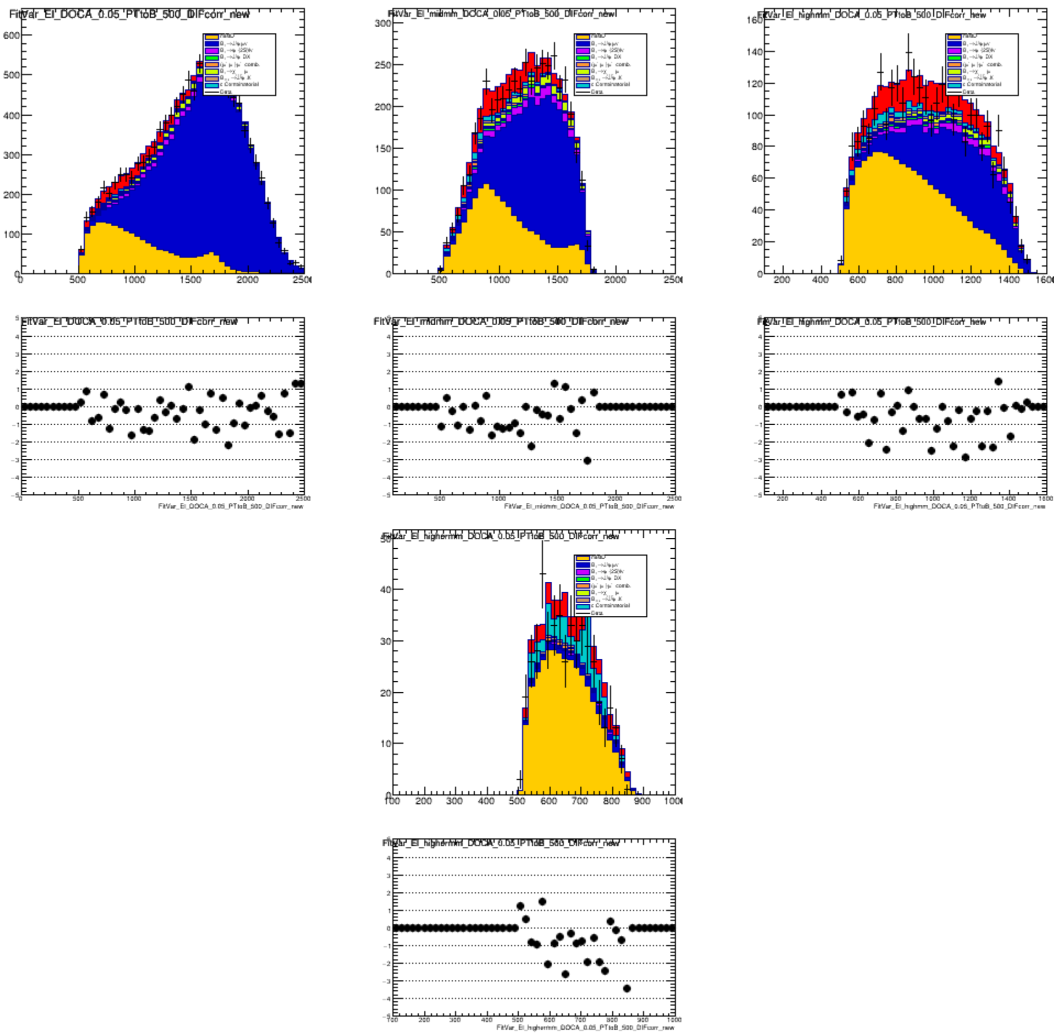


Figure 56: Comparison between data and simulation for the center of mass energy in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2018.

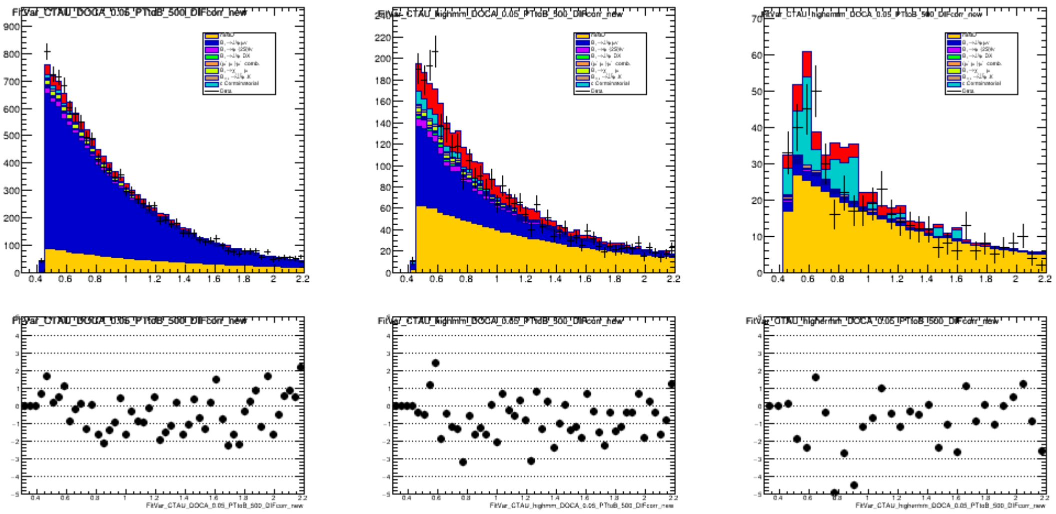
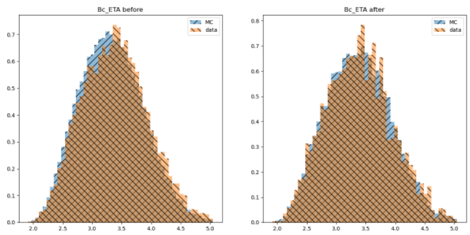
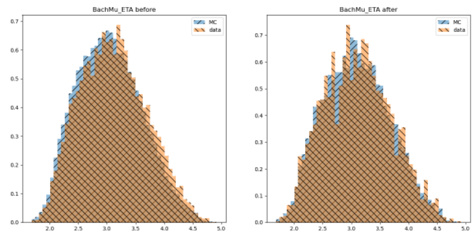


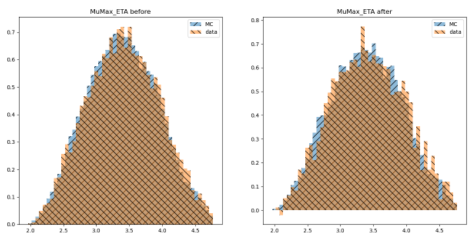
Figure 57: Comparison between data and simulation for the lifetime in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2018.



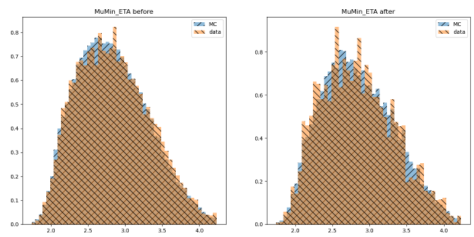
(a) η of B_c candidate



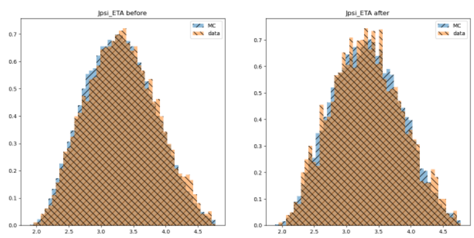
(b) η of unpaired muon candidate



(c) η of muon daughter candidate of J/ψ candidate with larger p_T

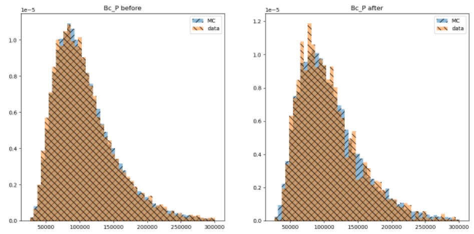


(d) η of muon daughter candidate of J/ψ candidate with smaller p_T

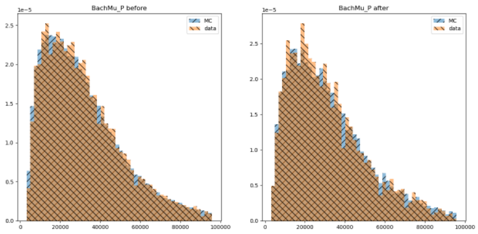


(e) η of J/ψ candidate

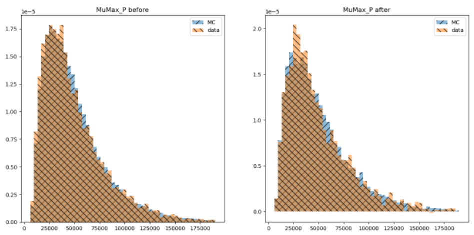
Figure 58: Before and after final reweighting comparison of MC cocktail to data in variable η. Left in each subfigure is before final reweighting, and right is after final reweighting. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.



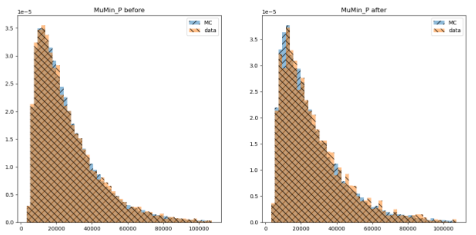
(a) p of B_c candidate



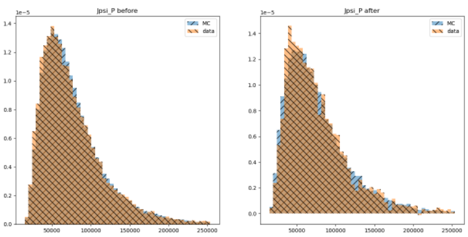
(b) p of unpaired muon candidate



(c) p of muon daughter candidate of J/ψ candidate with larger p_T

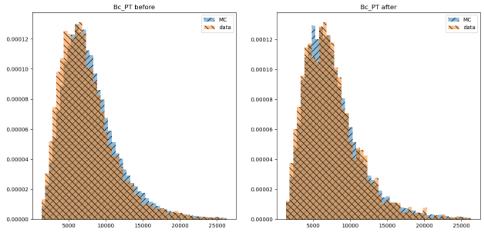


(d) p of muon daughter candidate of J/ψ candidate with smaller p_T

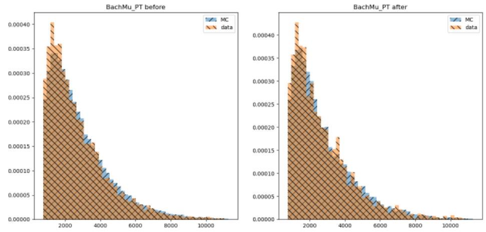


(e) p of J/ψ

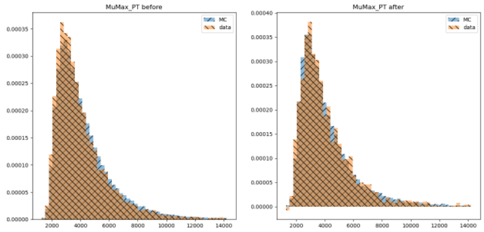
Figure 59: Before and after final reweighting comparison of MC cocktail to data in transverse momentum. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.



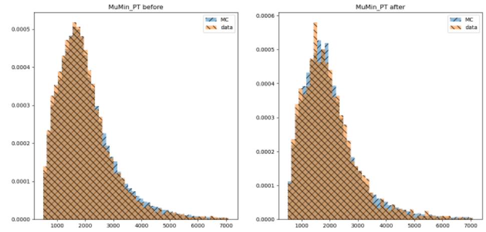
(a) p_T of B_c candidate



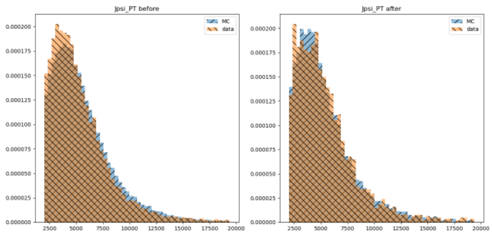
(b) p_T of unpaired muon candidate



(c) p_T of muon daughter candidate of J/ψ candidate with larger p_T

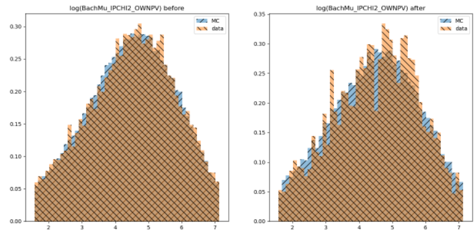


(d) p_T of muon daughter candidate of J/ψ candidate with smaller p_T

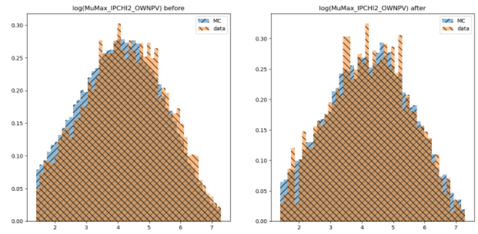


(e) p_T of J/ψ

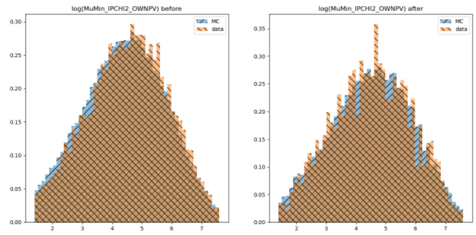
Figure 60: Before and after final reweighting comparison of MC cocktail to data in transverse momentum. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.



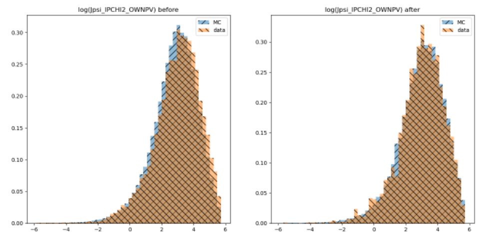
(a) $\log IP\chi^2$ of unpaired muon candidate



(b) $\log IP\chi^2$ of muon daughter candidate of J/ψ candidate with larger p_T

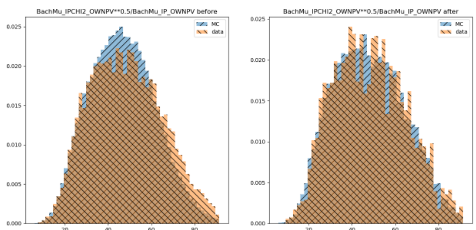


(c) $\log IP\chi^2$ of muon daughter candidate of J/ψ candidate with smaller p_T

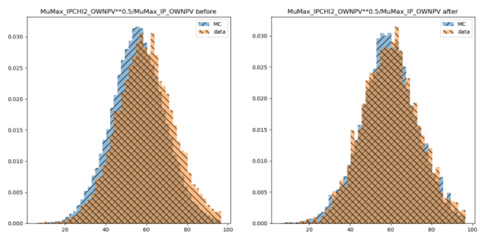


(d) $\log IP\chi^2$ of J/ψ candidate

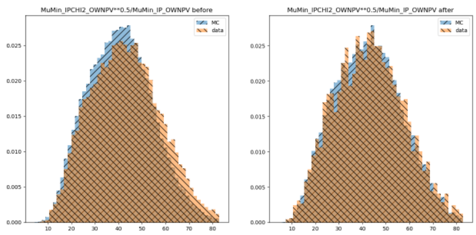
Figure 61: Before and after reweighting comparison of MC cocktail to data in $\log IP\chi^2$. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.



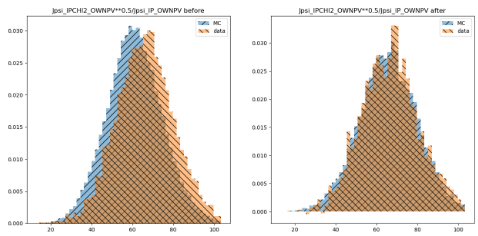
(a) $IP\chi^2/IP$ of primary vertex of unpaired muon candidate



(b) $IP\chi^2/IP$ of primary vertex of muon daughter candidate of J/ψ candidate with larger p_T

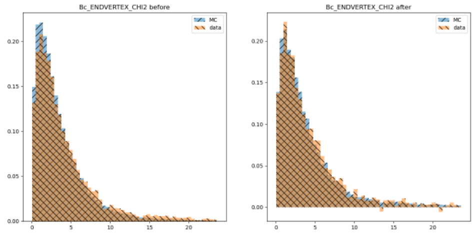


(c) $IP\chi^2/IP$ of primary vertex of muon daughter candidate of J/ψ candidate with smaller p_T

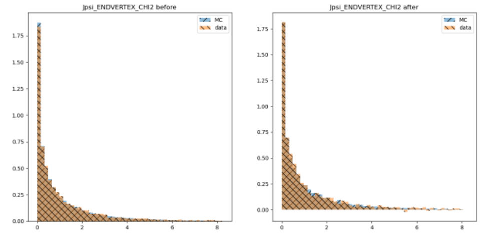


(d) $IP\chi^2/IP$ of primary vertex of J/ψ candidate

Figure 62: Before and after final reweighting comparison of MC cocktail to data in $IP\chi^2/IP$ of primary vertex. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.



(a) χ^2 of the end vertex of B_c



(b) χ^2 of the end vertex of unpaired muon

Figure 63: Before and after final reweighting comparison of MC cocktail to data in χ^2 of the end vertex. The MC distribution is shown in blue, and data is shown in orange. The training and testing data samples are plotted on the left and right hand plots, respectively.

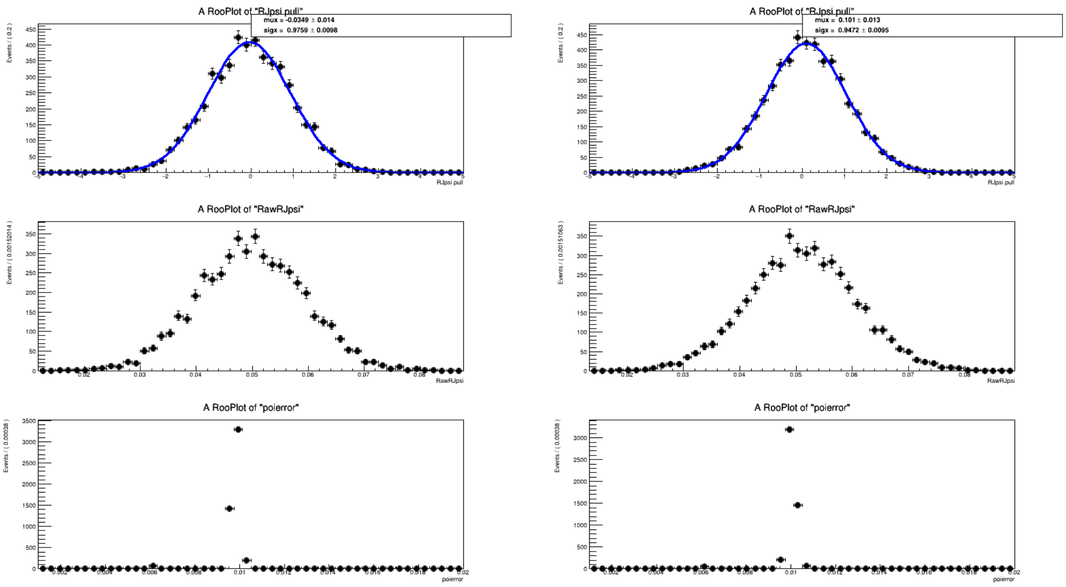


Figure 64: Bias investigations using 5000 toys with fits using UMD (top) and UCAS (bottom) mis-ID configurations. No significant bias in the $R(J/\psi)$ is observed.

17 Systematic uncertainties

The sources of systematic uncertainties in this study are discussed in this section. While the value of $R(J/\psi)$ remains blinded to avoid introducing bias before finalizing the measurement, the systematic uncertainties can be estimated with the error of R_{yield}, the ratio of signal vs normalization yields, as well as the discrepancy between different fit values of $R(J/\psi)$, since the blinded offset remains constant. The systematic uncertainties on R_{yield} are summarized in Table 24, with the statistical uncertainty listed for comparison. Each source of systematic uncertainty is described in greater detail below. The statistical uncertainty from the efficiency ratio is derived from the limited sample size of the MC used to evaluate efficiencies, and the branching fraction uncertainty is taken from the PDG [39]. Fits resulting from evaluating systematic uncertainties are relegated to Appendix 21.

Table 24: Summary of systematic uncertainties on the ratio of signal vs normalization yields, compared with the statistical uncertainty. These are scaled by the efficiency ratio and the branching fraction of $\tau \to \mu\nu_\mu\overline{\nu}_\tau$ and summed in quadrature to obtain the final value.

Source of uncertainty on $R(J/\psi)$	Value ($\times 10^{-2}$)
simulation statistical uncertainty	5.85
$B_c^+ \to \psi(2S)\ell\nu$ modeling	2.62
simulation corrections	2.03
$B_c^+ \to J/\psi\,\ell^+\,\nu$ form factors	1.84
mis-ID composition estimation	1.69
$B_c^+ \to \chi_c\,\mu\nu$ scaling	1.52
efficiency ratio	1.32
combinatorial charm modeling	1.27
mis-ID decay-in-flight correction	1.08
$(J/\psi + \mu^+)$ combinatorial modeling	0.56
mis-ID decay-in-flight smearing	0.34
$B_c^+ \to \psi(2S)\,\ell\nu$ scaling	0.32
$\mathcal{B}(\tau \to \mu\nu_\mu\overline{\nu}_\tau)$	0.23
trimuon effect	0.09
Systematic uncertainty	7.66
Statistical uncertainty	11.94
Total uncertainty	14.19

17.1 MC statistical uncertainty

The templates derived from simulations have shape uncertainties due to the finite simulation samples. Guided by the Run-1 analysis, the simulations are produced with sufficient sample sizes to keep this systematic uncertainty significantly below with the expected statistical uncertainty.

The uncertainty coming from the limited size of the MC samples is evaluated using the Beeston-Barlow-light procedure [52], implemented in the HistFactory package, with a nuisance parameter reflecting the number of MC events in each bin. The systematic uncertainty is evaluated as the quadrature difference in fit uncertainties with and without the Barlow-Beeston method enabled. We take the difference in quadrature between the fit uncertainties with and without Beeston-Barlow enabled, and report an uncertainty of $5.85 \times 10^{-}2$. Although tracker-only simulation was produced for this analysis, we elected not to use it due to the relatively small contribution of this systematic uncertainty.

17.2 $B_c^+ \to \psi(2S)\ell\nu$ modeling

Two MC samples were produced for this background using the Kiselev and EFG models, which are described in greater detail in Section 2.2.4. The nominal fit uses the EFG model, which is modeled by a relativistic constituent quark model. The Kiselev model uses an NRQCD treatment (see a more extended discussion of the two models in Section 2.2.8). The alternative fit uses the Kiselev model, and half the shift in $R(J/\psi)$ is 2.62×10^{-2}, which we use as our systematic uncertainty.

17.3 Simulation corrections

A final data/MC reweighting was performed using a BDT reweighter to correct for differences in certain kinematic variables, with the procedure described in further detail in Section 16.2. To estimate a systematic uncertainty accounting for the limitations of these corrections, we retrained the final reweighting BDT using different training parameters, with 200 estimators, 120 as the minimum number of samples per leaf, and a subsample parameterof 0.3. Half of the difference in $R(J/\psi)$ between this fit with the alternative final reweighting and the nominal fit gives a systematic uncertainty of 2.08×10^{-2}.

17.4 $B_c^+ \to J/\psi$ form factors

The form factors are taken from the lattice, as described in Section 2.2.4. However, as explained in Section 14.1.2, the discrepancy between data and lattice results in the values of the $B \to D^*\ell\nu$ form factors exceeds the estimated error bars from lattice. Therefore, in the nominal fit we take a more conservative approach by loosening the constraints on these parameters in the fit to twice the given lattice error bars. We estimate the systematic uncertainty by performing an alternate fit with the Gaussian constraints on the form factors removed, and then taking half the difference in the value of $R(J/\psi)$ with respect to the nominal value. This results in a systematic of 1.84×10^{-2}. We also looked at the case where we take the lattice error bars at face value, and the alternative fit from this resulted in a change in $R(J/\psi)$ smaller than the one from removing the Gaussian constraints, as expected.

17.5 mis-ID composition estimate

An alternative estimation of the species composition of the $J/\psi X$ sample was done in Section 29.1. The fit was performed using the template derived using the full probability method. Half of the shift in $R(J/\psi)$ was taken as the systematic uncertainty. This systematic also includes any systematic uncertainty associated with the modeling of ghosts, as the alternative template does not include ghosts. The systematic uncertainty is found to be 1.69×10^{-2}. The final result averages the two values of $R(J/\psi)$ obtained from the disparate methods, but other systematic uncertainties are computed relative to the value of $R(J/\psi)$ from the fit which uses unfolding.

17.6 χ_c background contribution

Similarly to $\psi(2S)$, the $\chi_{c1,2}$ background contribution is relatively small. After correcting for the isolation efficiency, we expect the $\chi_{c1,2}$ background to contribute 3.2% relative to the normalization events. In the nominal fit, the proportion of $\chi_{c1,2}$ background floats freely, and the fit does not predict any $\chi_{c1,2}$ event candidates. As an alternative fit, we fix the number of $\chi_{c1,2}$ events relative to normalization, and take half the resulting change in $R(J/\psi)$ as the systematic uncertainty, which is 1.52×10^{-2}.

17.7 Combinatorial charm background modeling

The combinatorial charm background was first introduced in Section 10. The dependence of the fit variables on the ratio $|p_D|/|p_{B_c}|$, where p_D and p_B are the momenta of the D and B_c, was studied. Particularly of concern was the region of high p_D/p_{B_c}, which may be favored by other production models which assume stronger production correlation between the D and B_c. The slice forming the top 30% of the p_D/p_{B_c} distribution was taken to form a template by adding a cut requiring $p_D/p_{B_c} > 0.6$. A fit with this altered template representation changes $R(J/\psi)$ by 40% relative to the fit without the combinatorial charm template. We take half of the shift in $R(J/\psi)$ as the systematic uncertainty for a value of 1.27×10^{-2}, as the top 30% is a harsh cut which is understood to be an upper bound. More details are given in Appendix 25.

To validate our uncertainty estimation, a toy study was performed by generating 4000 toys using alternative c combinatorial templates. The alternative templates used the high p_D/p_{B_c} template and the high $\cos\theta_{DB_c}$ c combinatorial templates. The toys were fitted using the nominal template and fit configurations, and the bias was assessed. The pulls of raw $R(J/\psi)$ are -0.16 ± 0.05 and -0.25 ± 0.05 respectively (see Figure 65), and the RMS of the pulls is very similar to the initial assessment, giving confidence in the uncertainty estimation.

17.8 mis-ID decay-in-flight correction

To account for a systematic uncertainty associated with different choices of decay-in-flight corrections, a systematic uncertainty is obtained by taking half the shift in $R(J/\psi)$ relative to a fit where the PIDCalib efficiencies are uncorrected for decay in flight. In the nominal fit, the fit is allowed to interpolate between the corrected and uncorrected templates. For 2016 and 2017, the fit chooses α such that the interpolated template shape is close to that of the corrected shape, while in 2018 the fit chooses a shape averaged between the

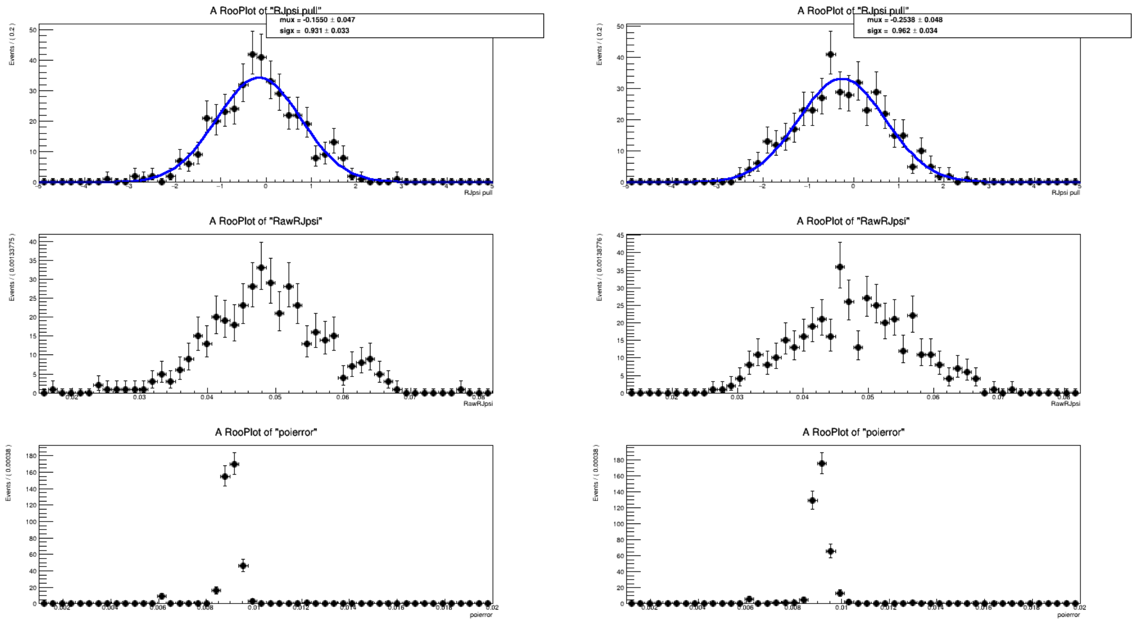


Figure 65: Toy studies generated using high p_D/p_{B_c} and high $\cos\theta_{DB_c}$ c combinatorial templates (left and right), fitted with nominal configuration. The pulls of $R(J/\psi)$ are -0.16 ± 0.05 -0.25 ± 0.05, respectively.

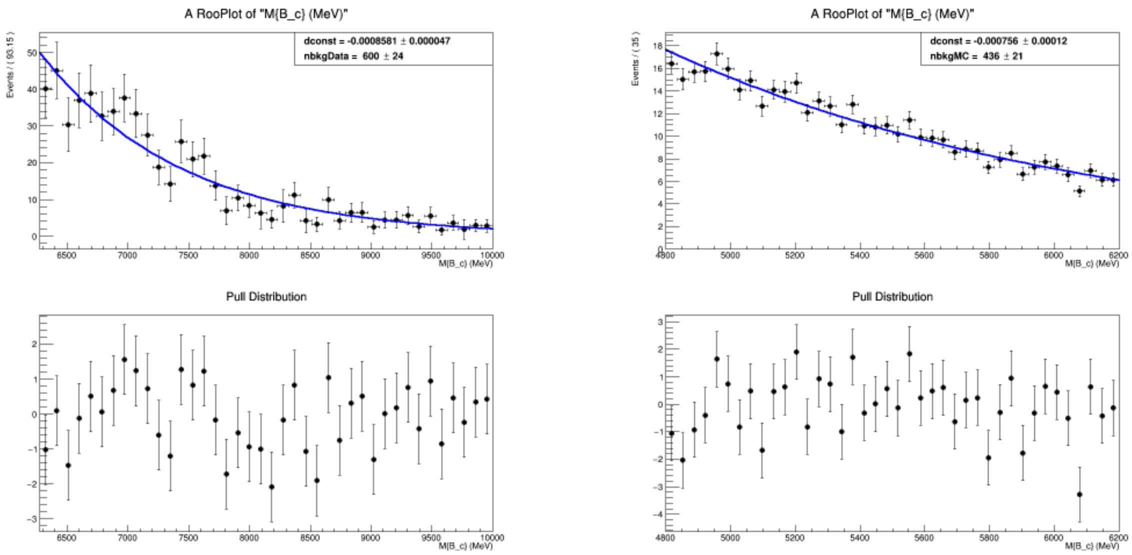
corrected and uncorrected shape. The correction also improves fit quality. For this reason, we believe the correction to be the correct strategy, and the deviation with the fit using the uncorrected template represents a maximum shift, justifying taking half the shift as the uncertainty. The uncertainty associated with the decay in flight efficiency correction is 1.08×10^{-2}.

17.9 Combinatorial $J/\psi\mu$ background modeling

The combinatorial $J/\psi\mu$ background is simulated using a cocktail of $B_{d,s} \to J/\psi X$ decays. This background is difficult to model, as there is no data-driven way to simulate it (see Section 9). In the $R(D^{(*)})$ analysis, the model was corrected by reweighting the distribution of combinatorial candidates as a function of the $J/\psi\mu$ mass, using the data above m_{B_c} as a guide.

We fit an exponential to the upper sideband of the $J/\psi\mu$ mass distribution. The final selection is still applied, but without the cuts on the $J/\psi\mu$ mass, on the DOCA between J/ψ and μ candidates, and the lower bound on the missing mass at -5 GeV$^2/c^4$. For data, this is done with the expected misID and fake J/ψ contributions weighted out. For the MC, we use a cocktail with B_d and B_s candidate contributions fixed to the expected proportions given by the fit. We take the upper sideband of the data, which is defined to be above the B_c mass and below 10000 MeV. There is a generator cut which removes the upper sideband of the B_c mass in the MC, so we instead fit to the mass region from 4800 MeV to 6200 MeV. This is removed enough from the region where the generator cuts are applied such that the slope in this region is unaffected by the upper sideband cut. Exponentials are fitted to both these regions, as shown in Fig. 66. Then, using the resulting fit parameters, a weight as a function of the B_c mass is produced, $e^{\lambda_{data}-\lambda_{MC}/m_{B_c}}$.

The alternative fit is performed while applying this weight to the combinatorial MC, and we take half the difference with the nominal value of $R(J/\psi)$ to obtain a systematic of 0.56×10^{-2}.



(a) Data

(b) Combinatorial MC

Figure 66: Exponential fit to B_c mass upper sideband for data and combinatorial MC.

17.10 Misidentification background smearing

The pion and kaon contributions to the misID background can decay in-flight to muons, necessitating a smearing procedure in order to account for the momentum misreconstruction (see Section 8.6 for more details). The nominal fit allows pions and kaons to decay in flight in different ratios. Pions and kaons occupy 23.9% and 40.3% of the misID sample, respectively (see Table 12). The misID rate of pions is about 0.695, and for kaons it is 0.137. The alternative fit fixes pions and kaons to have the same amount of smearing. Taking half the difference with the nominal $R(J/\psi)$ results in a systematic uncertainty of 0.34×10^{-2}.

17.11 $\psi(2S)$ background contribution

In the nominal fit, the number of $\psi(2S)$ background events is allowed to float. In 2016 and 2017, the fits have no $\psi(2S)$ event candidates, and in 2018, the fit finds about 12 $\psi(2S)$ event candidates. Theory predicts that the ratio between the branching fractions of $B_c \to \psi(2S)\mu\nu$ with $\psi(2S) \to J/\psi X$ and $B_c \to J/\psi\mu\nu$, including the effects of the isolation selection, is about 1.3% (see more details in Section 2.2.8). The alternative fit is performed fixing the ratio between the number of $\psi(2S)$ events and normalization events to the theoretical prediction (see more details in Section 13.1). Potential contributions from $X(3872)$ are considered but ultimately not used (see Section **??**). We take half the difference between the alternative $R(J/\psi)$ and the nominal $R(J/\psi)$ to get a systematic uncertainty of 0.32×10^{-2}.

17.12 Trimuon effect contribution

Studies in Run-1 have found a momentum dependent increase in the efficiency of the muon PID variable `IsMuon`, theorized to be from the J/ψ daughter muons creating hits falsely identified as belonging to the bachelor muon when closely aligned. This effect was seen again in Run-2, although with some inconsistencies between pions and kaons that are not yet well understood (Tables 28, 29). Attempts to incorporate an efficiency correction into the analysis resulted in a worse fit (see more in Section 26). Because of these caveats indicating potentially other effects going on not related to the trimuon effect, we have elected not to incorporate the correction into the nominal fit. Instead, to eliminate the need for this correction, we have added a selection requiring a minimum angle between the bachelor muon and the same sign muon in the non-bending plane. This minimum angle (about 15 mrad) was set by requiring the tracks be spaced apart by the size of a logical pad in M2. The fit was performed with this additional selection and half the difference with the nominal value of $R(J/\psi)$ was taken as the systematic. After accounting for efficiencies, this systematic was 0.09×10^{-2}.

18 Prospects

The analysis framework for $R(J/\psi)$ measurement with the LHCb Run-2 data set from 2016, 2017, and 2018 has been developed in this study.

The LHCb Run-1 data set was recorded at center-of-mass-energies of 7 and 8 TeV for an integrated luminosity of $3.2\,\mathrm{fb}^{-1}$. The Run-2 data contains a much larger sample of B mesons, at the center of mass energy of 13 TeV for an integrated luminosity of $5.6\,\mathrm{fb}^{-1}$. Because the b-quark production cross-section at 13 TeV is approximately double that at 7 TeV [53], the Run-2 data set should have about $2 \times \frac{5.6\,\mathrm{fb}^{-1}}{3.2\,\mathrm{fb}^{-1}} = 3.5$ times the Run-1 sample size. As a result, the statistical uncertainty of the Run-2 measurement is expected to be approximately half ($1/\sqrt{3.5} \approx 0.53$) that of the Run-1 measurement.

The systematic uncertainties were dominated by form factors and limited simulation sample size in the Run-1 measurement. The LQCD form factor results and the much larger simulation sample size will significantly improve these systematic uncertainties in the Run-2 measurement.

Therefore, the Run-2 $R(J/\psi)$ measurement will be an important update to the tests of lepton flavor universality in B_c^+ decays.

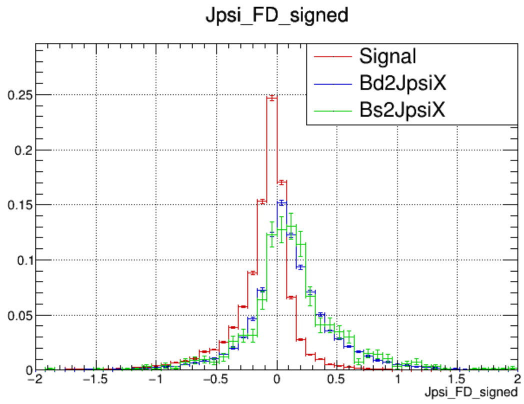


Figure 67: The signed reconstructed flight distance of the J/ψ, where positive sign is associated with the J/ψ candidate being ahead of the B_c candidate along the z-direction. The signal distribution is in red, while the B_d and B_s combinatorial backgrounds are given in blue and green, respectively.

19 Appendix A: $J/\psi\mu$ combinatorial background correction validation

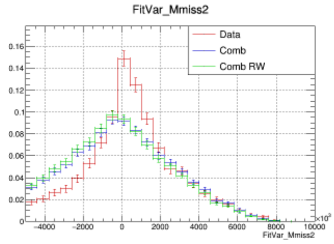
The systematic uncertainty associated with the choice of $J/\psi\mu$ combinatorial background model is estimated by applying a correction to the $B_{d,s} \to J/\psi X$ tuples, generated from the upper sideband of the B_c mass distribution in data (see Section 17.9 for a more in depth description). Several investigations were done to ensure the validity of this correction. Events with large positive values of the signed J/ψ flight distance (which is the displacement between the vertex from the two muon candidate daughters of the J/ψ and the vertex from all three muon candidates in the event) are enriched in combinatorial background, as seen in Figure 67.

The assumption is that the combinatorial MC cocktail should match the data for the fit variables in this large J/ψ flight distance region, and that the reweighting should improve the agreement if valid. We weighted out the data-driven backgrounds and generated a cocktail of the $B_{d,s} \to J/\psi X$ tuples representing the combinatorial background. The comparison of the fit variable distributions is shown in Figure 68. The lifetime distribution agrees fairly well (Figure 68b), but the missing mass agreement is problematic. The reweighting strategy also does not appear to significantly change the distribution in the rest frame variables

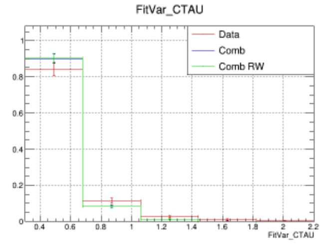
Tightening the flight distance cut would purify the number of combinatorial candidates in the data sample, but also reduce statistics. The fit variable distribution with a tighter flight distance cut at 1 mm is shown in Figure 69. This appears to somewhat improve the agreement in the missing mass but does not fix the situation entirely.

The shape of the missing mass peak in the data could imply that normalization still remained in the data sample despite the flight distance cut. A version of the distribution was plotted with the normalization weighted out of the data sample in Figure 70. This rids the data of its sharp peak in missing mass but also has a distribution that is shifted to the right of the combinatorial missing mass distribution (Figure 70a). It also detracts from the agreement in the lifetime and Z distributions (Figures 70b and 70c).

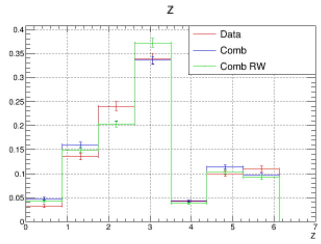
The geometric pointing resolution scales by $1/p_T$ of the bachelor muon candidate. To



(a) Missing mass.

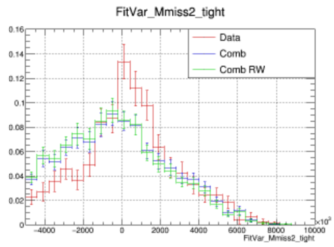


(b) Lifetime.

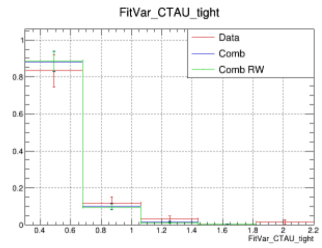


(c) $Z(q^2, E_\ell^*)$.

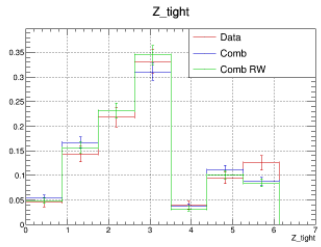
Figure 68: Fit variable distribution comparison between the original combinatorial MC cocktail (blue), the reweighted combinatorial MC cocktail (green), and data with data-driven backgrounds weighted out (red). The top row shows the original distribution and the bottom row shows the distribution after the combinatorial The comparison is done in a combinatorial enriched region with the signed flight distance of the J/ψ greater than 0.5 mm.



(a) Missing mass.

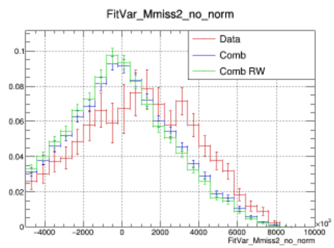


(b) Lifetime.

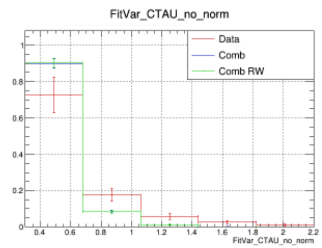


(c) $Z(q^2, E_\ell^*)$.

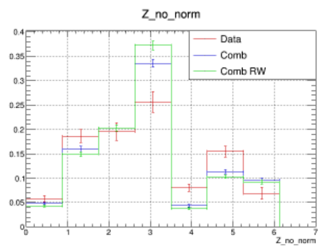
Figure 69: Fit variable distribution comparison between combinatorial MC cocktail (blue), the reweighted combinatorial MC cocktail (green), and data with data-driven backgrounds weighted out (red). The top row shows the original distribution and the bottom row shows the distribution after the combinatorial The comparison is done in a combinatorial enriched region with a tighter cut requiring the signed flight distance of the J/ψ greater than 1 mm.



(a) Missing mass.



(b) Lifetime.



(c) $Z(q^2, E_\ell^*)$.

Figure 70: Fit variable distribution comparison between combinatorial MC cocktail (blue), the reweighted combinatorial MC cocktail (green), and data with data-driven backgrounds and normalization weighted out (red). The top row shows the original distribution and the bottom row shows the distribution after the combinatorial The comparison is done in a combinatorial enriched region with the signed flight distance of the J/ψ greater than 0.5 mm.

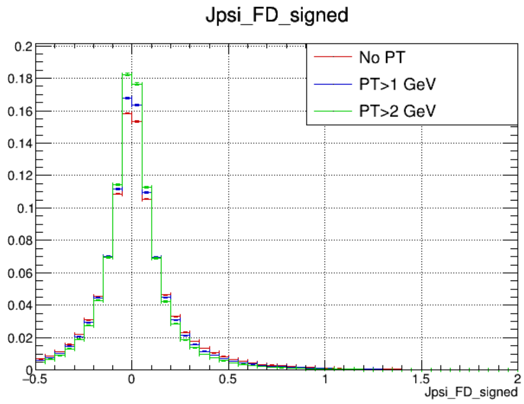
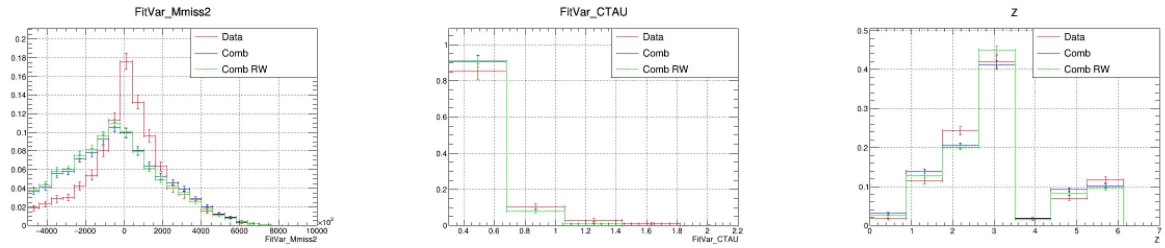


Figure 71: The reconstructed J/ψ flight distance distribution of the normalization MC. In red is the distribution with the nominal cut of $p_T > 750$ MeV, in blue a slightly harder cut of $p_T > 1$ GeV, and in green an even harder cut of $p_T > 2$ GeV.



(a) Missing mass.

(b) Lifetime.

(c) $Z(q^2, E_\ell^*)$.

Figure 72: Fit variable distribution comparison between combinatorial MC cocktail (blue), the reweighted combinatorial MC cocktail (green), and data with data-driven backgrounds weighted out (red). The top row shows the original distribution and the bottom row shows the distribution after the combinatorial The comparison is done in a combinatorial enriched region with the signed flight distance of the J/ψ greater than 0.5 mm and the transverse momentum of the bachelor muon greater than 1.2 GeV.

test the possibility of normalization events with poor resolution passing the flight distance cut and skewing the missing mass distribution, the J/ψ flight distance distribution of the normalization is plotted with several different p_T cuts. The discrepancy between harder and softer p_T cuts is insignificant in the upper tail of the flight distribution, suggesting this is not a significant effect.

This is validated by the distributions of the fit variables requiring the bachelor muon p_T to be greater than 1.2 GeV showing the same issue with sharply peaked missing mass.

20 Appendix B: New Run-2 selections

A number of new selections were added in Run-2 to reduce backgrounds that were large or hard to model using data, and thus improve fit and postfit agreement between data and model. Further details on and validation of these cuts are given in this section.

20.1 DIRA

In the version of the fit using selections from Run-1 (Figure 76), the low missing mass region as well as the high Z region showed large disagreement between the fit and data. This issue was consistent between all years.

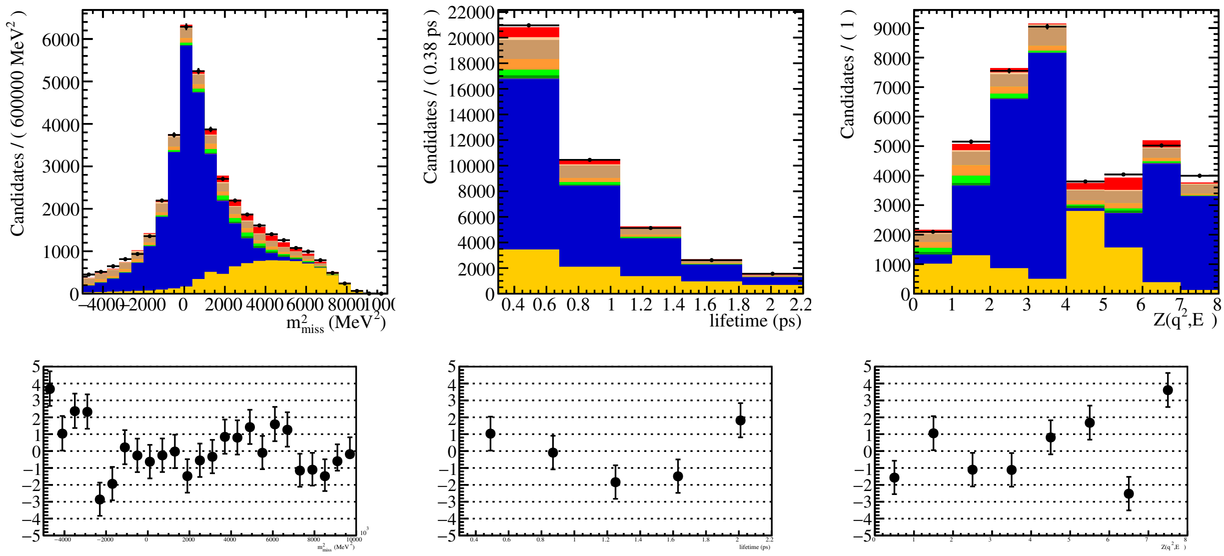


Figure 73: 2016 fit without cut.

A promising variable we investigated was the cosine of the angle between the $J/\psi\mu$ system and the B_c. A cut requiring $\cos\theta > 0.9998$, where θ is the angle between the $J/\psi\mu$ system and the B_c. reduces the combinatorial background by about 50%. When this angle is large, the missing mass is small. The distribution of the cosine of this angle (Figure 74) shows that the combinatorial occupies a much higher proportion of the low cosine region of the plot.

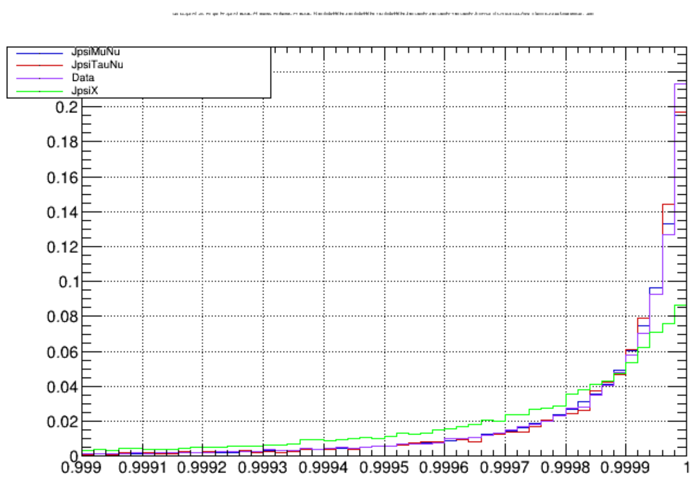


Figure 74: The distribution of the cosine of the angle between B_c and the $J/\psi\mu$ system for normalization (blue), signal (red), data (purple), and combinatorial background (green).

Because other investigations have shown that the low missing mass region and the high Z bin are correlated, we expected a cut that affected one variable would also affect the other. A cut requiring $\cos\theta > 0.9998$ reduces the combinatorial background by about 50%. In particular, the combinatorial is disproportionately reduced in the low missing mass area, as seen in 75.

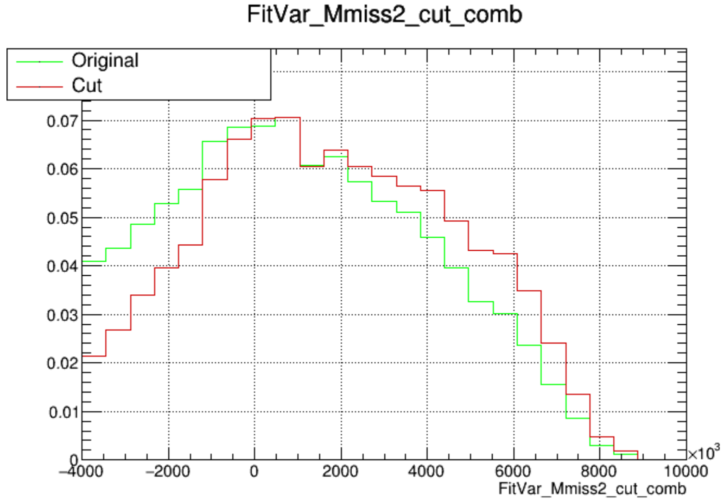


Figure 75: The missing mass distribution before (green) and after (red) cutting on $\cos\theta$, both normalized to 1.

About 72% of the signal and normalization are retained. The cut indeed reduces the discrepancy at low missing mass and high Z, as shown in the final fit. A cut using the equivalent species in the $B_0 \to J/\psi K^*$ decay, with the kaon used instead of the μ and the B^+ instead of the B_c, in order to keep the lifetime acceptance corrections consistent with the rest of the analysis.

A study was done loosening the cut to $\cos\theta > 0.9995$. This improved the issue with the low missing mass region, but did not fully fix it, resulting in the choice of the more aggressive cut at 0.9998.

Another check was done to ensure that adding this cut did not bias the other fit variables or the muon BDT. These plots can be seen in Figs. 77, 78, 79, 80, 81 for signal, normalization, and data.

20.2 DOCA

A DOCA cut between the J/ψ and bachelor muon is used to reduce combinatorial background. Since Run-1, this cut was tightened.

The DOCA considered here is between the J/ψ and the bachelor muon and is signed according to whether the J/ψ is ahead of the B_c along the beam line or not (+: J/ψ ahead, −: J/ψ behind). This variable is extremely effective in eliminating combinatorial backgrounds with relatively little cost to signal statistics (see Figure 82). In particular, the $B_{d,s} \to J/\psi X$ combinatorial background has no data-driven model and thus is a suspect for contributing to the poor agreement in the regions of very large absolute value of DOCA (Figure 83).

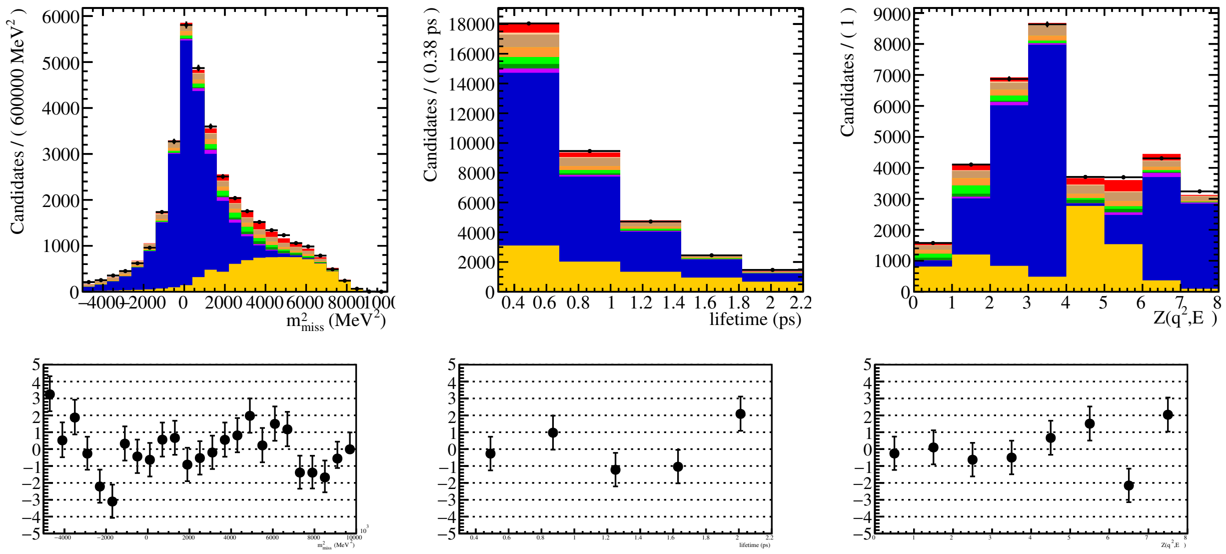


Figure 76: 2016 fit with cut at $\cos\theta < 0.9995$.

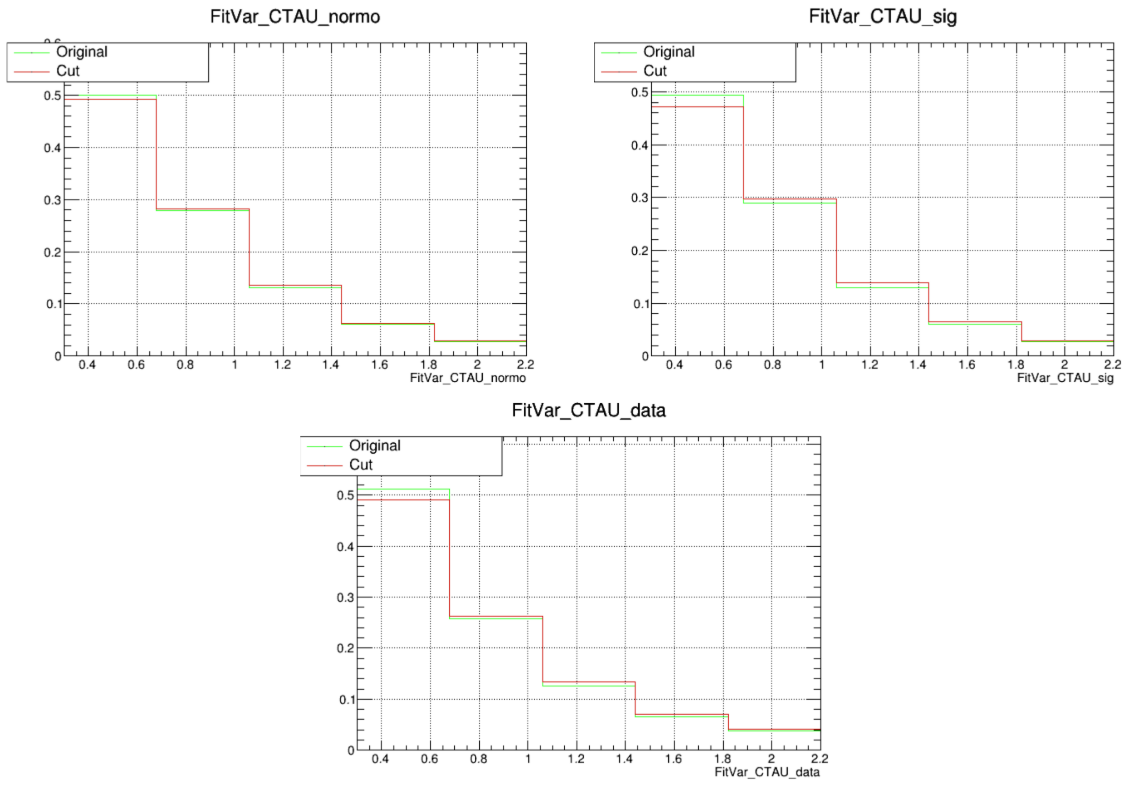


Figure 77: Plots of templates for normalization, signal, and data in lifetime before (green) and after (red) applying the $\cos\theta < 0.9998$ cut.

We have further tightened the cut from $\mathrm{DOCA}_{B_c} < 0.07$ to $-0.13 < \mathrm{DOCA}_{B_c} < 0.05$, which does not increase the statistical uncertainty or non-negligibly modify the efficiency ratio but significantly improves the overall fit quality and the postfit agreement in the low lifetime region, likely due to the reduction of combinatorial background.

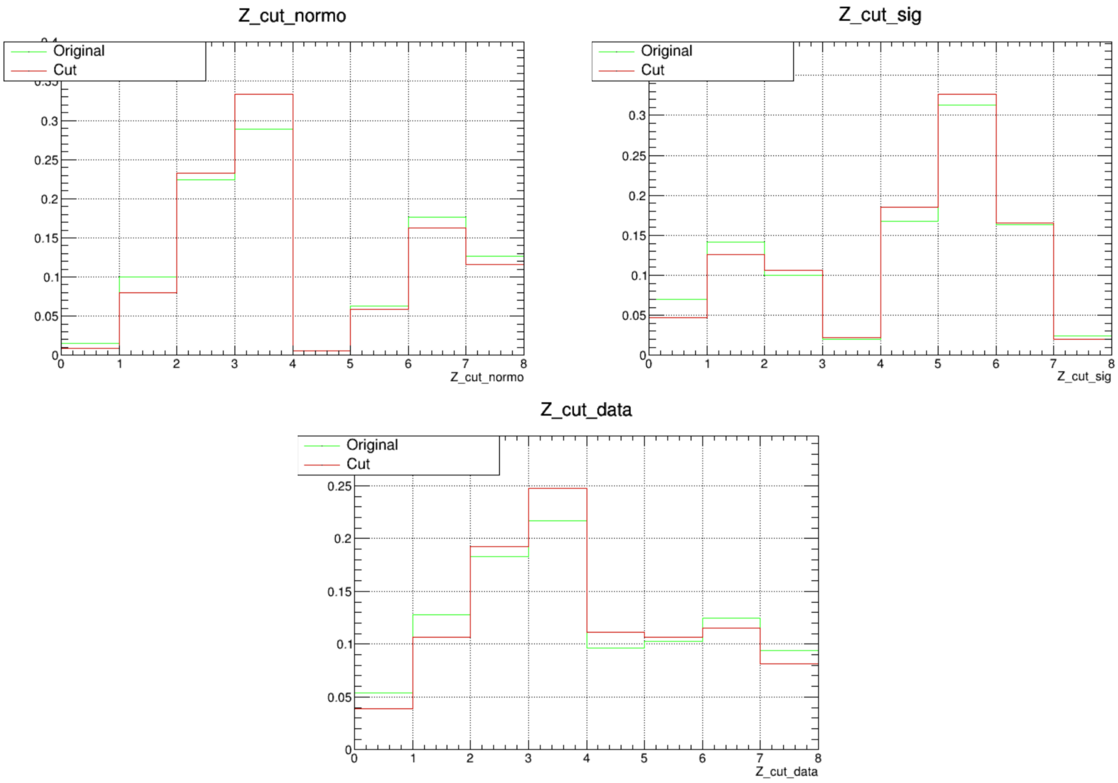


Figure 78: Plots of templates for normalization, signal, and data in Z before (green) and after (red) applying the $\cos\theta < 0.9998$ cut.

20.3 Lifetime

In past meetings, we have discussed our rationale for tightening the lower lifetime cut from 0.3 to 0.45 ps. This rationale is briefly reviewed here.

The lowest lifetime region is heavily populated by the poorly modeled backgrounds and suffer from acceptance effects. Backgrounds such as the charm combinatorial cannot be modeled in a data-driven way, and as discussed in previous meetings by Bo, acceptance effects at the very low lifetime region are not well modeled in MC. The shift from 0.3 ps to 0.45 ps as the lower lifetime cut helps reduce the impact of these effects.

A strong indicator of these mismodeled acceptance effects was the asymmetry of the B_c DOCA variable, which was not well modeled in MC, with the worst effect in 2018 and the mildest in 2016. The lifetime selection removes the region of disagreement. The variation of the mismodeling of this asymmetry across data-taking periods aligns with other observations of data/MC differences arising from the VELO error parameterization ([54]).

20.4 Perpendicular momentum

Initial postfit investigations showed poor agreement in the fit variable E_ℓ in the high missing mass region, where mis-ID is concentrated (Figure 85 bottom).

A new variable defined as the component of the bachelor muon momentum perpendicular to the B_c flight direction, $p_\mu^\perp$, was introduced as a Lorentz invariant alternative to

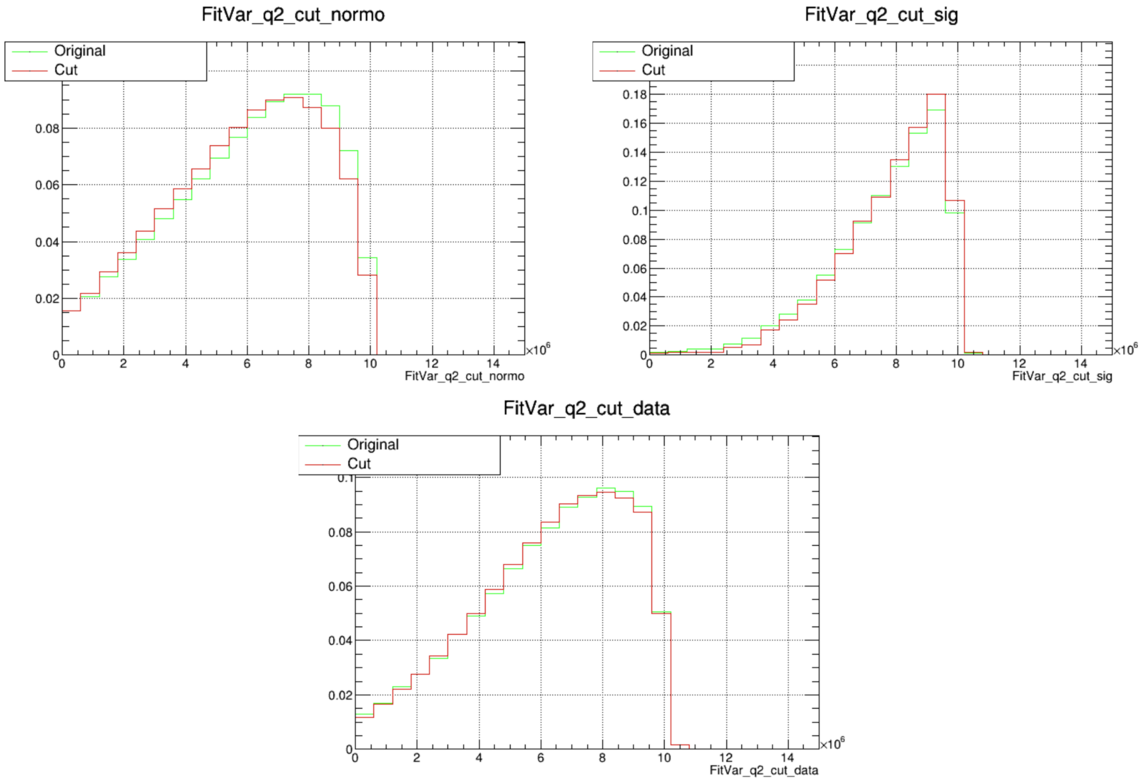


Figure 79: Plots of templates for normalization, signal, and data in q^2 before (green) and after (red) applying the $\cos\theta < 0.9998$ cut.

investigate potential issues with the rest frame approximation. The distribution of this variable in a few key components is shown in Figure 86.

Postfit plots of this variable also showed similar discrepancies in the high missing mass region (Figure 87), reducing the likelihood of rest frame approximation issues. Figure 86 shows this variable to be an excellent discriminator of mis-ID against signal and normalization. One of our primary goals for improvement over Run-1 was to decrease the amount of mis-ID background. This was partially achieved through the incorporation of the uBDT for PID, but further improvements could be made in suppressing the mis-ID contribution. Since many of the postfit issues appeared to arise in mis-ID enriched regions, this variable appeared to be a powerful handle for removing a large amount of mis-ID component with little impact on the uncertainty of $R(J/\psi)$. With a cut at $p_{\mu}^{\perp} > 500MeV$, the postfit agreement in the same high missing mass region improves significantly (compare top and bottom in Figures 85 and 87). Although the efficiency ratio decreases by about 15%, the statistical uncertainty remains unchanged and the large reduction in background, improvement in fit quality, and improvement across many different postfit variables is a worthy tradeoff.

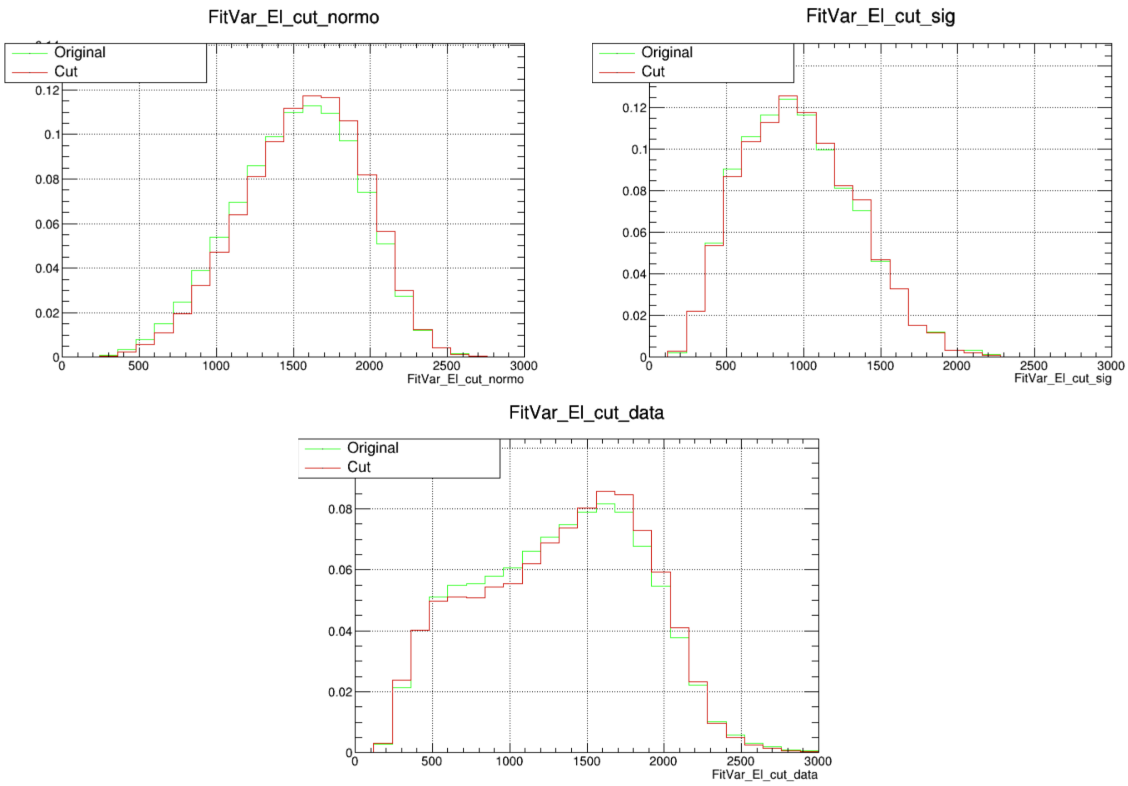


Figure 80: Plots of templates for normalization, signal, and data in E_ℓ before (green) and after (red) applying the $\cos\theta < 0.9998$ cut.

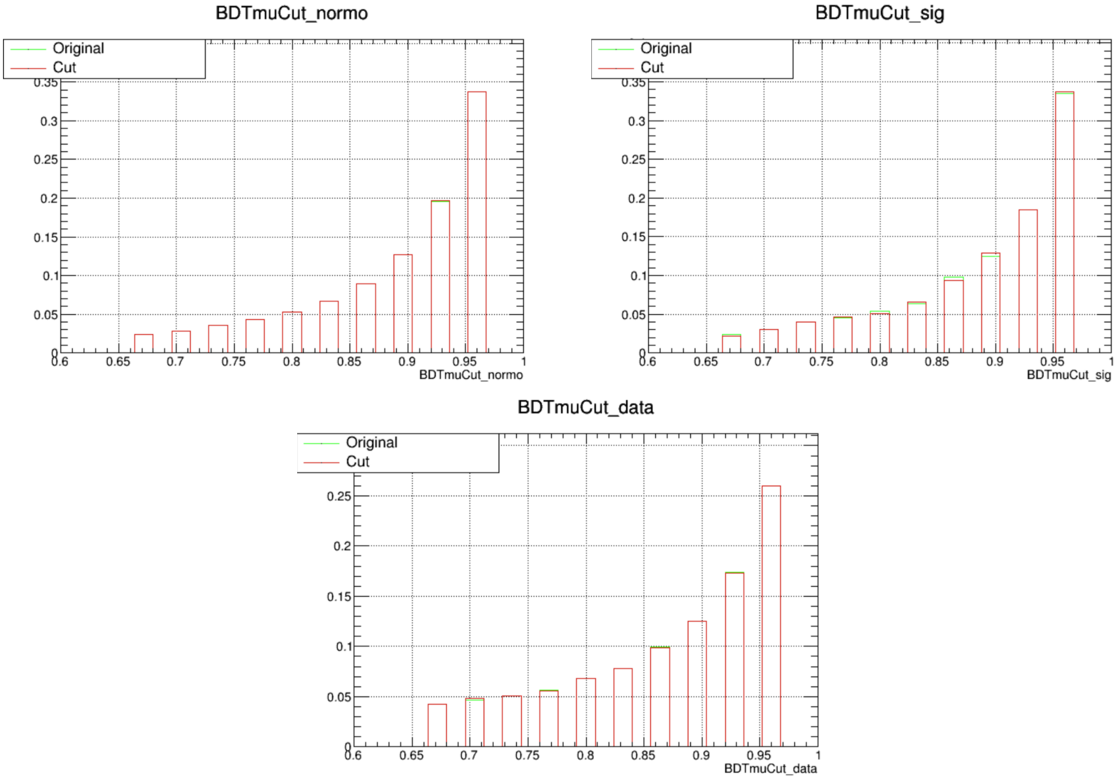


Figure 81: Plots of templates for normalization, signal, and data in the muon BDT before (green) and after (red) applying the $\cos\theta < 0.9998$ cut.

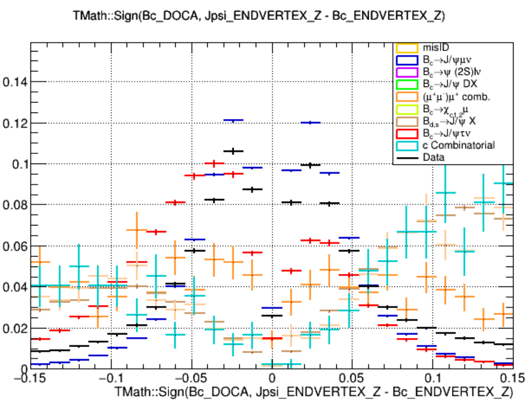


Figure 82: Distribution of the signed B_c DOCA variable in signal, normalization, data, and the various combinatorial backgrounds.

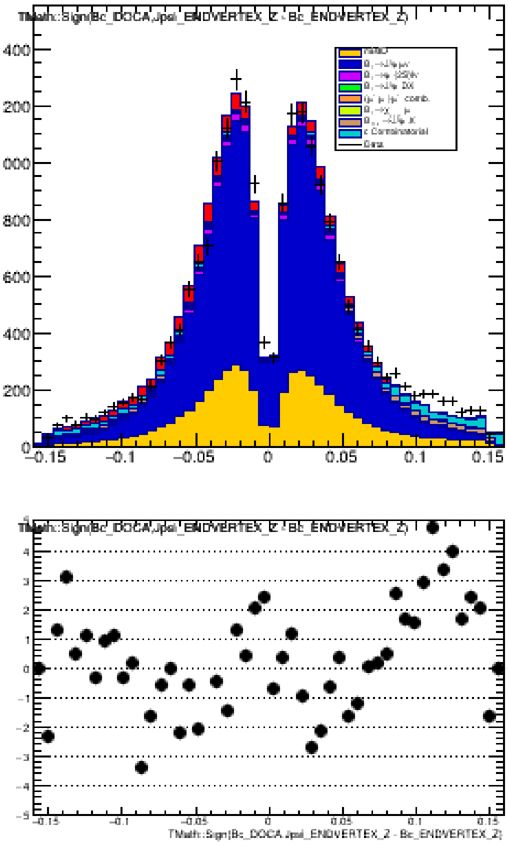


Figure 83: Postfit plot of the signed B_c DOCA distribution before new selections.

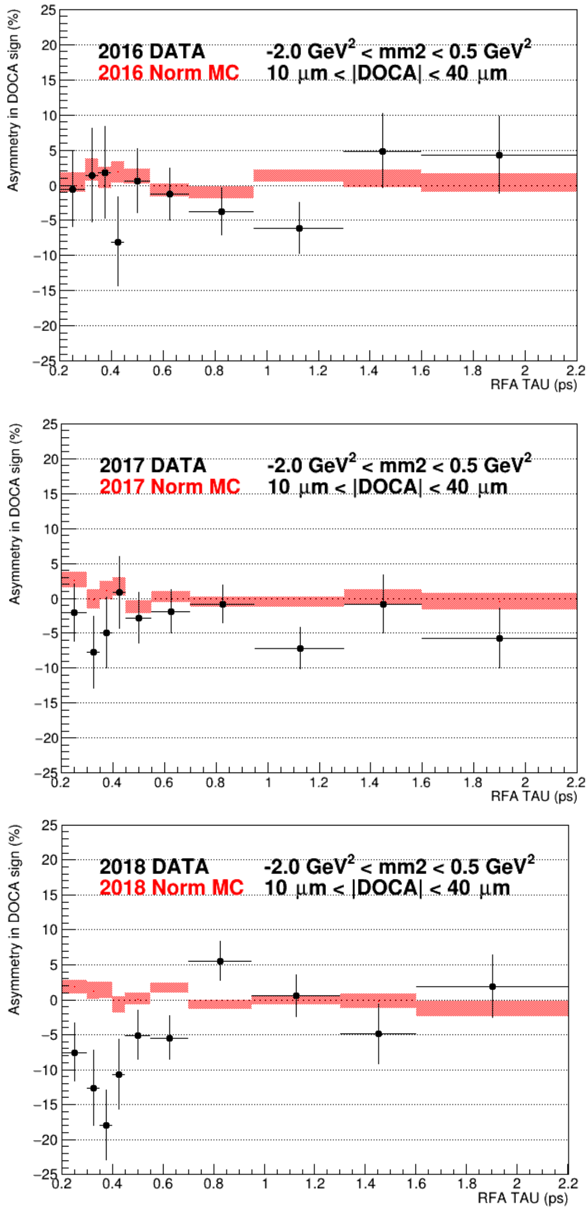


Figure 84: Asymmetry of signed B_c DOCA variable in all three years as a function of lifetime, going from 2016-2018 top to bottom.

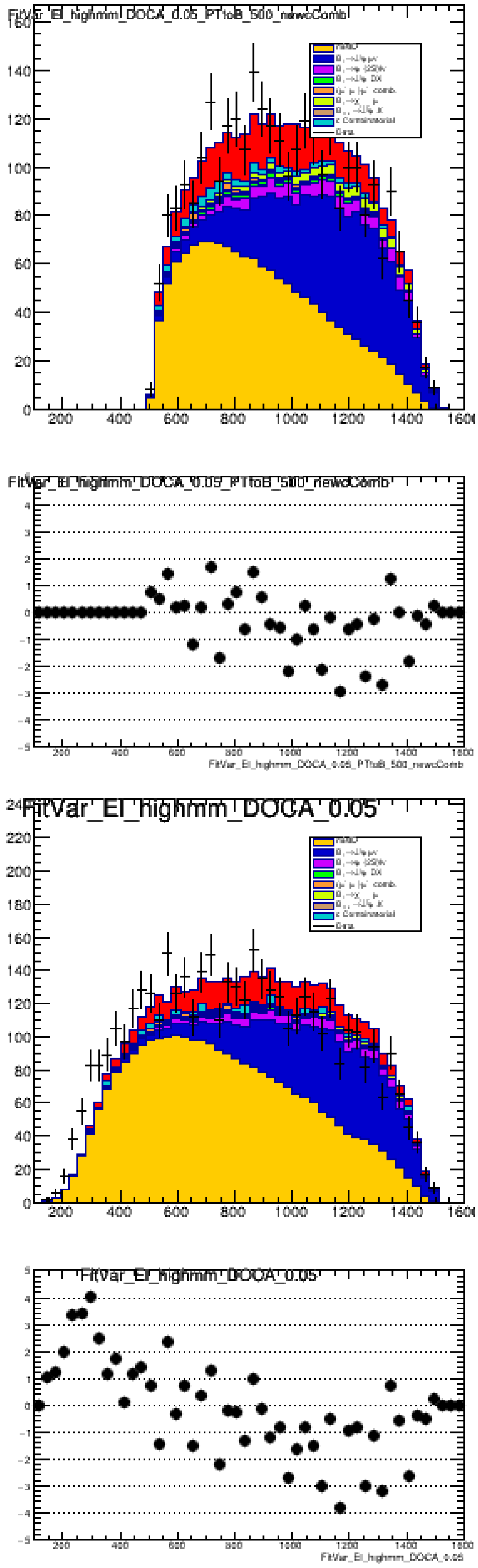


Figure 85: E_ℓ postfit with perpendicular momentum cut (top) versus without (bottom).

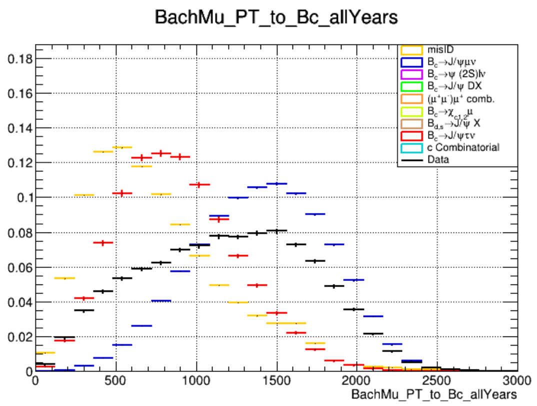


Figure 86: Shows the perpendicular momentum distributions of signal, normalization, data, and misID. MisID clearly shows the lowest perpendicular momentum distribution.

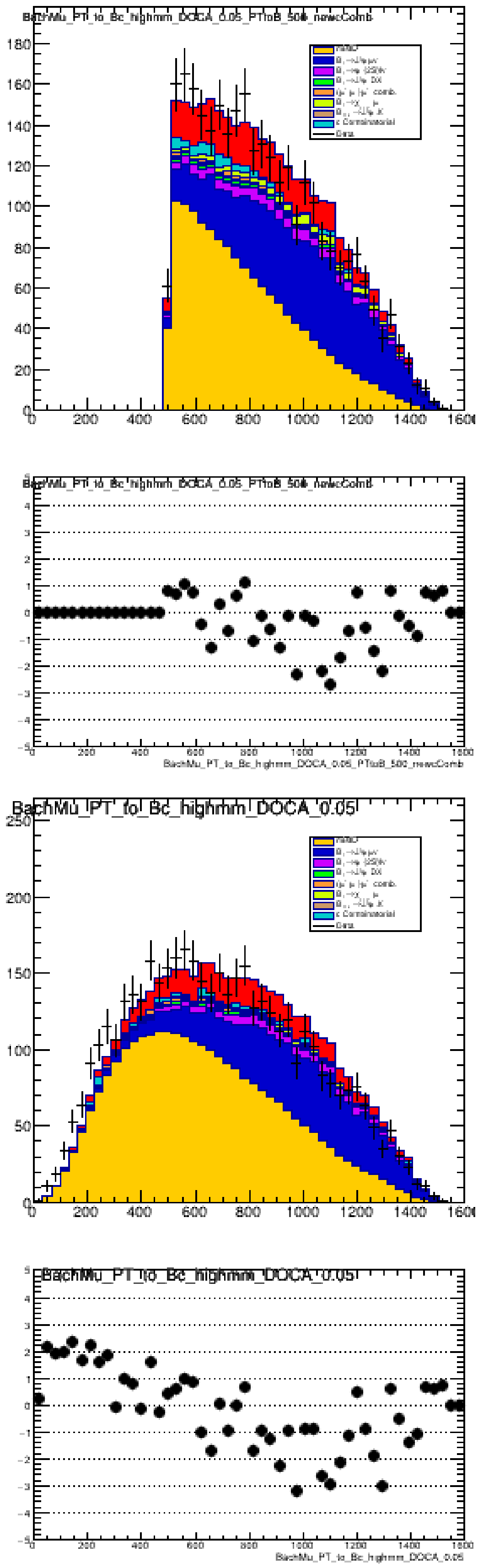


Figure 87: Perpendicular momentum postfit distribution with perpendicular momentum cut (top) versus without.

21 Appendix C: Systematic uncertainty fits

This section gives the fits associated with the alternative fits used to estimate systematic uncertainties. Fits are given for each year, in order (2016 top, 2016 bottom).

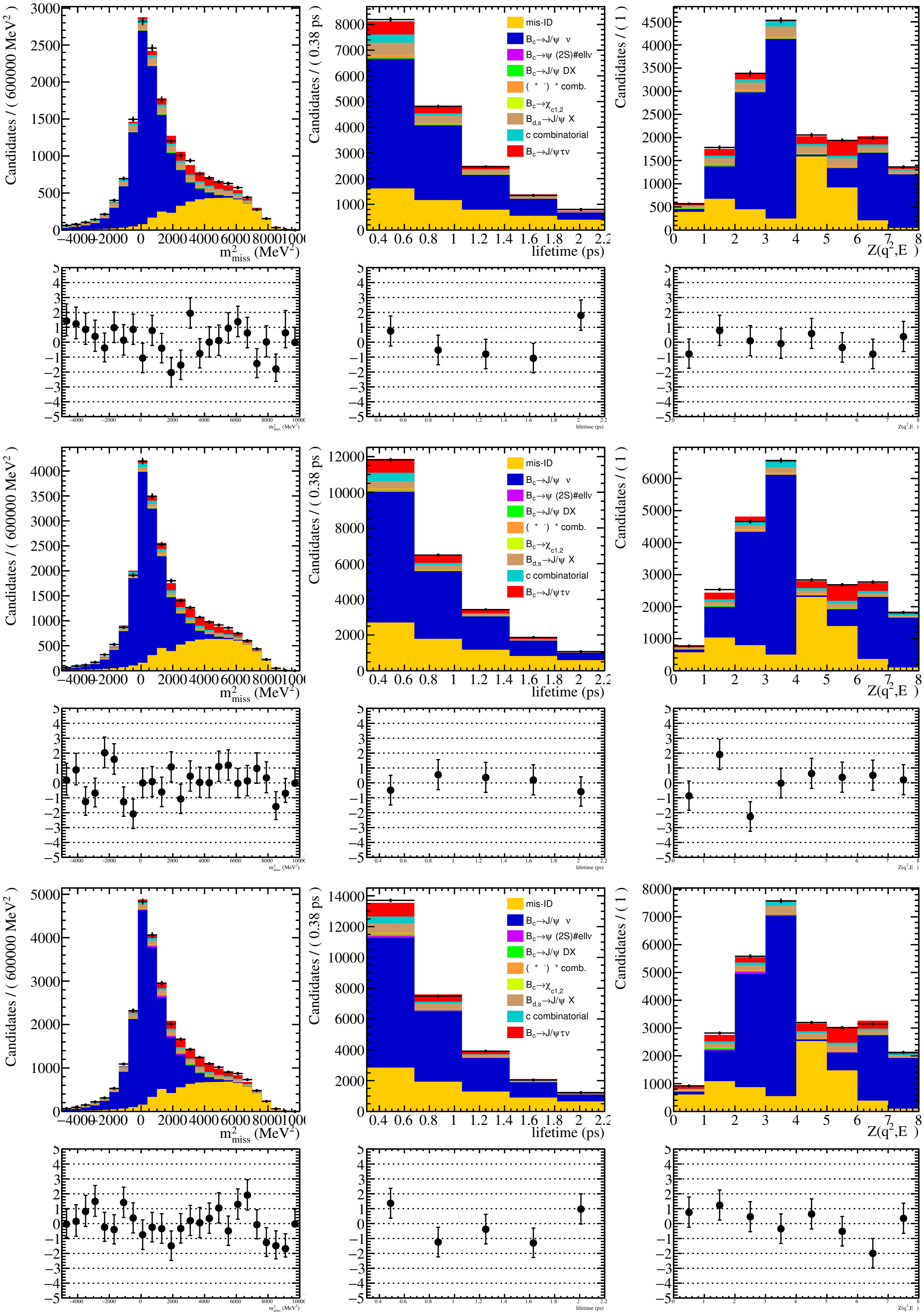


Figure 88: The results of the fit with form factor constraints removed.

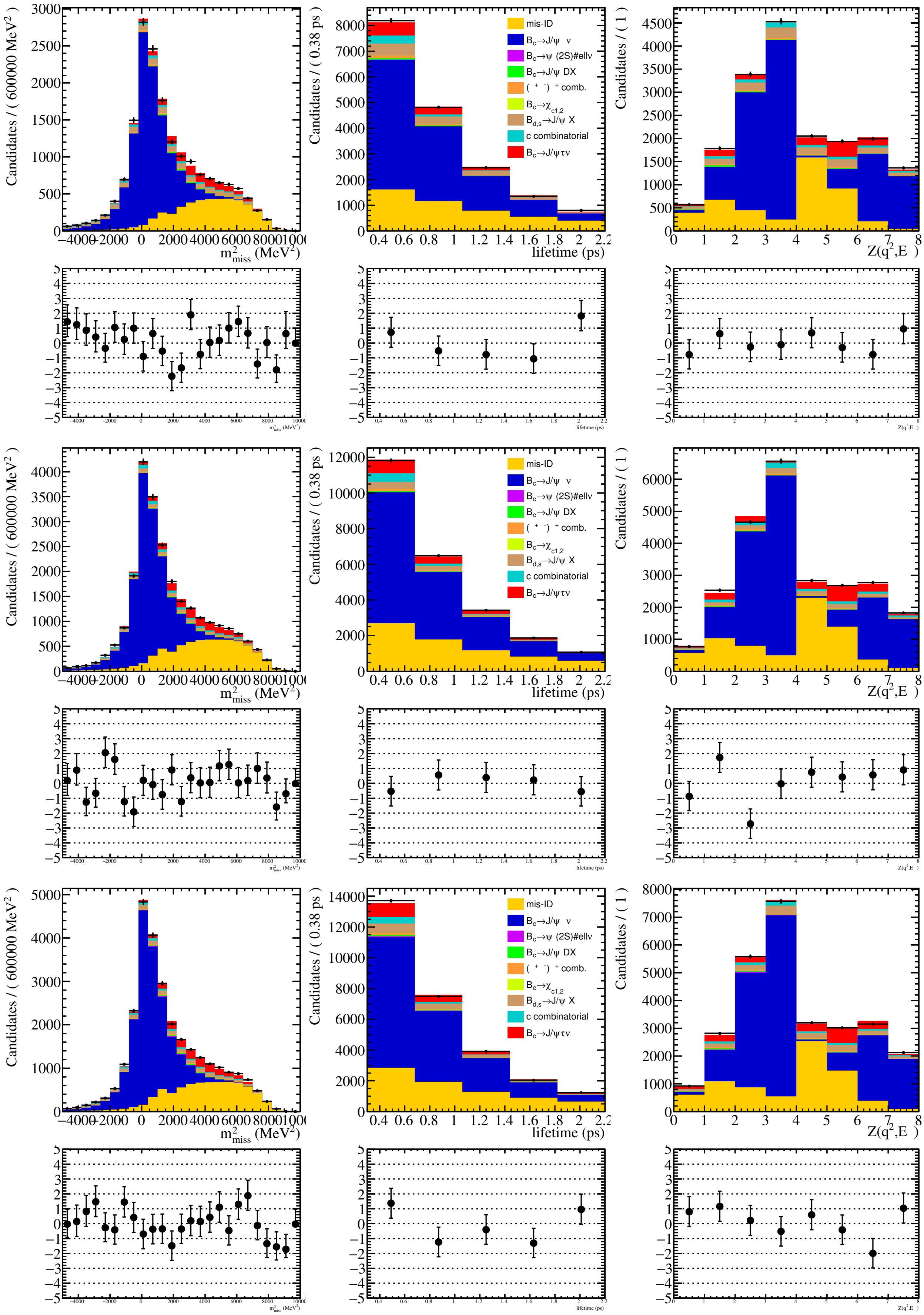


Figure 89: The results of the fit with form factors fixed to the value chosen by the nominal fit. Not used for estimating systematics, just a check for expected behavior.

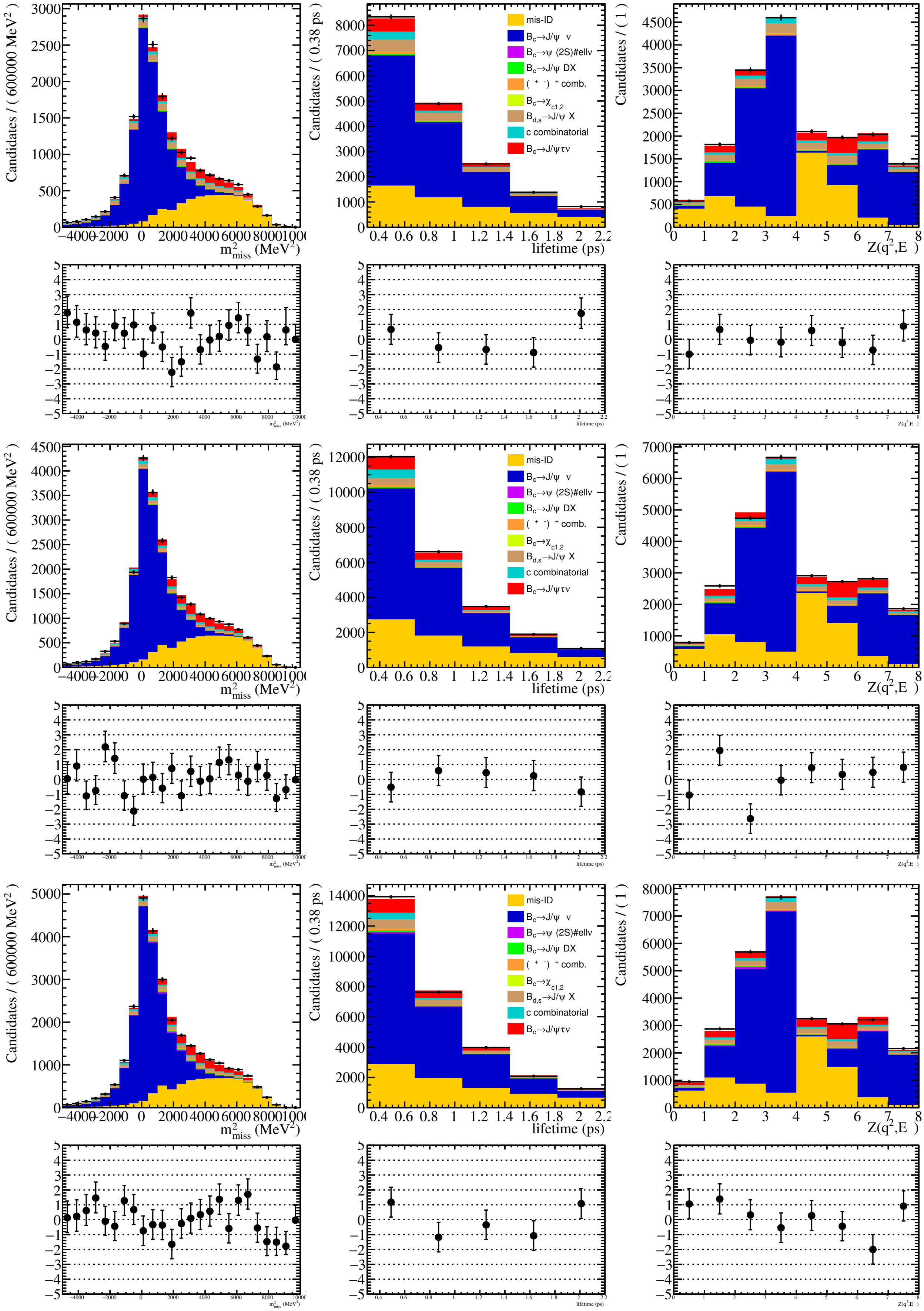


Figure 90: The results of the fit with the decay-in-flight contributions of kaons and pions allowed to float separately. Used to estimate the systematic uncertainty associated with misID decay-in-flight smearing.

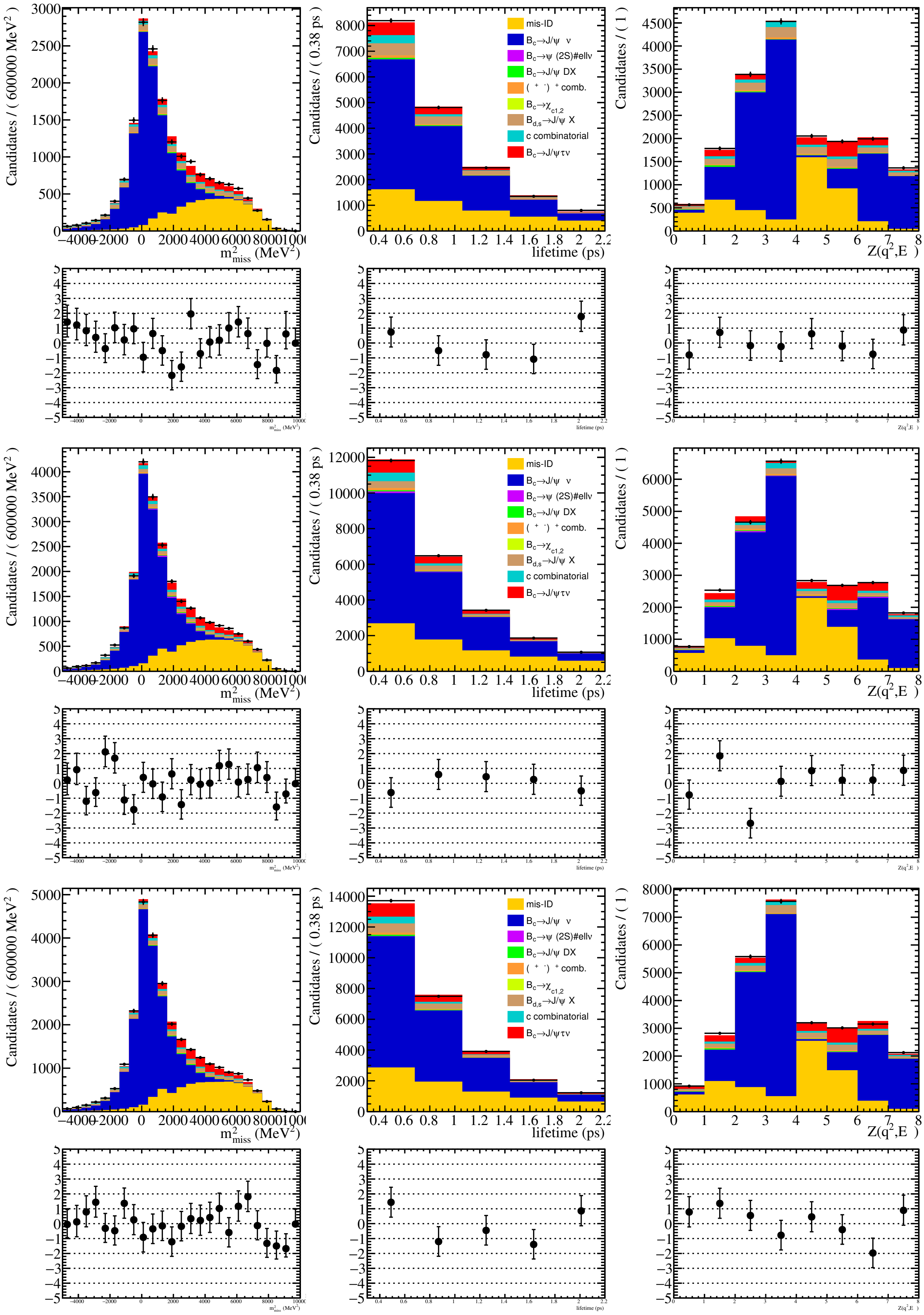


Figure 91: The results of the fit using the Kiselev model for the $\psi(2S)$ feed down background contribution, used to estimate systematic uncertainty.

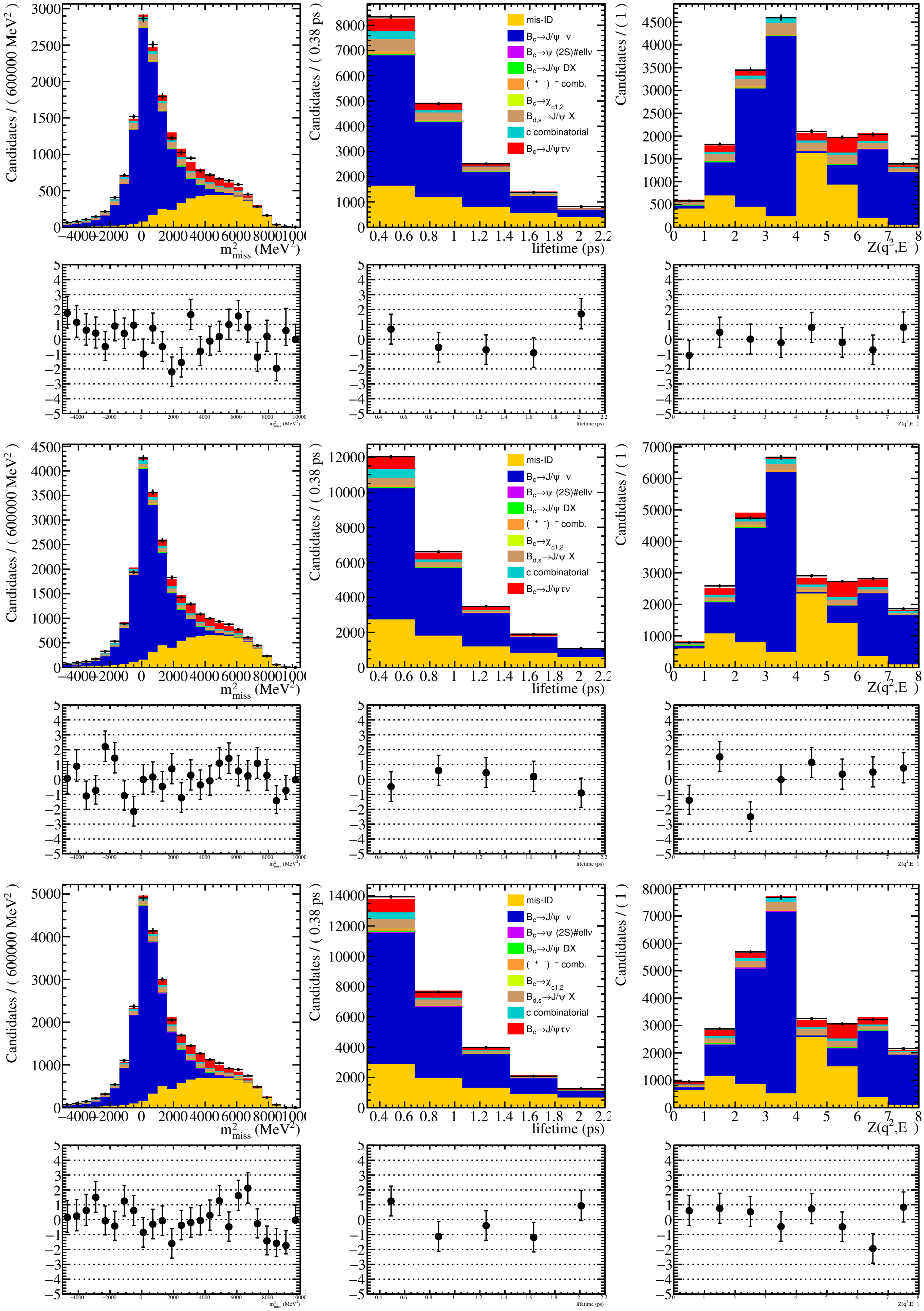


Figure 92: The results of the fit using alternative ghost efficiencies, used to estimate the systematic uncertainty associated with misID ghost identification.

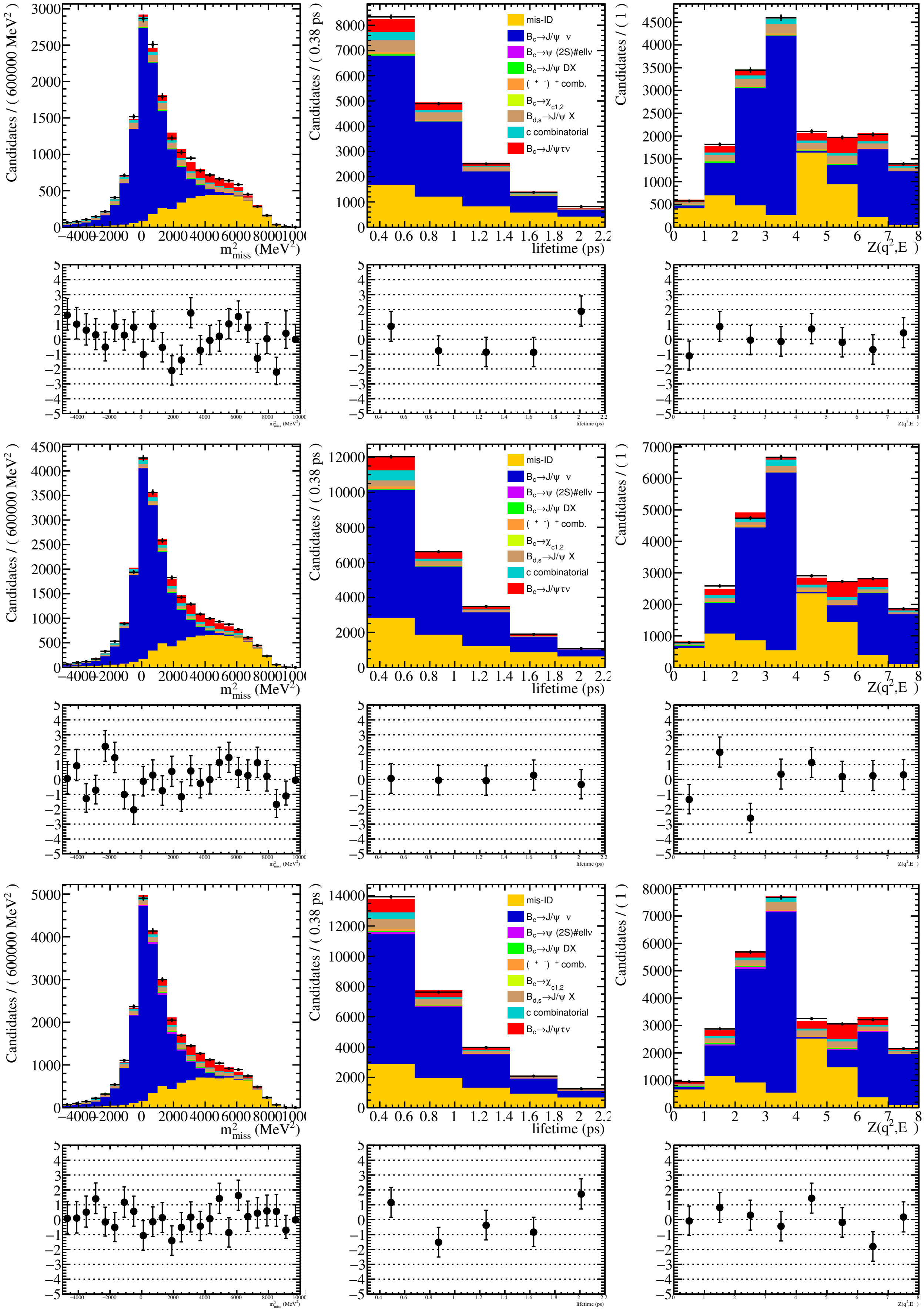


Figure 93: The results of the fit using an alternative BDT for final reweighting, used to estimate the systematic uncertainty associated with the final data/MC correction.

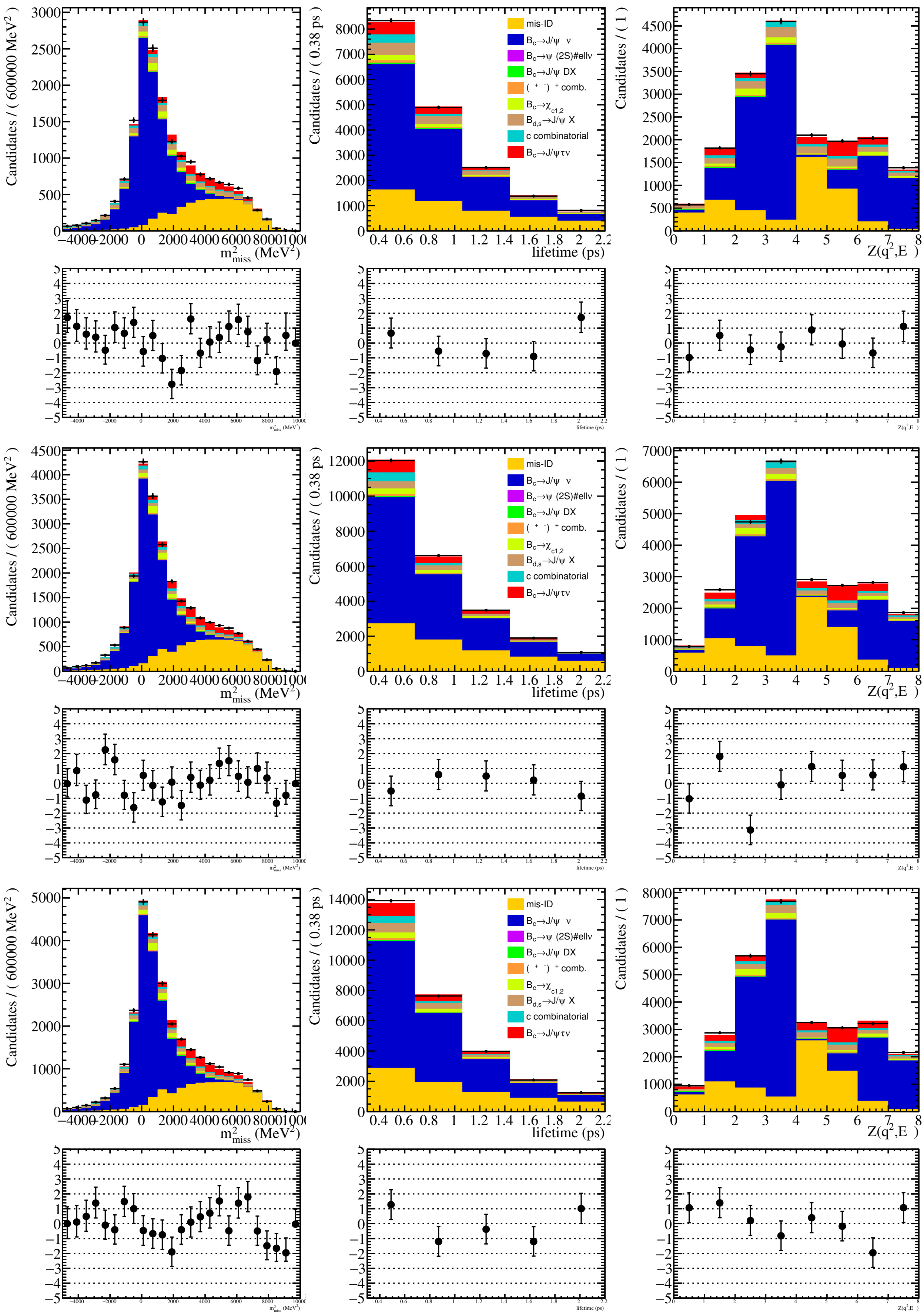


Figure 94: The results of the fit with $\chi_{c1,2}$ contributions fixed to theoretical predictions, used to estimate the systematic uncertainty associated with the scaling of this background.

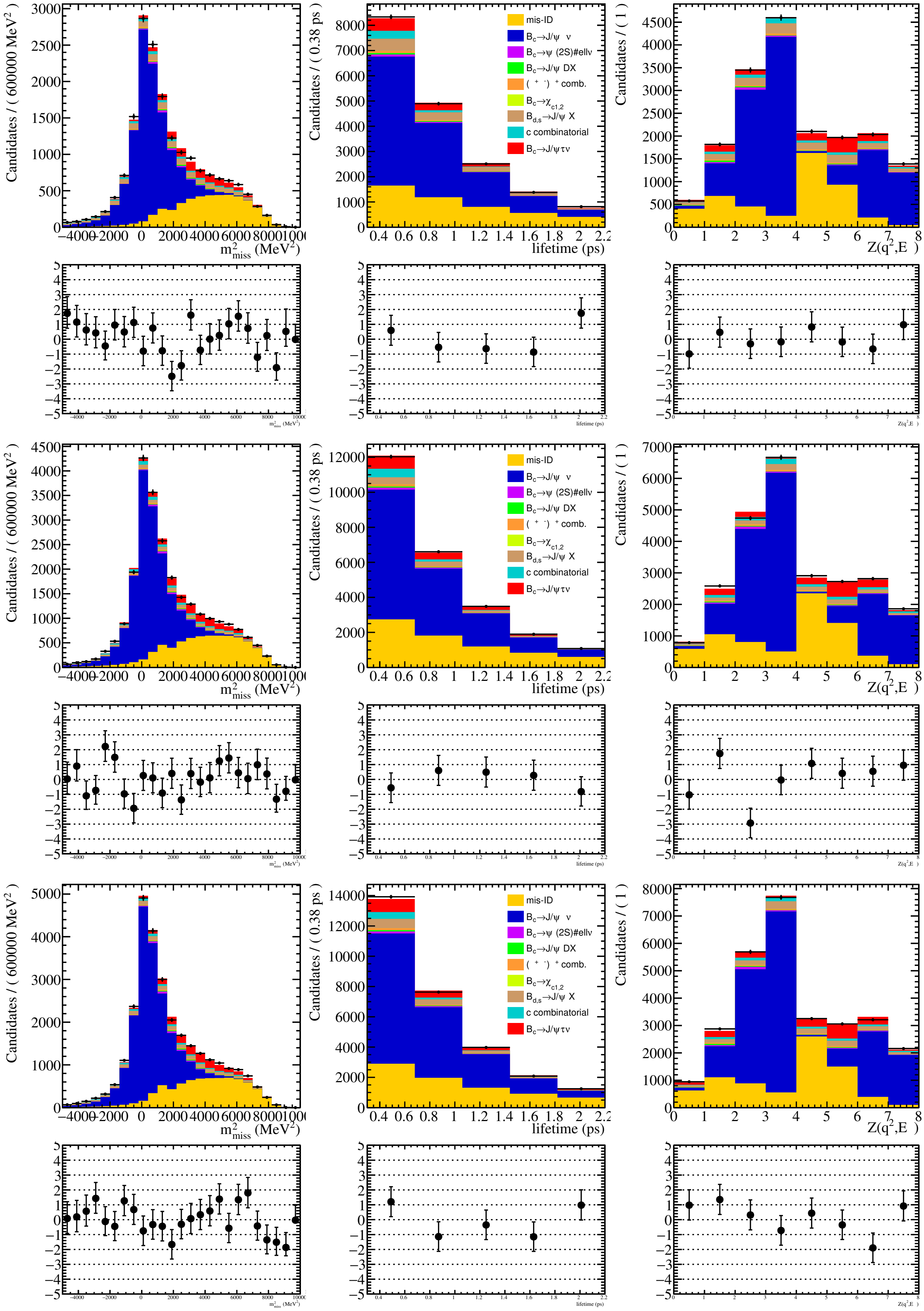


Figure 95: The results of the fit with $\psi(2S)$ contributions fixed to theoretical predictions, used to estimate the systematic uncertainty associated with the scaling of this background.

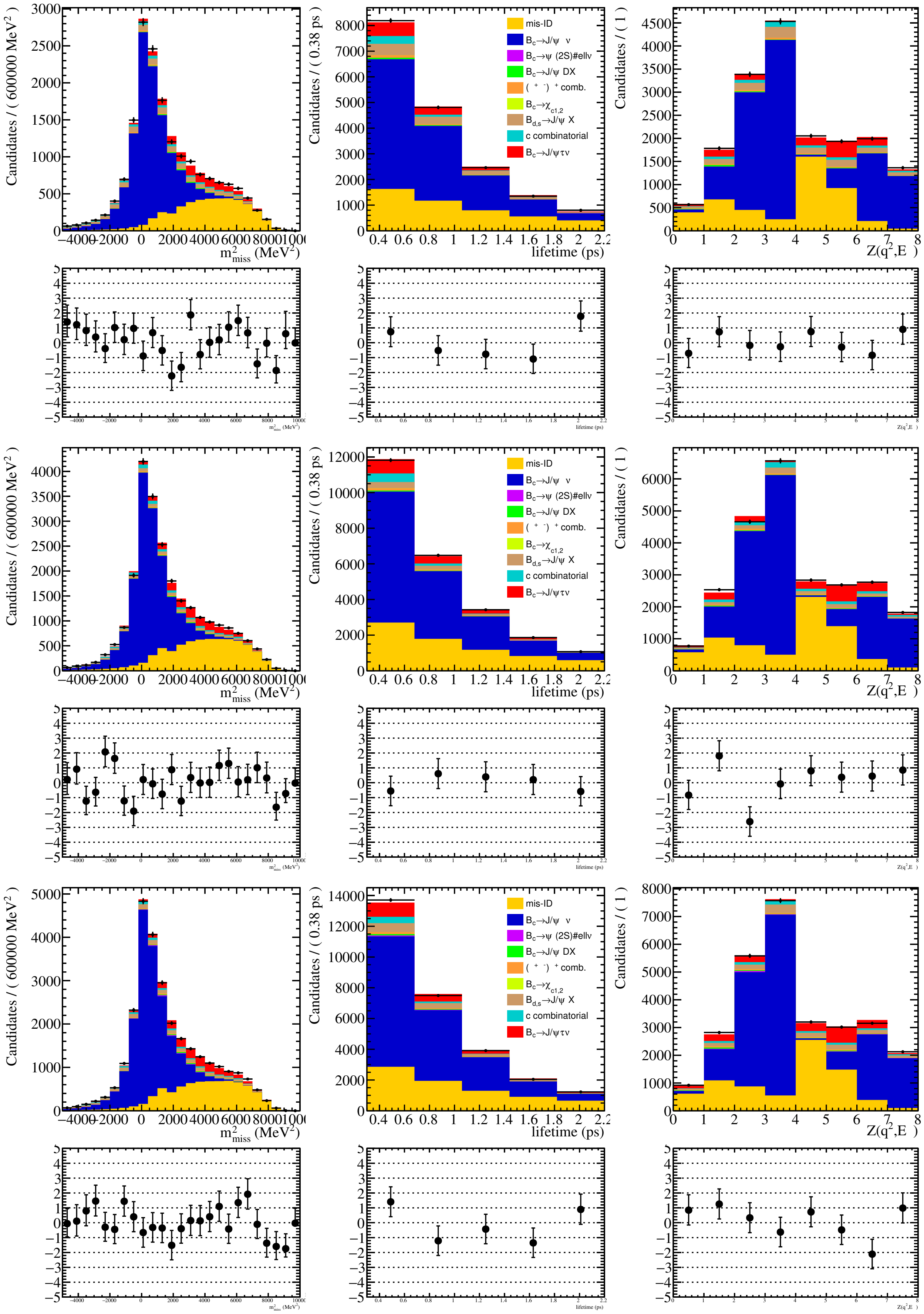


Figure 96: The results of the fit with the $J/\psi\mu$ combinatorial reweighted to match the upper sideband of the B_c mass in data, used to estimate the systematic uncertainty of our $J/\psi\mu$ combinatorial background modeling.

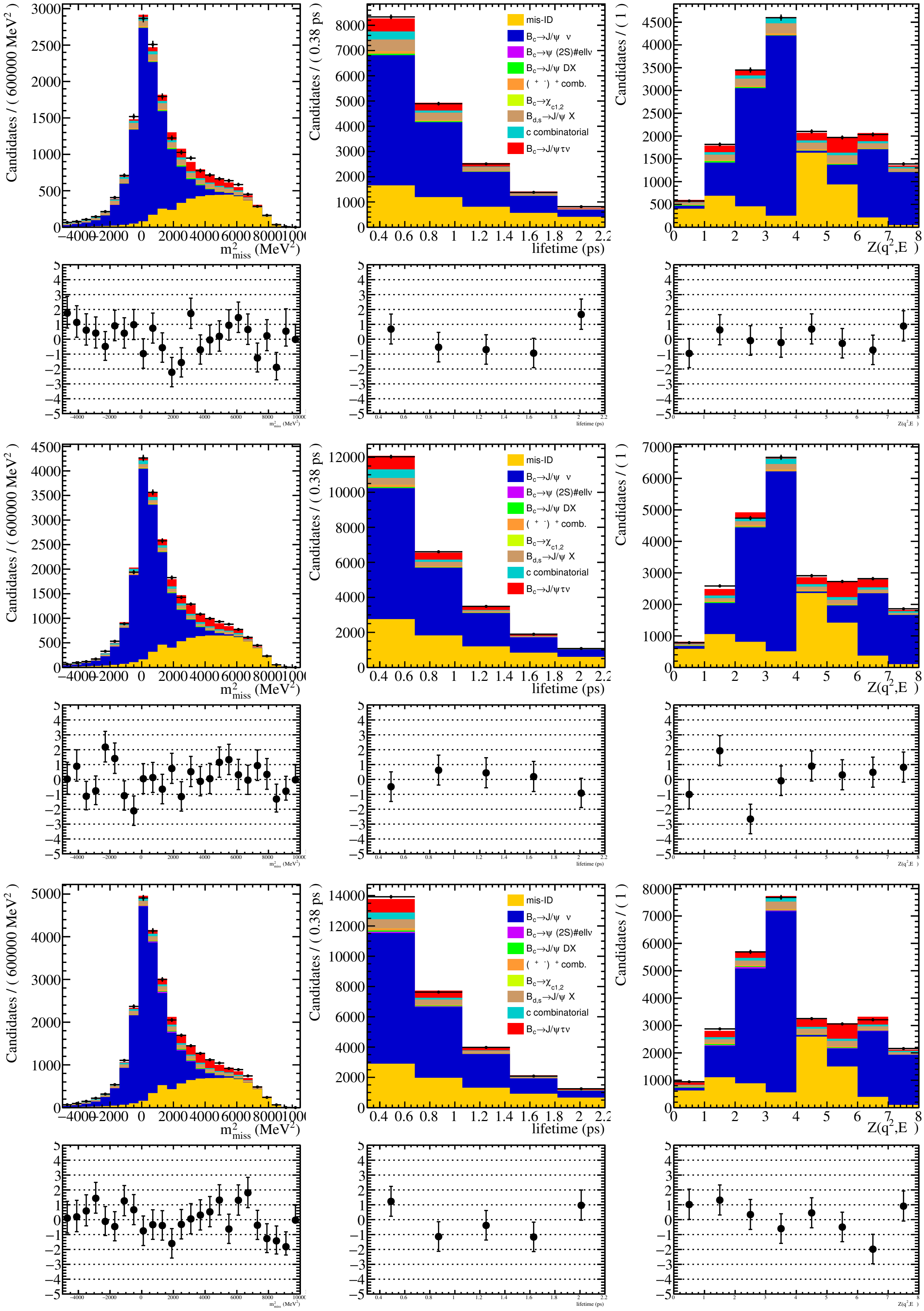


Figure 97: The results of the fit with the number of fake J/ψ allowed to vary, used in estimating the systematic uncertainty associated with the scaling of this background.

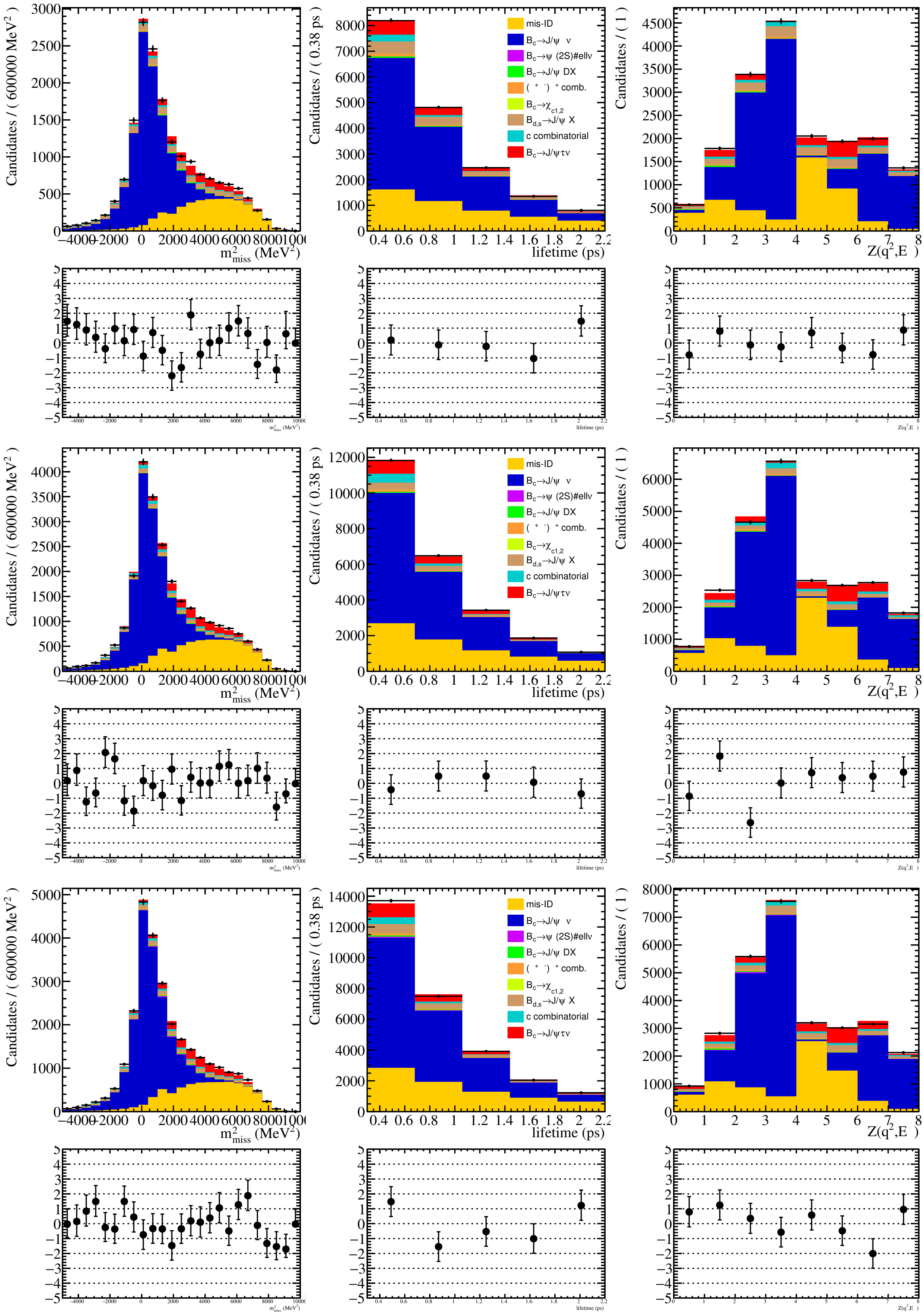


Figure 98: The results of one of the fits where the lifetime acceptance correction is resampled, used to estimate the systematic uncertainty associated with the lifetime acceptance correction.

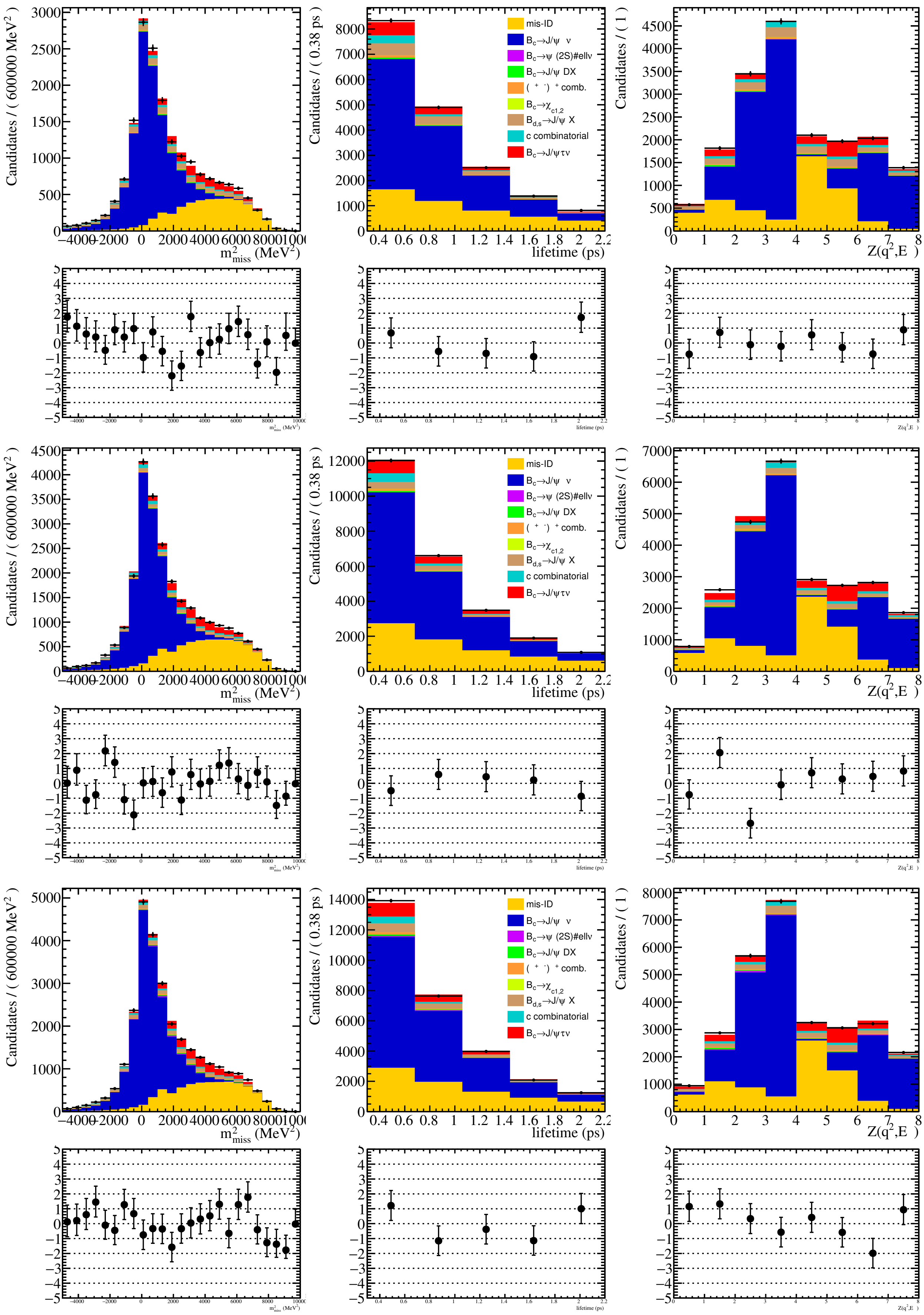


Figure 99: The results of one of the fits using the DIF efficiency correction with data/MC corrections using IsMuon and binned in nTracks.

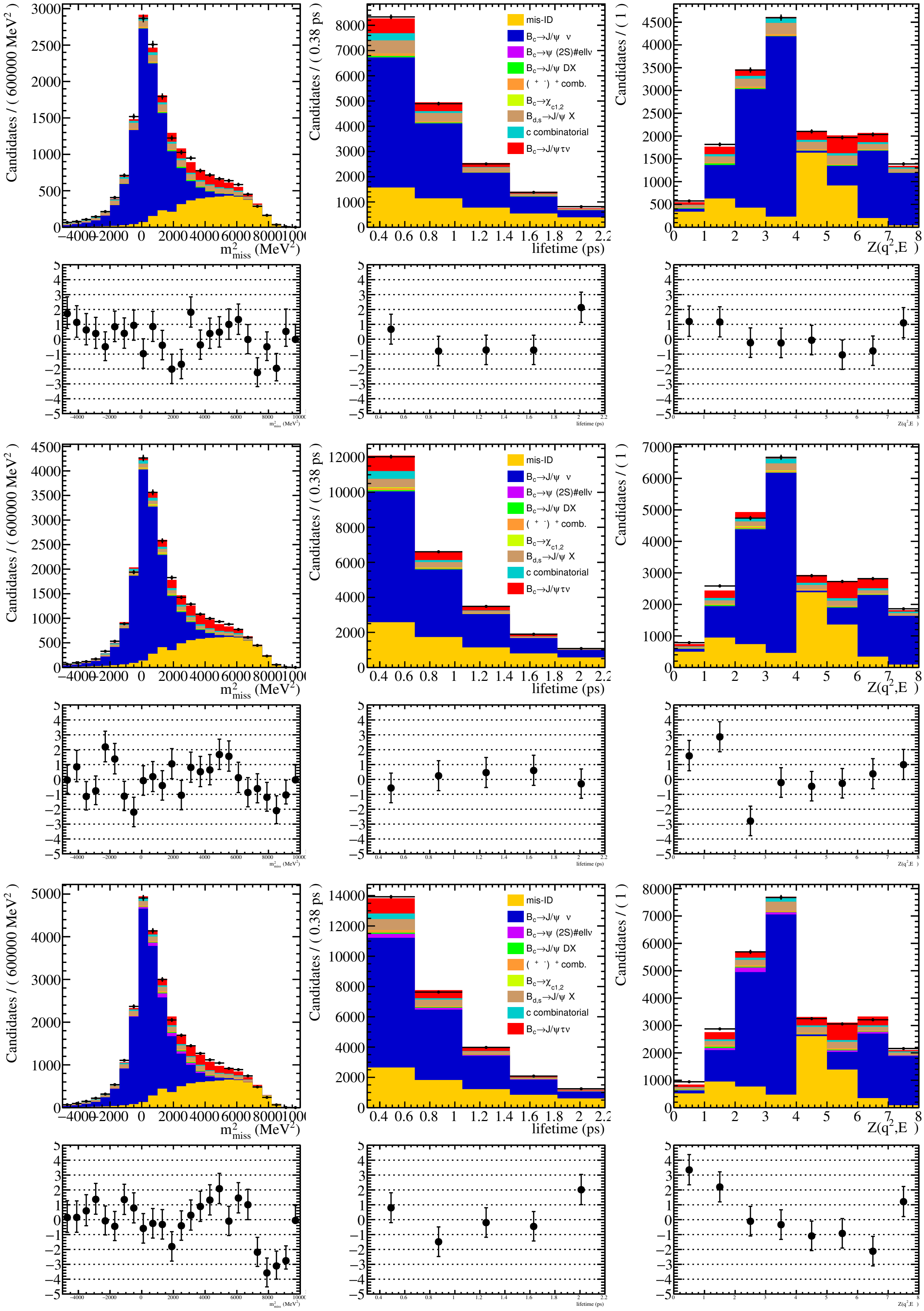


Figure 100: The results of one of the fits using the DIF efficiency correction with data/MC corrections using IsMuon and binned in the bachelor muon momentum.

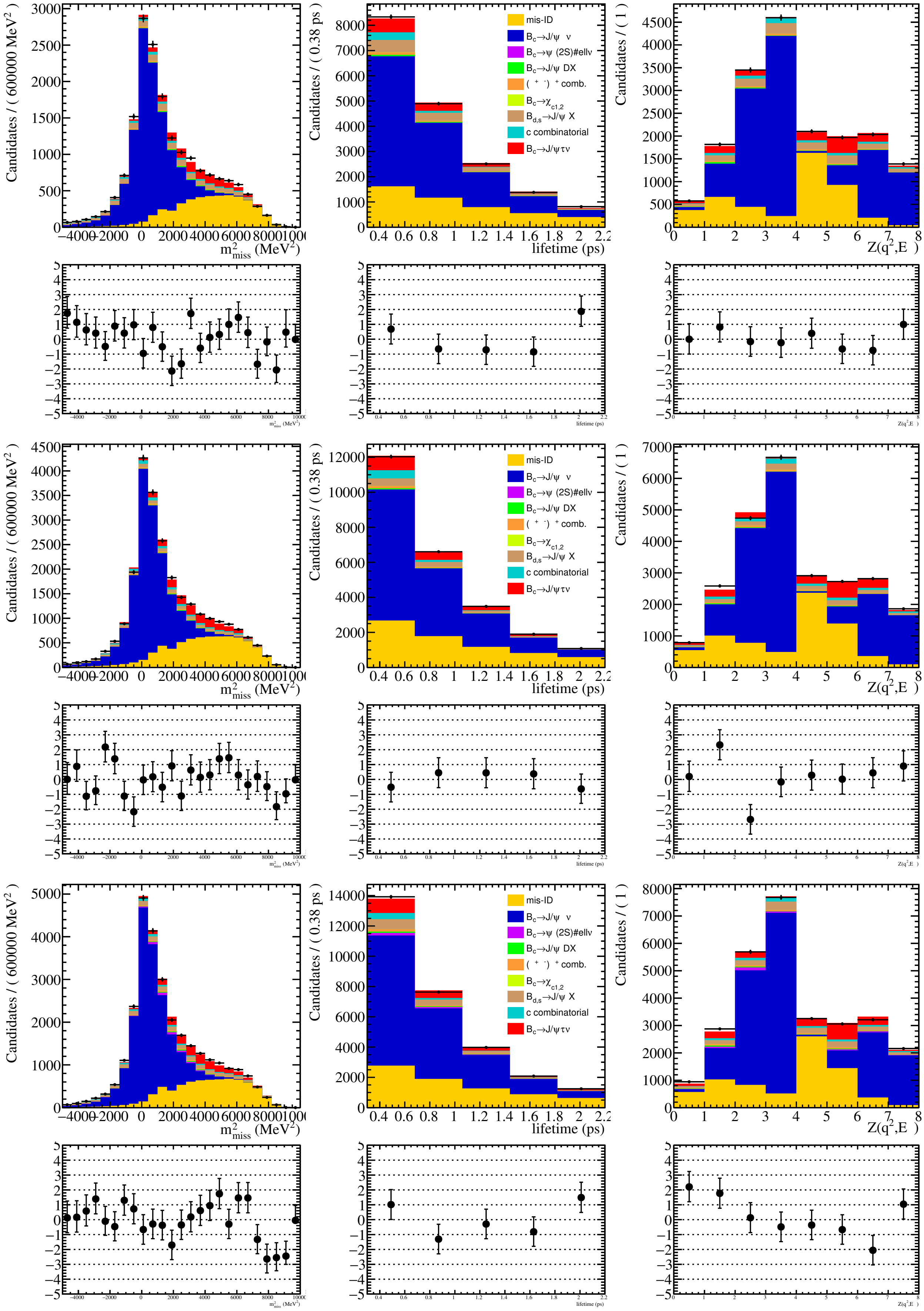


Figure 101: The results of one of the fits using the DIF efficiency correction with data/MC corrections using IsMuonTight and binned in the bachelor muon momentum.

22 Appendix D: Alternative J/ψ reconstruction

The J/ψ decays promptly and thus is non-flying. Using `StdLooseJpsi2MuMu` to select the J/ψ, however, results in the imposition of an additional constraint that the B_c vertex is located at the J/ψ vertex. This can reject otherwise usable events in the case of the signal channel, where the τ can fly and lead to a situation where the reconstructed J/ψ and B_c vertices do not align. To resolve this issue, we use a different neutral particle (K^0_S) and apply to it all the selections from `StdLooseJpsi2MuMu` in order to construct a J/ψ in DaVinci which is allowed to fly. After the final selections, this increases the signal efficiency by about 20%. However, making these modifications in the reconstruction also increased the proportion of combinatorial, which needed to be removed by tightening cuts on the χ^2 of the B_c vertex and the flight distance of the J/ψ. This improved the fit, but also slightly reduced statistics for the signal. After these modifications, the agreement between data and MC looked worse than without these modifications, especially in the low lifetime region.

23 Appendix F: HistFactory

HistFactory is a tool used to build probability distribution functions (PDFs) from data and MC templates.

The normalization of various templates is dealt with in HistFactory by `NormFactor` elements. All normalization systematics, known as `OverallSys` elements, have Gaussian constraints. There are high and low attributes corresponding to when the systematic is at $\pm 1\sigma$, with the high attribute corresponding to η^+_{ps} and $\alpha_p = 1$, and the low attribute corresponding to η^-_{ps} and $\alpha_p = -1$, where α_p is the nuisance parameter in Table 22. The nominal value is $\eta^0_{ps} = 1$. Shape systematics are accounted for using `HistoSys` elements, which are able to extrapolate between variational histograms σ^+_{psb} and σ^-_{psb}, which correspond to the nuisance parameters $\alpha_p = \pm 1$.

HistFactory is allowed to interpolate between different templates and is able to handle correlations between uncertainties. The default interpolation strategy is known as the piecewise-linear strategy. It is defined as

$$\eta_s(\alpha) = 1 + \sum_{p \in Syst} I_{lin.}(\alpha_p; 1, \eta^+_{sp}, \eta^-_{sp}) \tag{71}$$

and for shape interpolation it is

$$\sigma_{sb}(\alpha) = \sigma^0_{sb} + \sum_{p \in Syst} I_{lin.}(\alpha_p; \sigma^+_{psb}, \sigma^+_{psb}, \sigma^-_{psb}) \tag{72}$$

with

$$I_{\text{lin.}}\left(\alpha; I^0, I^+, I^-\right) = \begin{cases} \alpha\left(I^+ - I^0\right) & \alpha \geq 0 \\ \alpha\left(I^0 - I^-\right) & \alpha < 0 \end{cases}. \tag{73}$$

This is the default convention for `HistoSys`. $\alpha = 0$ is the nominal value of the parameter, and $\sigma^{0,\pm}_{sb}$ represent the nominal shape and the $\pm$ variations of the shape. $\eta^{0,\pm}_{ps}$ represent the nominal prediction and the variations of normalization related systematics.

The default strategy for normalization related uncertainties $\eta_s(\alpha)$, known as `OverallSys`, is the piecewise exponential interpolation strategy. It is defined as

$$\eta_s(\alpha) = \prod_{p \in Syst} I_{exp.}(\alpha_p; 1, \eta^+_{sp}, \eta^-_{sp}). \tag{74}$$

The default strategy for shape interpolation is

$$\sigma_{sb}(\alpha) = \sigma^0_{sb} + \prod_{p \in Syst} I_{exp.}(\alpha_p; \sigma^+_{psb}, \sigma^+_{psb}, \sigma^-_{psb}) \tag{75}$$

with

$$I_{\text{exp.}}\left(\alpha; I^0, I^+, I^-\right) = \begin{cases} (I^+/I_0)^\alpha & \alpha \geq 0 \\ (I^-/I_0)^{-\alpha} & \alpha < 0 \end{cases}. \tag{76}$$

Different interpolations have different advantages and disadvantages which are further described in the HistFactory manual [48].

The Gaussian constraint corresponds to a likelihood of

$$G\left(a_p \mid \alpha_p, \sigma_p\right) = \frac{1}{\sqrt{2\pi\sigma_p^2}} \exp\left[-\frac{\left(a_p - \alpha_p\right)^2}{2\sigma_p^2}\right]. \tag{77}$$

Other possible constraints include Poisson and log-normal.

24 Appendix G: Post-fit comparisons for 2016 and 2017

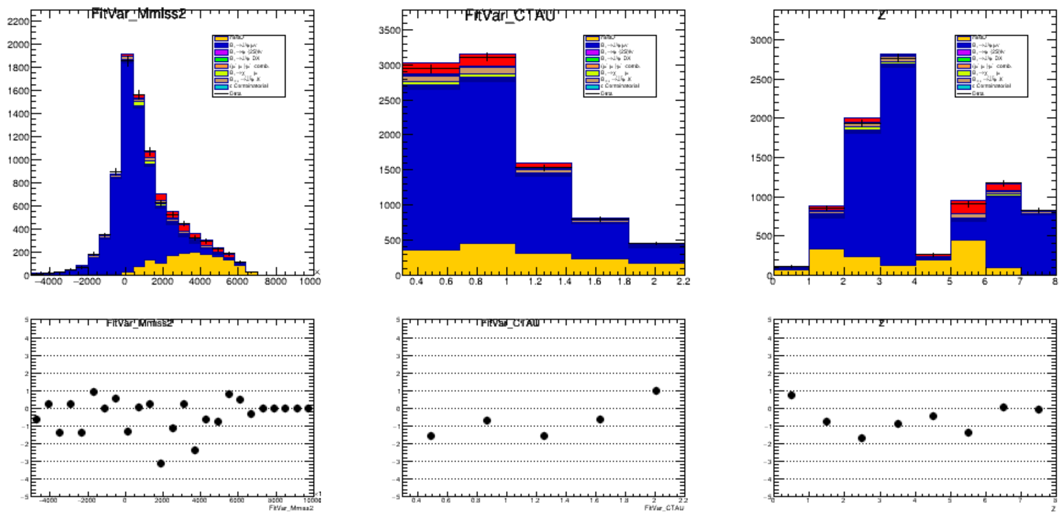


Figure 102: Plots for the fit variables with MC yields fixed from fit in 2016.

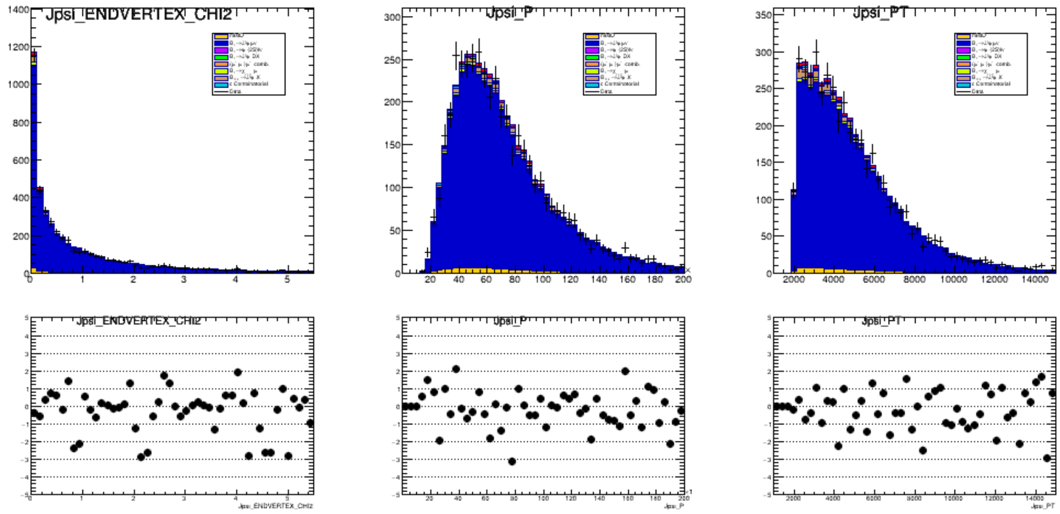


Figure 103: Comparison between data and simulation for the J/ψ candidate's vertex χ^2 along with the momentum, transverse momentum, and eta in 2016.

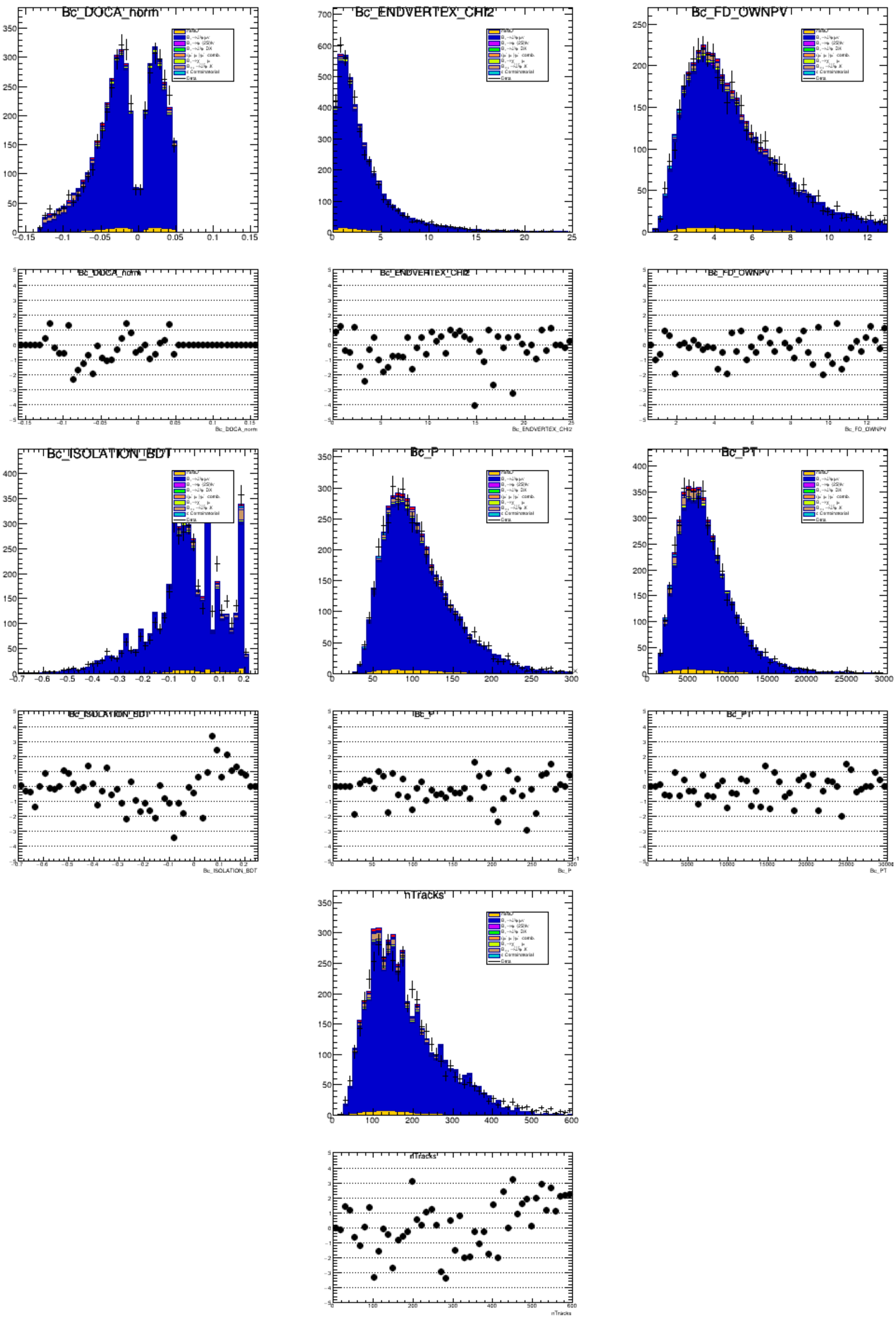


Figure 104: Comparison between data and simulation for the $J/\psi\mu^+$ system's DOCA, vertex fit χ^2, flight distance from the PV, momentum, transverse momentum, and nTracks in 2016.

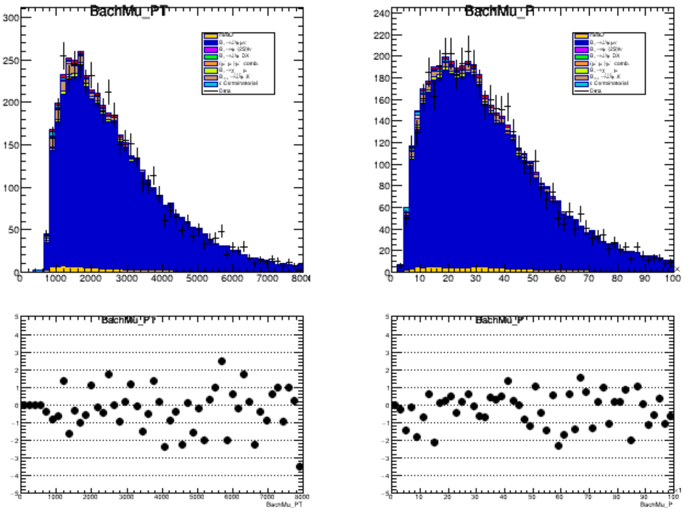


Figure 105: Comparison between data and simulation for the unpaired muon's momentum and transverse momentum in 2016.

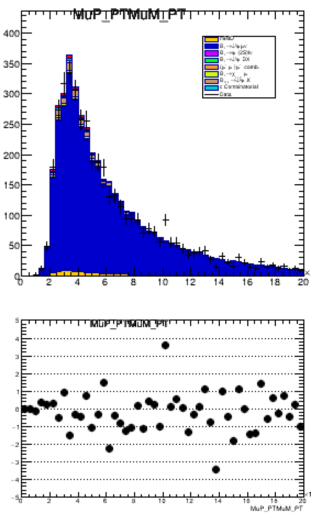


Figure 106: Comparison between data and simulation for the product of the PT of the dimuon system in 2016.

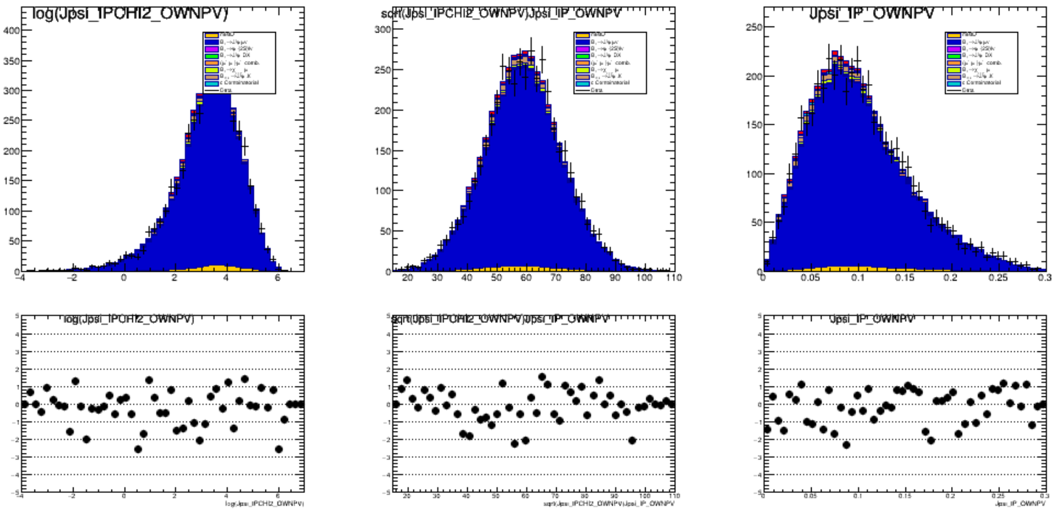


Figure 107: Comparison between data and simulation for the J/ψ's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2016.

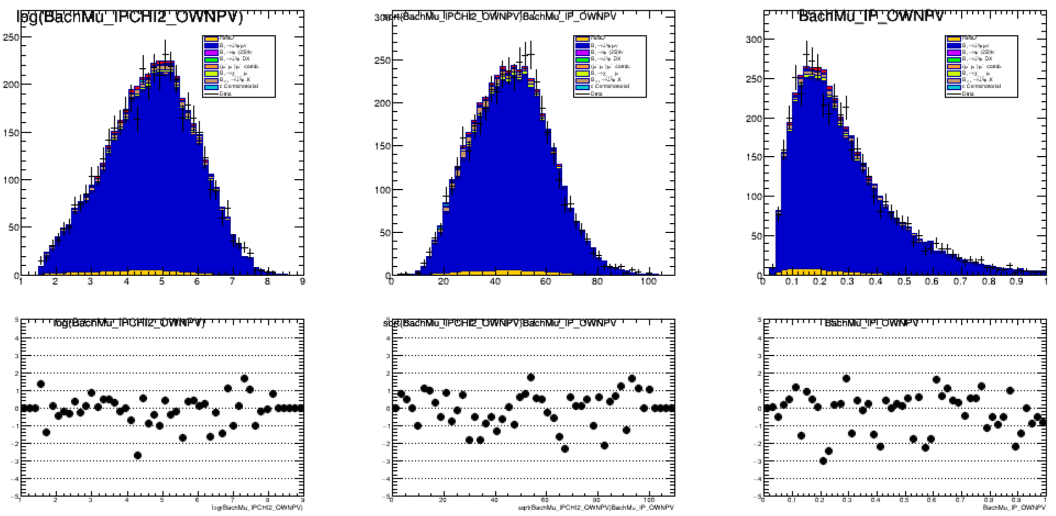


Figure 108: Comparison between data and simulation for the unpaired muon's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2016.

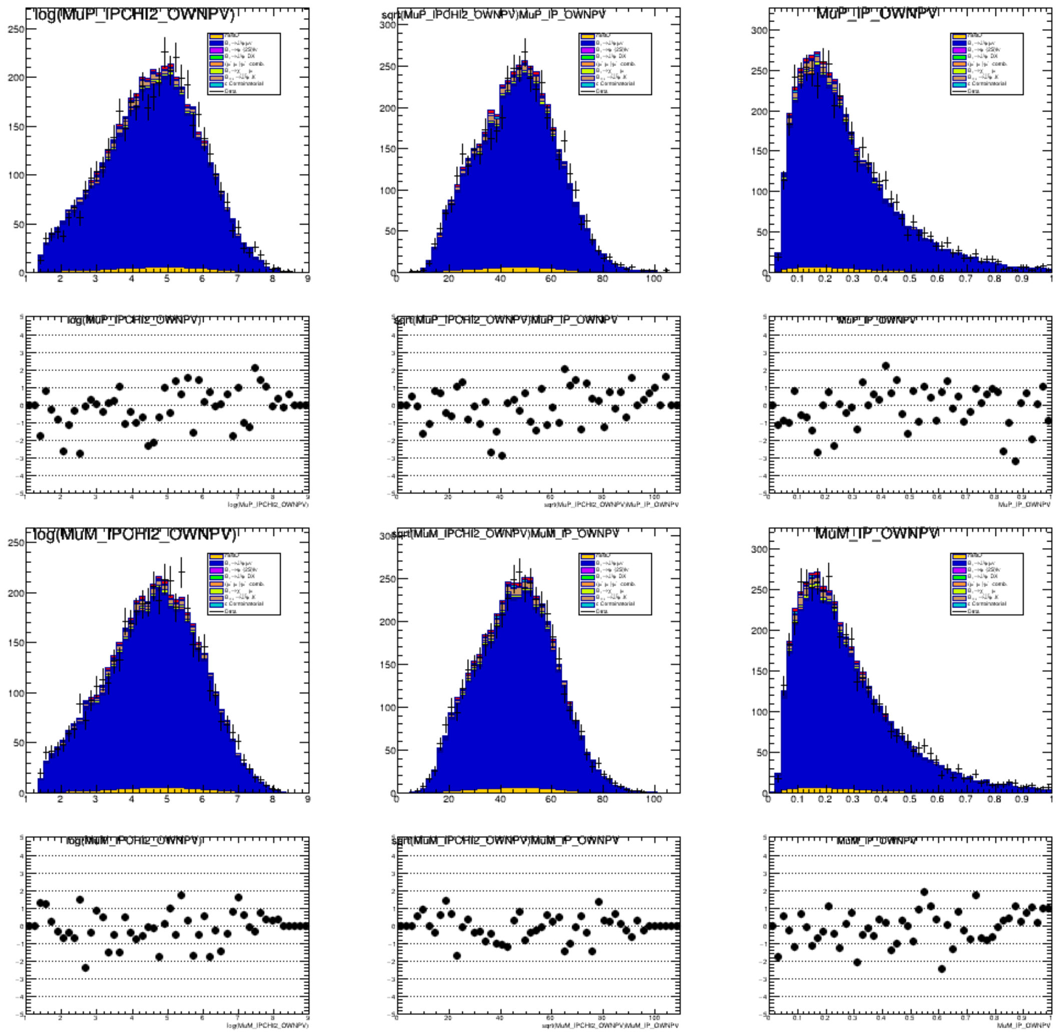


Figure 109: Comparison between data and simulation for the dimuon system's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2016.

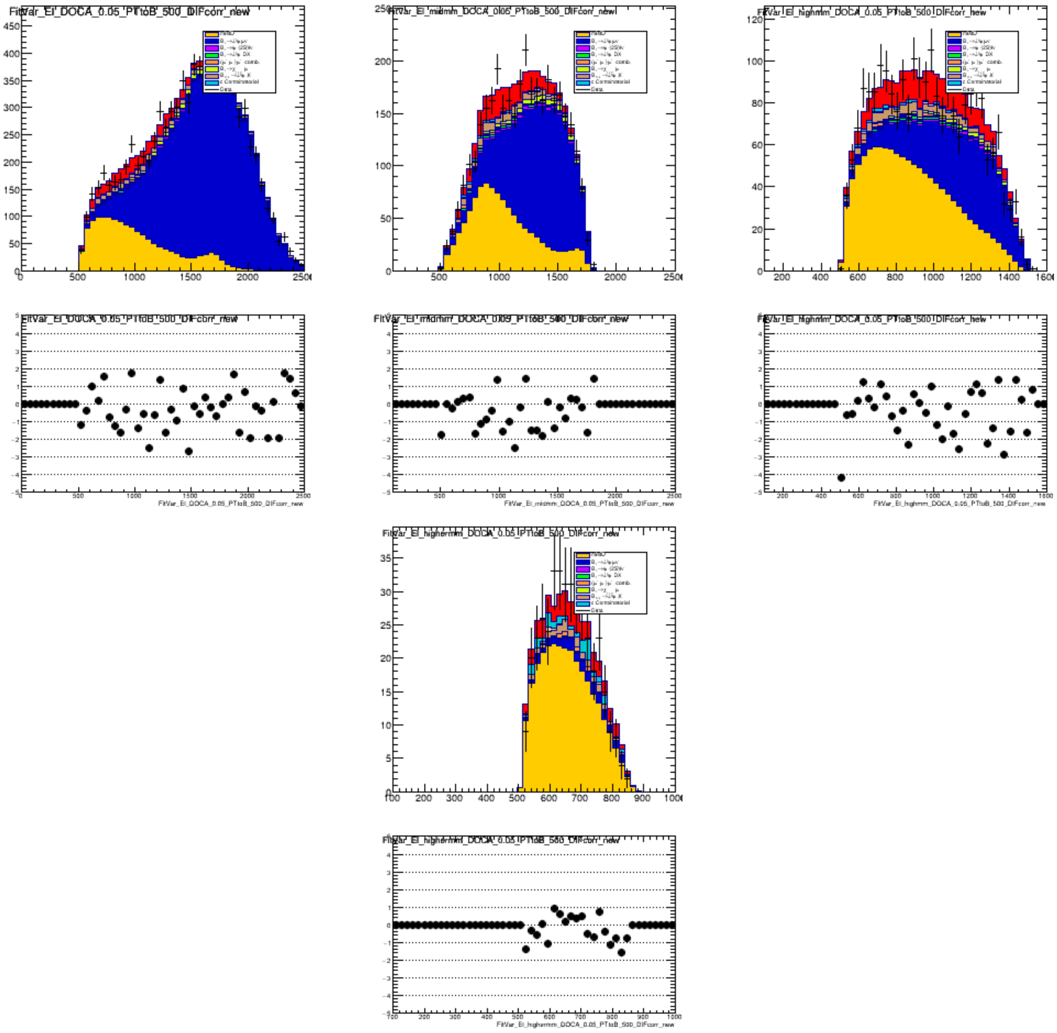


Figure 110: Comparison between data and simulation for the center of mass energy in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2016.

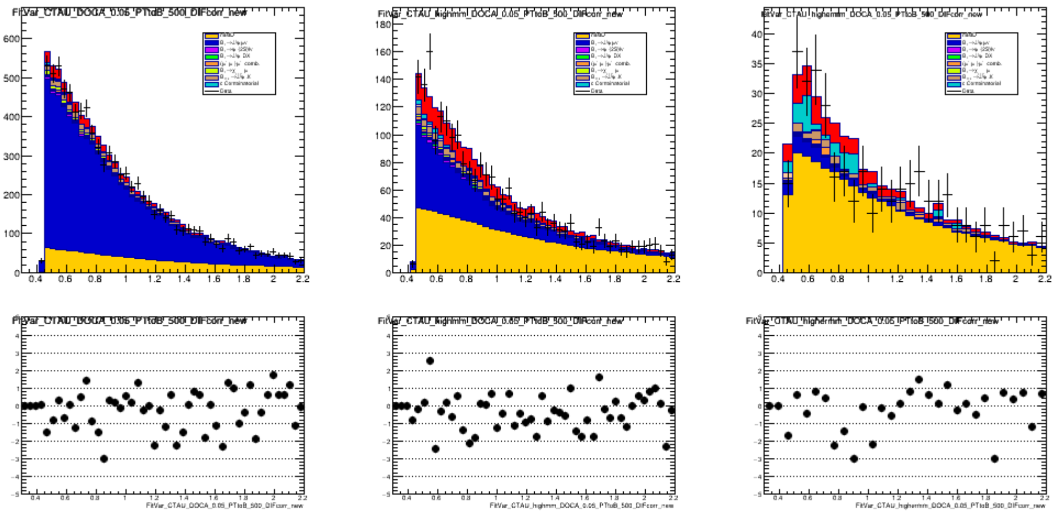


Figure 111: Comparison between data and simulation for the lifetime in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2016.

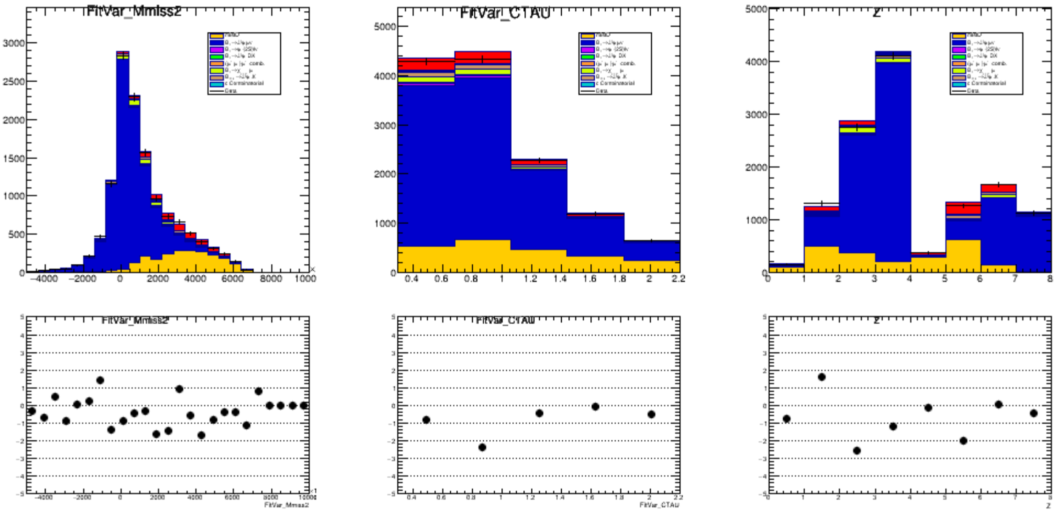


Figure 112: Plots for the fit variables with MC yields fixed from fit in 2017.

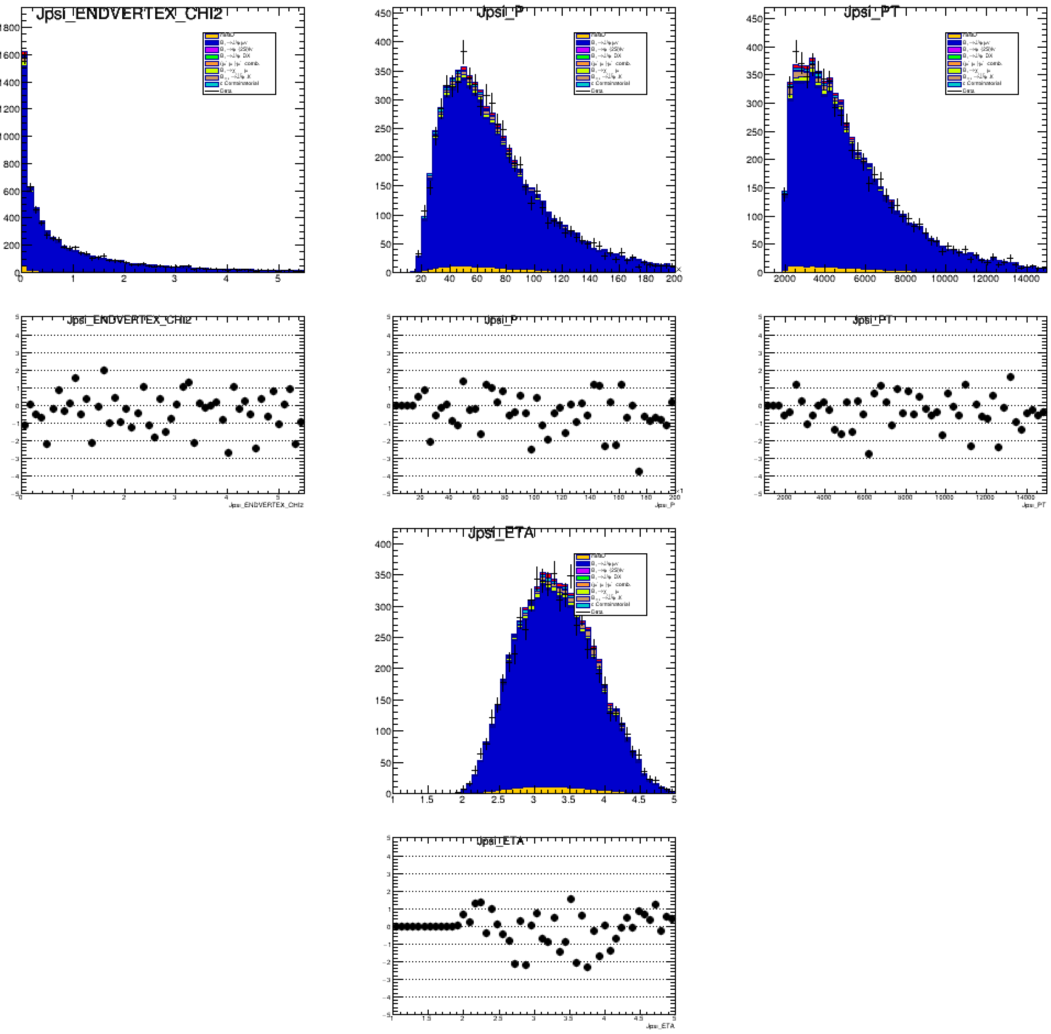


Figure 113: Comparison between data and simulation for the J/ψ candidate's vertex χ^2 along with the momentum, transverse momentum, and eta in 2017.

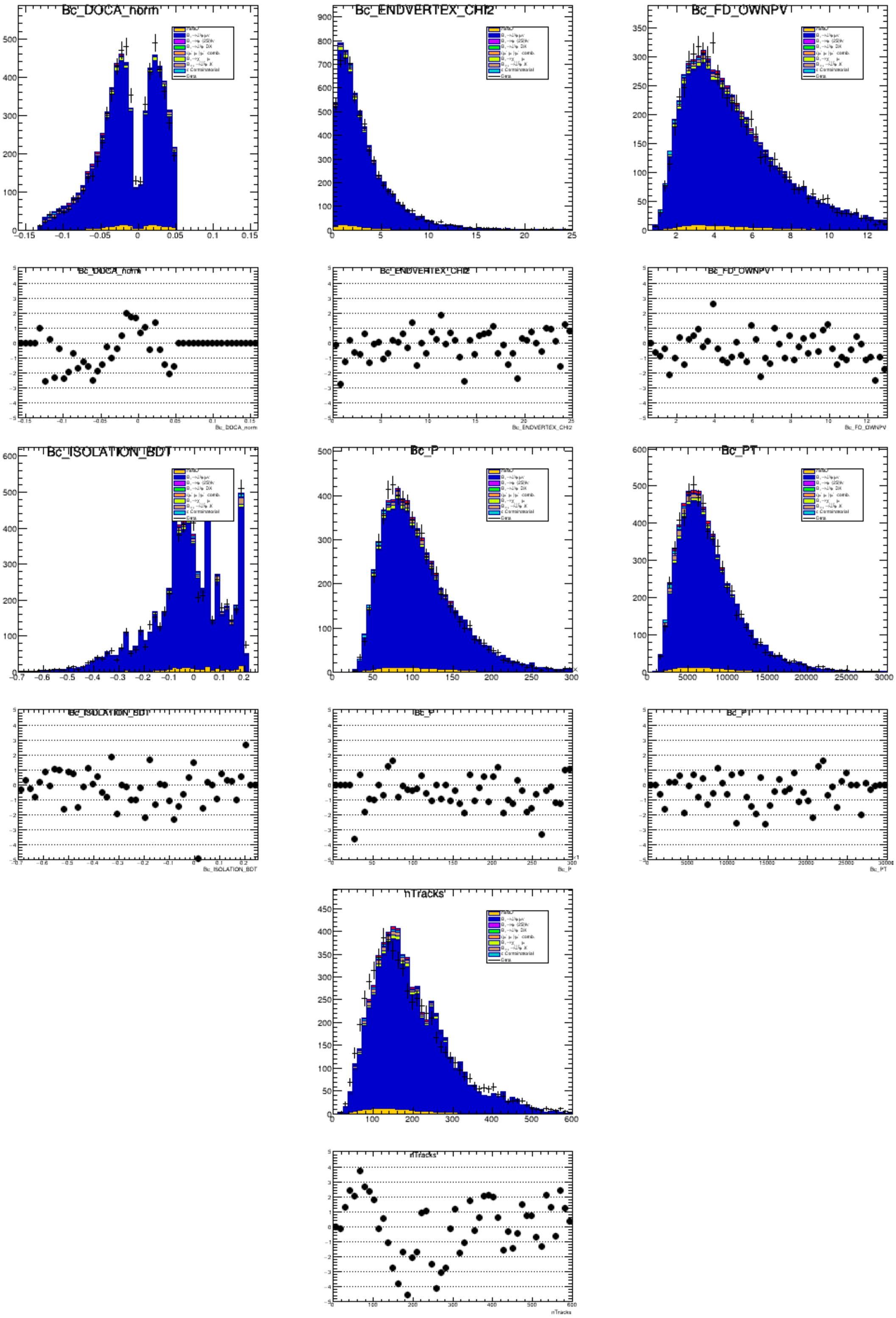


Figure 114: Comparison between data and simulation for the $J/\psi\mu^+$ system's DOCA, vertex fit χ^2, flight distance from the PV, momentum, transverse momentum, and nTracks in 2017.

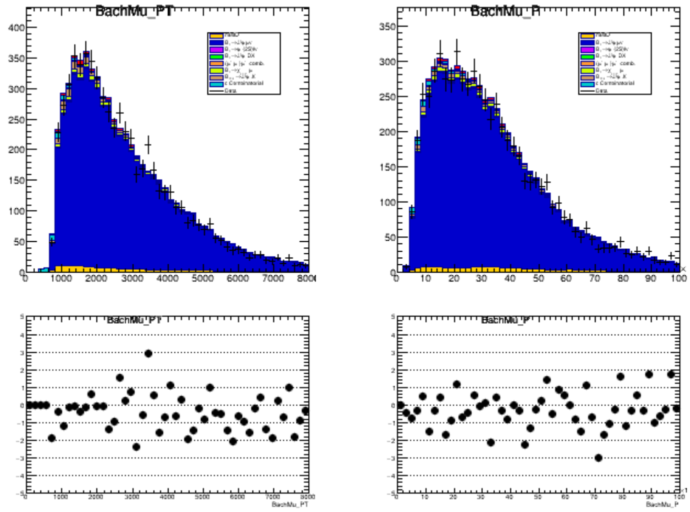


Figure 115: Comparison between data and simulation for the unpaired muon's momentum and transverse momentum in 2017.

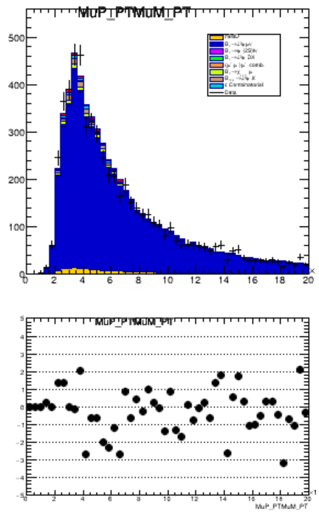


Figure 116: Comparison between data and simulation for the product of the PT of the dimuon system in 2017.

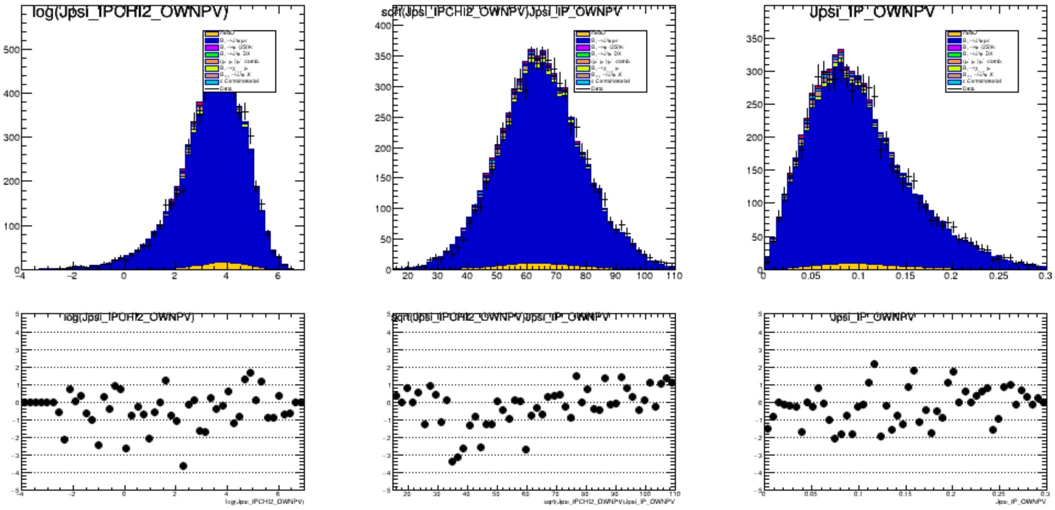


Figure 117: Comparison between data and simulation for the J/ψ's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2017.

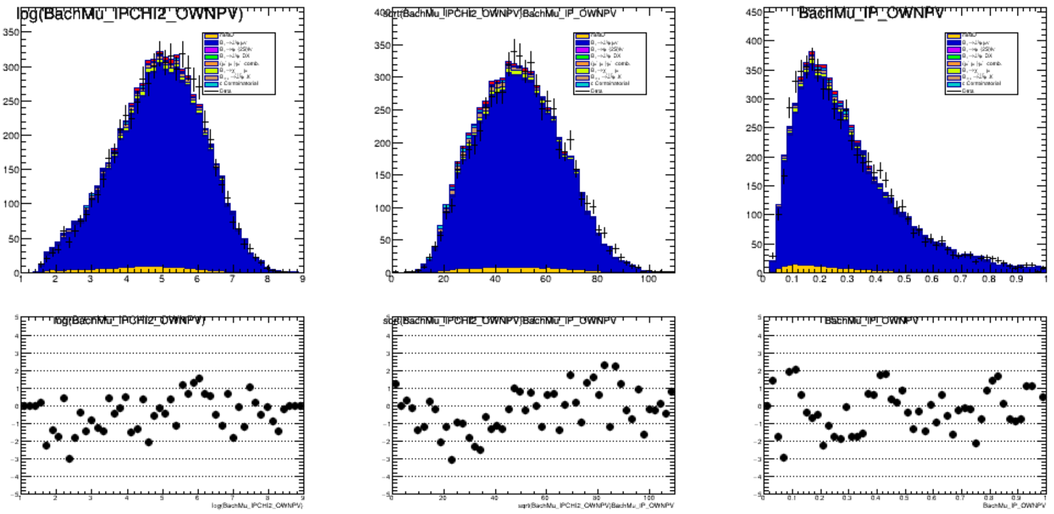


Figure 118: Comparison between data and simulation for the unpaired muon's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2017.

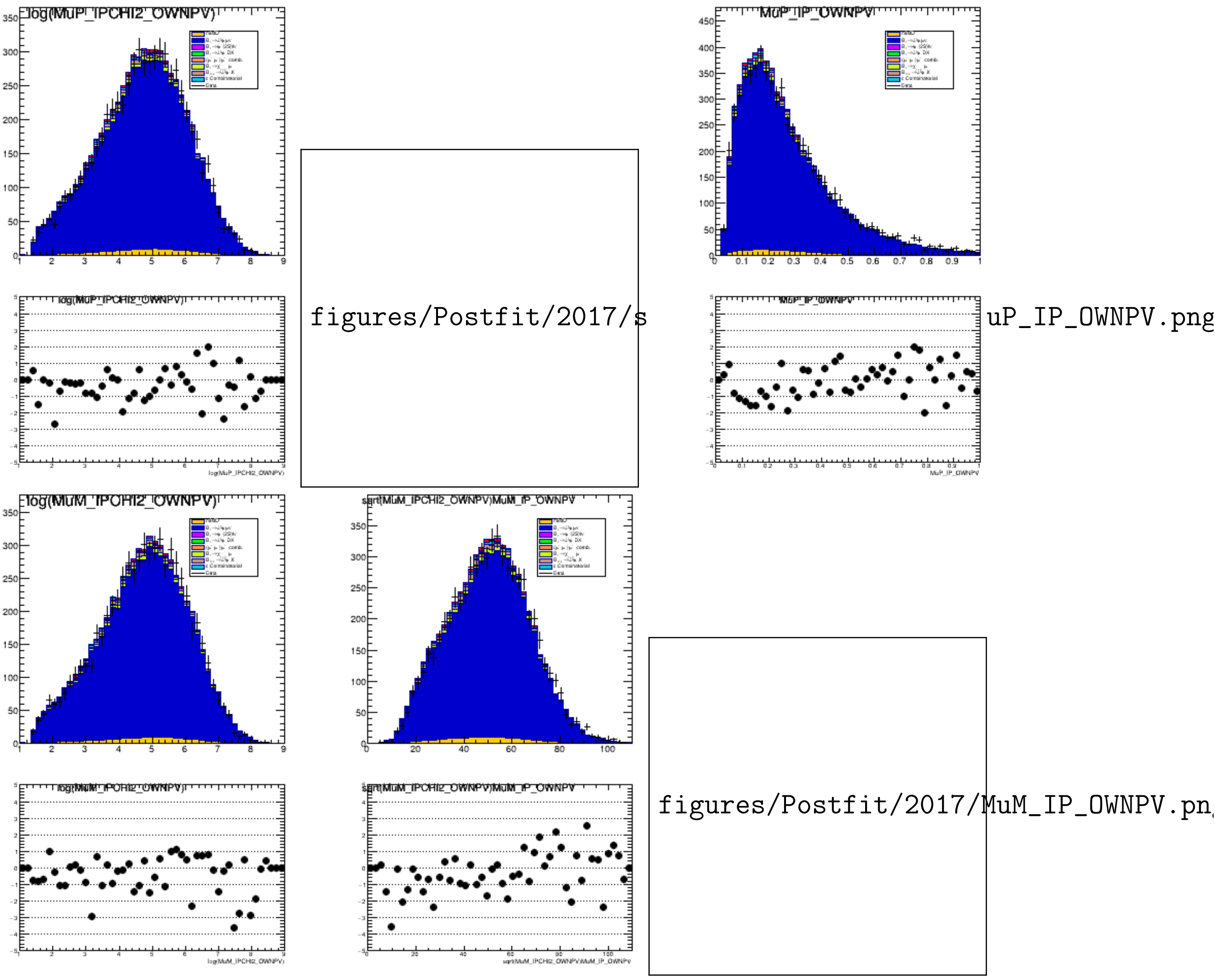


Figure 119: Comparison between data and simulation for the dimuon system's IP related variables (the log of the IP χ^2 with respect to the PV, the square root of the IP χ^2 divided by the IP, and the IP for 2017.

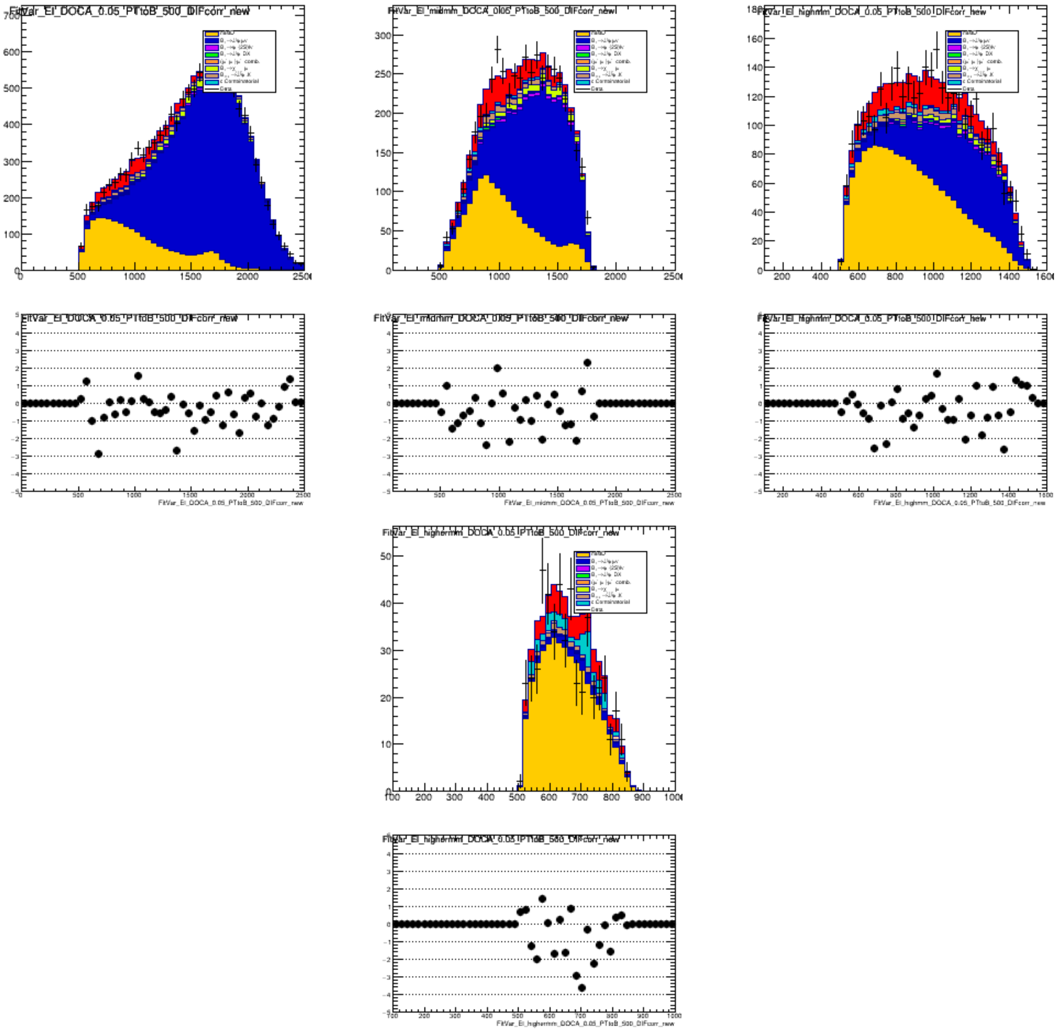


Figure 120: Comparison between data and simulation for the center of mass energy in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2017.

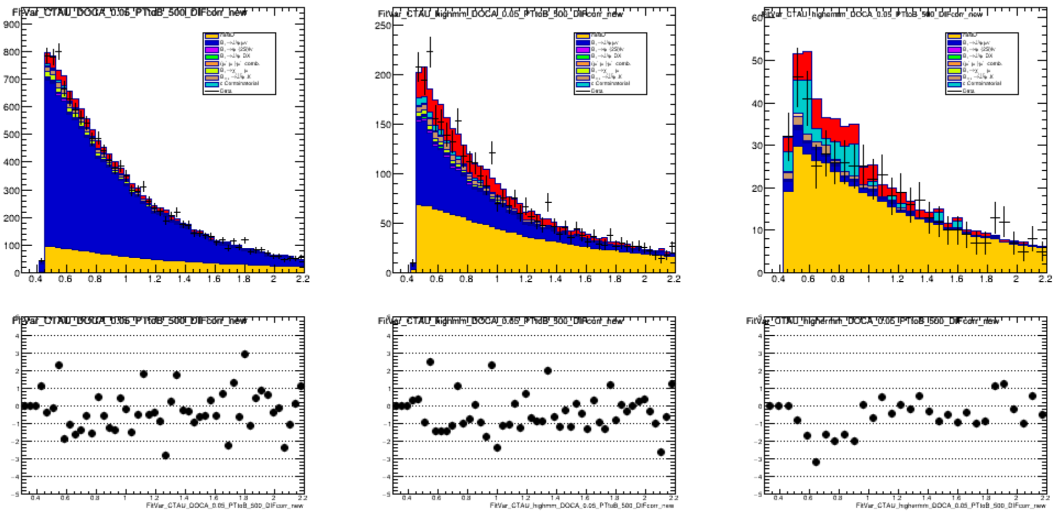


Figure 121: Comparison between data and simulation for the lifetime in the full signal region, $m^2_{miss} > 1\,\mathrm{GeV}^2$, $m^2_{miss} > 2\,\mathrm{GeV}^2$, and $m^2_{miss} > 5\,\mathrm{GeV}^2$, for 2017.

25 Appendix H: Charm companion combinatorial background

Recently, it was noted that backgrounds from spurious combinations in real B_c events could be enhanced due to kinematic correlations between the B_c meson and the associated charm hadron produced in the fragmentation process from the other half of the $c\bar{c}$ pair providing the charm half of the B_c^+ meson [40]. The contribution from this background is referred to as the combinatorial charm background hereafter.

When the associated charm hadron decays (semi)muonically, the resulting muon can be wrongly combined with a J/ψ from the associated B_c^+ decay, hence the term combinatorial charm background. The topology of this background is shown in Figure 20.

In the following, the general features of this background are briefly introduced before arriving at a rough estimate of its expected size from the yields observed in `BcVegPy` [28] events relative to the normalization channel. A set of templates is developed for different kinematic regions and added to the fit, which allows the assessment of the potential impact of this background on the fit results, as well as the effect of kinematic variations which may arise from uncertainties in the B_c production mechanism. These include templates produced in slices of the ratio of the momenta of the D and B_c and the collinearity of the D and B_c (here and throughout this section D refers to any of D^0, D_s^+, D^+, Λ_c^+).

25.1 Yield estimate

25.1.1 BcVegPy expectation

To derive a rough estimate of the expected number of candidates from the combinatorial charm background present in the data sample, this background is modeled using MC taken from the normalization channel. The bachelor muon is required to have a parent which is identified as a D, and to not have the same parent as the J/ψ. After truth matching and reconstruction, this yields a ratio of $\frac{N_{c-comb}}{N_{B_c \to J/\psi \mu\nu}} \approx 0.03$. Following the application of the nominal selection, the ratio drops to $\frac{N_{c-comb}}{N_{B_c \to J/\psi \mu\nu}} \approx 0.001$. These studies show that the selections of note in reducing the combinatorial charm background are the DIRA (Figure 122) and bachelor muon transverse momentum (Figure 123) selections. The DIRA selection has an efficiency of roughly 73% for normalization, but 51% for the combinatorial charm. More substantially, the bachelor muon p_T had an efficiency of 94% for the normalization, but 38% for the combinatorial charm. The DIRA cut was originally added to remove combinatorial background from $B_{d,s}$. The combinatorial charm is predicted by `BcVegPy` [28] to have a soft p_T spectrum due to the large mass of the B_c^+, leading to a soft p_T for the D and its child muon. It has been suggested that the actual charm spectra and their angular correlation relative to the B_c may differ from `BcVegPy`, resulting in harder charm momentum spectra. As stated above, the effect of these deviations from the `BcVegPy` -based simulations on the measurement of $R(J/\psi)$ will be explored by performing fits using templates representing the combinatorial charm background in various kinematics and angular regions.

The isolation was initially expected to drastically reduce this background–however, it has a 90% efficiency for normalization and a 75% efficiency for the combinatorial charm. Various studies were attempted to investigate the efficacy of the isolation in

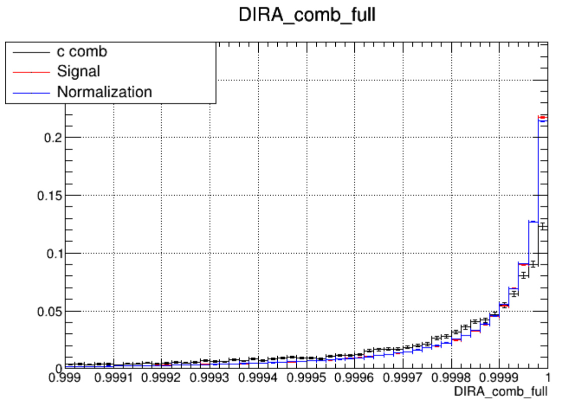


Figure 122: The DIRA distributions in combinatorial charm (teal), signal (red), and normalization (blue) before selection. The signal and normalization are more sharply peaked at high DIRA.

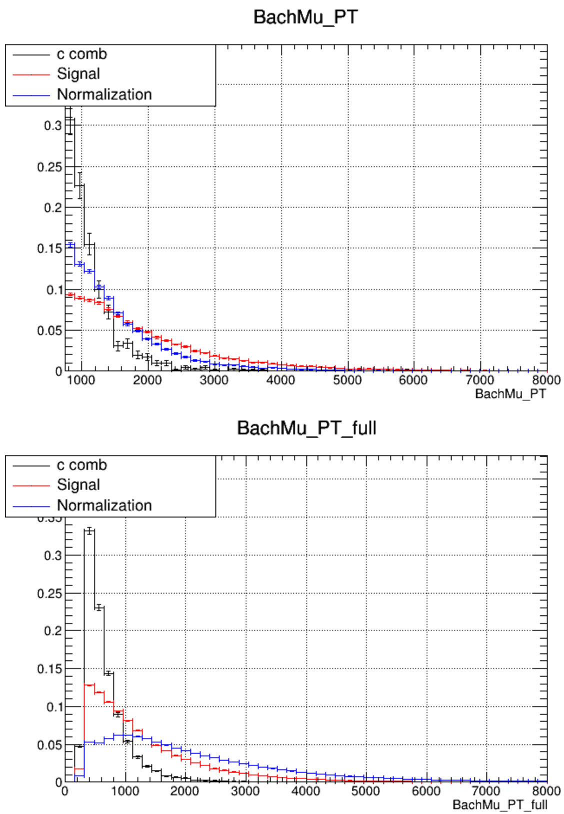


Figure 123: The distributions of the bachelor muon p_T in combinatorial charm (teal), signal (red), and normalization (blue). Left shows the distribution after the nominal selection, while right shows the distribution after the full offline selection. The combinatorial charm distribution is notably peaked at low momentum.

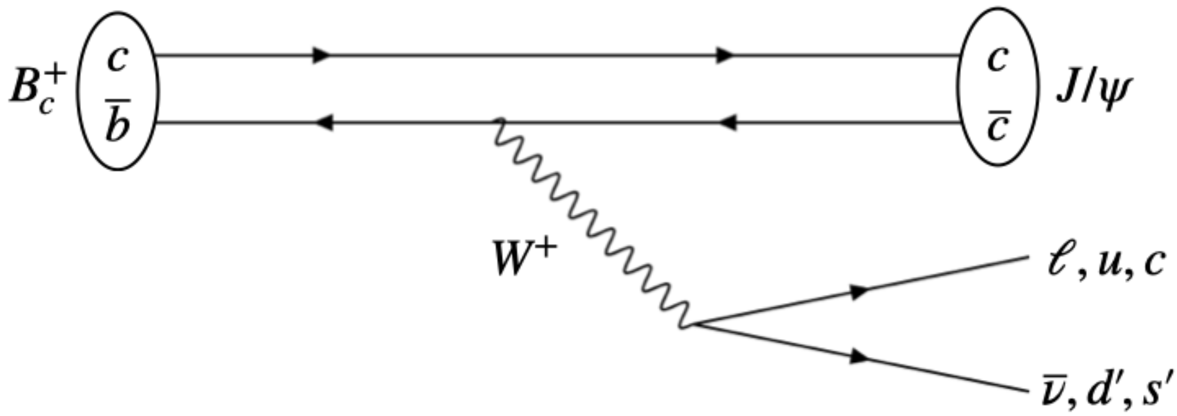


Figure 124: Depiction of the Feynman diagram associated with this background, where a B_c is produced and decays into a J/ψ and several possible configurations. The charm in the B_c is produced in association with an anti-charm at the primary vertex due to quark fragmentation and goes on to form a charm hadron which produces a muon.

other $B_c \to J/\psi X$ systems with higher charge multiplicity, for example in $B_c \to J/\psi DK$, $B_c \to \psi(2S)\mu\nu$, and $B_c \to J/\psi\pi\pi\pi$, but these yielded similar results. It is expected that the somewhat reduced discriminating power from the isolation is due to the short lifetime of the B_c–in the analysis, whereas the focus of the isolation cut is the reduction of mis-ID from the lighter Bs which, on average, have a significantly higher lifetime.

With the contribution from the normalization events calculated, the contributions coming from other $B_c \to J/\psi X$ decays can be estimated. This is estimated through a simple counting method. In addition to the muonic channel, contributions from $B_c \to J/\psi e\nu$ and hadronic channels exist, where the hadronic contributions arise from the two possible quark configurations (the W decaying to u, d' or c, s') with 3 color possibilities (see Figure 124).

This leads to 7 additional possible channels, which corresponds to a factor of 8 for scaling the combinatorial charm from only the normalization channel to all possible contributions from $B_c \to J/\psi X$ decays for a total of 0.8% of normalization channel events. If production is `BcVegPy`-like, this is an upper bound on the overestimation of other fitted yields, since as yet, no use has been made of of the kinematic separation power between signal and combinatorial charm background, which will be reflected in the derived templates, as discussed in detail below.

25.1.2 Estimate from multiple candidate events

To obtain a more data-driven estimate of the expected size of this background, we consider the class of events where a real $B_c^+ \to J/\psi\mu X$ decay was reconstructed in addition to a charm combinatorial candidate, resulting in a multiple-candidate event. We observe 65 events with multiple candidates *of any kind* summed over all years after all selections are applied. These events are filtered by requiring that the multiple candidates in an event share the same reconstructed J/ψ candidate, determined according to daughter track keys, and have different track keys for the third muon. To account for reconstruction abnormalities it was checked that the same candidates were selected by requiring only that the two J/ψ candidates have equal momentum to 1-10 MeV precision. This selection reduces the total number of events selected to 22.

For real charm companions from correlated production, the sign of the muon should be opposite to the sign of the muon from the $B_c^+ \to J/\psi\mu$. An opposite-sign requirement on the third muon from each candidate reduces the yields to a total of 17 candidates. The selection is summarized in Table 25

To estimate the expected number of charm combinatorial events from these yields, we require two factors: one accounting for the efficiency for finding both candidates in a given event with a real $B_c^+ \to J/\psi\mu X$ decay and one to scale from $B_c^+ \to J/\psi\mu X$ to inclusive $B_c^+ \to J/\psi X$. For the latter, the factor of 8 above is used, and for the former, we use simulation to compute

$$\frac{\varepsilon(c - \mathrm{comb})}{\varepsilon(B_c^+ \to J/\psi\mu\nu \text{ AND } c - \mathrm{comb})} \approx 5.$$

Note that, in the case of independent efficiencies, this would correspond to just $(\varepsilon(B_c^+ \to J/\psi\mu\nu))^{-1}$. Given the variation in $B_c^+ \to J/\psi\mu X$ efficiency already known between $B_c^+ \to J/\psi\mu\nu$ and $B_c^+ \to J/\psi\mu\nu$ this factor could be larger, and so we consider that the product of these two factors could be in the range of 40-60. This gives the estimated yields in the last column of Table 25.

Data sample	Total multiples	Shared J/ψ	Anticorrelated charge	Estimated c-comb yield
2016	15	6	5	$250 \pm 110 \pm 50$
2017	23	5	3	$150 \pm 90 \pm 30$
2018	27	11	9	$480 \pm 150 \pm 120$

Table 25: Yields of multiple candidates and derived estimate for c-comb yeilds, where in the last column the first uncertainty is statistical and the second is from the range in the transfer factor.

25.2 Kinematic and topological properties

The shape of the fit variables for the combinatorial charm background can be derived from the simulation as well. In order to enhance the statistics of the simulation samples, the normalization, signal and $B_c \to \chi_{c1,2}\mu\nu$ MC samples for the 2016, 2017, and 2018 productions were combined. The template distributions are shown in Figure 21. Notably, the missing mass distributions of the combinatorial charm and that of the $B_d \to J/\psi X$ combinatorial share similarities, while the lifetime distribution of the charm combinatorial is heavily shifted towards low lifetime, far lower than any other template used in this fit. Later sections will show that this feature appears robust against different production kinematics.

The peaking near zero missing mass is determined to be primarily due to the candidates with lowest lifetime, for which the B_c direction used in calculating the rest frame variables is poorly determined. Figure 125 shows the distribution at long and short B_c flight distance, clearly indicating that the peak is diminished at long flight distances.

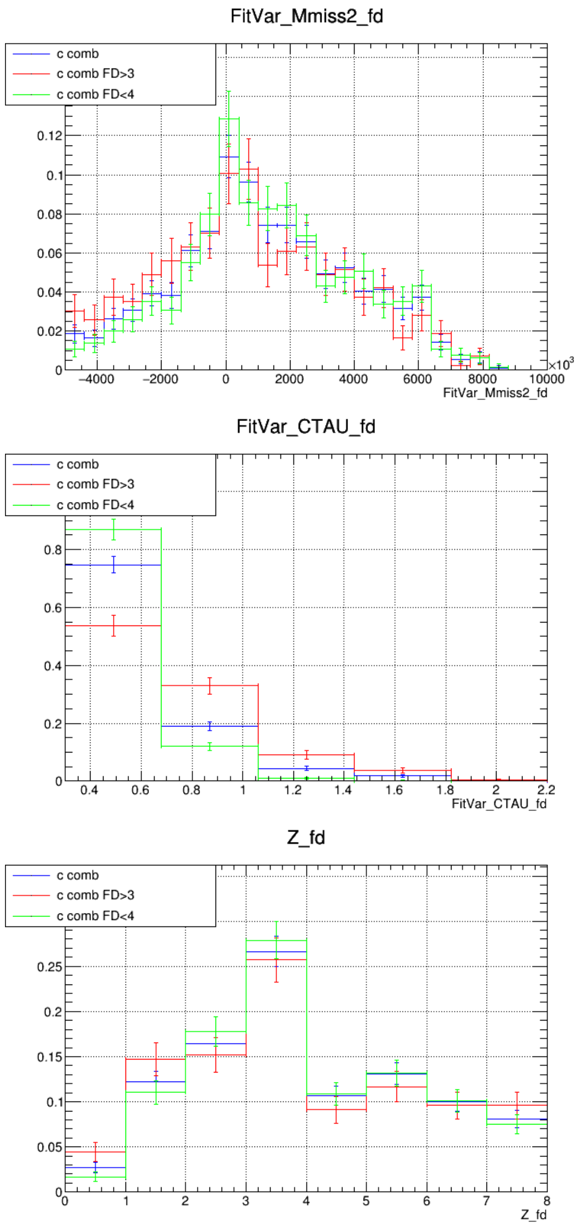


Figure 125: The distributions of the rest frame variables of the combinatorial charm, with the nominal selection in blue, the B_c flight distance less than 4 mm in green, and the B_c flight distance greater than than 3 mm in red.

The peak is replicated in the combinatorial charm derived from signal MC (Figure 126), again supporting the notion that the origin of this peak is not mis-identified normalization events due to a failure in the MC association, but rather a feature of the combinatorial charm background primarily arising from the specific geometric distribution of the events and the rest frame approximation. In Figure 126, it can be seen that in addition to the lifetime effect, the DIRA cut also appears to enhance the peak.

About half the events in this background appear to originate from a D which has some species of B as an ancestor. These are not the associated charm hadrons, but are still a legitimate background that need to be included in the $B_c \to J/\psi X$ combinatorial background. The distribution of the templates for the background originating from Bs

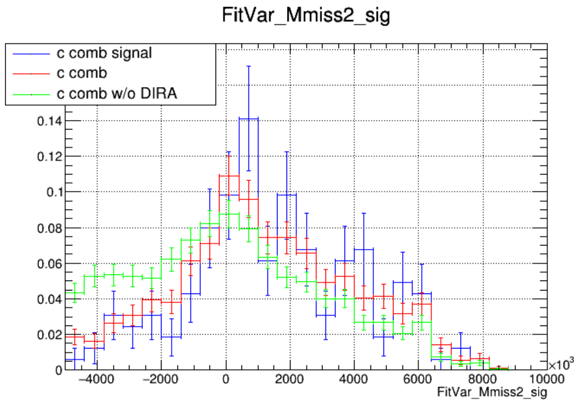


Figure 126: The missing mass variable is depicted, with red showing the nominal combinatorial charm, blue showing the combinatorial charm derived from $B_c \to J/\psi\tau\nu$ MC, and green showing the combinatorial charm distribution without the selection $DIRA < 0.9998$.

does not differ significantly from that of the main combinatorial charm template.

25.3 Impact on the fit results

Adding the combinatorial charm template, which passes the nominal selection, to the fit gives the results shown in Figure 127 with yields given in Table 26. Notably, adding the new template does not significantly affect the fit quality, assessed by the quality of the fit residuals. The signal yield changes by approximately $0.06\sigma_{stat}$. The main changes to the yields were reductions in the $B_{d,s} \to J/\psi X$ combinatorial and the double charm background, which makes sense given the similarities in the template distributions. The MC for this nominal fit is produced with `BcVegPy` and the effect of the kinematic deviations are discussed below.

25.4 *D* kinematic dependence

The dependence of the fit variables on the ratio $|p_D|/|p_{B_c}|$, where p_D and p_B are the momenta of the D and B_c, was studied. Particularly of concern was the region of high p_D/p_{B_c}, which may be favored by other production models. The slice forming the top 30% of the p_D/p_{B_c} distribution was taken to form a template by adding a cut requiring $p_D/p_{B_c} > 0.6$. A fit with this altered template representation changes $R(J/\psi)$ by 50% of the statistical uncertainty relative to the fit with the combinatorial charm template derived using the nominal selection, and by 40% relative to the fit without the combinatorial charm template. The templates corresponding to the slice of events with the top 30% of the $|p_D|/|p_{B_c}|$ distribution is shown next to the nominal combinatorial charm distribution in Figure 128, and the fit results are shown in Figure 129 with yields given in Table 27. Notably, the template remains heavily populated in the short lifetime region, providing significant separation from other background components.

The dependence on the p_T of the bachelor muon is also examined. The p_T distribution was weighted higher or lower by 30% and 20%. This did not appreciably change $R(J/\psi)$ or the template distributions, with shifts of $R(J/\psi)$ less than 5% of the statistical uncertainty for all configurations of p_T reweighting.

Table 26: The raw numbers of events for each component from the fit to data for each year including the combinatorial charm template. Uncertainties are statistical.

Component	2016	2017	2018
$B_c^+ \to J/\psi\, \tau^+\, \nu$	N/A	N/A	N/A
$B_c^+ \to J/\psi\, \mu^+\, \nu$	10350 ± 153	14713 ± 169	17017 ± 188
$B_c^+ \to J/\psi\, D$	190 ± 125	0 ± 124	104 ± 133
$B_c^+ \to J/\psi\, D^{**}$	53 ± 35	0 ± 35	29 ± 37
$B_c^+ \to J/\psi\, D\, K$	19 ± 13	0 ± 13	10 ± 14
$B_c^+ \to \psi(2S)\, \mu^+\, \nu$	0 ± 46	0 ± 57	0 ± 137
$B_c^+ \to \psi(2S)\, \tau^+\, \nu$	0 ± 46	0 ± 57	0 ± 137
$B_c^+ \to \chi_c\, \mu^+\, \nu$	0 ± 84	0 ± 97	0 ± 85
misID bkg.	4657 ± 137	7430 ± 169	8311 ± 184
Comb. $(J/\psi + \mu^+)$ bkg. $(B_{u,d} \to J/\psi X)$	917 ± 153	580 ± 160	927 ± 179
Comb. $(J/\psi + \mu^+)$ bkg. $(B_s \to J/\psi X)$	111 ± 19	70 ± 20	112 ± 23
Comb. c bkg.	437 ± 133	698 ± 158	572 ± 176
Comb. $(\mu^+ + \mu^-)\mu^+$ bkg.	235	255	210

The angular distribution of the D relative to the B_c was also investigated in order to determine the effect of potentially underestimating the correlation in direction of the D and B_c. Following a similar strategy to that in D momentum dependence studies, the top 30% of the events which had the most highly collinear D and B_c flight directions were selected for by applying a cut requiring $\cos(D, B_c) > 0.9996$. This resulted in the rest frame variable distributions shown in Figure 130. The fit variable distributions do not change appreciably in the lifetime, although changes are seen in the missing mass and Z. The value of $R(J/\psi)$ shifts by less than 5% of the statistical uncertainty relative to the fit with the combinatorial charm template from the nominal selection (Figure 131).

A reweighting was performed for the nominal fit using the results described in [**?**], with weights given by

$$e^{-0.171245+1.61134\phi}, \tag{78}$$

where ϕ is the angle between the D and the B_c in the XY plane. The fit variable distributions before and after reweighting are shown in Figure 132. The yield of the charm combinatorial and the measurement of $R(J/\psi)$ is not significantly affected by the angular reweighting.

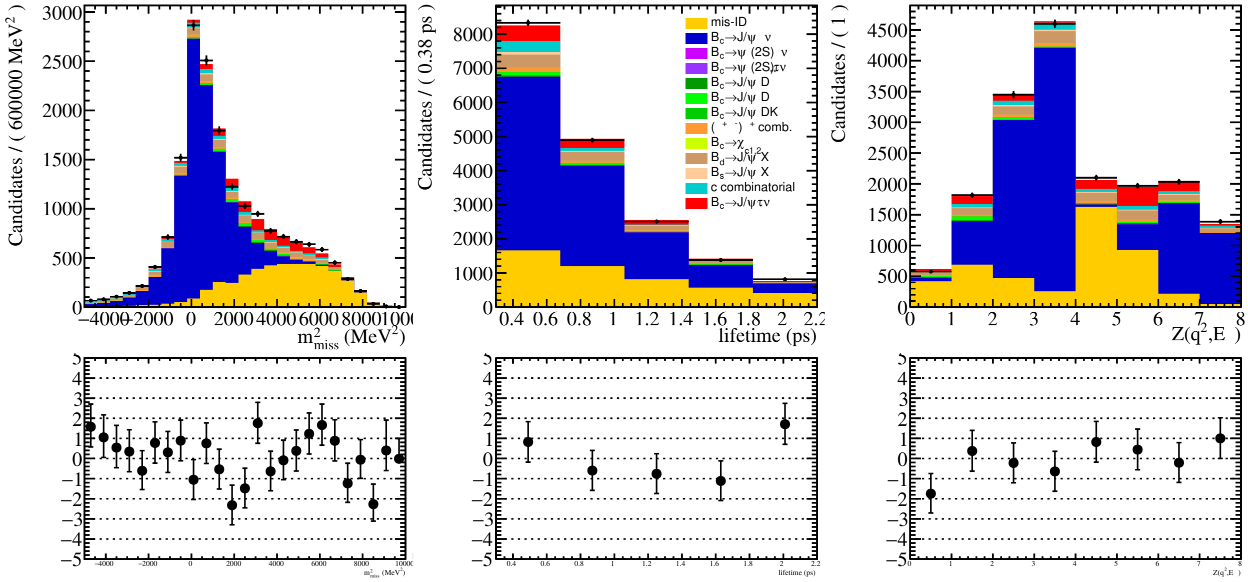
25.5 Summary

This study considered the effect of combinatorial backgrounds originating from charmed hadrons accompanying the B_c meson. A rough estimate of the expected yield of this background, based on the predictions of `BcVegPy`, is derived and found to be about

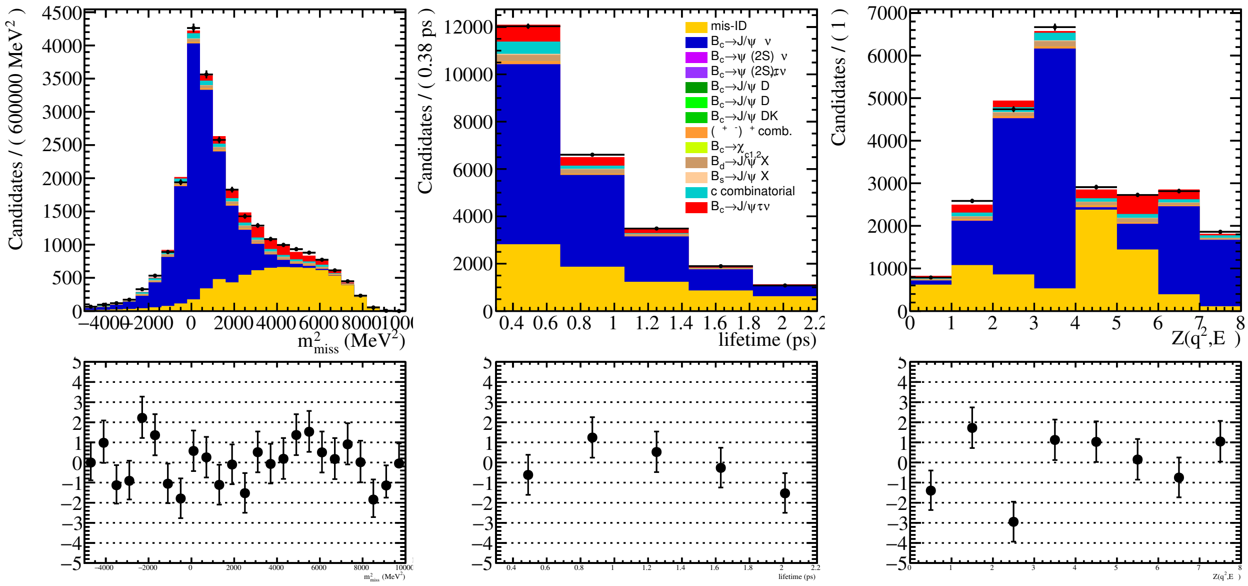
Table 27: The raw numbers of events for each component from the fit to data for each year with the combinatorial charm template at high p_D/p_{B_c}. Uncertainties are statistical.

Component	2016	2017	2018
$B_c^+ \to J/\psi\,\tau^+\,\nu$	N/A	N/A	N/A
$B_c^+ \to J/\psi\,\mu^+\,\nu$	10406 ± 153	14867 ± 169	17106 ± 188
$B_c^+ \to J/\psi\,D$	215 ± 125	53 ± 124	136 ± 133
$B_c^+ \to J/\psi\,D^{**}$	60 ± 35	15 ± 35	38 ± 37
$B_c^+ \to J/\psi\,D\,K$	23 ± 13	5 ± 13	14 ± 14
$B_c^+ \to \psi(2S)\,\mu^+\,\nu$	0 ± 46	0 ± 57	0 ± 137
$B_c^+ \to \psi(2S)\,\tau^+\,\nu$	0 ± 46	0 ± 57	0 ± 137
$B_c^+ \to \chi_c\,\mu^+\,\nu$	0 ± 84	0 ± 97	0 ± 85
misID bkg.	4629 ± 137	7376 ± 169	8242 ± 184
Comb. $(J/\psi + \mu^+)$ bkg. $(B_{u,d} \to J/\psi X)$	1050 ± 153	891 ± 160	1204 ± 179
Comb. $(J/\psi + \mu^+)$ bkg. $(B_s \to J/\psi X)$	127 ± 19	108 ± 20	146 ± 23
Comb. c bkg.	162 ± 30	87 ± 71	102 ± 86
Comb. $(\mu^+ + \mu^-)\mu^+$ bkg.	235	255	210

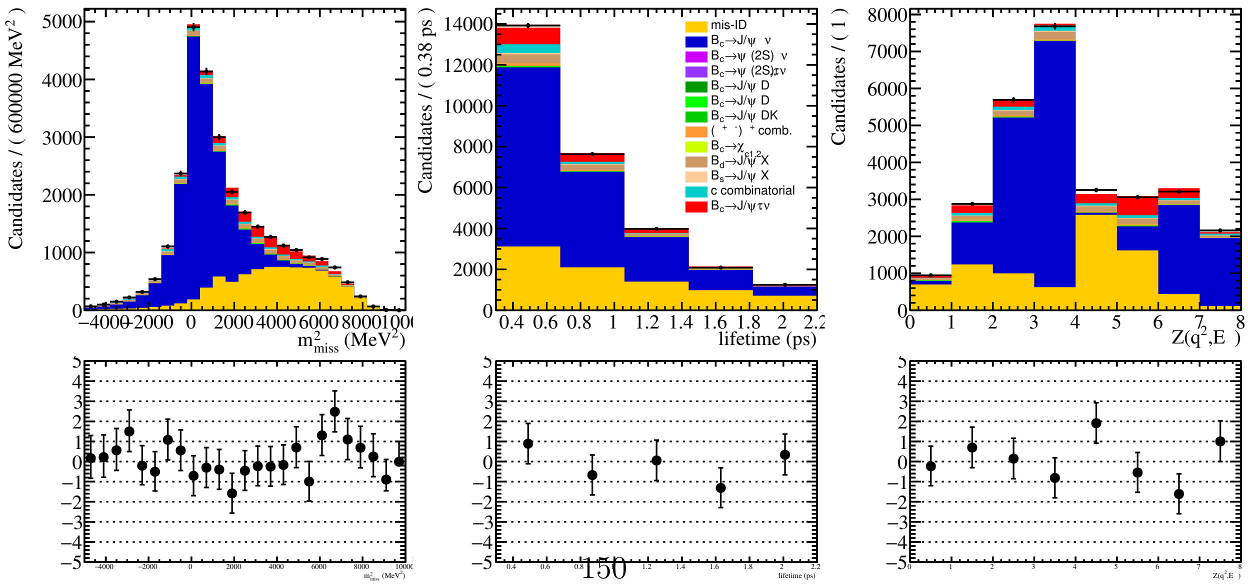
0.8% of the normalization channel. This estimate is used only to guide intuition, as the potential effect of this background on the $R(J/\psi)$ extraction is determined by introducing new templates in the fit which are derived from MC in different kinematic regions. The template distributions show some similarities with the combinatorial from the light Bs, but display a distinct lifetime distribution which is greatly enhanced in the lower lifetime region, providing excellent discrimination from signal and other backgrounds. This feature appears robust across sub-selections enhancing various kinematic correlations. The value of $R(J/\psi)$ changes by about 6% of the statistical uncertainty for the template created with the nominal selection. The largest differences in the fit yields were a decrease in the lighter B combinatorial background and the double charm background. The fit was also done including only candidates with the highest p_D/p_{B_c}, high and low bachelor muon transverse momentum, or small angle between the D and B_c to explore how stronger B_c^+-charm correlations and harder charm momentum spectra would affect the results. The largest observed shift in $R(J/\psi)$ is from the high p_D/p_{B_c} template, which resulted in a shift of 40% of the statistical uncertainty relative to the fit without the combinatorial charm template.



(a) 2016 fit.



(b) 2017 fit.



(c) 2018 fit.

Figure 127: The results of the fit to the full data, projected in the fit variables and including the combinatorial charm template. The lower panels show the residual distributions.

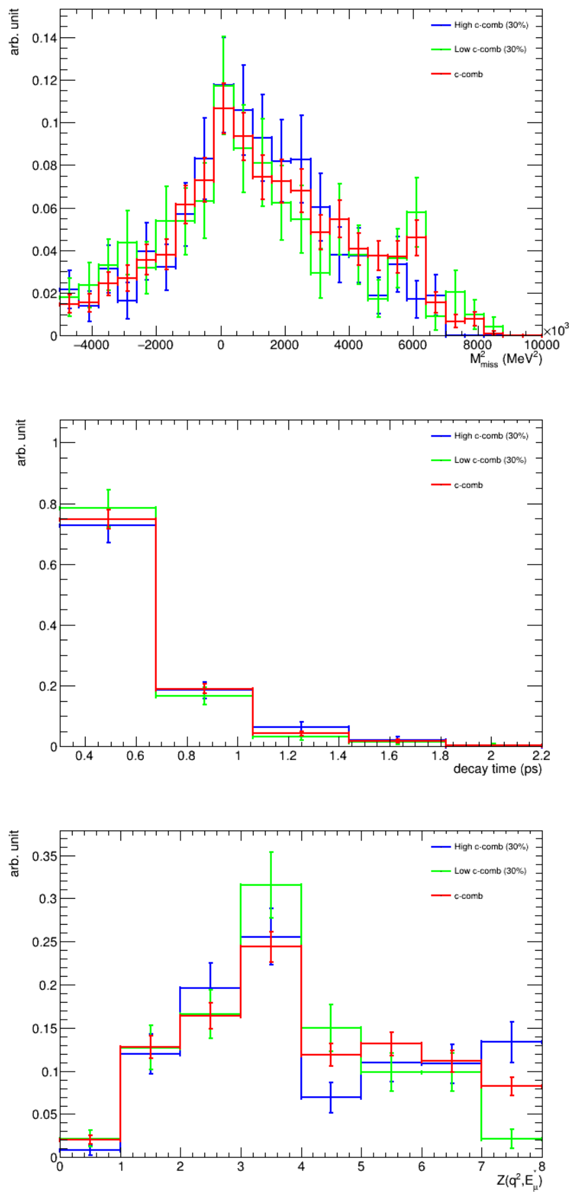
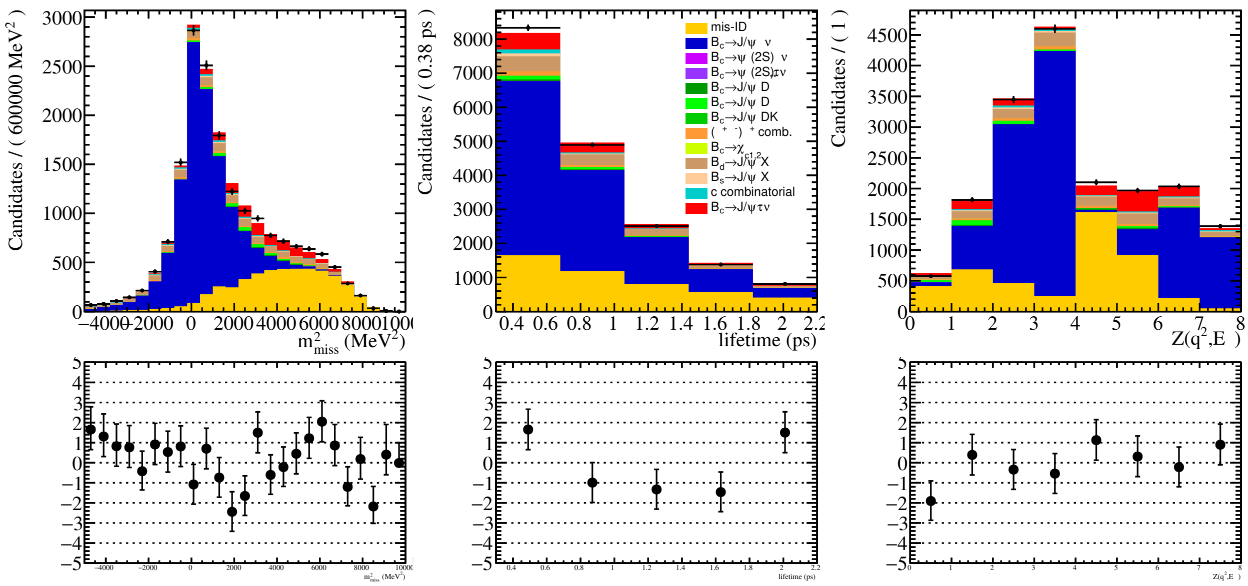
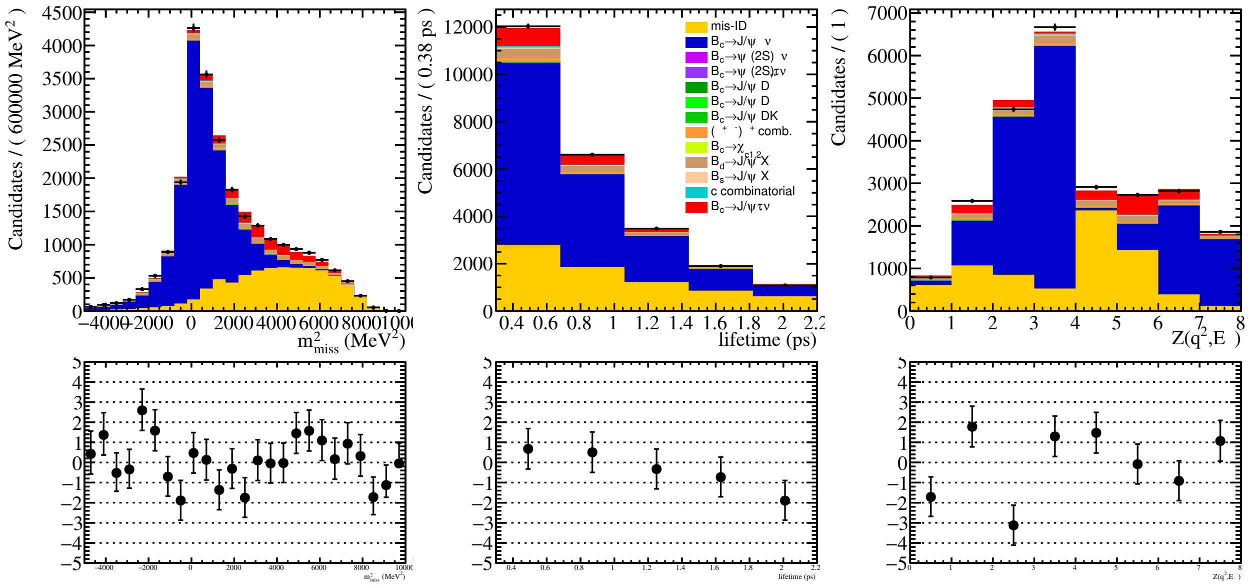


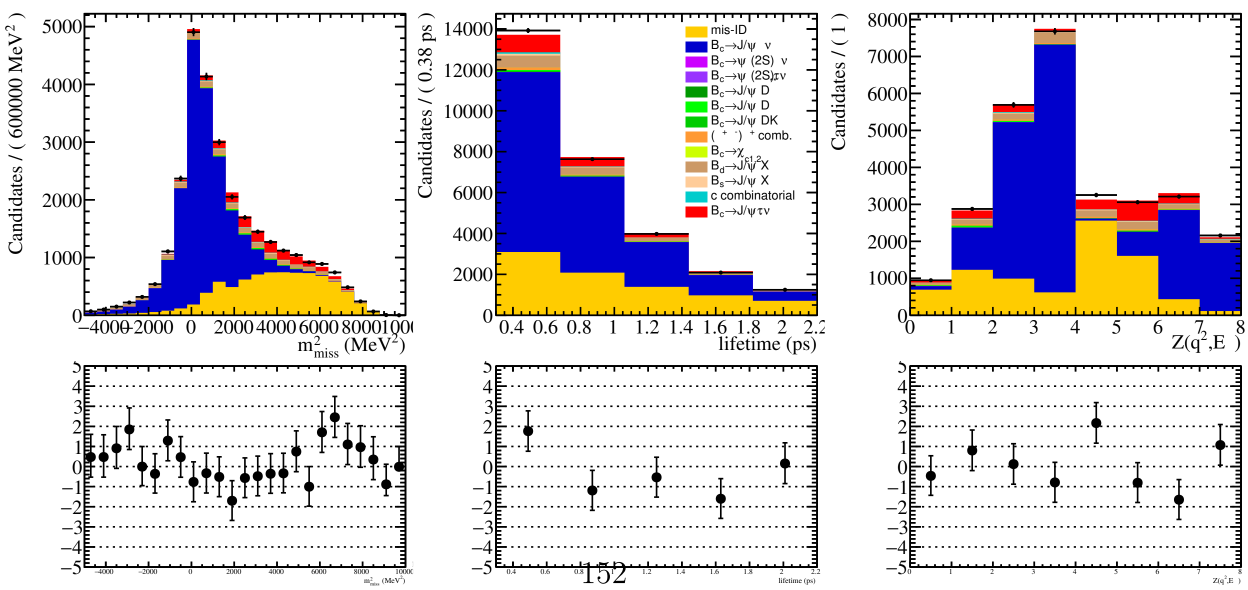
Figure 128: The distributions of the rest frame variables of the default combinatorial charm (red), the top 30% p_D/p_{B_c} (blue), and bottom 30% p_D/p_{B_c} (green).



(a) 2016 fit.



(b) 2017 fit.



(c) 2018 fit.

Figure 129: The results of the fit to the full data, projected in the fit variables and including the combinatorial charm template in teal. The lower panels show the residual distributions.

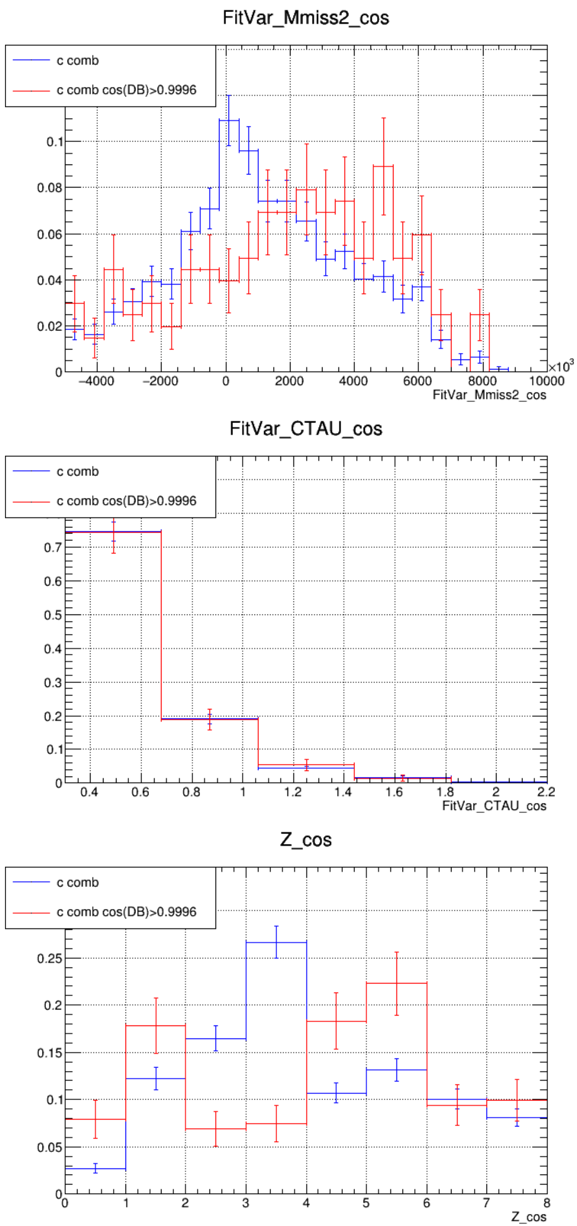
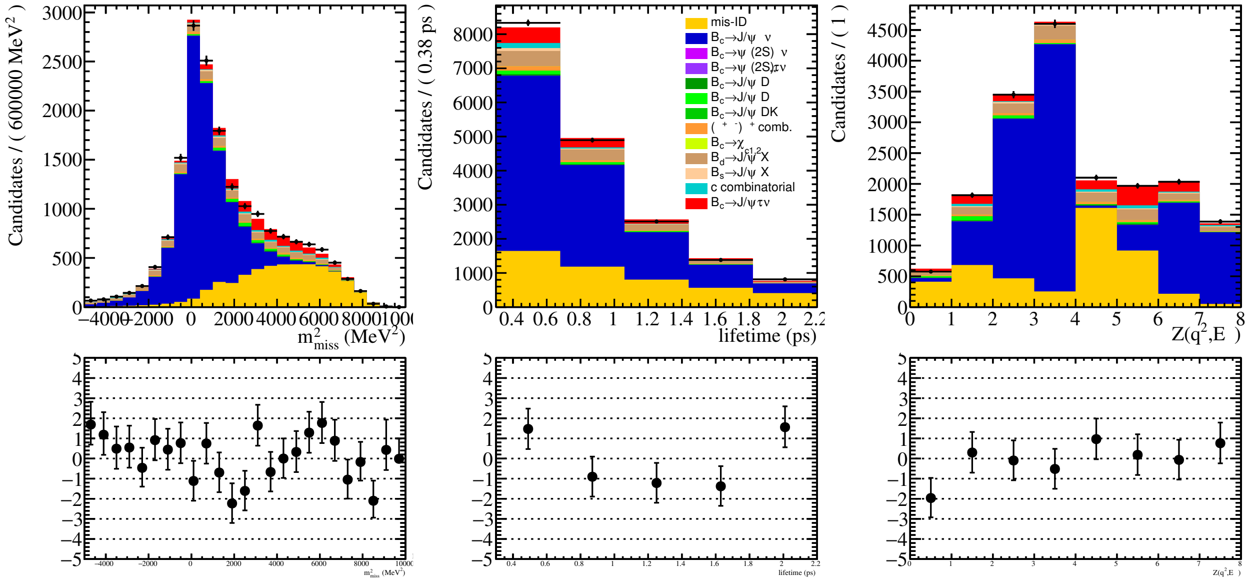
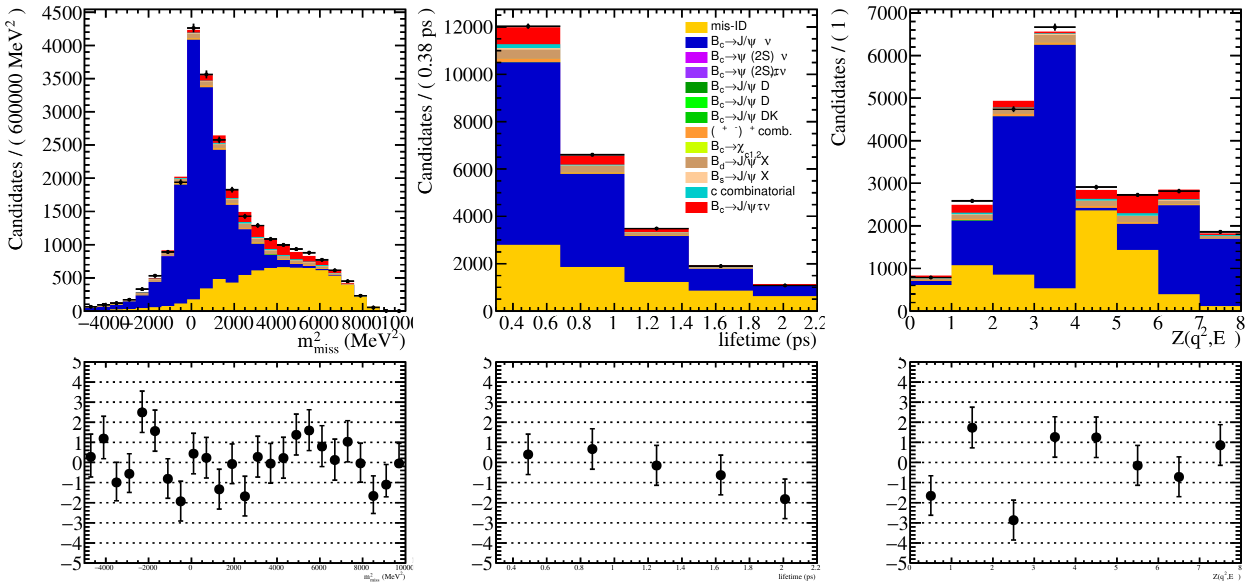


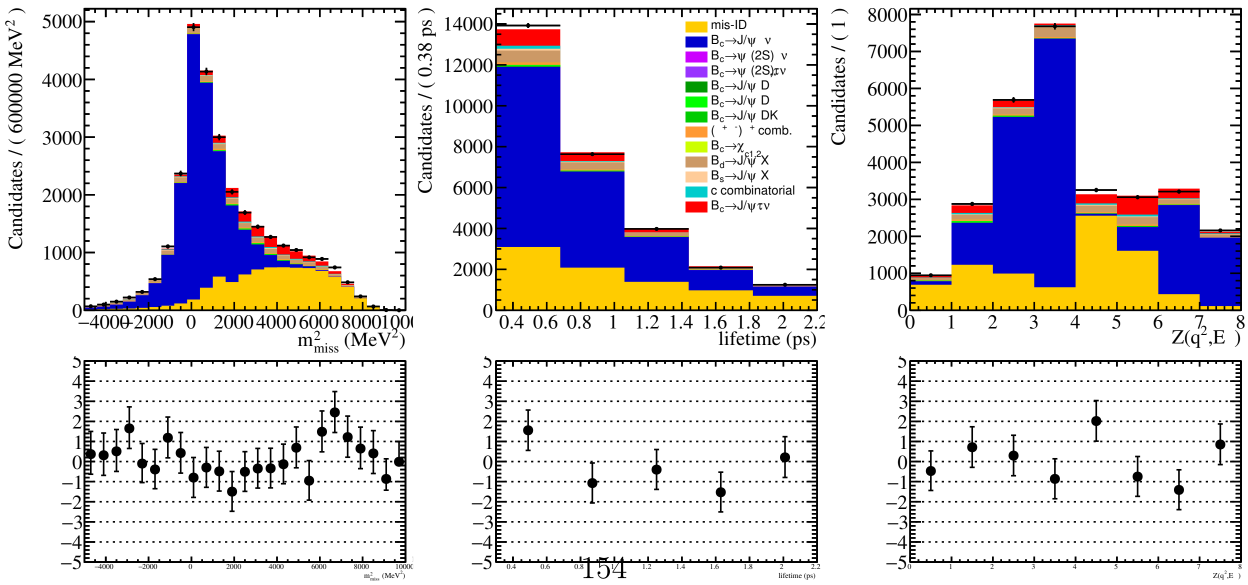
Figure 130: The distributions of the rest frame variables of the combinatorial charm before and after (blue and red) a cut requiring that $\cos\theta > 0.9996$, where θ is the angle between the B_c and the D.



(a) 2016 fit.

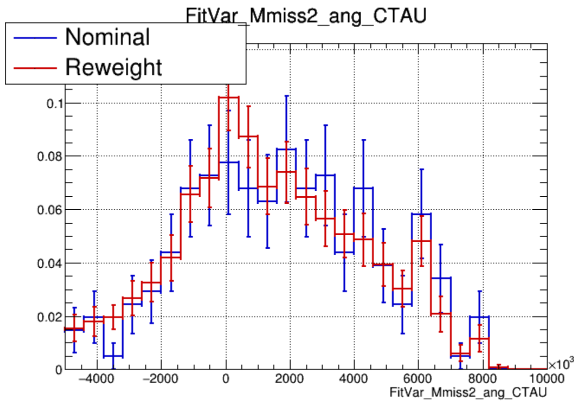


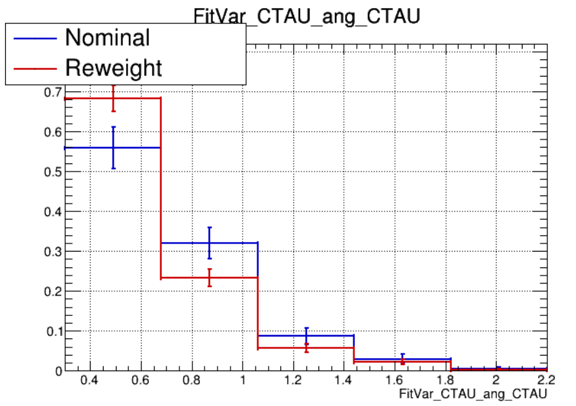
(b) 2017 fit.

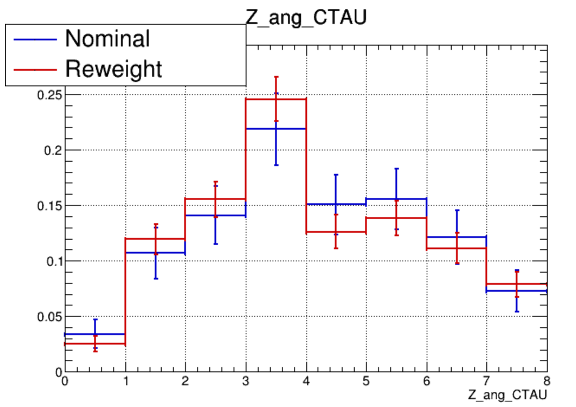


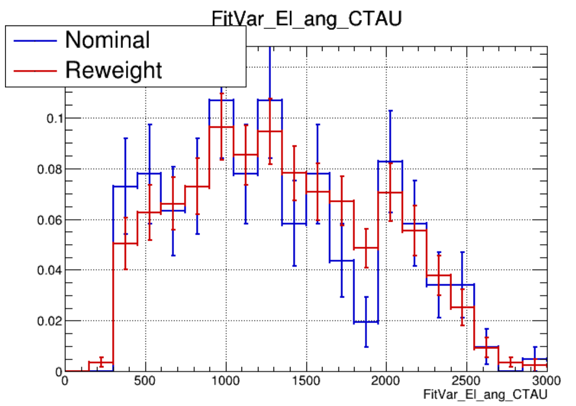
(c) 2018 fit.

Figure 131: The results of the fit to the full data, projected in the fit variables and including the combinatorial charm template in teal, where the combinatorial charm is also required to


FitVar_Mmiss2_ang_CTAU
Nominal
Reweight
FitVar_Mmiss2_ang_CTAU
×10^3


FitVar_CTAU_ang_CTAU
Nominal
Reweight
FitVar_CTAU_ang_CTAU


Z_ang_CTAU
Nominal
Reweight
Z_ang_CTAU


FitVar_El_ang_CTAU
Nominal
Reweight
FitVar_El_ang_CTAU

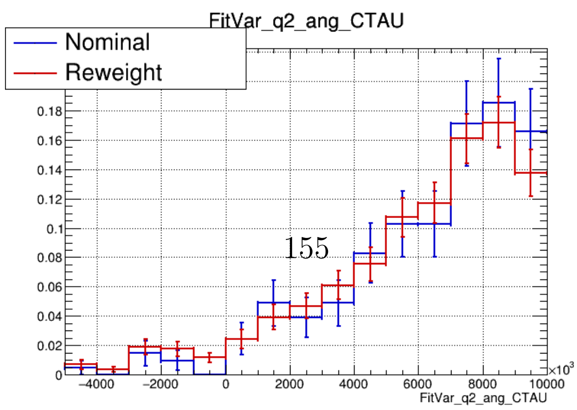

FitVar_q2_ang_CTAU
Nominal
Reweight
FitVar_q2_ang_CTAU
×10^3

Table 28: Ratio of $B \to J/\psi K^*$ sample to $D \to K\pi$ PIDCalib sample mis-ID rates of pions using 2016 data passing the selections applied to derive the lifetime acceptance correction. The different cuts are `IsMuon` & `NShared`=0, `IsMuon` & `NShared`=0 given that the non-bending opening angle is $\cos\theta_{YZ} < 0.99999$, and just `IsMuon`. Uncertainties are negligible except for in the highest momentum bin, which has approximately 50% uncertainty.

Cut	6-9.3 GeV	9.3-10 GeV	10-15.6 GeV	15.6-100 GeV	100-200 GeV
P(`IsMuon` & `NShared`=0)	0.99	1.11	1.58	2.93	5.75
P(`IsMuon` & `NShared`=0 | $\cos\theta_{YZ} < 0.99999$)	0.98	1.10	1.51	2.66	1.84
P(`IsMuon`)	1.59	1.79	2.76	5.01	12.9

Table 29: Ratio of $B \to J/\psi K^*$ sample to $D \to K\pi$ PIDCalib sample mis-ID rates of kaons using 2016 data passing the selections applied to derive the lifetime acceptance correction. The different cuts are `IsMuon` & `NShared`=0, `IsMuon` & `NShared`=0 given that the non-bending opening angle is $\cos\theta_{YZ} < 0.99999$, and just `IsMuon`. Uncertainties are negligible except for in the highest momentum bin, which has approximately 50% uncertainty.

Cut	6-9.3 GeV	9.3-10 GeV	10-15.6 GeV	15.6-100 GeV	100-200 GeV
P(`IsMuon` & `NShared`=0)	0.87	1.07	1.09	1.42	0.65
P(`IsMuon` & `NShared`=0 | $\cos\theta_{YZ} < 0.99999$)	0.85	1.05	1.06	1.34	0.86
P(`IsMuon`)	1.61	1.67	2.07	2.82	2.13

26 Appendix I: Trimuon corrections

At high momentum, the proportion of pion tracks passing `IsMuon` increases in samples with two other muons in the decay. With the additional two muons, tracks are more likely to share hits and pass `IsMuon`. However, this effect exists (but is decreased) even when requiring the tracks to have no shared hits in the muon chamber (see Table 28), suggesting that another effect may be at play. The $B \to J/\psi K^*$ sample used for this correction has the same selections applied to it as the tuple used for the lifetime correction (further explained in Section 28), except with a vertexing requirement added for pions in the reconstruction. Curiously, the kaons do not exhibit this effect using this selection (Table 29). However, when applying a different selection taken from Table 20 in the ANA note of the $B \to \mu^+\mu^-\mu^+\nu_\mu$ analysis (which also studies this effect), the effect appears to be similar for pions and kaons (Tables 30 and 31, with selections taken from [55]).

Additionally, the mis-ID rates were binned simultaneously in momentum and transverse momentum to explore any possible kinematic differences between the two samples (see Tables 32, 33). This binning improves the disagreement but does not eliminate it.

A cut on the opening angle between the track we are measuring the mis-ID rate of and the daughter muons from the J/ψ could reduce the effect of misattributed hits in the

Table 30: Ratio of $B \to J/\psi K^*$ sample to $D \to K\pi$ PIDCalib sample mis-ID rates of pions using 2016 data passing the selections taken from the $B^+ \to \mu^+\mu^-\mu^+\nu_\mu$ analysis (appendix of [55]). The mis-ID rate is for the selection `IsMuon` & `NShared`=0. Uncertainties are negligible except for in the highest momentum bin, which has approximately 50% uncertainty.

Cut	6-9.3 GeV	9.3-10 GeV	10-15.6 GeV	15.6-100 GeV	100-200 GeV
P(`IsMuon` & `NShared`=0)	1.12	1.15	1.77	3.16	1.43

Table 31: Ratio of $B \to J/\psi K^*$ sample to $D \to K\pi$ PIDCalib sample mis-ID rates of kaons using 2016 data passing the selections taken from the $B^+ \to \mu^+\mu^-\mu^+\nu_\mu$ analysis (appendix of [55]). The mis-ID rate is for the selection `IsMuon` & `NShared`=0. Uncertainties are negligible except for in the highest momentum bin, which has approximately 50% uncertainty.

Cut	6-9.3 GeV	9.3-10 GeV	10-15.6 GeV	15.6-100 GeV	100-200 GeV
P(`IsMuon` & `NShared`=0)	0.92	0.93	1.22	1.70	2.22

p lower bound (MeV) \ p_T upper bound (MeV)	0	1000	1500
0	0.61	0.61	3.9
6000	0.92	1.11	1.39
15600	1.83	2.18	2.16

Table 32: The ratio of pion misID efficiencies between PIDCalib samples and $B \to J/\psi K^*$ sample, binned in 2D. Very high numbers are a result of low bin statistics in high momenta.

p lower bound (MeV) \ p_T upper bound (MeV)	0	1000	1500
0	0.73	0.68	0
6000	0.85	1.02	0.82
15600	1.18	1.31	1.44

Table 33: The ratio of kaon misID efficiencies between PIDCalib samples and $B \to J/\psi K^*$ sample, binned in 2D. Very high numbers are a result of low bin statistics in high momenta.

muon chamber. However, this cut may not account for tracks which are bent away from one another. Cutting on the opening angle in the non-bending plane, the y-z plane, may be a solution for this. This does drastically decrease the difference in ratios by more than a factor of two, however, it does not rid us of the effect completely.

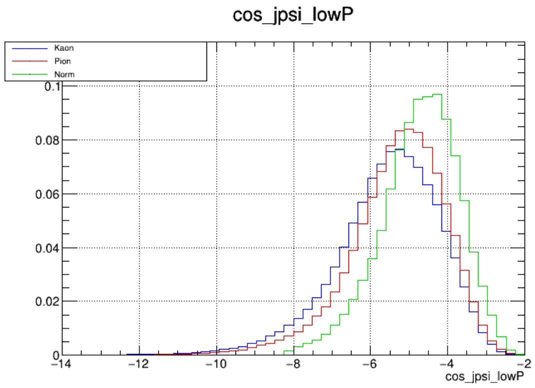
We also investigated the possibility of using `IsMuonTight` instead, but found this did not improve the agreement by much. Requiring `NShared==0` in the nominal fit improves the agreement at low momentum in the bachelor muon in the postfit comparisons.

The efficiency of `IsMuon` is measured using $B \to J/\psi K^*$ for the pion and kaon samples. The proton is treated with a separate sample because it does not decay in-flight. We used $\Lambda_b^0 \to J/\psi p^+ K^-$. The selections for this process are taken from the preselections in [56], with muons required to have $p_T > 550$ MeV/c and hadrons having $p_T > 250$ MeV/c. The J/ψ candidate must have a vertex χ^2 of less than 9. The sum of the kaon and proton $p_T >$ should be less than 1 GeV/c. To remove the $\overline{B}^0 \to J/\psi K^- \pi^+$ mis-ID background, we veto the mass region $m(J/\psi K^- \pi^+) \in [5250, 5310]$ MeV/c^2 where the proton is reconstructed as a pion. A cut is also added on the DIRA of the Λ_b^0. The electron and pion mis-ID efficiencies do not require a trimuon correction to their misID efficiencies, since they are estimated using the inclusive J/ψ MC which has two extra muons in the decay.

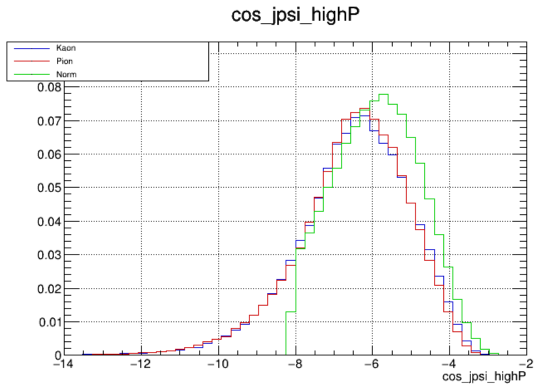
The correction was implemented by modifiying the efficiencies, with the efficiency from IsMuon and NShared taken from the relevant non-PIDCalib sample. The fit quality was worsened after implementing this correction. A few variations of this correction were implemented, with the correction combined or separated for the three years, with separate or combined smearing for all years, with the electron sample combined or separated for all years, or with separate corrections for low and high E_ℓ^*, but none of these configurations yielded a fit that had similar quality to the uncorrected one.

Further investigation found that the distribution of the angle between the same sign muon daughter of the J/ψ and the bachelor muon, in the case of the normalization sample, or the pion/kaon, in the case of the $B \to J/\psi K^*$ sample, did not align (see Figure 133). Additional investigations using the normalization-rich region of data confirmed that this effect was not due to data/MC differences. The kaon and pion from the $J/\psi K^*$ sample being distributed more colinearly with the same-sign muon at high momentum may result in an artificially inflated `IsMuon` efficiency. Both the $J/\psi K^*$ samples for the lifetime acceptance correction and for the mis-ID efficiency correction were reweighted to match the angular distribution of the normalization MC. The fit was performed with these reweighted corrections, which did not provide an improvement over the original corrections.

Due to the issues deriving the efficiency correction from data samples, a functional correction was attempted for pions and kaons. This correction assumes a linear increase in efficiency as a function of the track momentum, derived from an average of the values in Tables 29 and 28. The proton correction remains derived from data. The fit was allowed to float between an uncorrected and a nominally corrected template, and settled on nuisance parameters of $-0.67, -0.46, 0.30$ for 2016, 2017, and 2018 respectively, with -1 being uncorrected and 1 being the nominal correction. This did not improve the fit, and so was ultimately discarded.



(a) Low momentum of the track in question (<1.56 GeV).



(b) High momentum of the track in question (>1.56 GeV).

Figure 133: $\log(1 - \cos\theta)$, where θ is the angle between same-sign muon and track with the bachelor muon from normalization and the pion/kaon from $B \to J/\psi K^*$. The $J/\psi K^*(\to X)$ tracks are distributed more colinearly at higher momentum.

27 Appendix J: Decay in-flight efficiency corrections

In this appendix, the plots from the D^0 mass and $\delta m = m_{D^*} - m_{D^0}$ fits in each kinematic bin are shown.

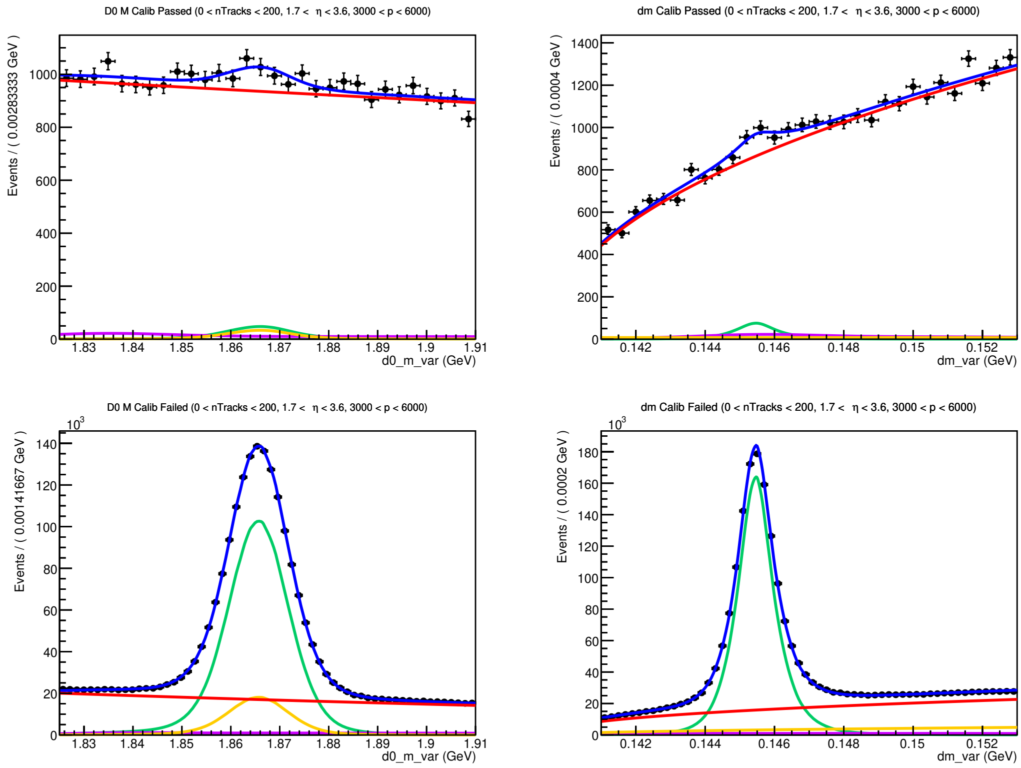


Figure 134: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

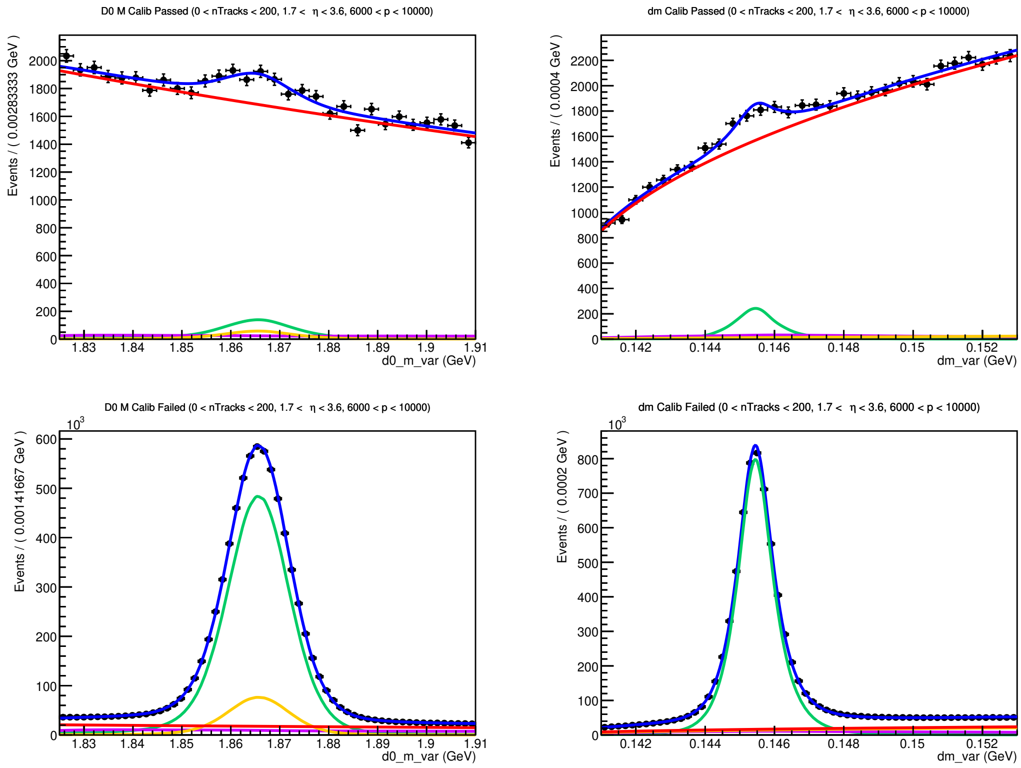


Figure 135: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

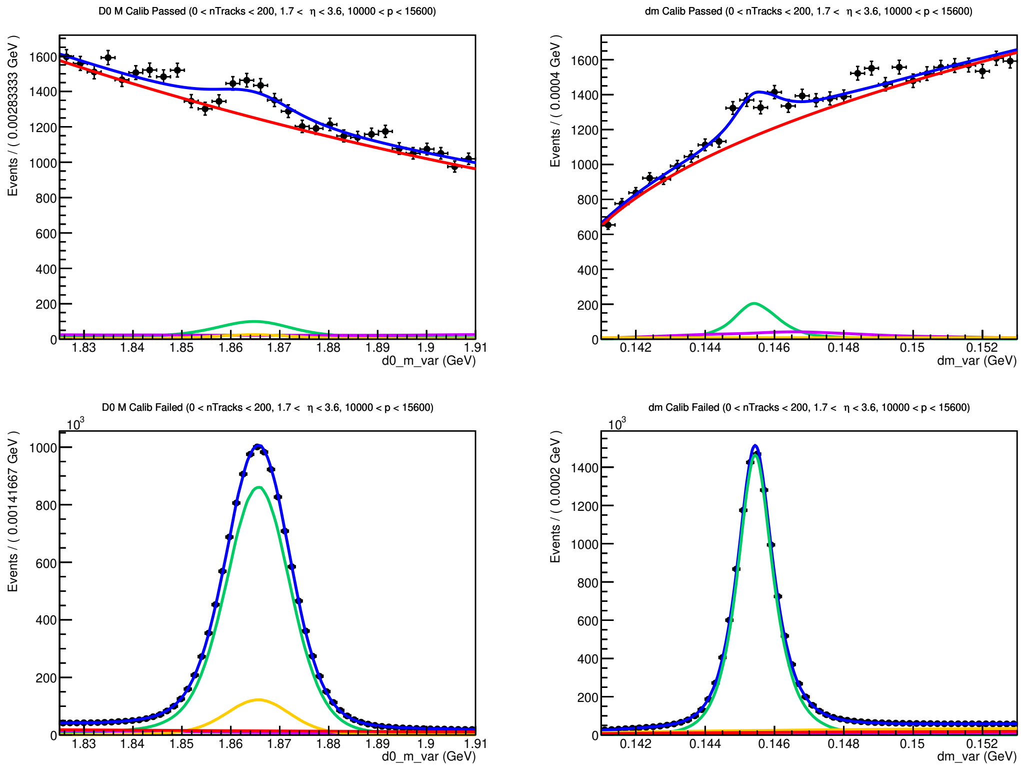


Figure 136: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

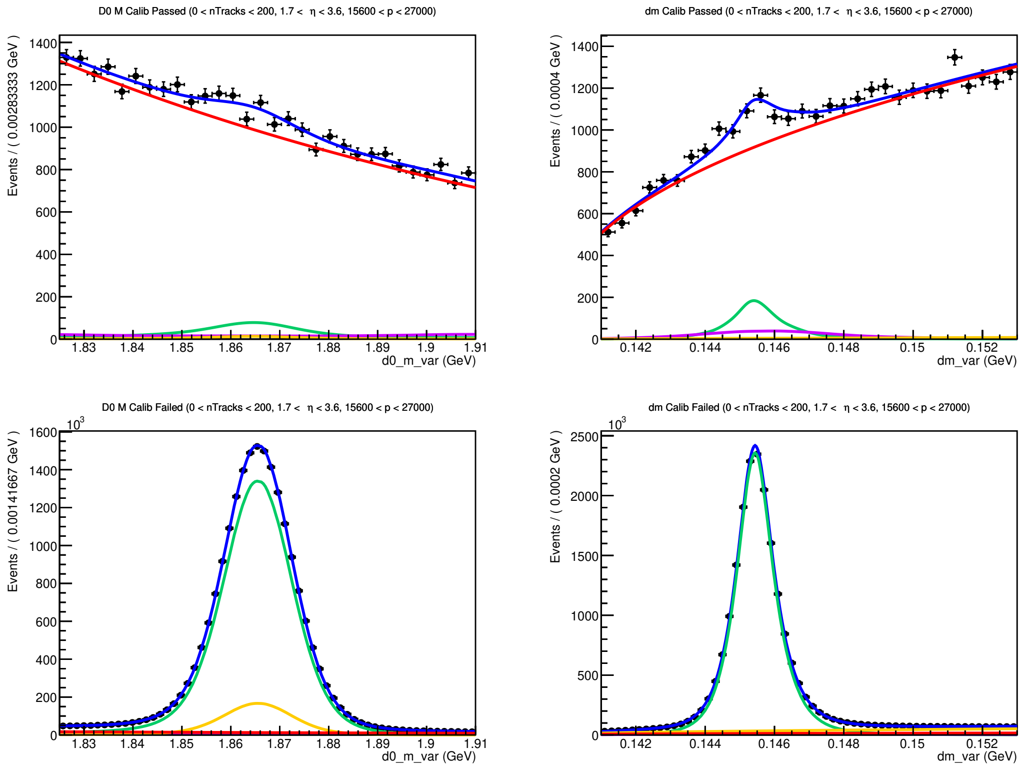


Figure 137: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

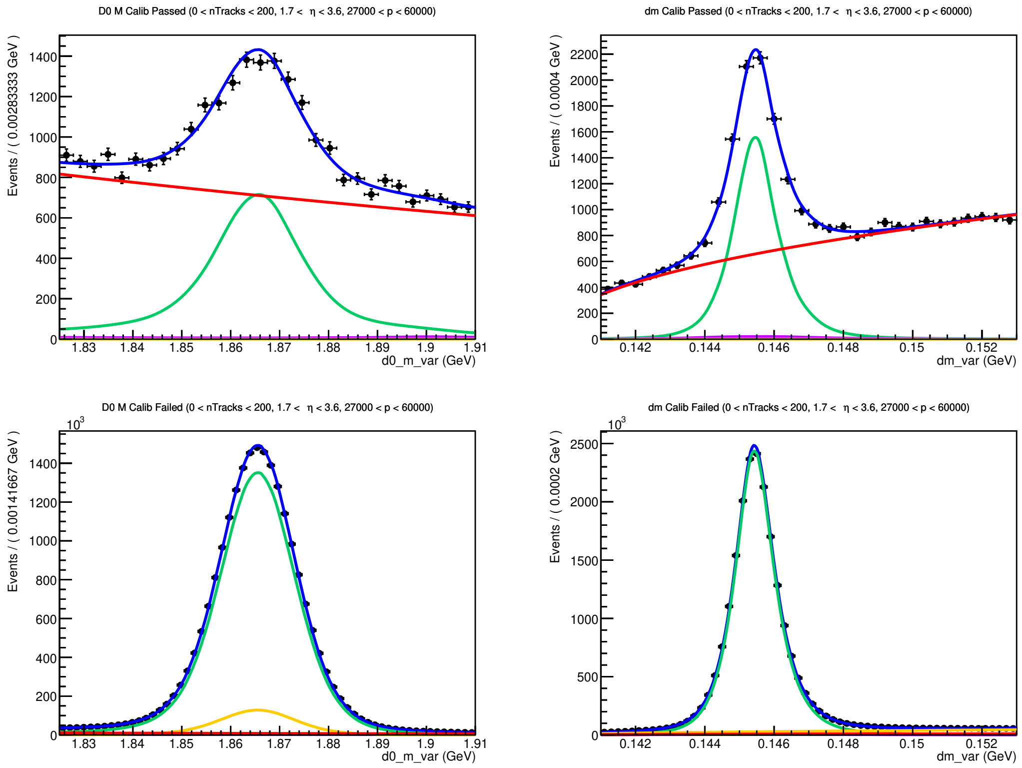


Figure 138: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $27000\,\text{MeV} < p < 60000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

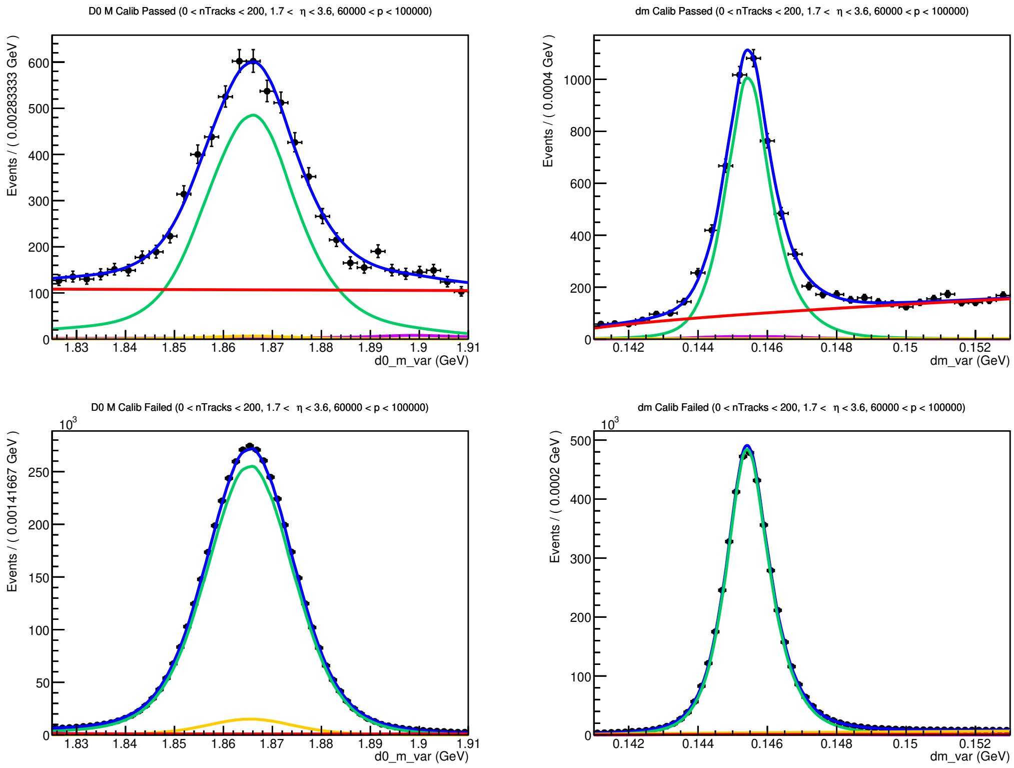


Figure 139: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

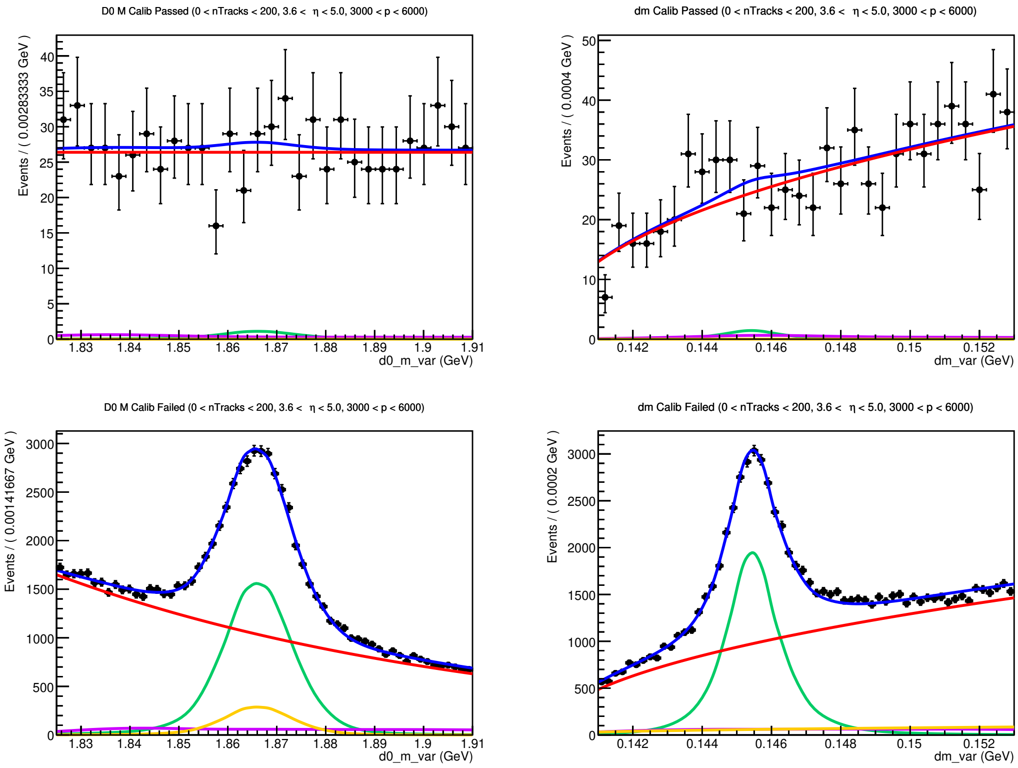


Figure 140: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

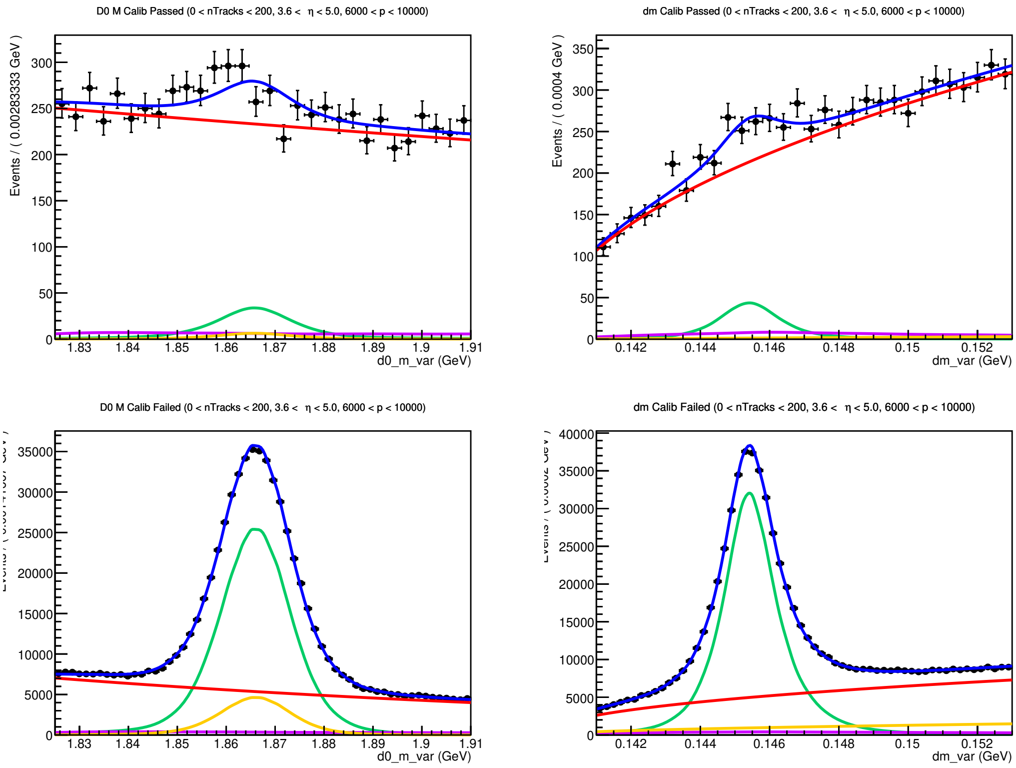


Figure 141: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

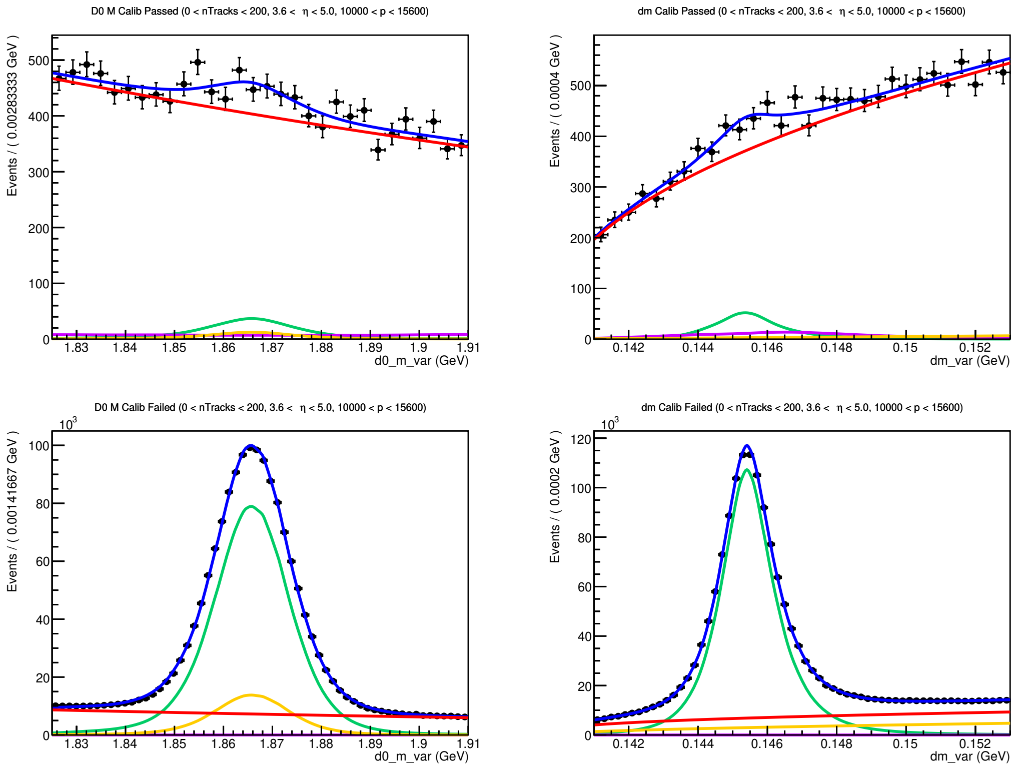


Figure 142: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

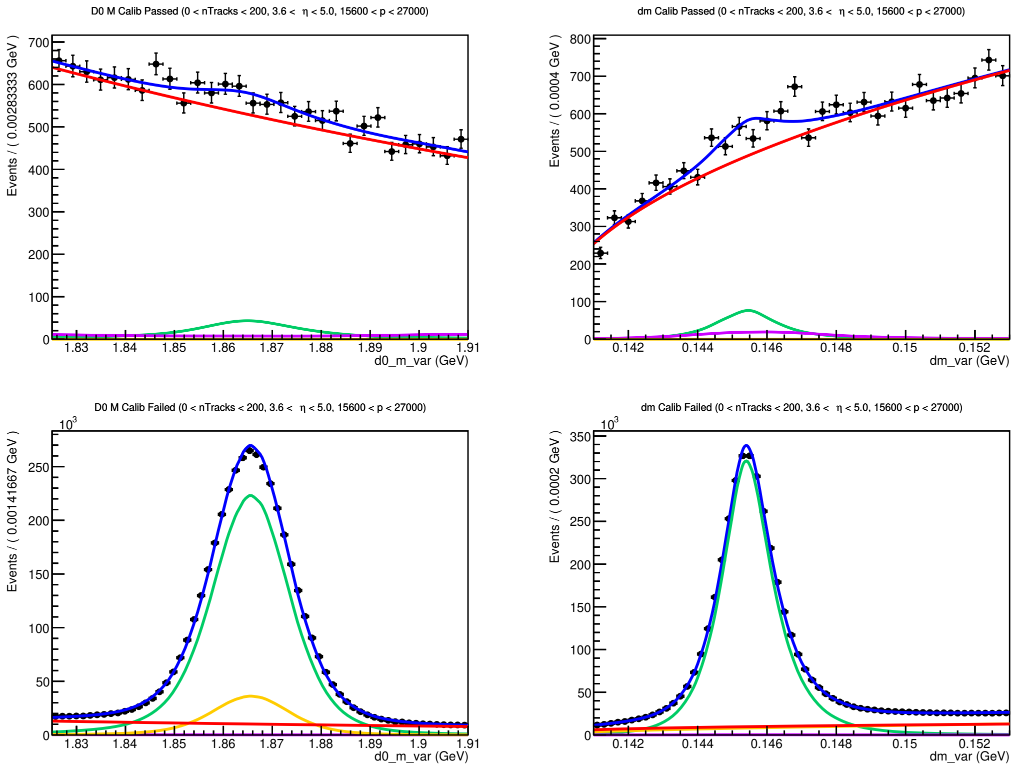


Figure 143: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

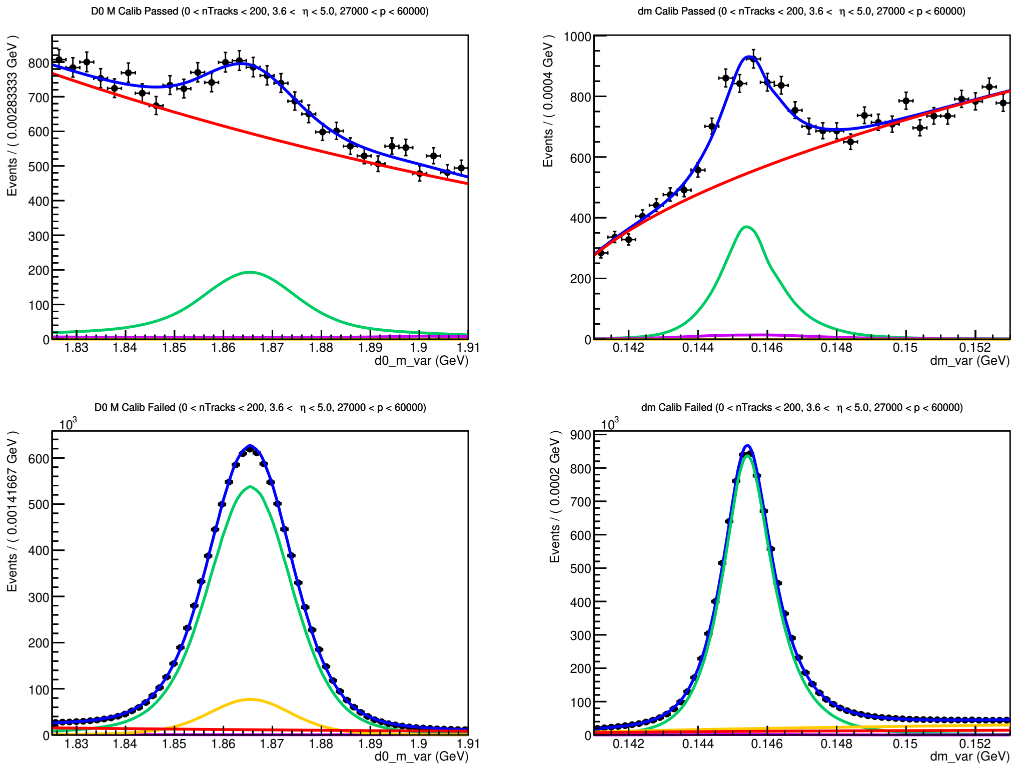


Figure 144: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $27000\,\mathrm{MeV} < p < 60000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

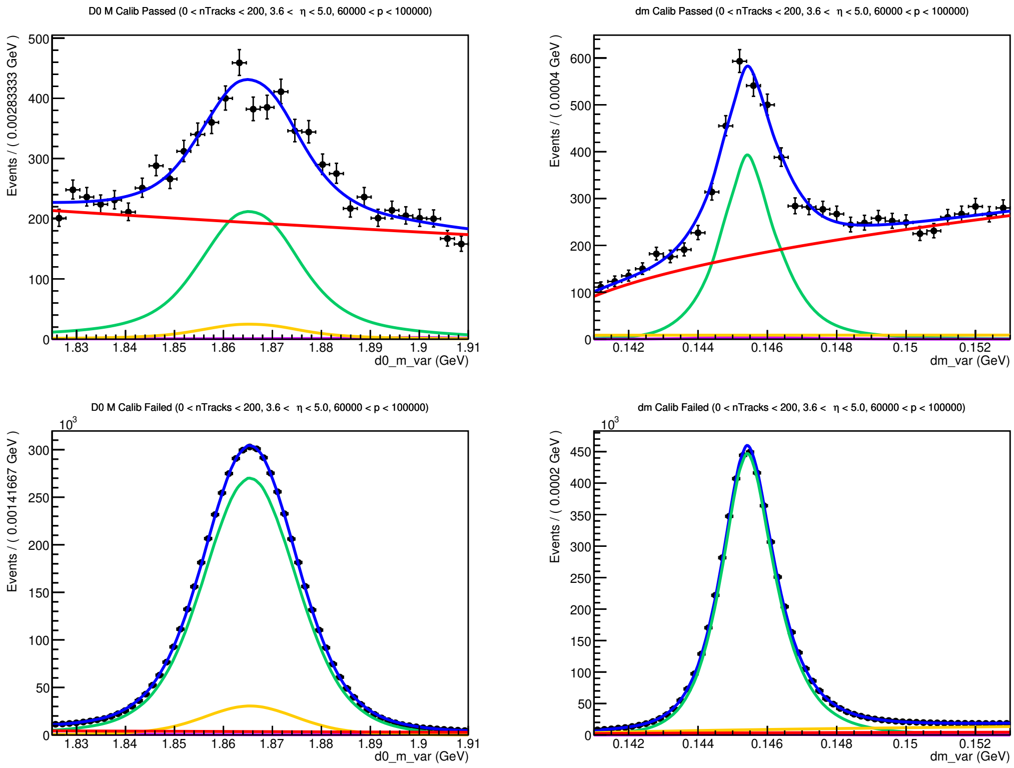


Figure 145: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

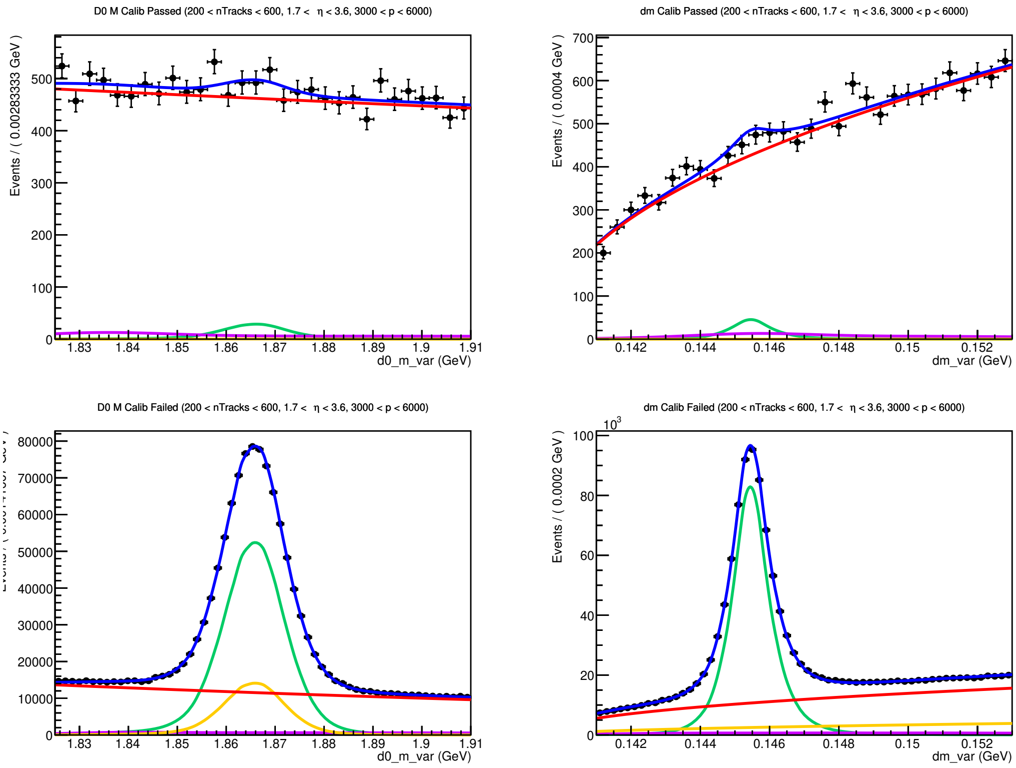


Figure 146: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

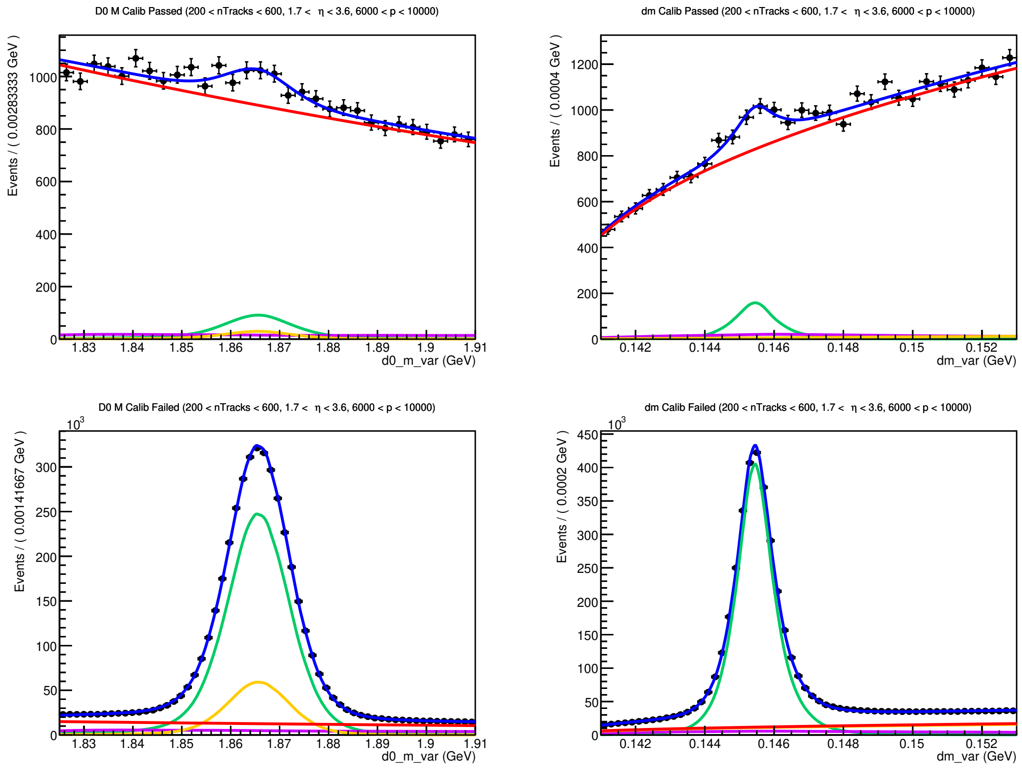


Figure 147: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

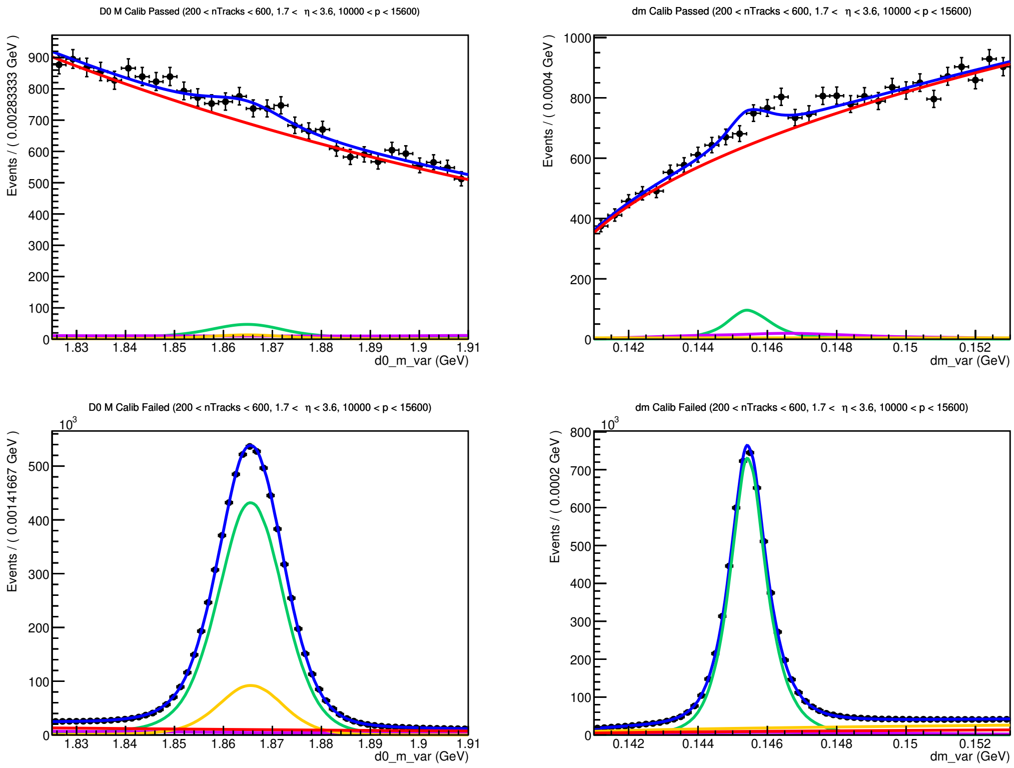


Figure 148: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

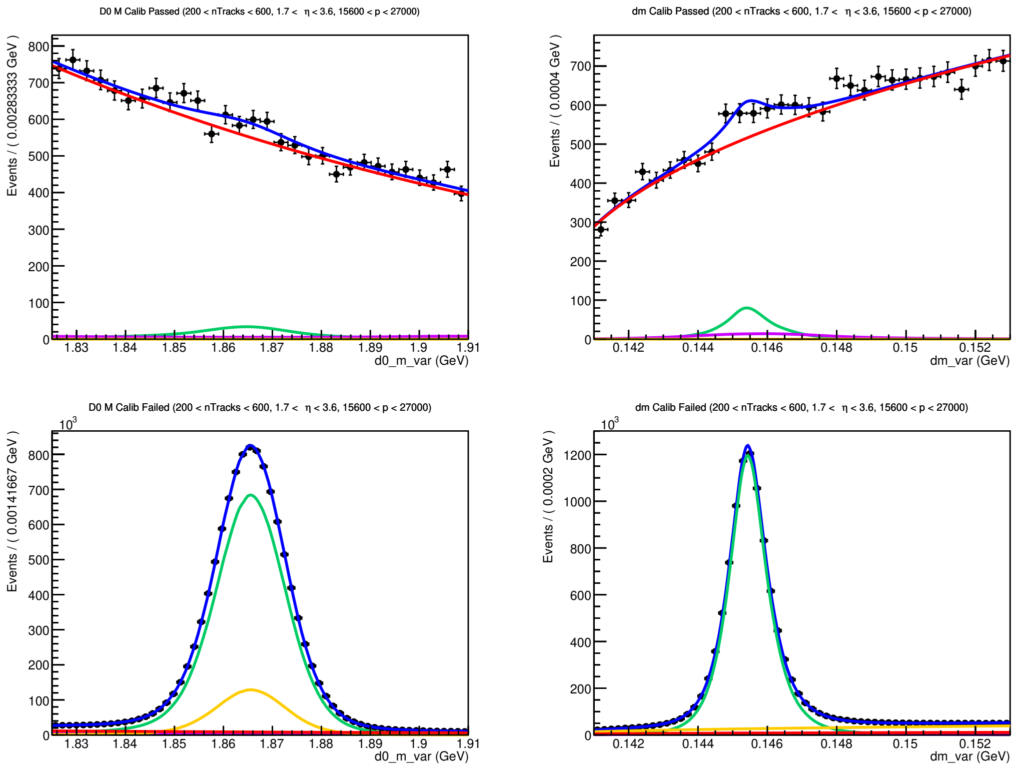


Figure 149: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

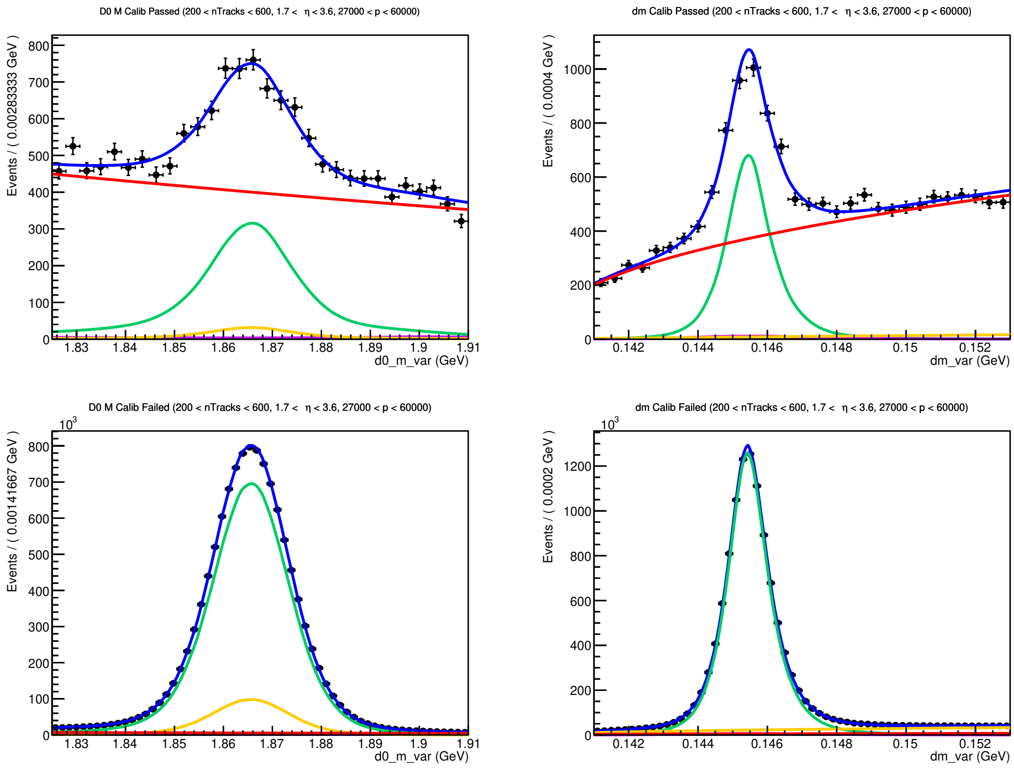


Figure 150: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and 27000 MeV $< p <$ 60000 MeV for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

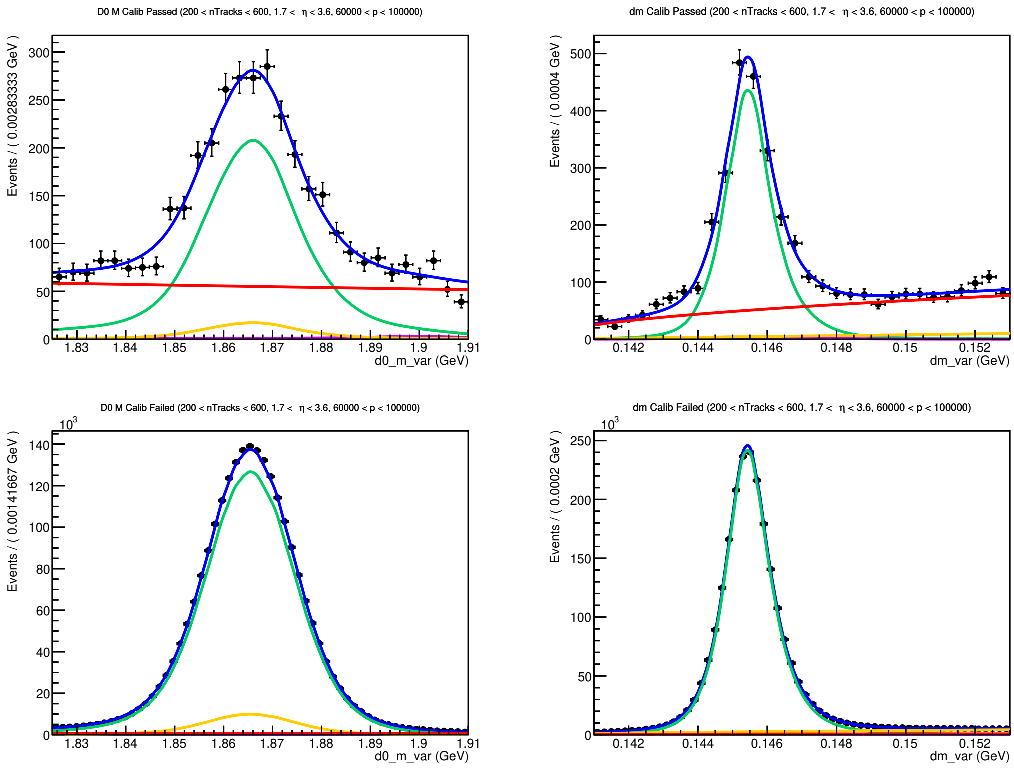


Figure 151: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

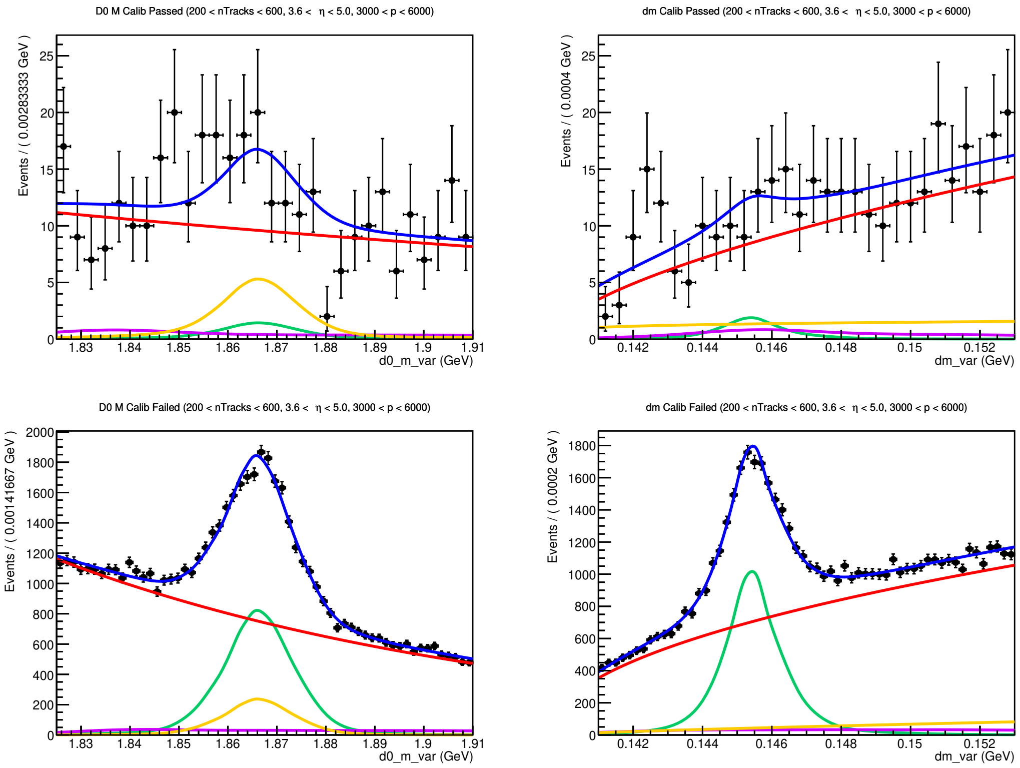


Figure 152: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $3000\,\text{MeV} < p < 6000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

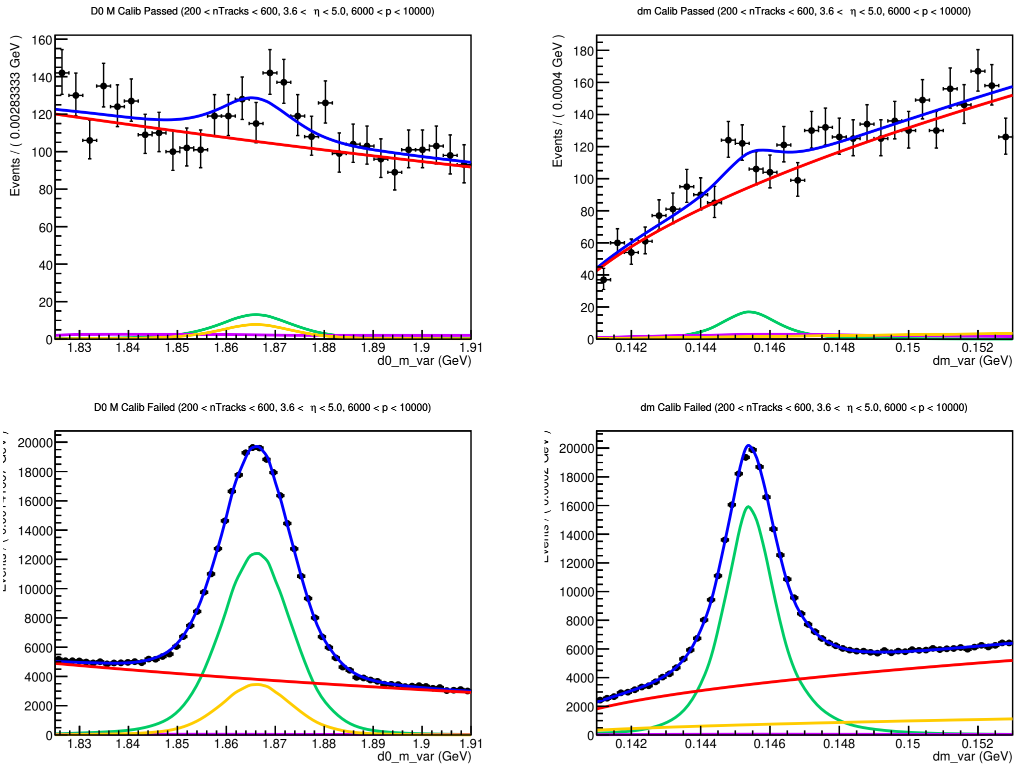


Figure 153: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

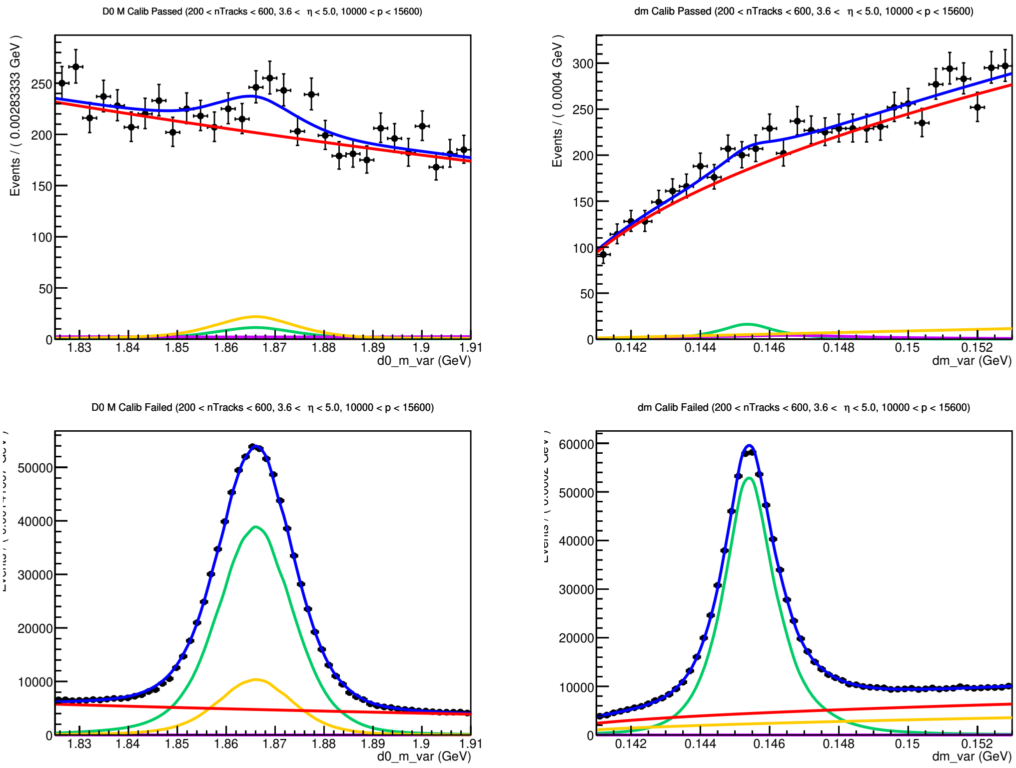


Figure 154: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

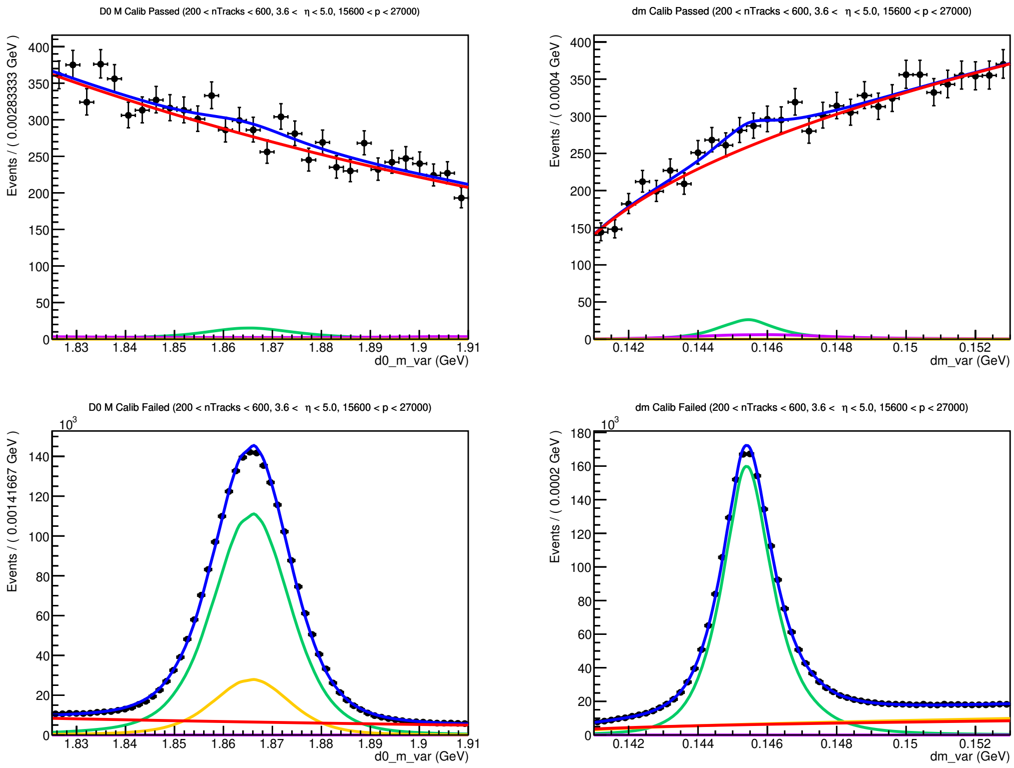


Figure 155: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

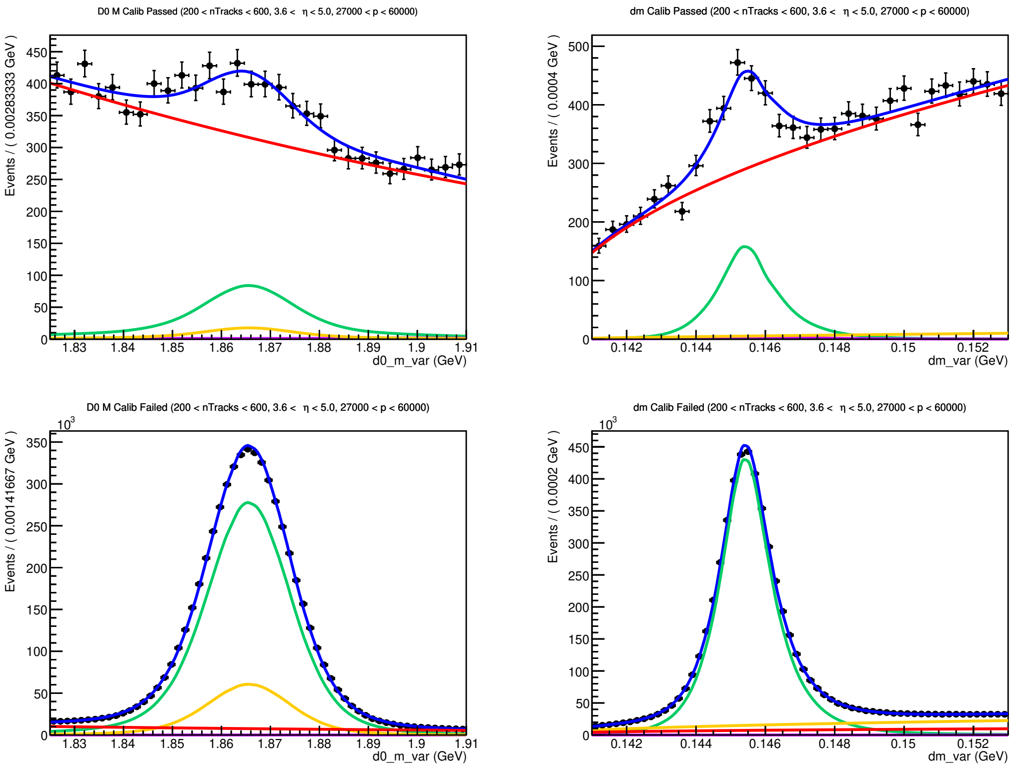


Figure 156: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $27000\,\text{MeV} < p < 60000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

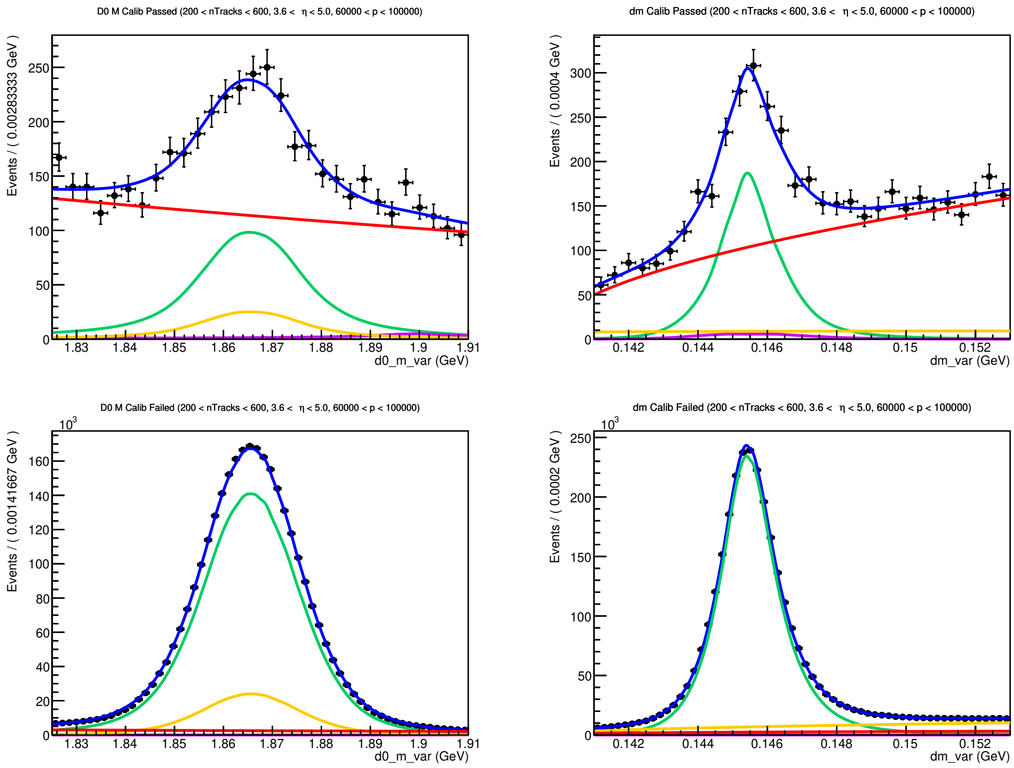


Figure 157: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for kaons in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

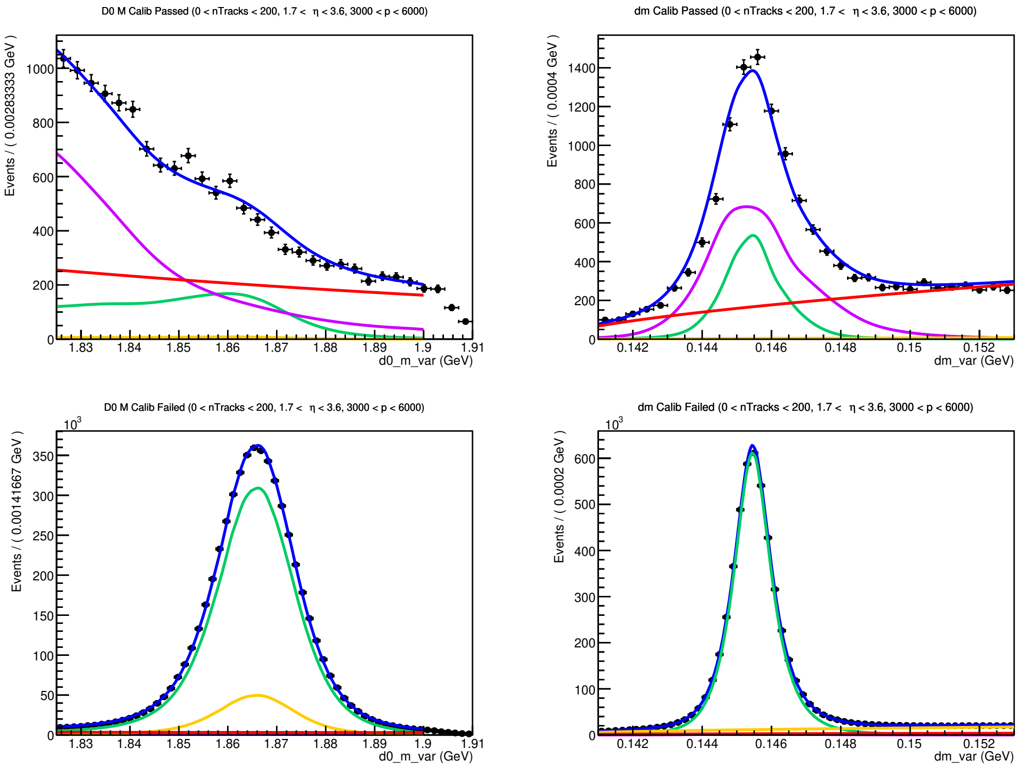


Figure 158: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

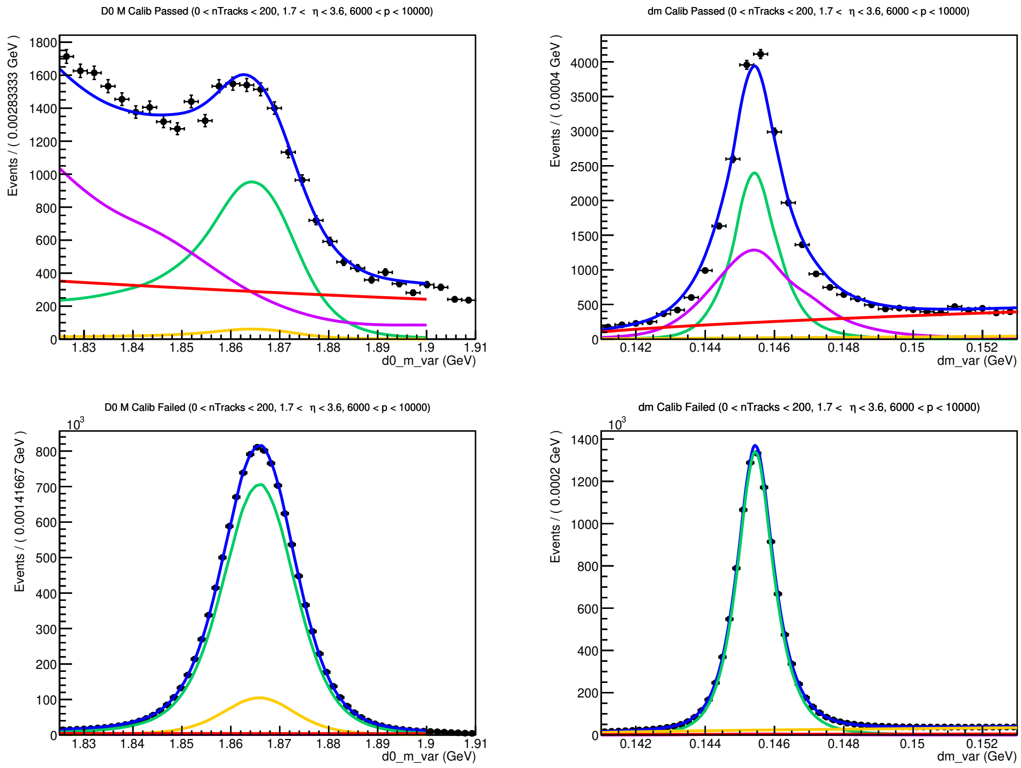


Figure 159: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

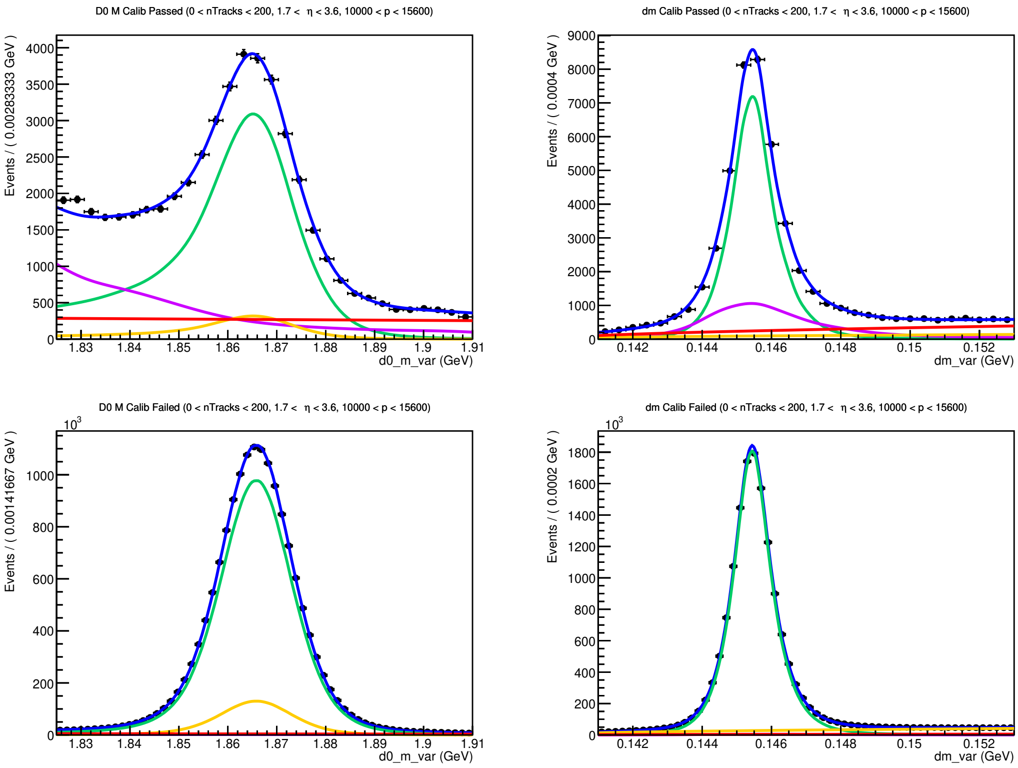


Figure 160: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

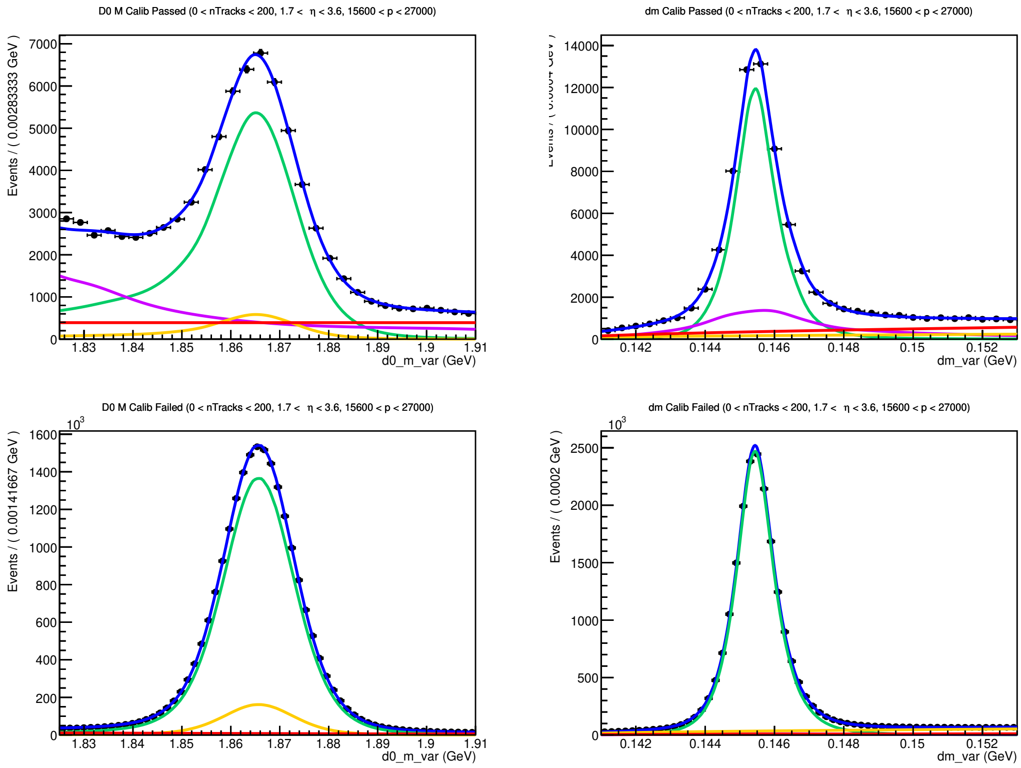


Figure 161: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

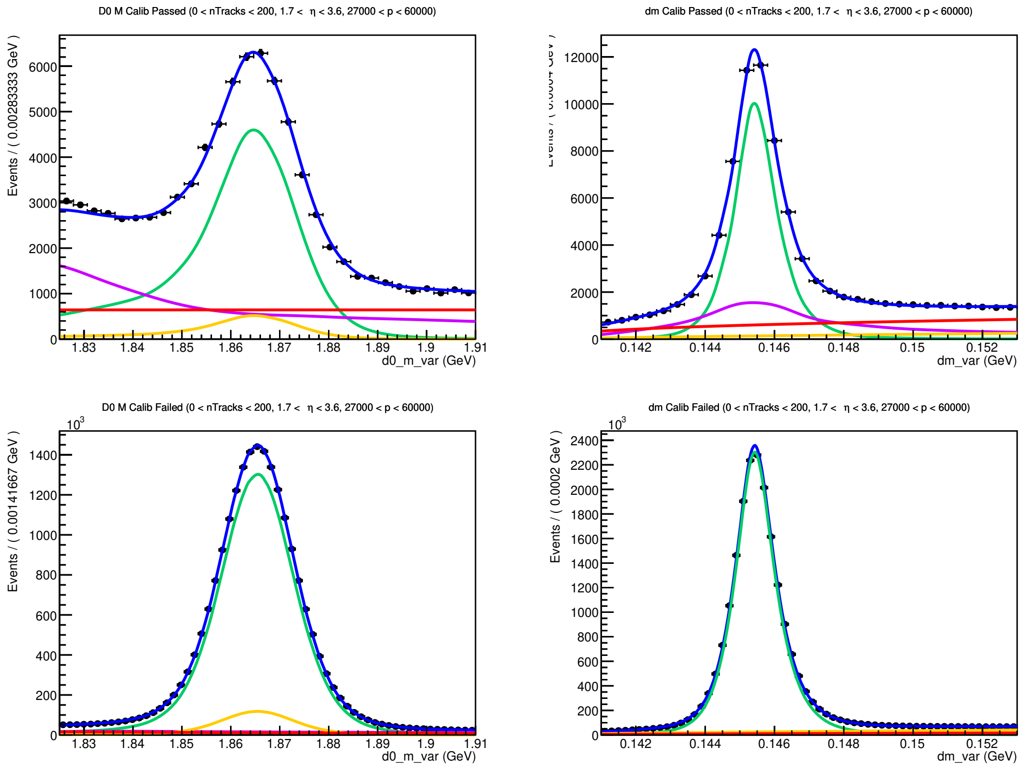


Figure 162: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $27000\,\mathrm{MeV} < p < 60000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

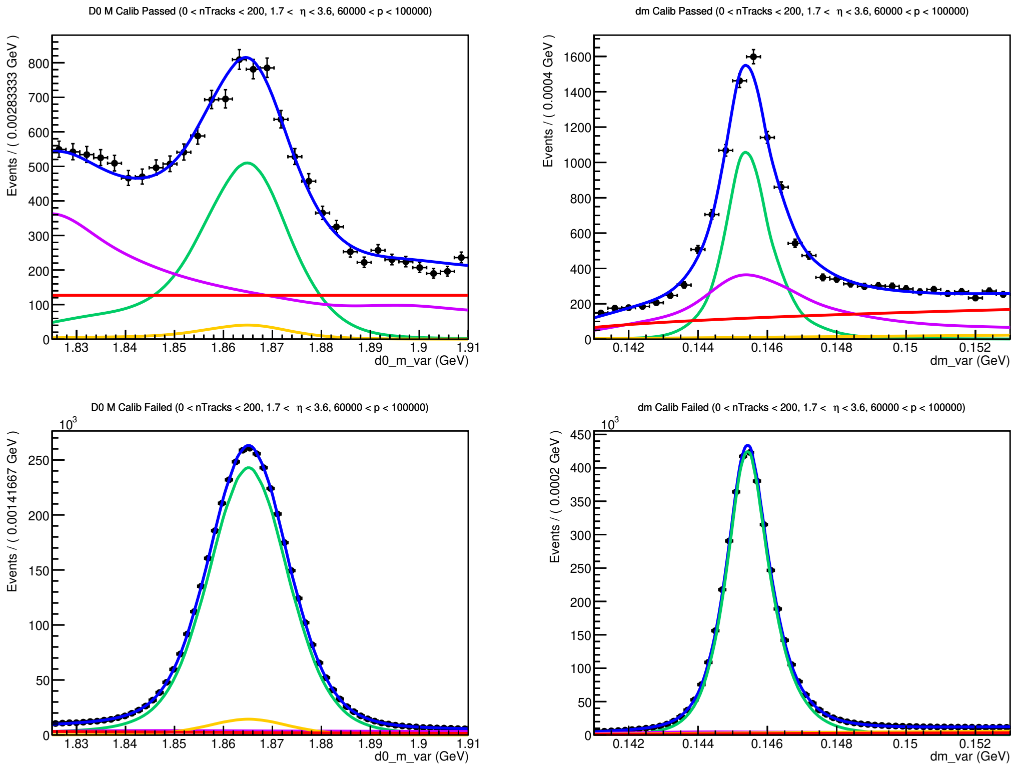


Figure 163: Fit to D^0 mass and δm in nTracks<200, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

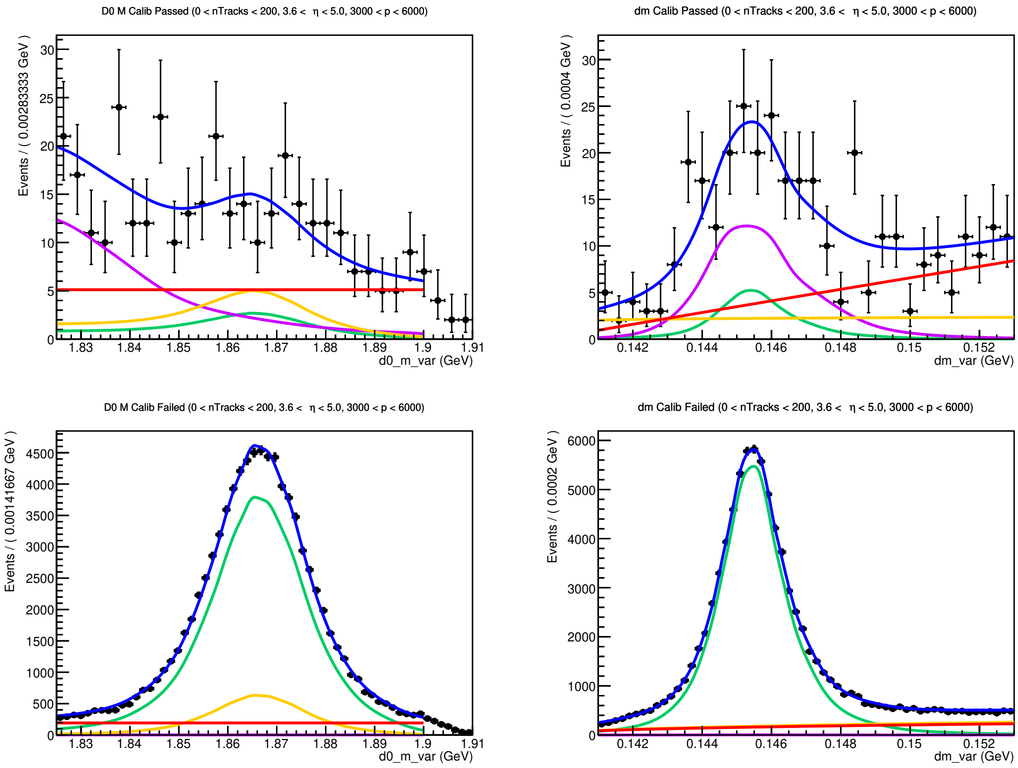


Figure 164: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $3000\,\text{MeV} < p < 6000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

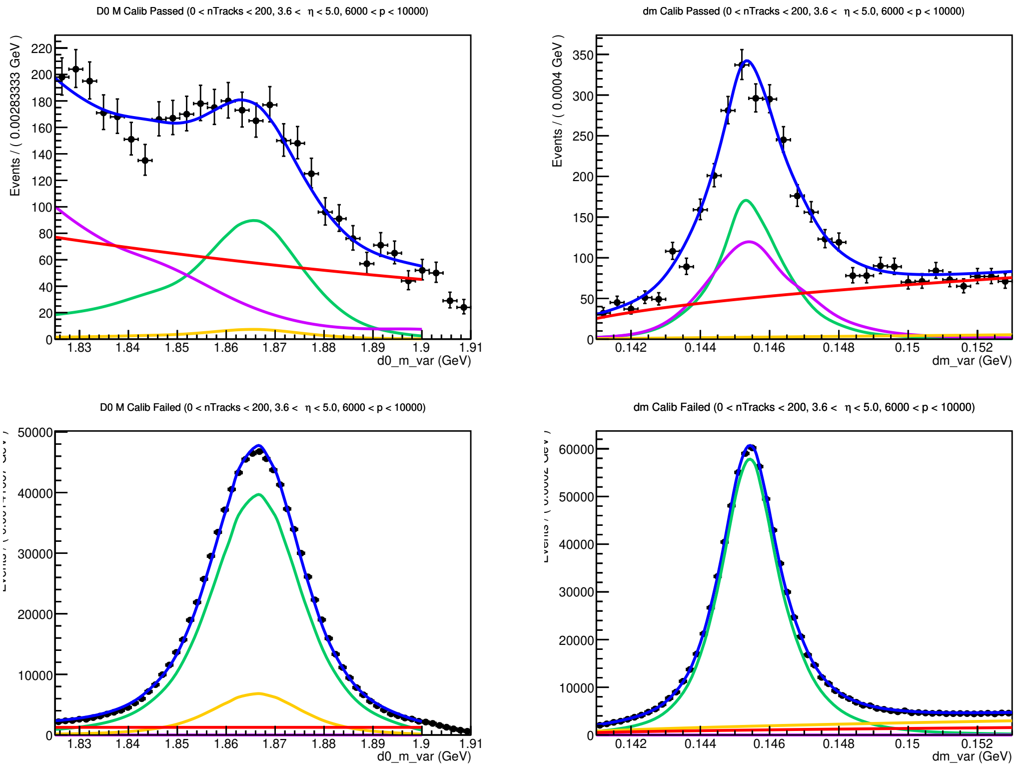


Figure 165: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

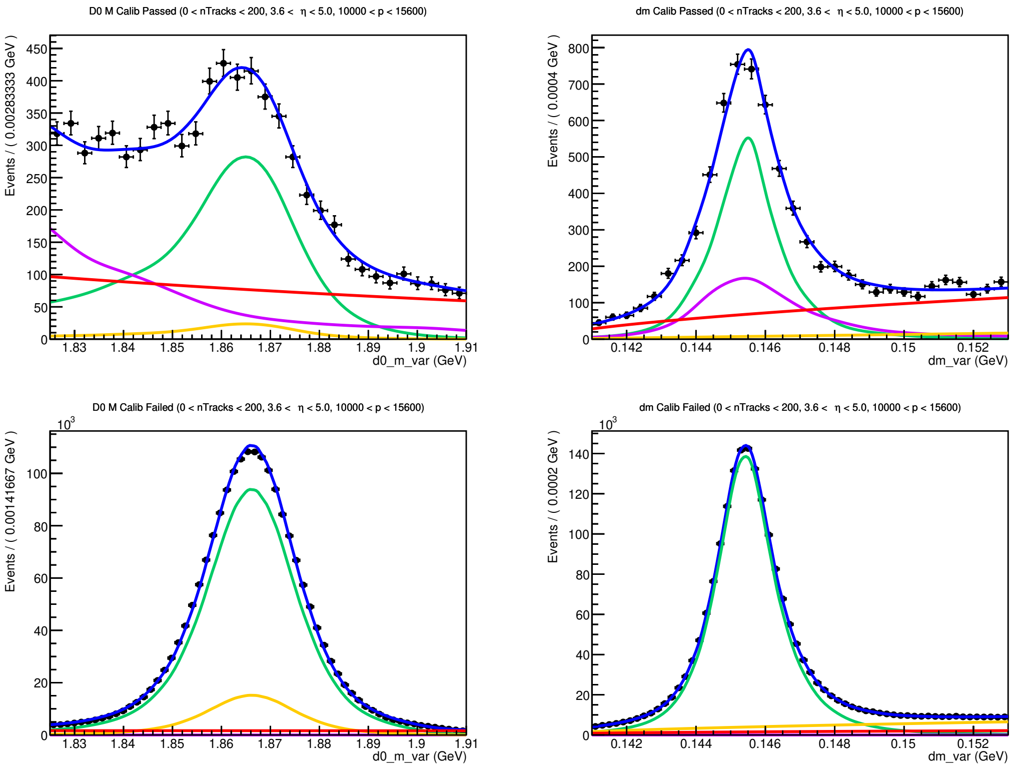


Figure 166: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

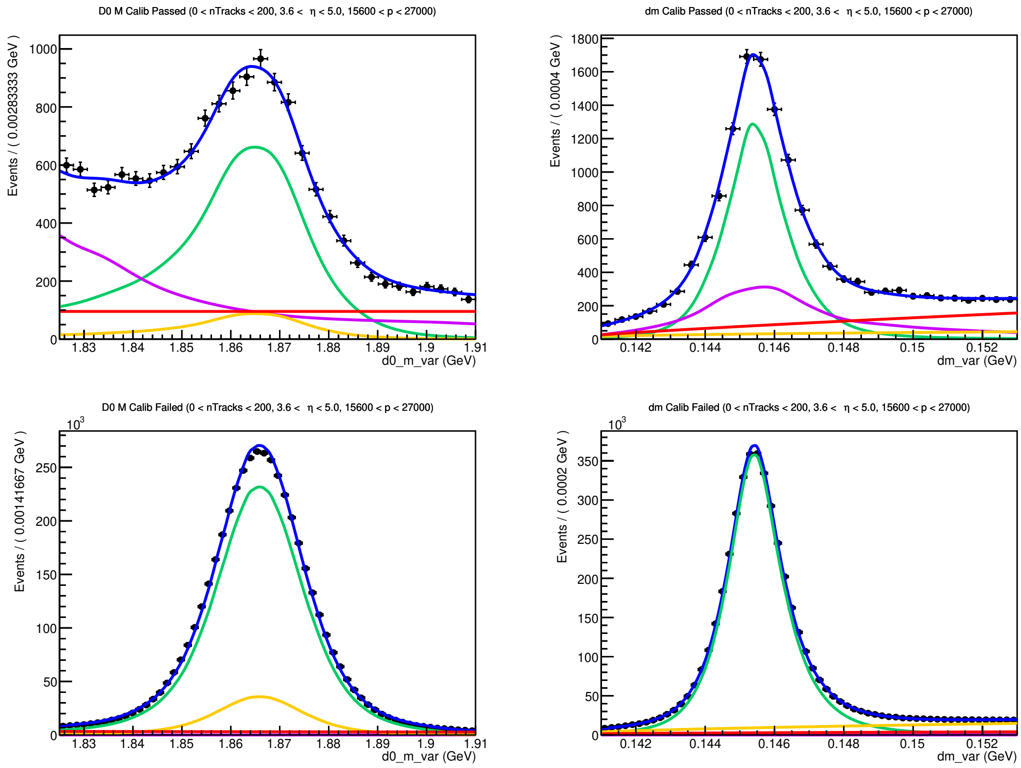


Figure 167: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

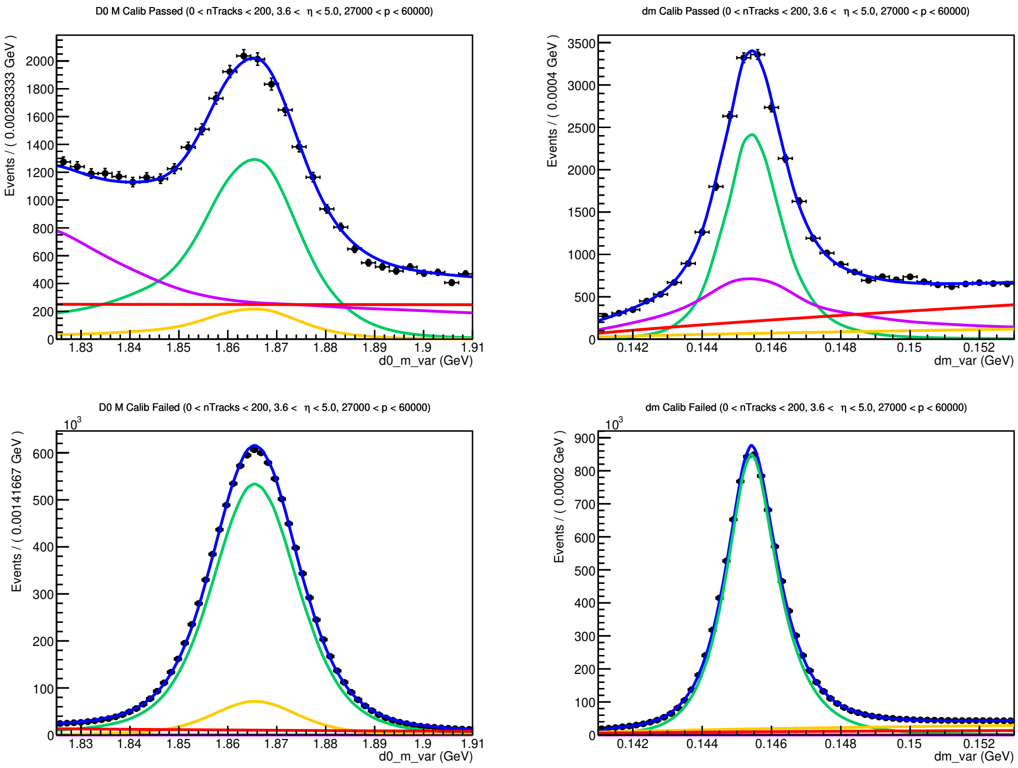


Figure 168: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $27000\,\text{MeV} < p < 60000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

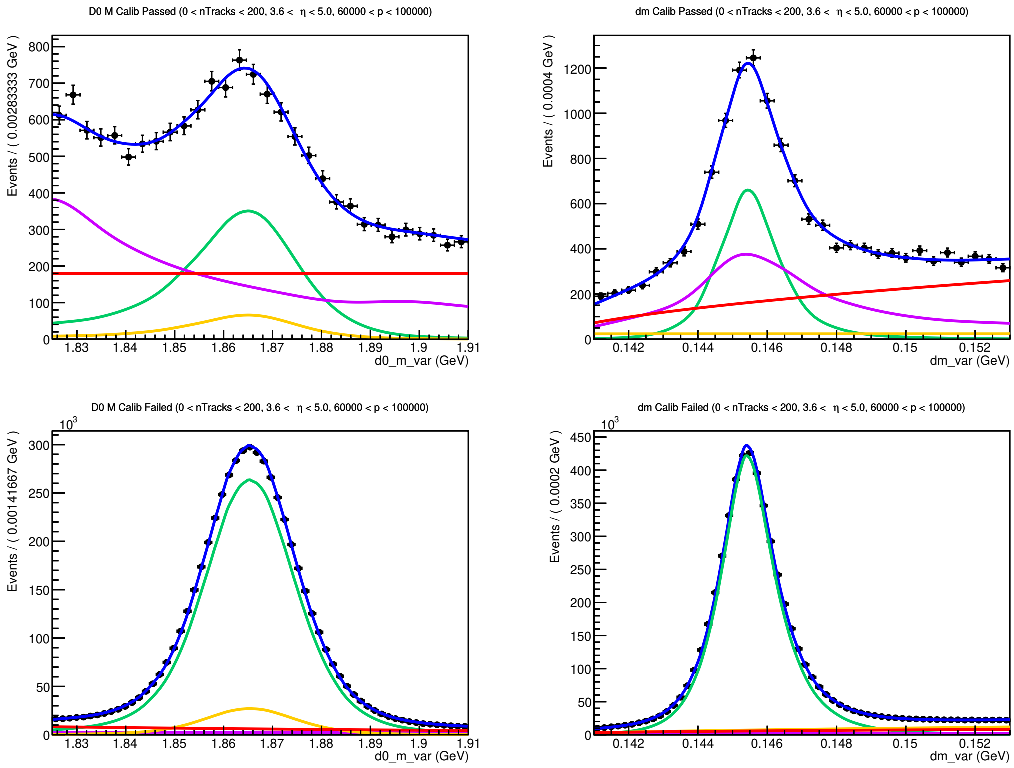


Figure 169: Fit to D^0 mass and δm in nTracks<200, $3.6 < \eta < 5$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

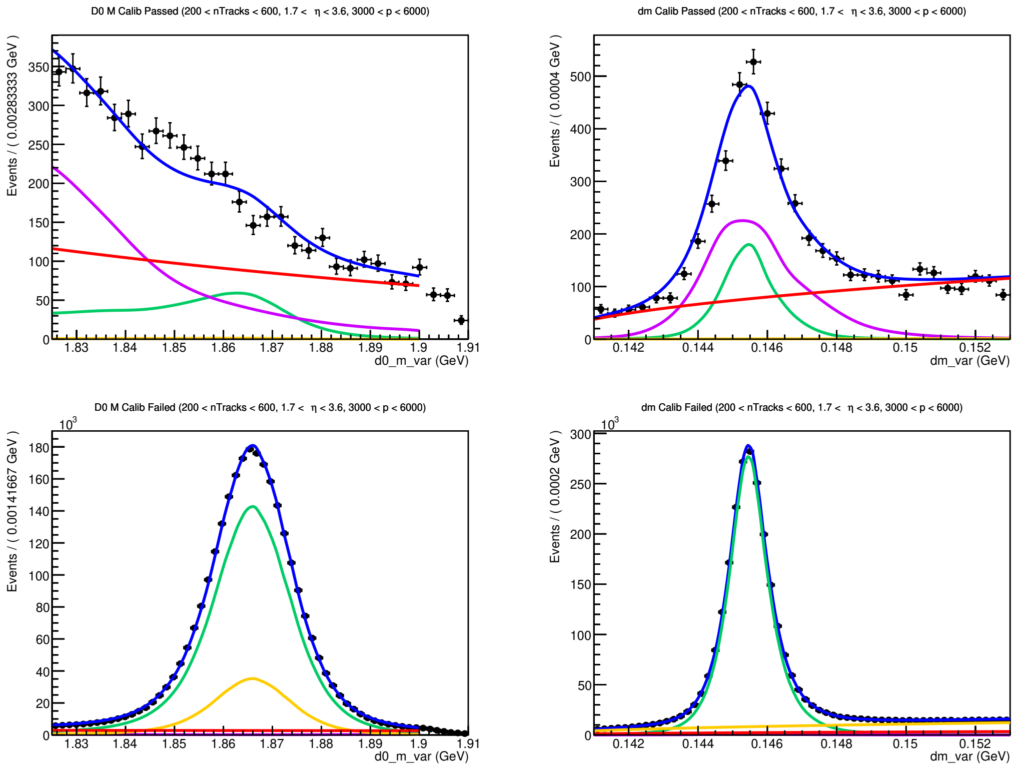


Figure 170: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

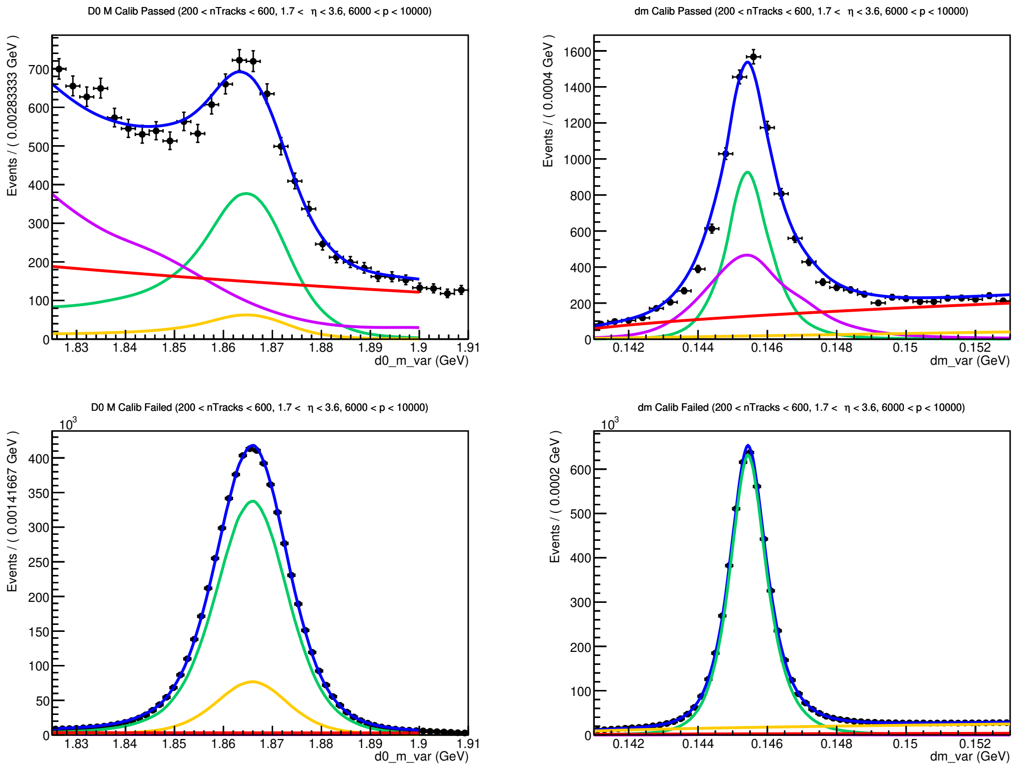


Figure 171: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

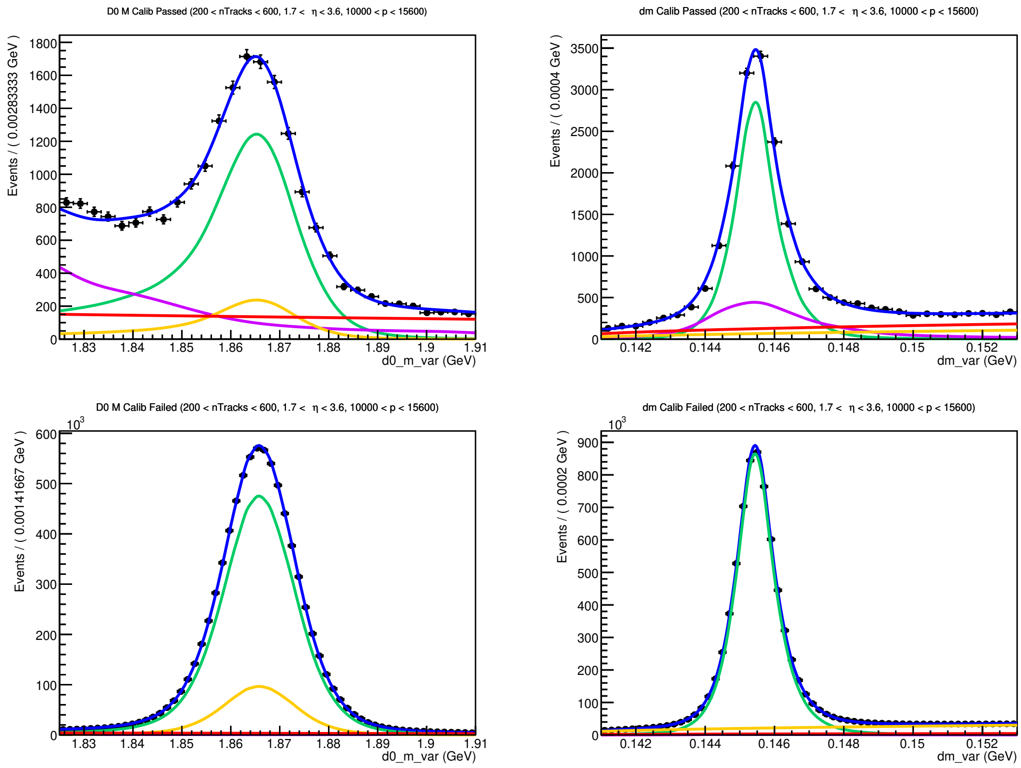


Figure 172: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $10000\,\mathrm{MeV} < p < 15600\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

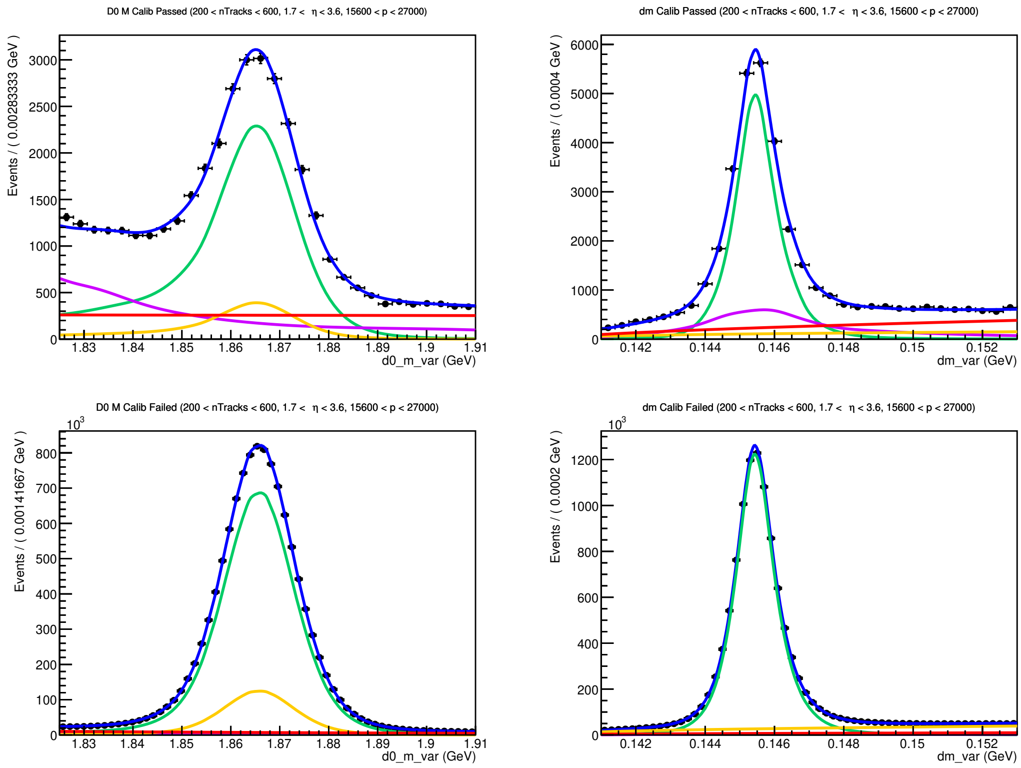


Figure 173: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $15600\,\text{MeV} < p < 27000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

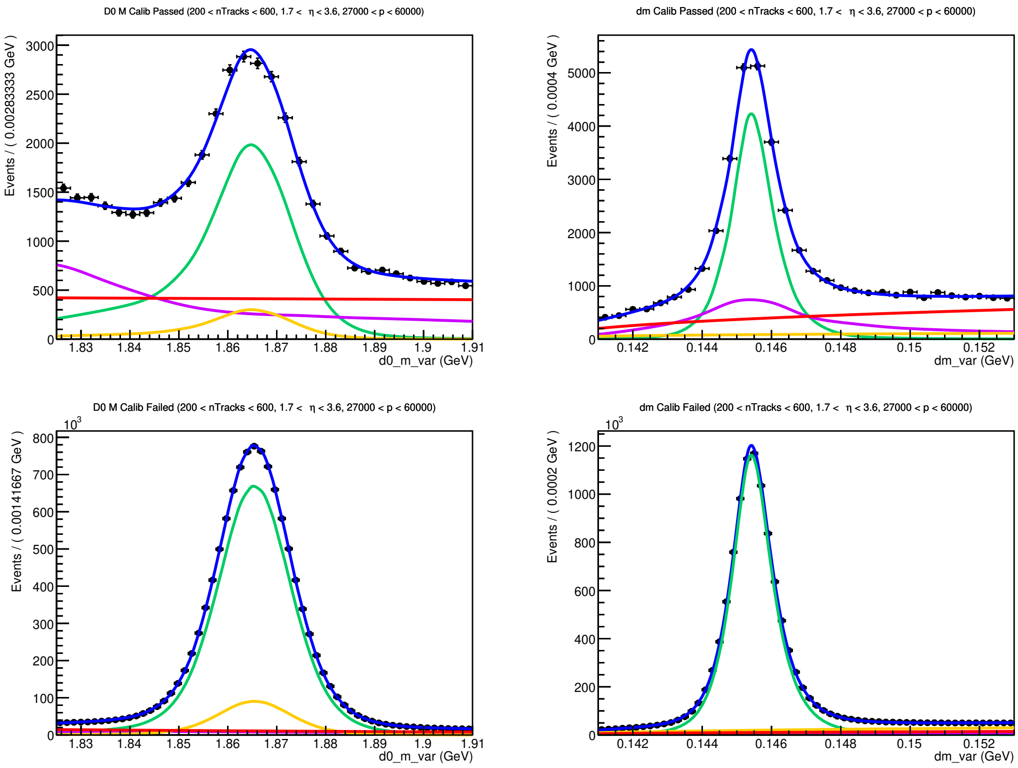


Figure 174: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $27000\,\mathrm{MeV} < p < 60000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

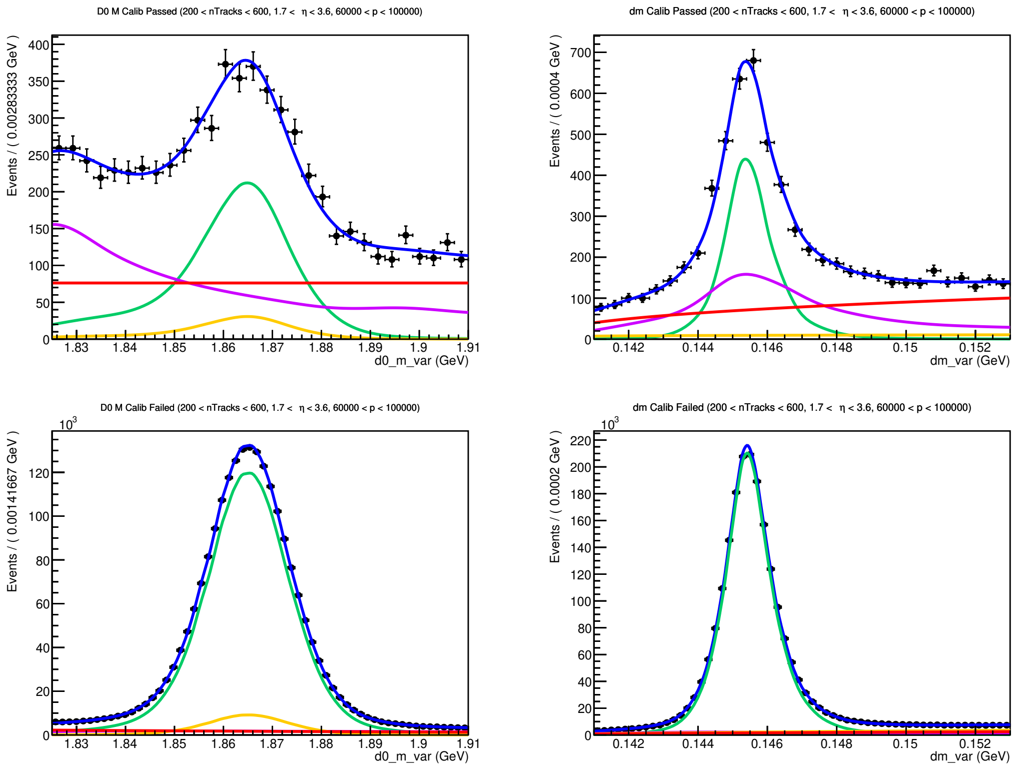


Figure 175: Fit to D^0 mass and δm in 200<nTracks<600, $1.5 < \eta < 3.6$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

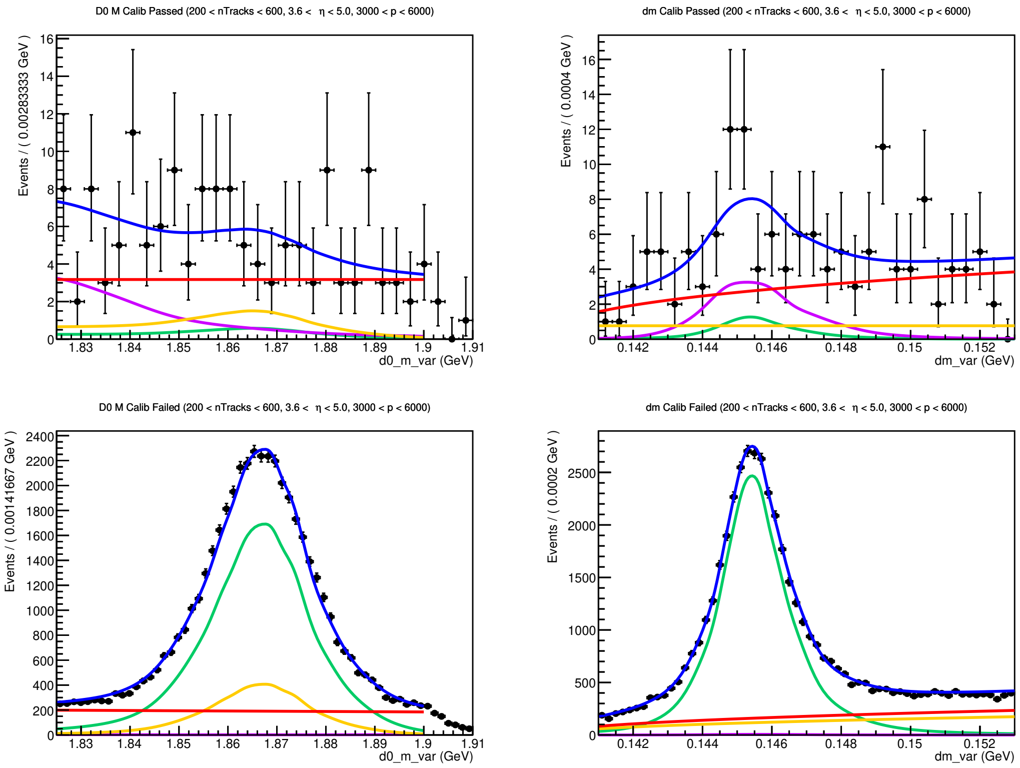


Figure 176: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $3000\,\mathrm{MeV} < p < 6000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

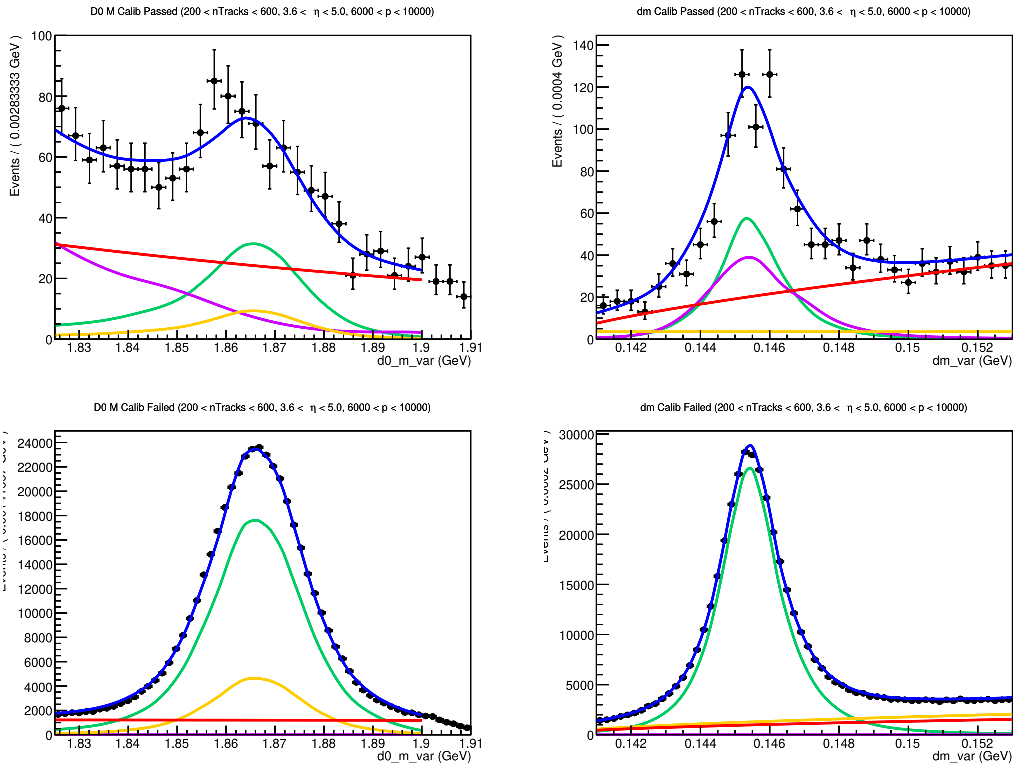


Figure 177: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $6000\,\text{MeV} < p < 10000\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

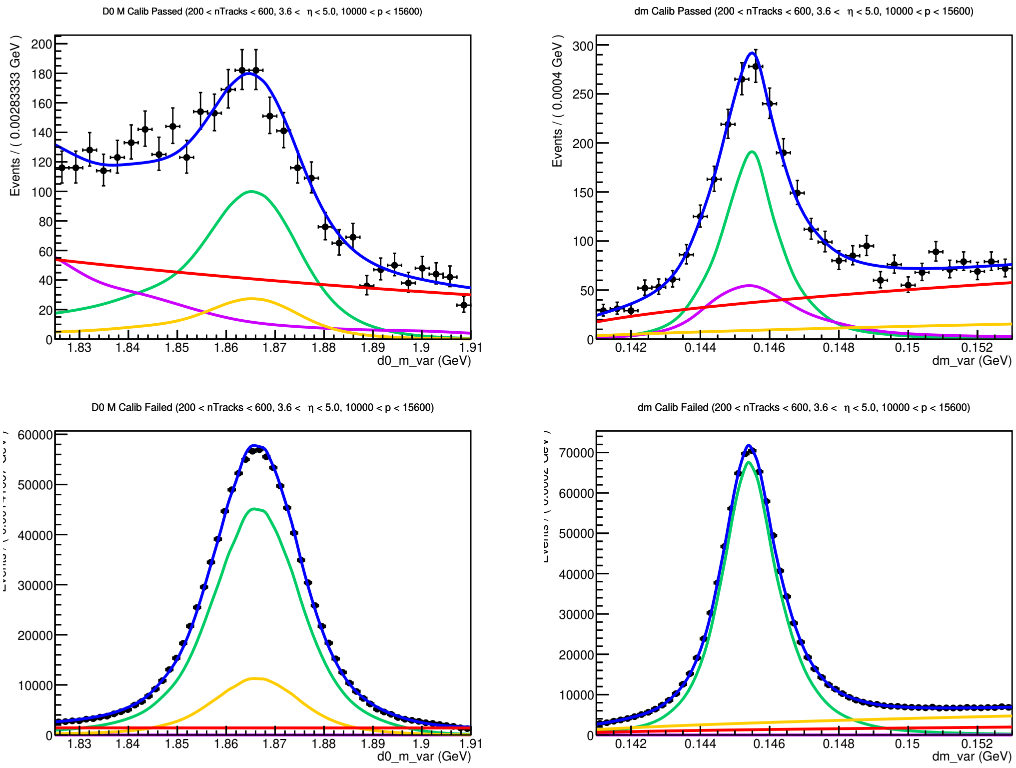


Figure 178: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $10000\,\text{MeV} < p < 15600\,\text{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

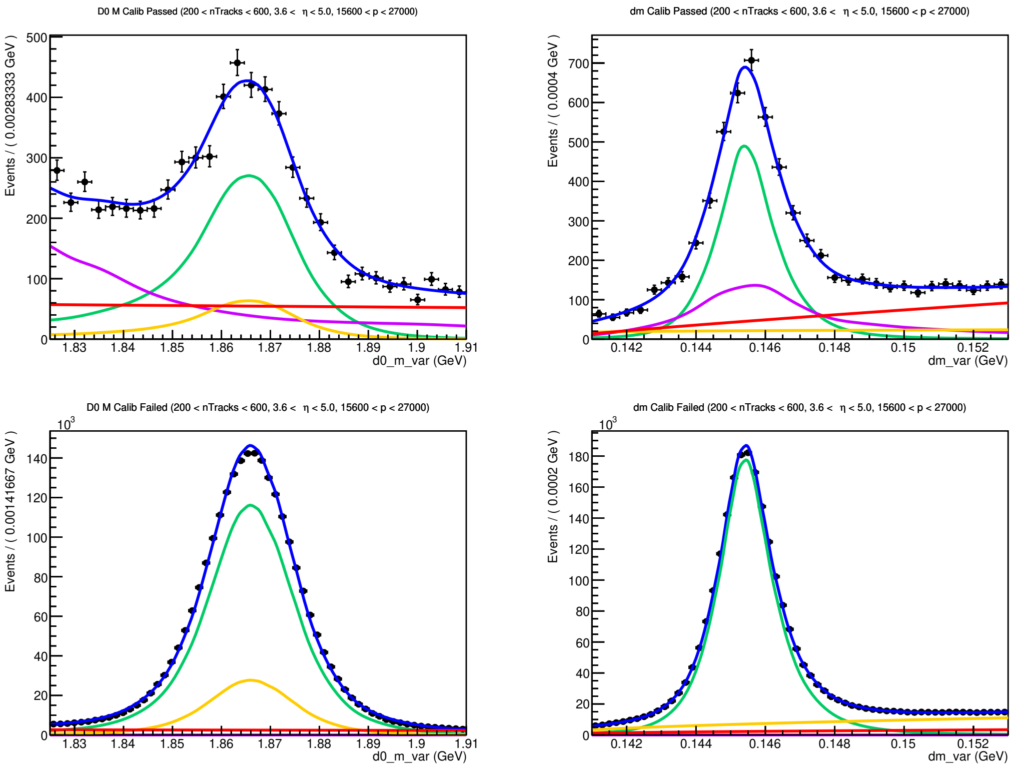


Figure 179: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $15600\,\mathrm{MeV} < p < 27000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

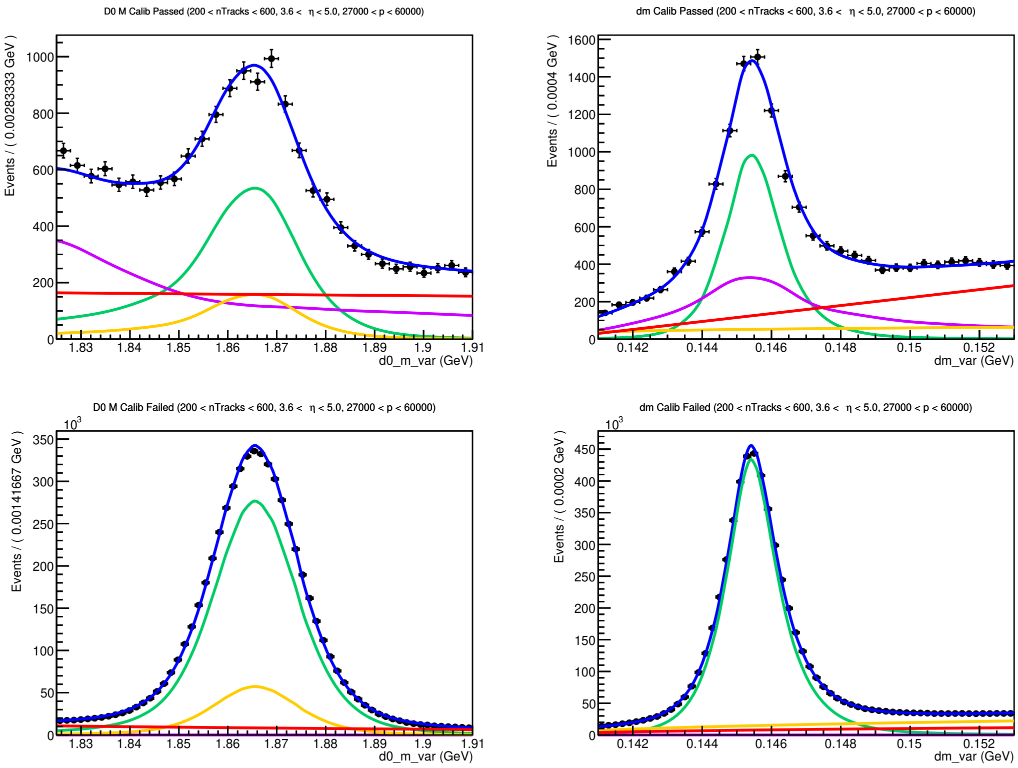


Figure 180: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $27000\,\mathrm{MeV} < p < 60000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

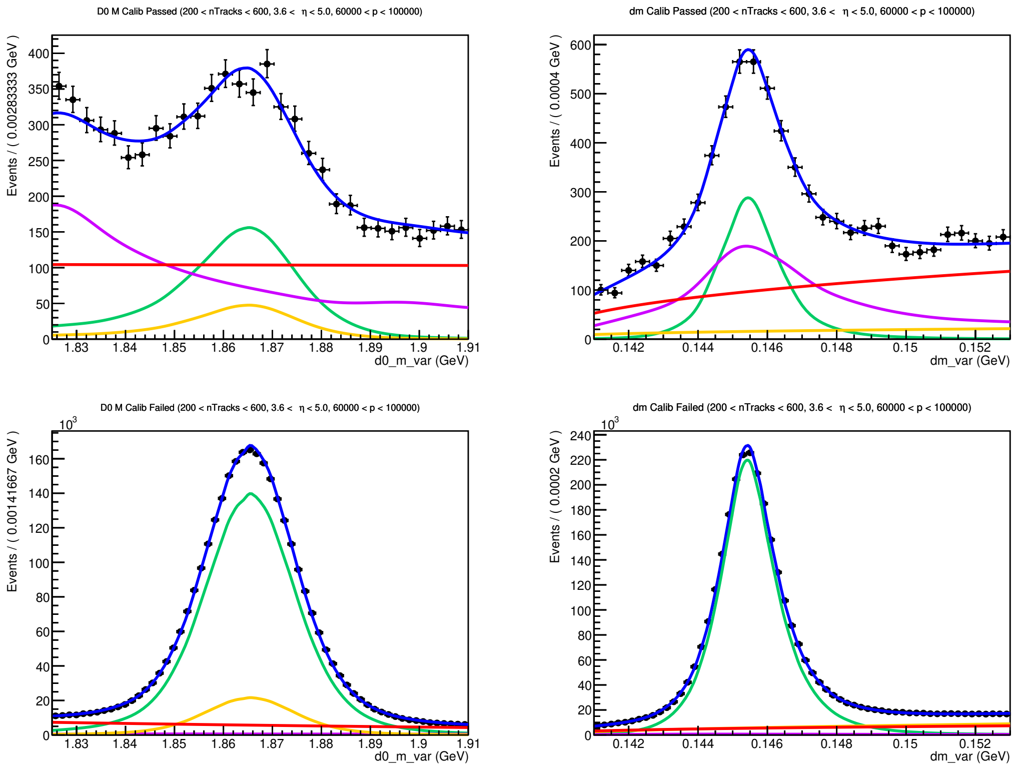


Figure 181: Fit to D^0 mass and δm in 200<nTracks<600, $3.6 < \eta < 5$, and $6000\,\mathrm{MeV} < p < 10000\,\mathrm{MeV}$ for pions in MC (left) and PIDCalib (right) in the events which passed (top) and failed (bottom) PID cuts.

28 Appendix J: Lifetime acceptance corrections

28.1 Description of initial correction attempt

The lifetime acceptance of the LHCb detector is mismodeled in MC simulations. Any selections requiring a minimum IP χ^2 cause the efficiency to be much lower for shorter lifetimes due to the difficulty of distinguishing the primary and secondary vertices. This results in an efficiency of near 0 at very short lifetimes, and a near perfect efficiency at 1 ps. At longer lifetimes, there is a linear trend in the reconstruction efficiency. To capture the correlations between lifetime acceptance and missing momentum in the partially reconstructed decay, a correction is made using data and MC samples of partially reconstructed $B^0 \to J/\psi\,(K^* \to K^+\pi^-)$ decays, where the pion track is not used in reconstructing the candidate. The $J/\psi K^*$ candidate pair is reconstructed with the same kinematic and vertexing requirements as in the decay $B^+ \to J/\psi K^+$. The pion is selected without any vertexing requirements in order to model the missing momentum, and we require the invariant mass of the $(J/\psi\,K^+\pi^-)$ system to be within the B^0 mass window.

For the data sample, shown in 182, the invariant mass distribution of the B^0 candidate is modeled with a double-sided Crystal Ball function, and the background is modeled with an exponential function.

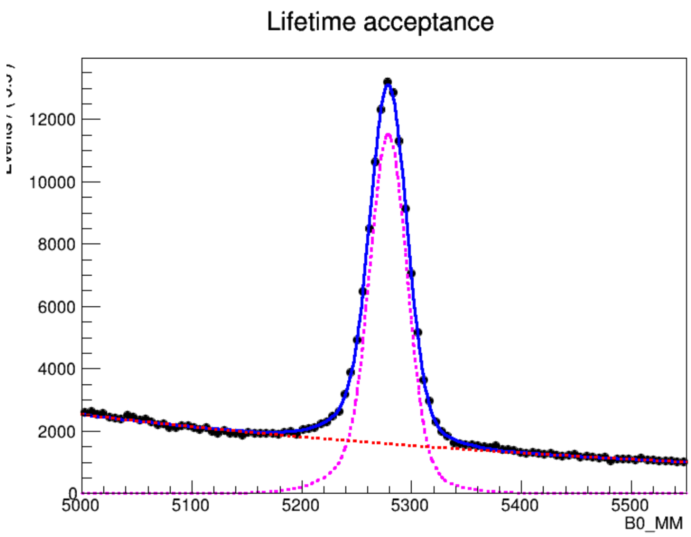


Figure 182: Fit to the invariant mass spectrum of $(J/\psi\,K^+\pi^-)$ to extract $B^0 \to J/\psi\,(K^* \to K^+\pi^-)$ decays

The sPlot method is used to generate weights from the fit to statistically subtract out the background. With the background-subtracted data sample and the simulation sample, we produce a histogram correcting the decay time by 4 bins of $p(\pi^-)/p(B^+)$ from 0 to 0.5, analogous to $p_{miss}/p(B_c)$ in the B_c decays, where p_{miss} is the unreconstructed momentum. A 2D histogram of decay time correction factor and momentum fraction is produced as shown in Figure 183, which shows the region of short lifetime where the correction is needed in green–however, the correction is applied to all lifetimes.

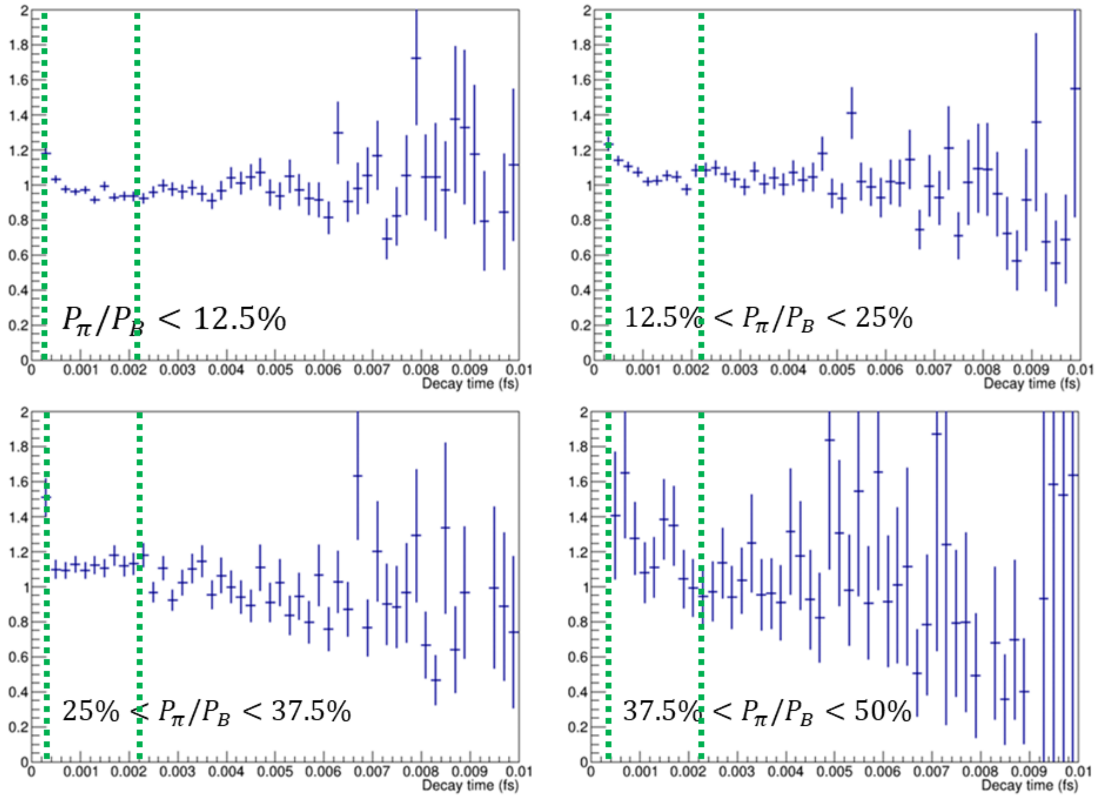


Figure 183: Lifetime acceptance correction factors as a function of decay time in 4 bins of missing momentum fraction (with the relevant range of short lifetime shown in the green band).

28.2 Reasoning for discontinuation

A bug was discovered where selections were inconsistently applied to the $B \to J/\psi K^*$ tuples, which resulted in artificial data/MC differences. Upon fixing the bug, the data/MC differences in the $B \to J/\psi K^*$ tuples are negligible in the fit region with lifetime above 0.45 ps (see Figure 184).

To compute the lifetime of a given sample, an exponential function is fitted to the range 0.6-2.2 ps. Additionally, the fitted value of the lifetime in the mis-ID sample agrees with the known B_d lifetime within uncertainties. This shows that for B_d decays, the lifetime is well modeled.

No pure B_c data sample is available to compare with MC, so we make do by comparing within the normalization-rich region. In this region, the normalization MC and the data agree with both each other and with the known B_c lifetime within uncertainties. During this investigation, we realized that the estimated lifetime shows a missing mass dependency, with lower missing mass corresponding to lower lifetimes. This effect is replicated in both data and MC, and the lifetime computed within a given sub-range of missing mass agreed for data and MC in every such range tested. Signal MC is also tested in the higher missing mass region where there are enough statistics to fit a function, and the computed lifetime also agrees with that extracted from data and normalization.

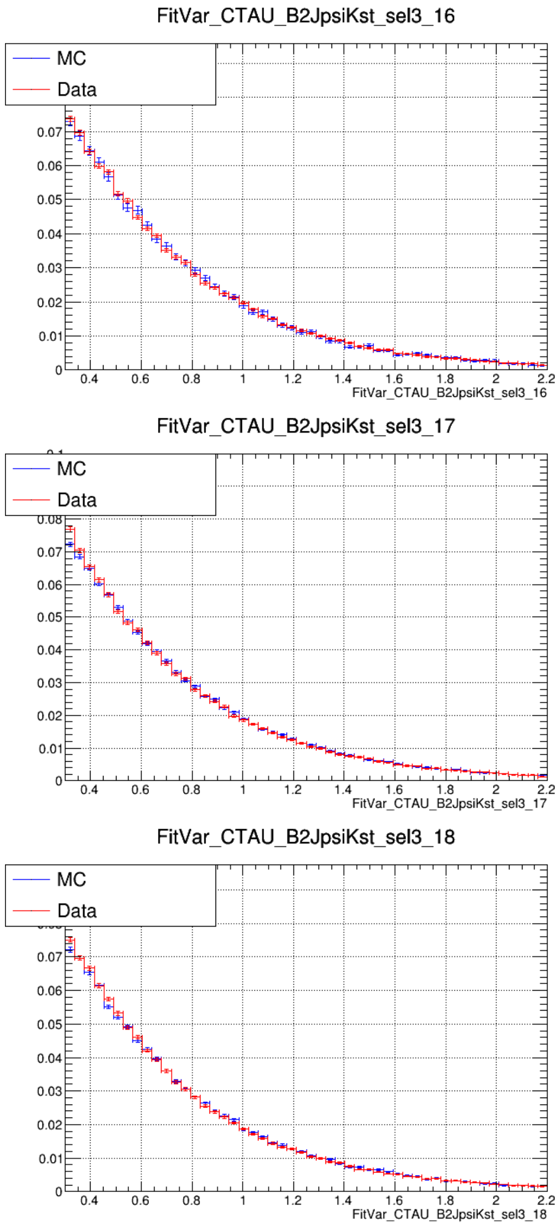


Figure 184: Data/MC comparison of the lifetime distribution of the $B \to J/\psi K^*$ candidates for 2016-2018, from top to bottom. Compatible in fit region.

29 Appendix: Cross-checks

This section summarizes the cross-checks performed to provide constraints on the inputs of the nominal fit. The studies presented here independently validate the modeling of key signal and background components.

- Event selection and relative-efficiency cross checks.
- Background modeling updates and validation.
  - The misID background cross-check is detailed in Sec. 29.1.
  - The double charm background cross-check is detailed in Sec. 29.2.

- The feed-down background cross-check is detailed in Sec. 29.3.
- The J/ψ–μ combinatorial background cross-check was not included. An event-mixing technique was investigated as an alternative modeling strategy. 29.4 However, since the fit observables depend explicitly on the reconstructed B_c vertex, a consistent re-evaluation of these quantities for mixed events is not feasible. Therefore, the event-mixing approach was not adopted, and the nominal background model is retained.

- Reweighting studies
  - Study of VELO error issue. 29.5
  - B_c^+ life time correction. 29.6

29.1 Cross-check of the misID background

29.1.1 Method overview

The mis-identification (misID) background refers to events in which a J/ψ candidate is combined with a charged hadron that is misidentified as a muon and therefore passes the muon PID requirements. Such hadrons cannot be distinguished from true muons under the applied PID selection and consequently contribute to the selected signal sample.

This background is estimated using a data-driven, event-by-event weighting approach. The method combines particle-dependent mis-identification efficiencies with per-event PID information, allowing both the yield and the kinematic distributions of the misidentified background to be evaluated.

29.1.2 Template-based particle composition determination

To estimate the relative contributions of different hadron species in the $J/\psi + X$ sample before the muon PID requirement, a template-based, data-driven method is employed. The particle composition is determined by fitting PID-variable distributions using templates obtained from standard PID calibration samples.

A two-dimensional template fit in the variables DLLp and DLLK is performed to separate π, K, and p contributions. The fits are carried out in bins of the kinematic and event variables of the track X, including the momentum p, pseudorapidity η, impact-parameter significance $\chi^2_{\rm IP}$, and detector multiplicity. This binning ensures adequate modeling of PID response variations across the relevant phase space.

The kinematic space is divided into bins based on momentum p, pseudorapidity η, impact parameter significance χ^2_{IP}, and detector multiplicity (nSPDHits) of the X particle. Since the momentum range from 3 to 10 GeV/c was introduced at a later stage (exemplified by the 2016 data-taking period), the analysis is split accordingly:

- For $p \in (3, 10)$ GeV/c, the ranges $\eta \in (2, 5)$, $\chi^2_{IP} \in (4.8, 200000)$, and nSPDHits $\in (0, 900)$ are used, resulting in $2 \times 4 \times 4 \times 2 = 64$ bins with a non-uniform partition.

- For $p \in (10, 100)$ GeV/c, the same ranges of η, χ^2_{IP}, and nSPDHits apply, yielding $16 \times 4 \times 4 \times 2 = 512$ bins.

This binning scheme is designed to ensure sufficient statistics for stable unbinned template fits, with approximately 6000 events per particle species (π, K, and p) in each bin. The bin definitions are further optimized based on the actual event counts available in each data-taking year.

The fit performance is illustrated using 2016 data. Figure 185 shows the fit results in a low kinematic bin, while Figure 186 corresponds to a high kinematic bin. Each figure contains four panels: the upper-left panel displays the two-dimensional data distribution; the upper-right panel shows the corresponding PID templates for π (red), K (green), and p (blue) derived from standard calibration samples. The lower panels present one-dimensional projections onto the DLLp and DLLK axes, where the data points (black dots) are compared with the fitted template shapes (colored lines). The results demonstrate good separation between particle species, and the templates show excellent agreement with data across both kinematic regions, confirming the stability and reliability of the fitting procedure.

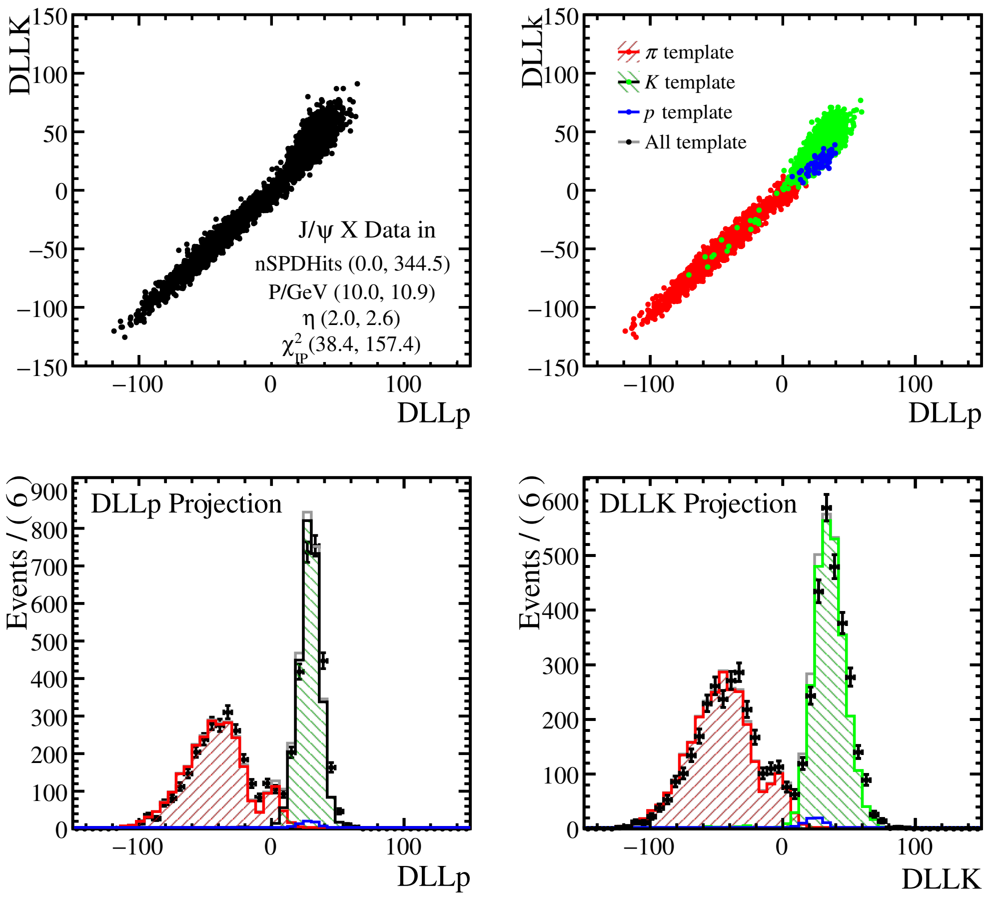


Figure 185: Fit results of particle abundances in a low kinematic bin in 2016 data. The upper-left panel shows the two-dimensional data distribution; the upper-right panel shows the corresponding templates for π (red), K (green), and p (blue) from standard PID samples. Lower panels show one-dimensional projections on DLLp and DLLK, respectively, where black dots represent data and colored lines represent templates. The fitted template shapes agree well with data.

After obtaining the PID-based fit to separate π, K, and p, the core challenge remains how to accurately calculate the final particle abundances. This conversion from fitted

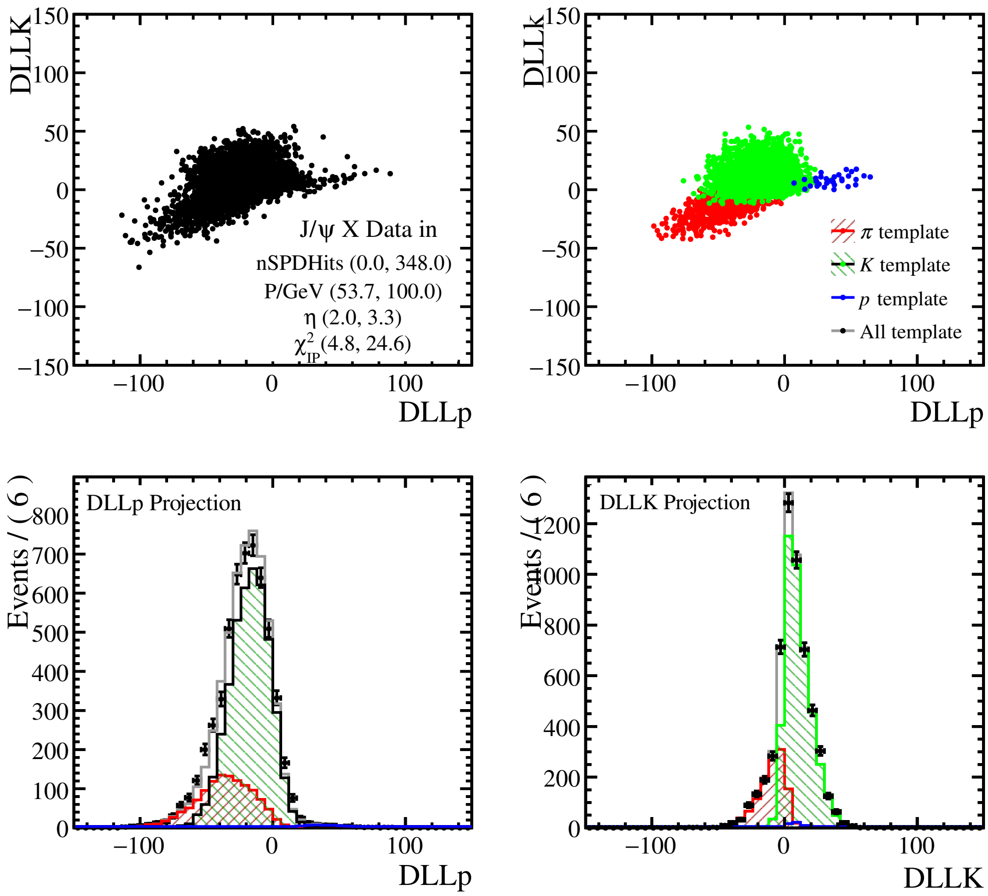


Figure 186: Fit results of particle abundances in a high kinematic bin.

yields to physical abundances is non-trivial and is rigorously investigated using dedicated Monte Carlo (MC) samples, where truth information is available.

In this study, the fitting method applied to MC is the same as for data, with two notes: the MC sample is an inclusive $B_{(s)}^{(+0)} \to J/\psi X$ mixture (with relative π, K, p abundances differing from data), and a coarser kinematic binning is used due to MC statistics being about one-fifth of data. The PID templates are taken from MC analogues of the standard calibration decays ($D^{*+} \to (D^0 \to K^-\pi^+)\pi^+$ for π/K, $\Lambda^0 \to p\pi^-$ for p).

The truth-matched π, K, and p distributions in this MC sample are shown in Figure 187, providing the benchmark for validation. An initial, simple abundance algorithm—which applies a uniform per-species weight to all events within a kinematic bin—results in significant K-π mis-identification cross-talk, as seen in the left panel of Figure **??**. This method, conceptually similar to those used in earlier studies such as Refs. [57, 58], fails to accurately reproduce the true kinematic distributions.

To overcome this limitation, this work introduces an improved abundance algorithm corrected by the local probability density function (PDF):

$$P_i(\text{DLLp}, \text{DLLK}) = \frac{n_i \cdot \text{pdf}i(\text{DLLp}, \text{DLLK})}{\sum hn_h \cdot \text{pdf}_h(\text{DLLp}, \text{DLLK})}, \quad (i = \pi, K, p), \tag{79}$$

where n_i is the fitted yield and pdf_i is the 2D probability density for species i. This

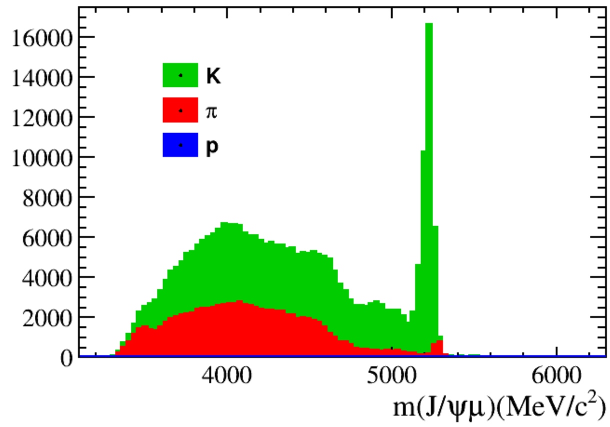
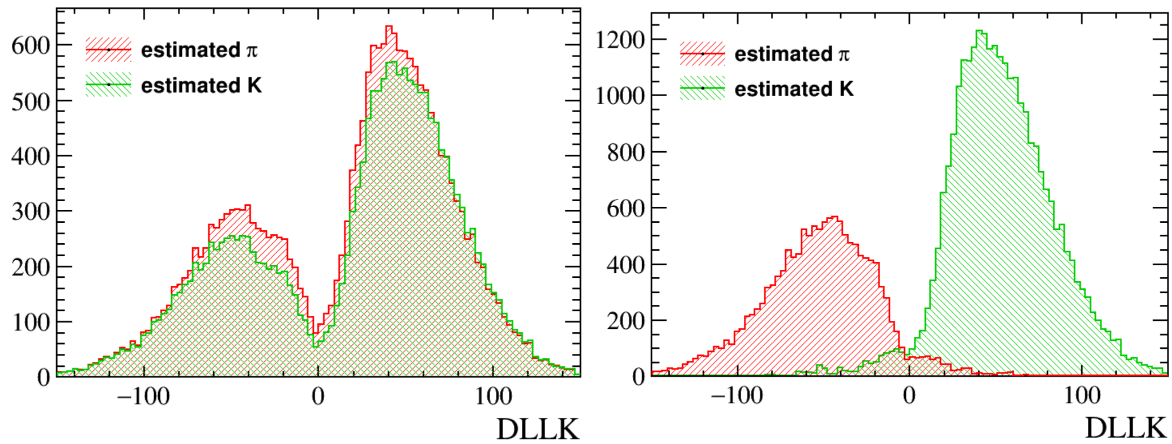


Figure 187: Truth-matched π, K, and p distributions in MC used to validate the abundance algorithm. Note the low statistics of protons (only 1043 instances). Peaks from $B^+ \to J/\psi K^+$ and $B^+ \to J/\psi \pi^+$ are visible; since all X particles are assigned the muon mass, peak positions are shifted.

method assigns event-dependent weights within a bin, leveraging the continuous PID information. As shown in the right panel of Figure **??**, it significantly improves separation and reduces cross-talk.

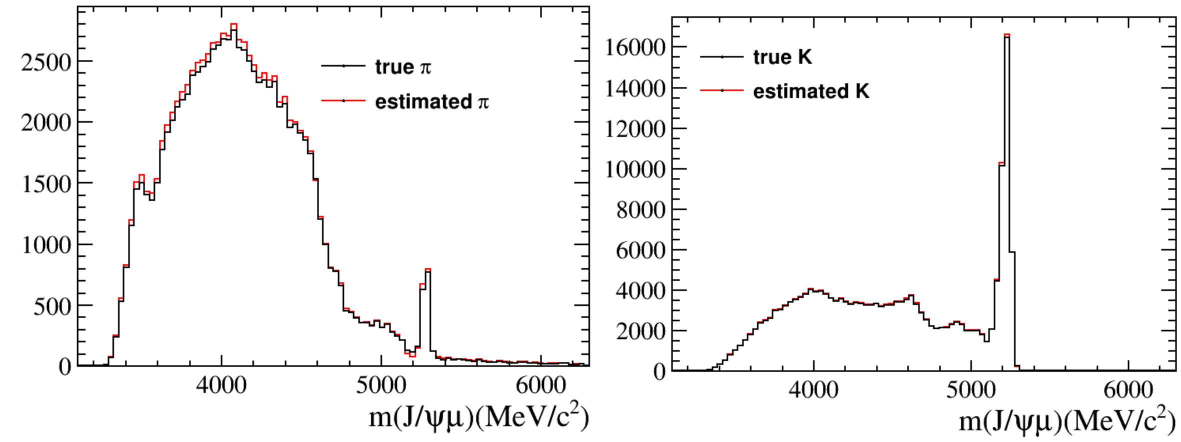


(a) (b)

Figure 188: Comparison of weights from two abundance algorithms within a single kinematic bin. Left: weights from the original algorithm showing significant K-π cross-talk. Right: weights from the PDF-corrected formula (Eq. 79) showing improved separation.

Finally, the sPlot technique is applied to obtain statistically optimal and decontaminated yields. Using the sWeights for π, K, and p from the fit, distributions are constructed that are disentangled from other species. The results, shown in Figures 189a and 189b, demonstrate excellent agreement with the true MC distributions, with residual relative discrepancies of only 2.2

In conclusion, the combination of the PDF-corrected abundance algorithm and the sPlot technique provides a robust and accurate method for extracting particle abundances from the PID template fit, successfully addressing the challenge of K-π mis-identification.



(a) (b)

Figure 189: Comparisons between reconstructed (using the sPlot method) and true particle distributions in MC. The sPlot method shows the best agreement.

The validation against MC truth confirms the method's reliability, with sub-percent to percent-level accuracy.

29.1.3 Mis-identification Efficiency

The mis-identification (misID) efficiencies used in this analysis are provided by the Maryland group. The derivation of these efficiencies is described in Sec. 8.4, while corrections accounting for decay-in-flight (DiF) effects are detailed in Sec. 8.6.2. These efficiencies serve as essential inputs for the estimation of backgrounds involving hadron-to-muon misidentification.

To model the misidentified background contribution, an event-by-event weighting procedure is applied. Dedicated calibration histograms provide the probability for a hadron (π, K, or p) to be misidentified as a muon as a function of the candidate track kinematics, namely the momentum p, pseudorapidity η, and the event track multiplicity n_{Tracks}.

For each event, the corresponding mis-identification efficiency

$$\epsilon_{h\to\mu}(p, \eta, n_{\text{Tracks}})$$

is extracted from the calibration histograms, where h denotes the hadron species. This efficiency is combined with the per-event PID weight to obtain the total mis-identification weight,

$$w_{\text{misID}} = w_{\text{PID}} \times \epsilon_{h\to\mu}(p, \eta, n_{\text{Tracks}}).$$

A weighted dataset is then constructed in which each candidate is retained and augmented with the relevant mis-identification weights. By summing these weights, the overall yield of the misidentified background is determined. This weighted sample is subsequently used as input for the background modeling in the analysis.

29.1.4 Smearing Method

In the cross-check analysis, the smearing procedure applied to the mis-identification templates follows the same approach as that adopted in the nominal fit. A detailed description of the smearing method can be found in Sec. 8.6.1.

29.1.5 Mis-identification Background Yield

The event-by-event weighting procedure described above is used to estimate the overall yield of the mis-identification background. Table 34 summarizes the contributions from the three relevant hadron species, π, K, and p, for each data-taking year.

As expected, the dominant contribution arises from pion mis-identification, primarily due to decay-in-flight and reconstruction effects. The kaon contribution is sub-dominant, while the proton contribution is found to be negligible within the statistical precision. The extracted yields are stable across different years, reflecting the robustness of the calibration inputs and the weighting procedure.

Table 34: Estimated yields of the mis-identification background for each data-taking year.

Year	estimated yields	nominal fit
2016	1723 ± 44	1518 ± 100
2017	1737 ± 46	2273 ± 119
2018	1980 ± 46	2074 ± 251

29.1.6 Summary of the MisID Background Cross-check

A data-driven cross-check of the mis-identification background has been performed using an independent event-by-event weighting method. The results of this study are incorporated into the nominal fit as part of the evaluation of systematic uncertainties associated with the mis-identification background modeling.

29.2 Cross check double charm background

29.2.1 Method overview

The double charm background arises from intermediate states decaying to μ, primarily from the $B_c^+ \to J/\psi D_{(s)}^{(*+0)} X$ processes, where $D_{(s)}^{(*+0)}$ mesons decay semileptonicly to final states containing μ. These processes are abundant and the branching fractions of charm-meson semileptonic decays to μ are also relatively high. These backgrounds are modeled with MC simulations of the corresponding two-body decays, quasi-two-body decays, and three-body decays. These decays are divided into three categories containing D^0, D^+ and D_s^+. This study implements a data-driven method to constrain their yield.

29.2.2 Data driven method

(1) The yield of target channel can be calculated by,

$$N(b \to J/\psi D(\to \mu X)Y) = \sigma(b\bar{b}) \cdot \mathcal{L} \cdot \mathcal{B}(b \to J/\psi DY) \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot\mathcal{B}(D \to \mu X) \cdot \varepsilon(b \to J/\psi D(\to \mu X)Y) \quad \{\mathrm{cut}R(J/\psi)\}, \tag{80}$$

where $\sigma(b\bar{b})$ means the production cross section of $b\bar{b}$ pairs, and $\mathcal{L}$ represents the integrated luminosity of data samples. $\mathcal{B}$ and ε are the branching fraction and efficiency of corresponding processes in the brackets. D refers to all D mesons, including D^0, D^+ and D_s^+ and their excited states.

(2) Inclusive control samples, which contain the same B decays but hadronic decays of charm mesons, can be used as normalization, and their yield can be calculated by similar equation,

$$N(b \to J/\psi D(\to h)Y) = \sigma(b\bar{b}) \cdot \mathcal{L} \cdot \mathcal{B}(b \to J/\psi DY) \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot\mathcal{B}(D \to h) \cdot \varepsilon(b \to J/\psi D(\to h)Y) \quad \{\mathrm{cut}b\} \tag{81}$$

(3) Dividing the above two equations, we can get,

$$\frac{N(b \to J/\psi D(\to \mu X)Y)}{N(b \to J/\psi D(\to h)Y)} = \frac{\mathcal{B}(D \to \mu X) \cdot \varepsilon(b \to J/\psi D(\to \mu X)Y)}{\mathcal{B}(D \to h) \cdot \varepsilon(b \to J/\psi D(\to h)Y)} \quad \{\frac{\mathrm{cut}R(J/\psi)}{\mathrm{cut}b}\} \tag{82}$$

Since we have tagged J/ψ (Tag-and-probe method), we assume that the fitted yield of selected hadronic D_s^+ decay events is equal to the yield of $N(b \to J/\psi D(\to \mu X)Y)$ multiplied by efficiency ratio. We also assume that the efficiencies of processed with two-body and three-body final states are equal ($\varepsilon(b \to J/\psi D(\to h)Y) = \varepsilon(b \to J/\psi D(\to h))$), which will be validated subsequently. $\mathcal{B}(D \to h)$ and $\mathcal{B}(D \to \mu X)$ are cite by PDG [39].There is no $D_s^+ \to \mu X$ in PDG, but $D_s^+ \to eX$ is $(6.33 \pm 0.15)\%$ and $D_s^+ \to \mu^+\nu_\mu$ is $(5.35 \pm 0.12) \times 10^{-3}$

(4) We calculate the yield of exclusive control channel using the aforementioned formula,

$$N(B_c^+ \to J/\psi D_s^+(\to KK\pi)) = \sigma(b\bar{b}) \cdot f(b \to B_c^+) \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+) \cdot \mathcal{L} \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot\mathcal{B}(D_s^+ \to KK\pi) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi)) \quad \{cutB_c^+\} \tag{83}$$

and compare it with the fitted yield to validate the correctness of this data-driven method.

29.2.3 Dataset and event selections

At least one of the J/ψ is required to fulfil tight selection cuts in order to serve as the so-called tag J/ψ. The inspiration for this method is derived from Ref. [59]. Candidate $B \to J/\psi\ (\to\mu^+\mu^-)D$ decays are reconstructed in events selected from

`StrippingFullDSTDiMuonJpsi2MuMuDetachedLine`

The stripping lines in the Dimuon stream are chosen with versions Stripping24r2, Stripping28r2, Stripping29r2, Stripping34 for 2015, 2016, 2017 and 2018, respectively. The software and stripping versions for the data set reconstruction in different years are listed in Table 35, and MC samples of double charm background are list in Table 36. The tag J/ψ selection criteria, which are summarised in Table 37, remain the same for all these versions. The J/ψ candidate is formed by two oppositely charged tracks consistent with μ hypotheses ($\mathrm{DLL}_{\mu\pi}$ >0) with p_{T} >550 MeV/c, selected on the basis of `StdLooseJpsi2MuMu`. The J/ψ candidate is required to have an invariant mass within ± 45 MeV/c^2 around the known J/ψ mass [33], and to have good vertex-fit quality χ^2_{vtx} <20. The J/ψ candidate is also required to be consistent with originating from b-hadron decays by restricting to its decay length significance (DLS) greater than 3. In addition the ghost probability is required to be less than 0.5.

Table 35: The versions of reconstruction software and Stripping lines for each year

Year	Reconstruction	Stripping Lines
2015	Rec15	Stripping24r2
2016	Rec16	Stripping28r2
2017	Rec17	Stripping29r2
2018	Rec18	Stripping34

Table 36: Monte Carlo samples of double charm background. The number of events in the table is represented in millions (M).

Decay channel	Event type	Sim versions	Gen	Function
$B_c^+ \to J/\psi D^0$	11145092	sim10c	2M(2018)	efficiency
$B_c^+ \to J/\psi D^+$	12145092	sim10c	2M(2018)	efficiency
$B_c^+ \to J/\psi D_s^+$	14175003	sim09b	1M(2016)	efficiency
$B_c^+ \to J/\psi K^+$	12143001	sim09l	20M(2016)	validate efficiency
$B_c^+ \to J/\psi K^*$	11144001	sim10a	20M(2016)	validate efficiency

Table 37: Tag J/ψ selections of double charm

Particle	Requirement
μ	$\mathrm{PID}_\mu > 0$
	$p_{\mathrm{T}} > 550\,\mathrm{MeV}/c$
	Prob ghost of track < 0.5
	$\chi^2/\mathrm{ndf} < 4$
J/ψ	$\chi^2_{FD} > 3$
	$m_{\mu^+\mu^-} \in m_{J/\psi}^{\mathrm{PDG}} \pm 45$ MeV/c^2
	$\chi^2_{\mathrm{vtx}}/\mathrm{ndf} < 20$

Only well reconstructed particles are used in these studies. Track quality is ensured by requiring that the track fit χ^2/ndf is less than four. The transverse momentum is required to be greater than 250 MeV/c for each hadron candidate. Good quality particle identification by the RICH detectors is ensured by requiring the momentum of the hadron candidates to be between 3.2 GeV/c and 100 GeV/c and the pseudo rapidity has to be in the range (2,5). To select well-identified kaons(pions), the corresponding difference in logarithms of the global likelihood of the kaon(pion) hypothesis provided by the RICH system with respect to the pion(kaon) hypothesis, is required to be greater than two(zero). These criteria from one side are tight enough to reduce significantly the mis-identification background, and on the other side loose enough to provide good agreement between data and simulation for chosen fiducial region. Pions and kaons, used for the reconstruction of long-lived charmed hadrons, are required to be inconsistent with being produced in proton-proton interaction vertices. Only particles with a minimal value of impact parameter χ^2_{IP} with respect to all reconstructed proton-proton collision vertices in excess of nine are considered for subsequent analysis. These selection criteria are summarized in Table 38.

Table 38: selection criteria for charged particles used for the reconstruction of charmed hadrons

Particle	Track Selection
	$2 \le \eta \le 5$
	$\chi^2_{\mathrm{IP}}/ndf > 9$
	$p_{\mathrm{T}} > 250\,\mathrm{MeV}/c$
$h^{\pm}$	Track χ^2/ndf < 4.0
	π, PIDK < -2
	K, PIDK > 2
	$3.2\,\mathrm{GeV}/c \le p \le 100\,\mathrm{GeV}/c$
	Track ghost probability of $h^{\pm}$ <0.5

The open charm hadrons are reconstructed in $D^0 \to K^-\pi^+$, $D^+ \to K^-\pi^+\pi^+$, $D_s^+ \to K^-K^+\pi^+$. The good quality of common two(three)-prong vertex is ensured by $\chi^2_{\mathrm{vtx}} <$ 9.0 (25.0) requirement. The mass intervals for selected charm hadrons are chosen to be inside $\pm 3\sigma$ of their known masses [33]. The selection criteria for open charm candidates are summarized in Table 39.

Table 39: selection criteria used for selection of charmed hadrons.

$D^0 \to K^-\pi^+$
$\chi^2_{\rm vtx} < 9.0$
$p_{\rm T} > 1\,{\rm GeV}/c$
$m_{D^0} \in (m^{\rm PDG}_{D^0} \pm 25\ )\,{\rm MeV}/c^2$
$D^+ \to K^-\pi^+\pi^+$
$\chi^2_{\rm vtx} < 25.0$
$p_{\rm T} > 1\,{\rm GeV}/c$
$m_{D^+} \in (m^{\rm PDG}_{D^+} \pm 20\ )\,{\rm MeV}/c^2$
$D^+_s \to K^-K^+\pi^+$
$\chi^2_{\rm vtx} < 25.0$
$p_{\rm T} > 1\,{\rm GeV}/c$
$m_{D^+_s} \in (m^{\rm PDG}_{D^+_s} \pm 20\ )\,{\rm MeV}/c^2$

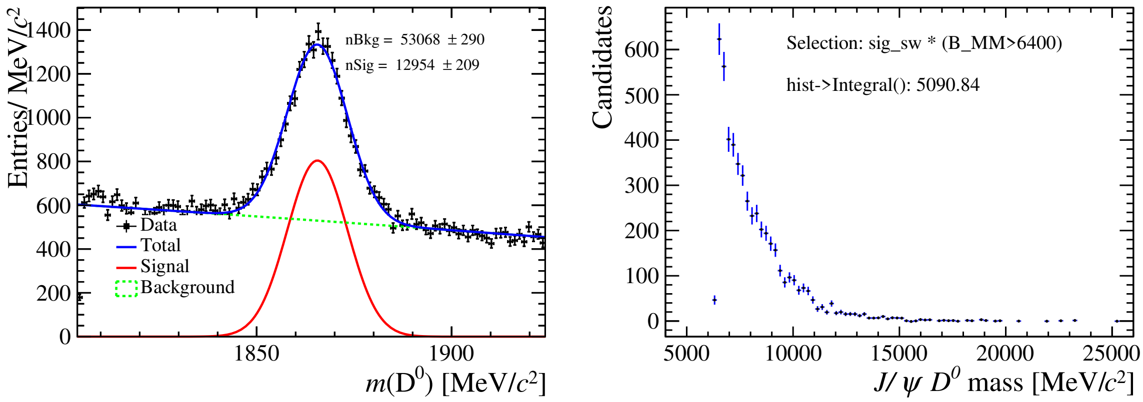
The B candidates are formed from $J/\psi D$ pairs with transverse momentum in excess of 1 GeV/c. A kinematic fit has been applied for B candidates. To improve the invariant mass and lifetime resolution, the primary vertex pointing constraint and the mass-constraints for intermediate J/ψ, D^0, D^+ and D^+_s states have been applied, and the reduced χ^2 from this fit is required to be less than five. The decay time of open charm hadrons $c\tau(D) > 75\,\mu{\rm m}$, extracted from this fit is required to satisfy and the corresponding signed significance is required to be in excess of three sigma. Finally the decay time of B^+_c candidates, $c\tau$ is required to exceed 75 μm, while in the validation procedure, an additional selection of $c\tau < 1\,{\rm mm}$ is required. The selection criteria for $B \to J/\psi D$ candidates are summarized in Table 40.

Table 40: Selection criteria for $B \to J/\psi D$

$B \to J/\psi D$ {cutb}	
$\chi^2_{\rm vtx}$ $(J/\psi D)$	< 16
$\chi^2_{\rm IP}$	< 9
$p_{\rm T}$	> 1 GeV/c
$c\tau(D)$	> 75 μm
$\mathcal{S}_{c\tau}(D)$	$> 3\sigma$
$\chi^2_{\rm fit}/ndf$	< 5
$B^+_c \to J/\psi D^+_s$ {cutB^+_c }	
$c\tau$	$> 75\,\mu$m
	$< 1\,$mm (only for validation)

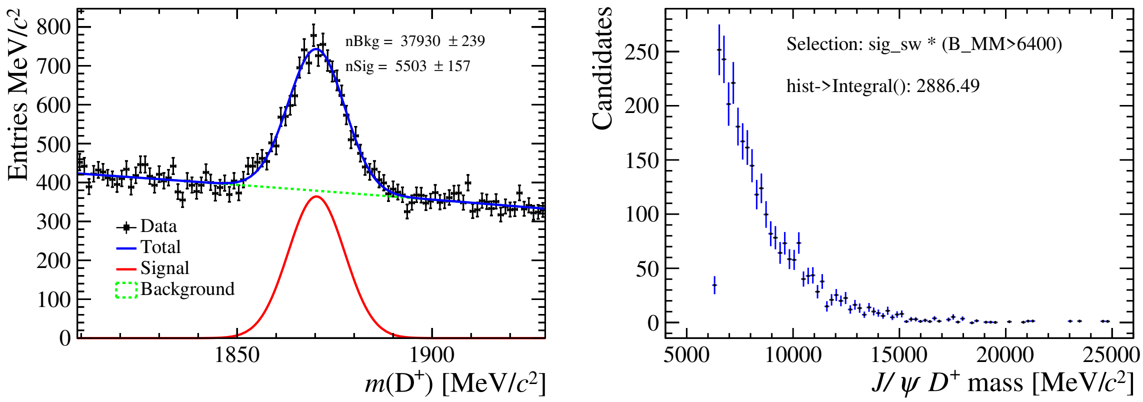
29.2.4 Target yield

In our study within the hadron channel, we need to determine our target yield. This paper uses the sPlot technique to distinguish between signal and background for the D meson. When obtaining the target yield, we found that, for the sweighted mass distribution of J/ψ D combination, there are still events with invariant mass of J/ψ D combination beyond $6400\,\mathrm{MeV}/c^2$. Therefore, we cannot simply assume that the yield of the D signal equal to that of the B signal. Consequently, we plan to subtract the background in the sweighted J/ψ D spectrum using an exponential function to achieve a more accurate yield.



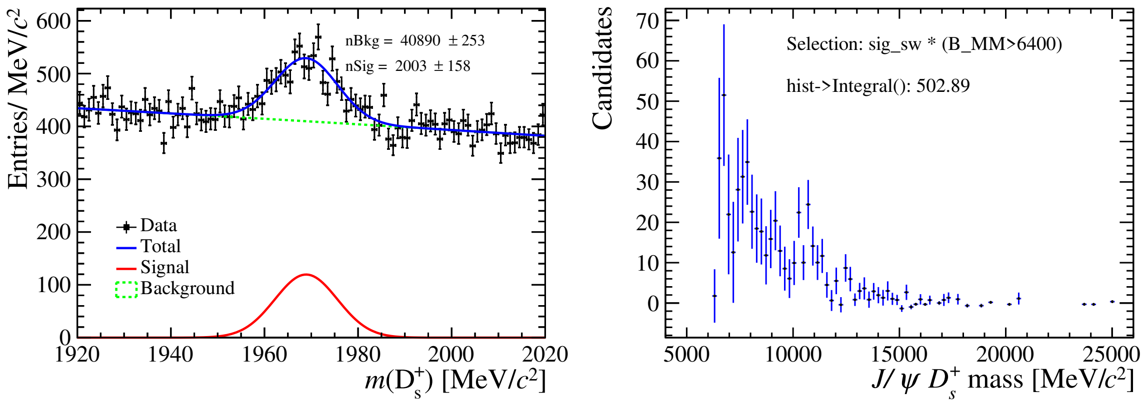
(a) (b)

Figure 190: The distributions of D^0 mass(left) and $J/\psi D^0$ mass(right).



(a) (b)

Figure 191: The distributions of D^+ mass(left) and $J/\psi D^+$ mass(right).



(a) (b)

Figure 192: The distributions of D_s^+ mass(left) and $J/\psi D_s^+$ mass(right).

The mass spectrum of the $J/\psi D$ combination is plotted, incorporating the signal weight for the D meson. The background in the mass region above 6400 MeV/c^2 is modeled using an exponential function. The slope of this exponential function is fixed and extended into the B meson signal region, allowing for the calculation of the yield within the B meson signal region. Additionally, the choice of tail range influences the fitting function, thereby affecting the fitting results. To assess this, the upper bound of the tail is extended to 10000 MeV/c^2 and 15000 MeV/c^2, with no significant impact on the yield result, indirectly supporting the reliability of the result.

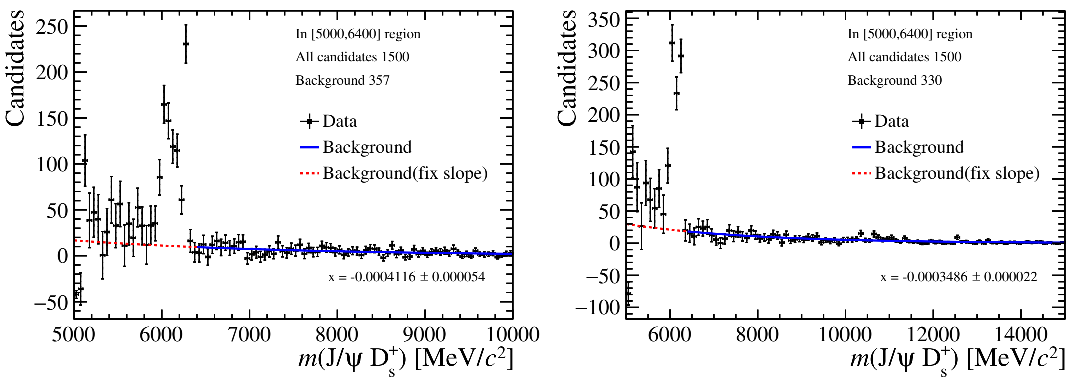


Figure 193: The mass distribution of the $J/\psi D_s^+$ (including the signal weight for the D_s^+ meson). The red dashed line represents the extension of the exponential function after fixing the slope.

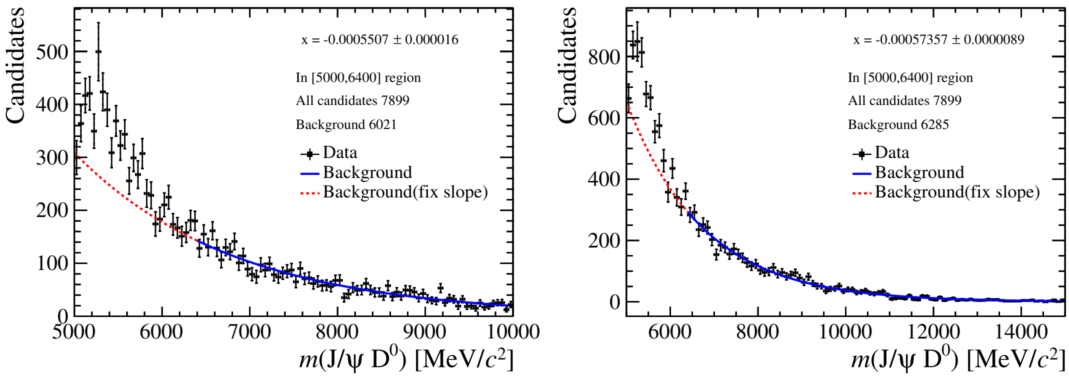


Figure 194: The mass distribution of the $J/\psi D^0$ (including the signal weight for the D^0 meson). The red dashed line represents the extension of the exponential function after fixing the slope.

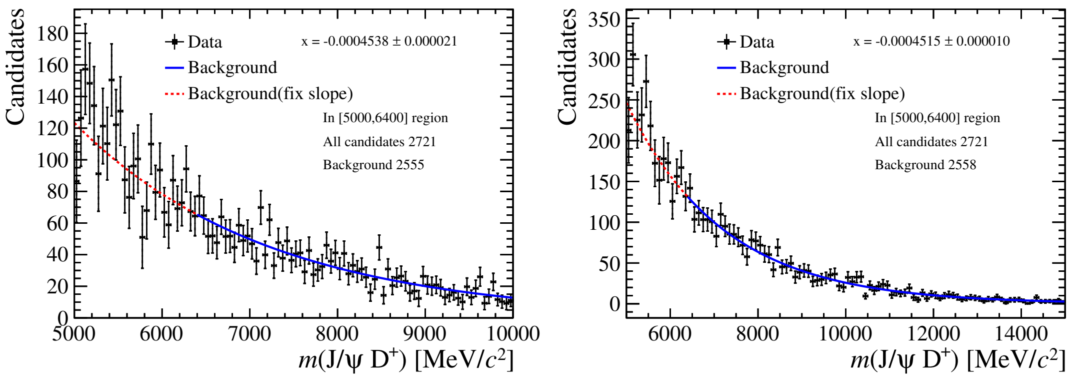


Figure 195: The mass distribution of the $J/\psi D^+$ (including the signal weight for the D^+ meson). The red dashed line represents the extension of the exponential function after fixing the slope.

29.2.5 Efficiency

Table 41: Content of decfile with EventType 14873607.

Double charm category	Decay of B_c^+	normalised BF (B_c^+)	Decay of charm	normalised BF(charm)	relative ratio
$B_c^+ \to J/\psi D^0$	$B_c^+ \to J/\psi D^{*+}$	0.03193	$D^{*+} \to D^0\pi^+$,$D^0 \to \mu X$	0.42227	0.01348
$B_c^+ \to J/\psi D^+$	$B_c^+ \to J/\psi D^{*+}$	0.03193	$D^{*+} \to D^+\pi^0$,$D^+ \to \mu X$	0.54911	0.01753
			$D^{*+} \to D^+\gamma$,$D^+ \to \mu X$	0.02862	0.00092
	$B_c^+ \to J/\psi D^+$	0.01503	$D^+ \to \mu X$	1.0	0.01503
$B_c^+ \to J/\psi D_s^+$	$B_c^+ \to J/\psi D_s^{*+}$	0.79608	$D_s^{*+} \to D_s^+\pi^0$,$D_s^+ \to \mu X$	0.058	0.04617
			$D_s^{*+} \to D_s^+\gamma$,$D_s^+ \to \mu X$	0.942	0.74991
	$B_c^+ \to J/\psi D_s^+$	0.15696	$D_s^+ \to \mu X$	1.0	0.15696

Table 42: Content of decfile with EventType 14873430.

Double charm category	Decay of B_c^+	normalised BF (B_c^+)	Decay of charm	normalised BF(charm)	relative ratio
$B_c^+ \to J/\psi D^0$	$B_c^+ \to J/\psi D^{*0} K^+$	0.3239	$D^{*0} \to D^0\pi^0, D^0 \to \mu X$	0.619	0.20049
			$D^{*0} \to D^0\gamma, D^0 \to \mu X$	0.381	0.12341
	$B_c^+ \to J/\psi D^{*+} K^0$	0.3847	$D^{*+} \to D^0\pi^+$, $D^0 \to \mu X$	0.42227	0.16245
	$B_c^+ \to J/\psi D^0 K^+$	0.0717	$D^0 \to \mu X$	1.0	0.0717
$B_c^+ \to J/\psi D^+$	$B_c^+ \to J/\psi D^{*+} K^0$	0.3847	$D^{*+} \to D^+\pi^0, D^+ \to \mu X$	0.54911	0.21124
			$D^{*+} \to D^+\gamma, D^+ \to \mu X$	0.02862	0.01101
	$B_c^+ \to J/\psi D^+ K^0$	0.2197	$D^+ \to \mu X$	1.0	0.21970

Table 43: Content of decfile with EventType 14873620.

Double charm category	Decay of B_c^+	normalised BF (B_c^+)	Decay of charm	normalised BF(charm)	relative ratio
$B_c^+ \to J/\psi D^0$	$B_c^+ \to J/\psi D_{s1}^{'+}$	0.0651	$D_{s1}^{'+} \to D^{*+}K^0, D^{*+} \to D^0\pi^+, D^0 \to \mu X$	0.6158*0.42227	0.01693
			$D_{s1}^{'+} \to D^{*0}K^+, D^{*0} \to D^0\pi^0, D^0 \to \mu X$	0.3842*0.619	0.01548
			$D_{s1}^{'+} \to D^{*0}K^+, D^{*0} \to D^0\gamma, D^0 \to \mu X$	0.3842*0.381	0.00953
	$B_c^+ \to J/\psi D_{s2}^{*+}$	0.0197	$D_{s2}^{*+} \to D^{*+}K^0, D^{*+} \to D^+\pi^+, D^0 \to \mu X$	0.0437*0.42227	0.00036
			$D_{s2}^{*+} \to D^{*0}K^+, D^{*0} \to D^0\pi^0, D^0 \to \mu X$	0.0273*0.619	0.00033
			$D_{s2}^{*+} \to D^{*0}K^+, D^{*0} \to D^0\gamma, D^0 \to \mu X$	0.0273*0.381	0.00020
			$D_{s2}^{*+} \to D^0K^+, D^0 \to \mu X$	0.6726	0.01325
$B_c^+ \to J/\psi D^+$	$B_c^+ \to J/\psi D_{s1}^{'+}$	0.0651	$D_{s1}^{'+} \to D^{*+}K^0, D^{*+} \to D^+\pi^0, D^+ \to \mu X$	0.6158*0.54911	0.02201
			$D_{s1}^{'+} \to D^{*+}K^+, D^{*+} \to D^+\gamma, D^+ \to \mu X$	0.6158*0.02862	0.00115
	$B_c^+ \to J/\psi D_{s2}^{*+}$	0.0197	$D_{s2}^{*+} \to D^+\pi^0, D^+ \to \mu X$	0.0437*0.54911	0.00047
			$D^{*+} \to D^+\gamma$, $D^+ \to \mu X$	0.0437*0.02862	0.00020
			$D_{s2}^{*+} \to D^+K^0, D^+ \to \mu X$	0.2564	0.00505
$B_c^+ \to J/\psi D_s^+$	$B_c^+ \to J/\psi D_{s1}^+$	0.8153	$D_{s1}^+ \to D_s^{*+}\pi^0$, $D_s^{*+} \to D_s^+X$, $D_s^+ \to \mu X$	0.80000*0.058	0.03783
			$D_{s1}^+ \to D_s^{*+}\pi^0$, $D_s^{*+} \to D_s^+\gamma$, $D_s^+ \to \mu X$	0.20000*0.942	0.61441
			$D_{s0}^{*+} \to D_s^+\gamma$, $D_s^+ \to \mu X$	0.2	0.16306
	$B_c^+ \to J/\psi D_{s0}^{*+}$	0.1000	$D_{s0}^{*+} \to D_s^+\pi^0$, $D_s^+ \to \mu X$	1.0	0.10000

The efficiency is obtained using MC listed in Table. 36. To obtain a more precise value of ε ($b \to J/\psi D$ ($\to\mu$ X)Y) in equation 82, the inclusive MC samples are categorized according to the D^0, D^+ and D_s^+ states. The relative ratio of each charm meson in the corresponding MC sample is first determined. As shown in Table 43, the normalized BF(B_c^+)is based on the values set in the decay file, while the normalized BF(charm) is obtained from the PDG [39]. Subsequently, normalization is performed using the yield($N(b \to J/\psi D(\to \mu X)Y)$) of inclusive MC(after cut$R(J/\psi)$). thereby providing a more accurate estimate of the efficiency. Each evevnt type for D^0, D^+, D_s^+ are shown in Table 43 42 41, respectively. Taking the D^0 sample as an example 84,

$$\varepsilon(b \to J/\psi D^0(\to \mu X)Y) = \frac{\sum_i^{i=3}(N_{bkk} \times \sum(D^0 \text{relativeratio}))}{\sum_i^{i=3}(N_{D^0} \times \sum(D^0 \text{relativeratio}))} \tag{84}$$

In this case, i represents each event type, N_{bkk} refers to the number of events for this sample in the bookkeeping, and $\sum(D^0$ relative ratio) denotes the sum of the relative ratios for the D^0 category in Table 43. N_{D^0} refers to the number of D^0 events remaining in the inclusive MC sample after classification. Similarly, the efficiency for the D^+ and D_s^+ samples can be obtained in the same manner.

The event selection for the B meson uses the cutb, as shown Table 40. The efficiency for the D hadron decay, as given in equation 82, can be obtained based on the calculations, which are summarized in table 44.

Table 44: Efficiency of muonic channel MC

Decay channel	total efficiency
ε ($b \to J/\psi D^0$ ($\to\mu$ X)Y)	$(0.035 \pm 0.003)\%$
ε ($b \to J/\psi D^+$ ($\to\mu$ X)Y)	$(0.036 \pm 0.003)\%$
ε ($b \to J/\psi D_s^+$ ($\to\mu$ X)Y)	$(0.209 \pm 0.004)\%$

Table 45: Efficiency of hadron channel MC

Decay channel	total efficiency
ε ($b \to J/\psi D^0$ ($\to$ h)Y)	$(5.55 \pm 0.02)\%$
ε ($b \to J/\psi D^+$ ($\to$ h)Y)	$(2.85 \pm 0.01)\%$
ε ($b \to J/\psi D_s^+$ ($\to$ h)Y)	$(1.38 \pm 0.01)\%$

After incorporating the obtained yield into Equation 82, we can derive the target yield for each type of background. Based on this strategy, we can calculate the target yield for each year. The results from Table 46 were used in the nominal fit to constrain the double charm yield.

Table 46: Yield of each year of double charm (fit range upper bound 10000 MeV/c^2)

Decay channel	2016	2017	2018
N($b \to J/\psi D^0$ ($\to\mu$ X)Y)	6 ± 1	6 ± 1	8 ± 2
N($b \to J/\psi D^+$ ($\to\mu$ X)Y)	1 ± 2	1 ± 2	1 ± 2
N($b \to J/\psi D_s^+$ ($\to\mu$ X)Y)	71 ± 11	64 ± 10	88 ± 13

29.2.6 Validation

At the beginning of this section, we assumed that $\varepsilon(b \to J/\psi D(\to h)Y) = \varepsilon(b \to J/\psi D(\to h))$. Now, we validate this assumption using $B_c^+ \to J/\psi K$ and $B_c^+ \to J/\psi K^*$ (892)($\to K\pi$) MC samples. In event selection, we applied identical selection criteria only to the J/ψ and K from both decays. From the results, it appears that our assumption holds true.

Table 47: The efficiency measure of K and K^*

All in %	$B_c^+ \to J/\psi K$	$B_c^+ \to J/\psi K^*$
Gen	25.59±0.02	22.63±0.02
Rec and pre-sel	80.39±0.04	80.60±0.04
L0	56.93±0.05	57.01±0.06
HLT1	88.34±0.05	88.29±0.05
HLT2	99.99±0.00	99.95±0.00
sel	98.74±0.02	98.27±0.02
Total	10.04 ±0.015	8.93±0.014

To validate our method for yield estimation, we utilized an exclusive channel for verification. Since we observed a signal for B_c^+ in the $B_c^+ \to J/\psi D_s^+$ decay process, using this process for validation carries significant persuasive weight.

$$\begin{aligned} N(B_c^+ \to J/\psi D_s^+(\to KK\pi)) = \sigma(b\bar{b}) \cdot f(b \to B_c^+) \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+) \cdot \mathcal{L} \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot \mathcal{B}(D_s^+ \to KK\pi) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi)) \quad \{cutB_c^+\} \end{aligned} \tag{85}$$

Firstly, we computed $\sigma(b\bar{b}) \cdot f(b \to B_c^+) \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+)$, where we normalized the branching ratios using known decays. The specific procedure is outlined in the following formula.

$$\begin{aligned} \frac{Br(B_c^+ \to J/\psi D_s^+)}{Br(B_c^+ \to J/\psi \pi^+)} = 2.8 \\ \frac{d\sigma(B_c^+)Br(B_c^+ \to J/\psi \pi^+)}{d\sigma(B^+)Br(B_c^+ \to J/\psi K^+)} = 0.698\% \end{aligned} \tag{86}$$

The above data are all taken from the PDG [33]. In conclusion, we can derive the following result.

$$\sigma(b\bar{b}) \cdot f(b \to B_c^+) \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+) = 0.86 \times 10^{-3}\,\mathrm{nb}^{-1} \tag{87}$$

Then, the $\mathcal{L}$ is $5.6\,\mathrm{fb}^{-1}$. $\mathcal{B}$ ($J/\psi \to \mu^+\mu^-$)and $\mathcal{B}$ ($D_s^+ \to KK\pi$) are from PDG. When we use event selection cutB_c^+, ε ($B_c^+ \to J/\psi D_s^+$ ($\to KK\pi$))= 1.27%. Finally, we get,

$$N(B_c^+ \to J/\psi D_s^+(\to KK\pi)) = 200 \tag{88}$$

The yield is calculated using the exclusive decay described in equation 83, and the result is consistent with the value of $N(B_c^+ \to J/\psi D_s^+(\to KK\pi))$ obtained from the data-196, demonstrating the validity of the method.

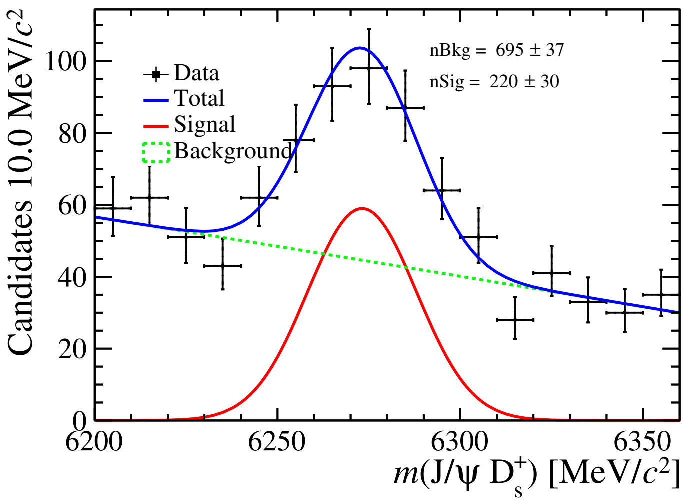


Figure 196: The mass distribution of the $J/\psi D_s^+$.

29.2.7 Yield of $B_c^+ J/\psi D_s^+$ Y(=nothing OR soft gamma from D_s^{*+} plus nothing) for nominal fit

(1) The yield of target channel can be calculated by,

$$N(B_c^+ \to J/\psi D_s^+(\to \mu X)Y) = \sigma(B_c^+) \cdot \mathcal{L} \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+ Y) \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot \mathcal{B}(D_s^+ \to \mu) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to \mu X)Y) \quad \{\text{cut}R(J/\psi)\}, \tag{89}$$

where $\sigma(B_c^+)$ means the production cross section of B_c^+ pairs, and $\mathcal{L}$ represents the integrated luminosity of data samples. $\mathcal{B}$ and ε are the branching fraction and efficiency of corresponding processes in the brackets. D refers to all D mesons, including D^0, D^+ and D_s^+ and their excited states.

(2) Inclusive control samples, which contain the same B decays but hadronic decays of charm mesons, can be used as normalization, and their yield can be calculated by similar equation,

$$N(B_c^+ \to J/\psi D_s^+(\to KK\pi)Y) = \sigma(B_c^+) \cdot \mathcal{L} \cdot \mathcal{B}(B_c^+ \to J/\psi D_s^+ Y) \cdot \mathcal{B}(J/\psi \to \mu^+\mu^-) \\ \cdot \mathcal{B}(D_s^+ \to KK\pi) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi))Y) \quad \{\text{cut}B_c^+\} \tag{90}$$

where Y=nothing OR soft gamma from D^* plus nothing.

(3) Dividing the above two equations, we can get,

$$\frac{N(B_c^+ \to J/\psi D_s^+(\to \mu X)Y)}{N(B_c^+ \to J/\psi D_s^+(\to KK\pi)Y)} = \frac{\mathcal{B}(D_s^+ \to \mu X) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to \mu X)Y)}{\mathcal{B}(D_s^+ \to KK\pi) \cdot \varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi)Y)} \quad \{\frac{\text{cut}R(J/\psi)}{\text{cut}B_c^+}\} \tag{91}$$

where Y=nothing OR soft gamma from D^* plus nothing.

Since we have tagged J/ψ (Tag-and-probe method), we assume that the fitted yield of selected hadron D_s^+ decay events is equal to the yield of $N(B_c^+ \to J/\psi D_s^+(\to \mu X)Y)$ multiplied by efficiency ratio. We also assume that the efficiencies of processed with two-body and three-body final states are equal ($\varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi)Y) = \varepsilon(B_c^+ \to J/\psi D_s^+(\to h))$), which will be validated subsequently. $\mathcal{B}(D_s^+ \to KK\pi)$ and $\mathcal{B}(D_s^+ \to \mu X)$ are cite by PDG [39].There is no $D_s^+ \to \mu X$ in PDG, but $D_s^+ \to eX$ is $(6.33 \pm 0.15)\%$ and $D_s^+ \to \mu\nu$ is $(5.35 \pm 0.12) \times 10^{-3}$.

Meanwhile, based on the calculated efficiency $\varepsilon(B_c^+ \to J/\psi D_s^+(\to KK\pi)) = 1.27\%$ and the fitted yield from Fig. 197, we obtain the yield of $N(B_c^+ \to J/\psi D_s^+(\to KK\pi))$. Then, we can estimate the yield of $N(B_c^+ \to J/\psi D_s^+(\to \mu X))$. Using the ratio $\mathcal{B}(B_c^+ \to J/\psi(1S)D_s^{*+})/\mathcal{B}(B_c^+ \to J/\psi(1S)D_s^+)$ from the PDG, we further calculate the yield of $N(B_c^+ \to J/\psi D^*(\to \mu X))$. The total yields are summarized in Table 48.

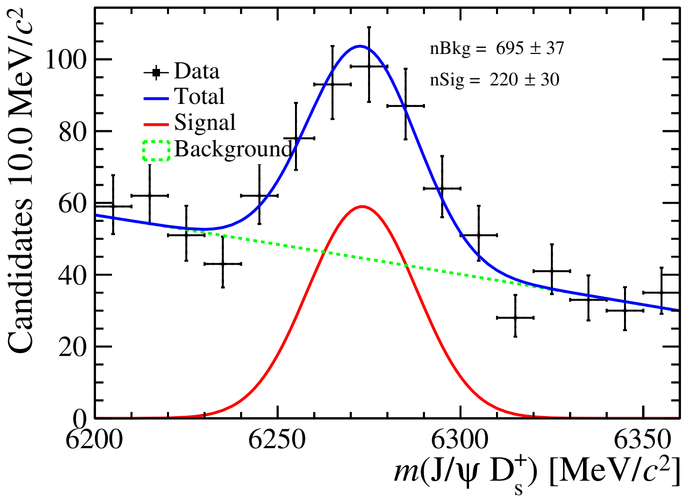


Figure 197: The mass distribution of the $J/\psi D_s^+$.

Table 48: Yield of each year of $B_c^+ J/\psi D_s^+$ Y

Decay channel	2016	2017	2018
N($B_c^+ \to J/\psi D_s^+$ ($\to\mu$ X)Y)	44 ± 5	40 ± 5	54 ± 7

29.2.8 Summary of the double charm background cross-check

This section presents a data-driven construction of the double charm background. The results, summarized in Table 49, demonstrate consistency with the total yield of the double charm background obtained in the nominal fit for each data-taking year. In addition, the $B_c \to J/\psi D_s$ decay channel, with $D_s \to KK\pi$, is observed in data and used as a normalization mode. Based on this channel, the yield of $B_c \to J/\psi D_s$, $D_s \to \mu X$ is computed. The yields provided in Table 48 serve as constraints in the nominal fit. Accordingly, the yield obtained from this data-driven method is then incorporated into the nominal fit as a constrained component (see Section 12.1), thereby providing a robust cross-check of the background modeling strategy.

Table 49: Yields of the double charm background obtained from the data-driven construction method. The results are consistent with those extracted from the nominal fit, thereby validating the background modeling strategy.

Year	Data-driven yield	Nominal fit yield
2016	78 ± 11	61 ± 8
2017	71 ± 10	56 ± 7
2018	97 ± 13	77 ± 10

29.3 Feed-down background

The feed-down background arises from semileptonic decays of the B_c^+ meson into excited charmonium states, which subsequently decay into a J/ψ and additional particles that are not reconstructed. The dominant contributions considered in this analysis originate from the channels $B_c^+ \to \psi(2S)\ell^+\nu_\ell$ and $B_c^+ \to \chi_{cJ}(1P)\ell^+\nu_\ell$ ($J = 0, 1, 2$).

The expected size of the feed-down contribution is estimated using theoretical predictions for the ratios of branching fractions with respect to the control mode $B_c^+ \to J/\psi\mu^+\nu_\mu$. Several theoretical approaches are available in the literature, including relativistic and non-relativistic quark models, QCD sum rules, and covariant light-front calculations. The corresponding predictions span a broad range, reflecting the sizable theoretical uncertainties associated with these decays. In the nominal fit, the covariant light-front model is adopted as a reference input.

The expected feed-down yields are obtained by scaling the observed yield of the control mode according to

$$\frac{N(B_c^+ \to H_c\,\ell^+\nu_\ell)}{N(B_c^+ \to J/\psi\,\mu^+\nu_\mu)} = \frac{\mathcal{B}(B_c^+ \to H_c\,\ell^+\nu_\ell) \times \varepsilon(B_c^+ \to H_c\,\ell^+\nu_\ell)}{\mathcal{B}(B_c^+ \to J/\psi\,\mu^+\nu_\mu) \times \varepsilon(B_c^+ \to J/\psi\,\mu^+\nu_\mu)}, \tag{92}$$

where H_c denotes the relevant excited charmonium state and ε represents the total selection efficiency.

For the $\psi(2S)\mu^+\nu_\mu$ channel, the relative efficiency with respect to the control mode is found to be approximately 60%, mainly driven by the isolation BDT requirement. Using representative theoretical inputs, the expected yield of this contribution is estimated to be at the percent level relative to the $B_c^+ \to J/\psi\mu^+\nu_\mu$ yield. For the corresponding $\psi(2S)\tau^+\nu_\tau$ channel, the expected contribution is strongly suppressed and is found to be negligible within the current statistical precision.

For the $B_c^+ \to \chi_{cJ}(1P)\mu^+\nu_\mu$ channels, the selection efficiencies are found to be close to unity relative to the control mode. Using the available theoretical predictions, the expected feed-down yields from these channels are at the level of a few percent of the $B_c^+ \to J/\psi\mu^+\nu_\mu$ yield. Due to the lack of generator-level efficiency information for some theoretical models, these estimates are considered illustrative.

The impact of the feed-down background is incorporated into the nominal fit using the chosen reference model. The spread of predictions among different theoretical approaches is used to assess the associated systematic uncertainty.

Table 50: Estimated feed-down yields from excited charmonium states for each data-taking year. The yields are obtained by scaling the nominal $B_c^+ \to J/\psi\mu^+\nu_\mu$ yield with theoretical branching fraction ratios and selection efficiencies. Numbers in square brackets indicate the branching fraction ratios relative to $B_c^+ \to J/\psi\mu^+\nu_\mu$ used in the calculation.

Decay channel	Estimated yield		
	2016	2017	2018
$B_c^+ \to J/\psi\mu^+\nu_\mu$ [nominal fit]	6425 ± 118	9240 ± 171	8520 ± 163
$B_c^+ \to \psi(2S)\mu^+\nu_\mu$ [0.9%]	58 ± 1	83 ± 2	77 ± 1
$B_c^+ \to \psi(2S)\tau^+\nu_\tau$ [1.2×10^{-4}%]	1 ± 0	1 ± 0	1 ± 1
$B_c^+ \to \chi_{c1}(1P)\mu^+\nu_\mu$ [3.0%]	193 ± 4	277 ± 5	256 ± 5
$B_c^+ \to \chi_{c2}(1P)\mu^+\nu_\mu$ [1.9%]	122 ± 2	176 ± 3	162 ± 3

29.4 Cross-check of the J/ψ–μ_b combinatorial background

During the application of the *sPlot* technique to subtract the fake J/ψ contribution, a correlation is observed between the shape of the J/ψ background and the reconstructed mass $m(J/\psi\mu_B)$, as illustrated in Fig. 198. To account for this effect, the $m(J/\psi\mu_B)$ spectrum is divided into three regions, with boundaries at 4400 MeV/c^2 and 5200 MeV/c^2, and an unbinned fit of the J/ψ mass distribution is performed independently in each region. The resulting sWeights are then combined and used as input for subsequent template-based studies.

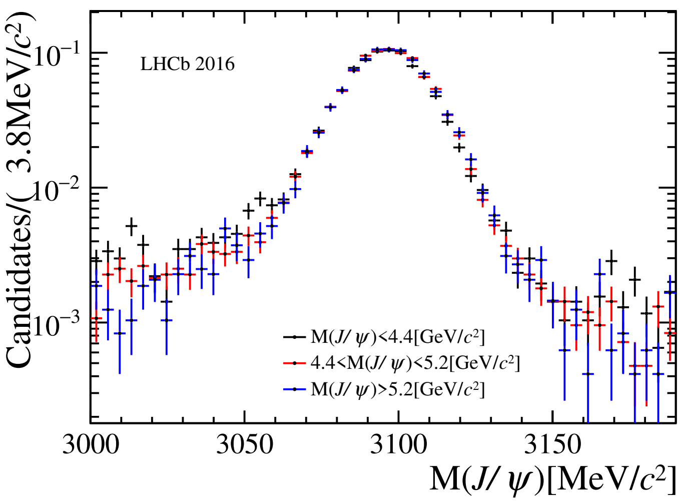


Figure 198: In Figure 198, the distribution of m(J/ψ) within three bins of $m(J/\psi\mu_B)$ is presented. The data points are color-coded as follows: black points correspond to $m(J/\psi\mu_B) < 4400$ MeV/c^2, red points correspond to $m(J/\psi\mu_B) \in [4400, 5200]$ MeV/c^2, and blue points correspond to $m(J/\psi\mu_B) > 5200$ MeV/c^2. Each of the distributions has been normalized for better signal-to-noise ratio comparison.

An event-mixing technique is investigated as an alternative, data-driven approach to model the $J/\psi + \mu_b$ combinatorial background. In this method, a J/ψ candidate from one event is combined with a bachelor track from a different collision event that satisfies the same selection requirements. Additional selections are applied to suppress overlap with mis-identified muon backgrounds and mis-reconstructed J/ψ contributions.

The validity of the mixed-event samples is assessed by comparing the distributions of the fit observables with data in the high-mass region $m_{B_c} > 6400\ \mathrm{MeV}/c^2$, which is dominated by combinatorial background. While several strategies are explored to improve the agreement, including kinematic reweighting and geometric transformations of the bachelor track, a significant discrepancy is consistently observed in the B_c^+ decay-time distribution. This discrepancy indicates that the event-mixing samples do not provide a reliable description of the combinatorial background once variables depending on the reconstructed B_c vertex are involved.

Figure 199 explicitly demonstrates this limitation by comparing the B_c proper decay time (B_τ) distribution between data and the event-mixed background. The event-mixed sample fails to reproduce the shape of the B_τ distribution, particularly at large decay times. This failure occurs because the calculation of B_τ requires knowledge of the B_c production and decay vertices, which cannot be consistently defined when the J/ψ and bachelor muon originate from separate events.

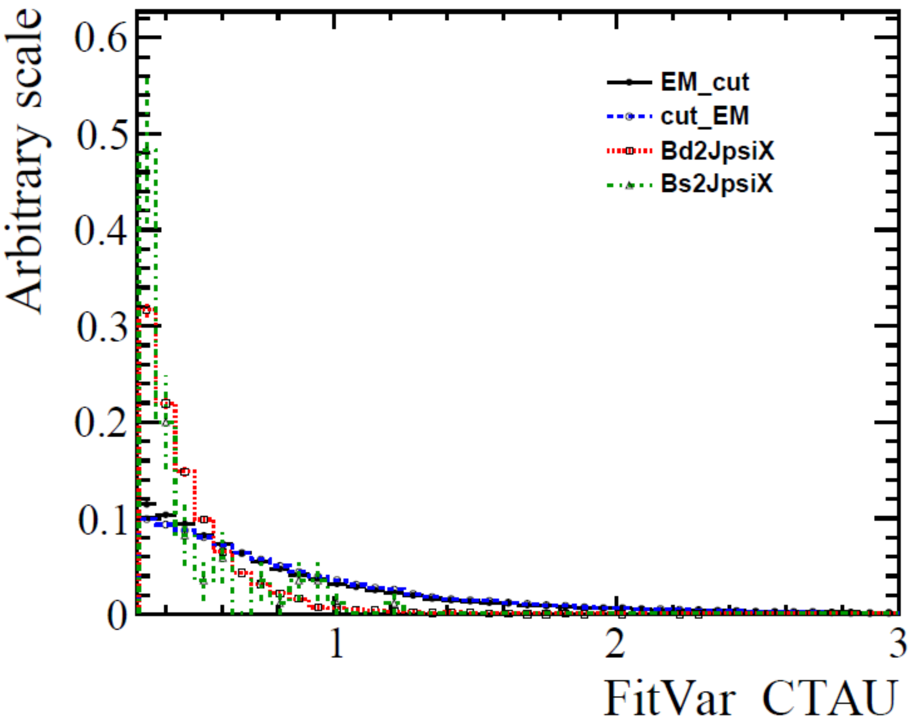


Figure 199: Comparison of the B_c proper decay time (B_τ) distributions. Two implementations of the event-mixing background are shown: the black markers corresponds to applying all selection cuts **after** the mixing procedure; the blue histogram (if present) corresponds to applying cuts **before** mixing. For reference, two inclusive Monte Carlo samples are also shown: the red line represents the nominal inclusive MC, and the green line represents an alternative inclusive MC sample. The event-mixed backgrounds fail to reproduce the B_τ distribution observed in data, particularly at large decay times, because the B_τ calculation requires well-defined B_c vertices, which are absent in the mixing approach. All backgrounds are normalized to the data in the displayed range.

29.4.1 Summary of the J/ψ–μ_b combinatorial background cross-check

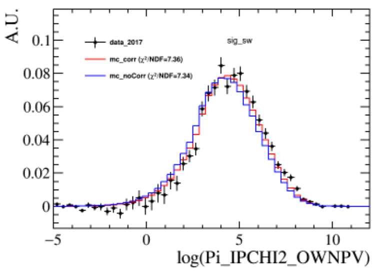
The event-mixing technique has been investigated as a possible data-driven method to model the $J/\psi + \mu_b$ combinatorial background. However, since the fit observables depend explicitly on the reconstructed B_c decay vertex, a consistent re-evaluation of these quantities is not feasible for mixed events. As a result, the event-mixing approach is not adopted in the analysis. Instead, the inclusive simulated samples used in the nominal fit are retained to describe the $J/\psi + \mu_b$ combinatorial background component.

29.5 VELO error parametrization cross-check

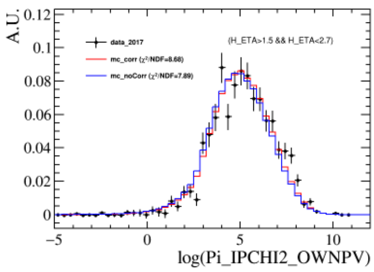
A potential mismodeling of the VELO track-parameter uncertainties has been reported in previous studies, particularly for the 2017 and 2018 data-taking periods. An updated parametrization of the VELO errors is available and has been investigated as part of this cross-check.

Following the prescription in Ref. [ANA-2024-078], we evaluated the impact of the updated VELO error correction by applying the corresponding scale factors to the relevant track parameters and comparing the resulting distributions between data and simulation. Specifically, we tested the correction using parameters derived from B-meson decays for both pions and protons.

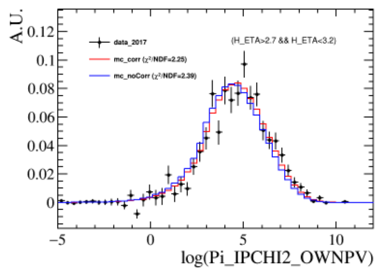
The comparison is performed using the $B_c^+ \to J/\psi \pi^+$ control channel. Figures 200 and 201 show the Data/MC agreement for the bachelor pion and proton impact parameter distributions in 2017, with similar observations for 2016 and 2018 (not shown). Only a limited improvement in the Data/MC agreement is observed after applying the correction.



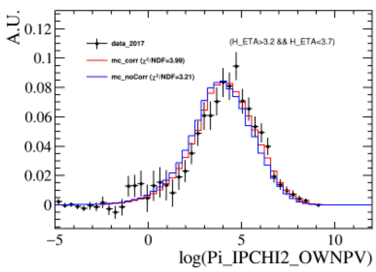
(a) Impact parameter distribution



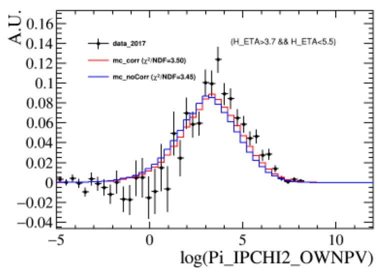
(b) η bin 0



(c) η bin 1

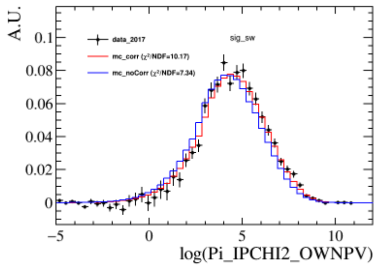


(d) η bin 2

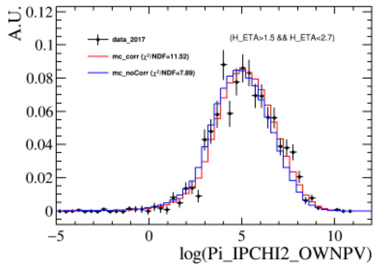


(e) η bin 3

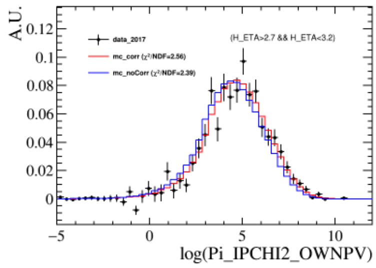
Figure 200: Data/MC comparison for pion distributions in 2017 with (red) and without (blue) the updated VELO error correction. The top row shows the full impact parameter distribution and the first two η bins; the bottom row shows the remaining η bins. Only minimal improvement is observed after applying the correction.



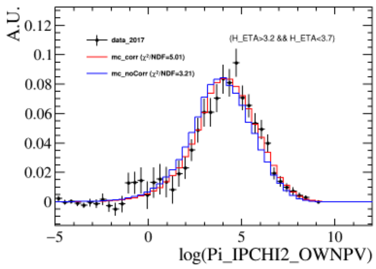
(a) Impact parameter distribution



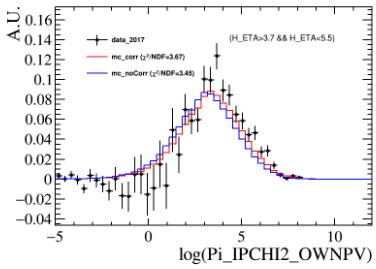
(b) η bin 0



(c) η bin 1



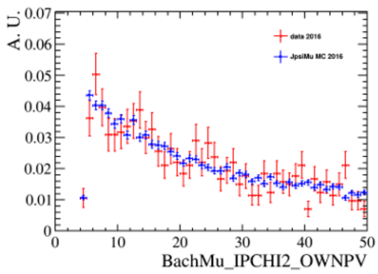
(d) η bin 2



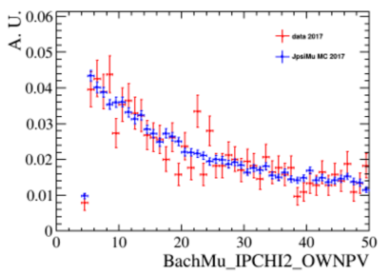
(e) η bin 3

Figure 201: Data/MC comparison for proton distributions in 2017 with (red) and without (blue) the updated VELO error correction. Similar to the pion case, the correction provides only limited improvement in Data/MC agreement across all η bins.

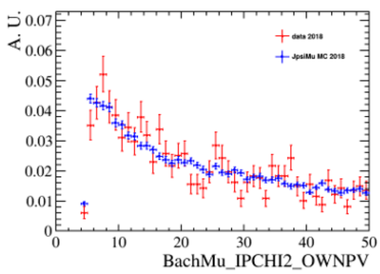
Data/MC comparison for the bachelor muon



(a) 2016 data



(b) 2017 data



(c) 2018 data

Figure 202: Data/MC comparison for the bachelor muon impact parameter distribution in 2016 (left), 2017 (center), and 2018 (right) data. The blue histogram shows the nominal simulation, while the red histogram shows the simulation with the updated VELO error correction applied. The bachelor muon kinematics show minimal sensitivity to the VELO error parametrization across all three years.

The effect of the VELO error correction is found to be small on average, with no significant difference observed between the 2016, 2017, and 2018 data-taking periods. This is further confirmed for the bachelor muon, a key component in the signal channel. As illustrated in Fig. 202, the comparison between data and simulation shows agreement for the muon impact parameter distribution within the current statistical precision. This minimal sensitivity is consistent with expectations, as high-momentum muon tracks are less affected by VELO alignment effects compared to hadronic tracks.

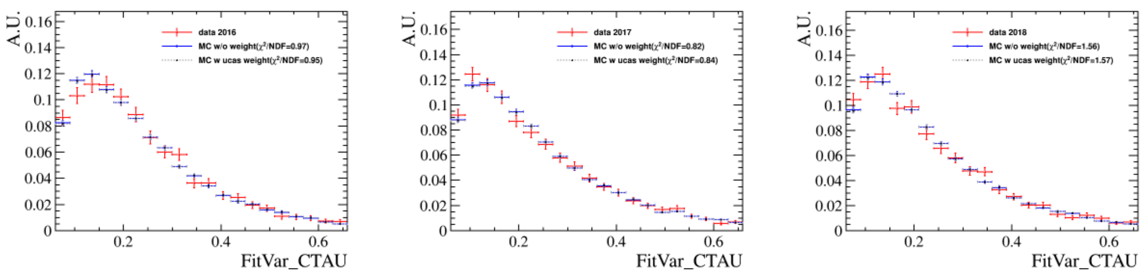
Since this dedicated study is based on hadronic final states (using $B_c^+ \to J/\psi\pi^+$ as a control channel), and the observed effect on the bachelor muon is negligible, we adopt a conservative approach and **do not apply** the VELO error correction to the semi-leptonic $B_c^+ \to J/\psi\mu^+\nu_\mu$ analysis. This avoids introducing additional, uncontrolled systematic effects. Any residual differences are covered by the assigned systematic uncertainties, and no significant impact on the nominal result is expected.

29.6 B_c^+ lifetime correction cross-check

To align the simulation with the world-average B_c^+ lifetime, each event in the nominal MC sample is assigned a weight defined as the ratio of the target distribution (using the PDG lifetime) to the native MC distribution (using the generator lifetime):

$$w_{\text{lifetime}} = \frac{\dfrac{1}{\tau_{\text{PDG}}}\, e^{-\tau_{\text{true}}/\tau_{\text{PDG}}}}{\dfrac{1}{\tau_{\text{MC}}}\, e^{-\tau_{\text{true}}/\tau_{\text{MC}}}} = \frac{\tau_{\text{MC}}}{\tau_{\text{PDG}}}\, e^{-\tau_{\text{true}}(1/\tau_{\text{PDG}}-1/\tau_{\text{MC}})}, \tag{93}$$

where τ_{true} is the true decay time of the MC particle, $\tau_{\text{PDG}} = 0.510$ ps is the B_c^+ lifetime from the Particle Data Group [39], and $\tau_{\text{MC}} = 0.507$ ps is the value used in the event generation of the nominal simulation. This weight adjusts the simulated decay-time spectrum to match the PDG lifetime without altering other kinematic correlations.



(a) 2016 (b) 2017 (c) 2018

Figure 203: Data/MC comparison of the decay time distribution for the $B_c^+ \to J/\psi\mu^+\nu_\mu$ candidates in 2016 (left), 2017 (center), and 2018 (right). The black points represent data, the blue histogram shows the nominal simulation, and the red histogram shows the simulation after applying the lifetime reweighting according to Eq. 93. No significant improvement is observed after reweighting.

The Data/MC agreement for decay-time–dependent observables is compared before and after applying the correction in Eq. 93. No significant improvement is observed; therefore, this parameterized lifetime reweighting is **not** introduced as an additional correction to the nominal analysis.

In addition, a lifetime-acceptance correction derived from the control channel $B^0 \to J/\psi K^{*0}$ has been introduced in the nominal fit. Details are described in Section 28.

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